STUDY GUIDE

Equity & Capital Markets

A complete, concept-first reference — prepared for Saikumar Kaleru.

certifications / study / Finance • 2026

Overview of Equity & Capital Markets

The Problem / Why this matters

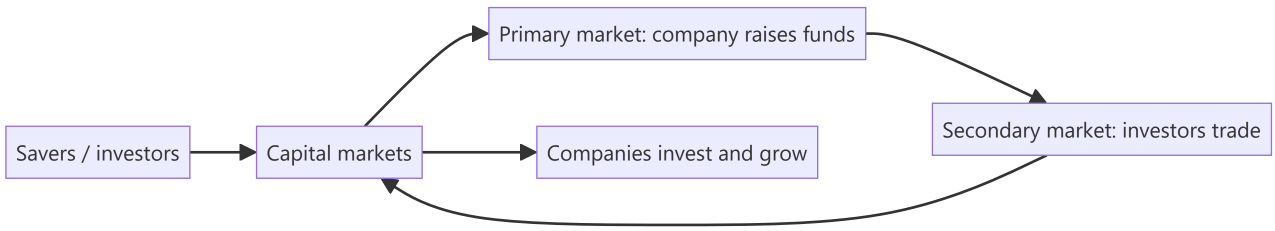
Companies need long-term capital to grow; savers need somewhere to put their money to earn a return. **Capital markets are the machinery that connects the two** — channelling household and institutional savings into productive businesses, and giving investors tradable claims on those businesses. Understanding how this system is structured — primary vs secondary, equity vs debt, the players and the plumbing — is the foundation for every equity, research, and markets role.

Core Idea

Capital markets are where **long-term funds** are raised and traded. The **primary market** is where new securities are issued and the company actually raises money; the **secondary market** is where those securities then trade among investors, providing the liquidity that makes the primary market work. Equity gives ownership; debt gives a creditor claim.

Why it works this way

Investors will only fund companies in the primary market if they believe they can **exit** later by selling to someone else — that exit is the secondary market. Liquidity in the secondary market lowers the return investors demand, which lowers the company's cost of capital. So the two markets are symbiotic: primary raises capital, secondary provides the liquidity that makes raising capital cheap.



Full technical content

Money markets vs capital markets:

	MONEY MARKET	CAPITAL MARKET
Maturity	Short-term (< 1 yr)	Long-term (> 1 yr)
Instruments	T-bills, CP, CDs, repo	Equities, bonds
Purpose	Liquidity management	Long-term funding/investment

Equity vs debt:

	EQUITY	DEBT
Claim	Ownership, residual	Creditor, fixed
Return	Dividends + capital gains, uncapped	Interest, capped
Risk	Last in line, higher risk	Senior, lower risk
Control	Voting rights	Usually none

Primary vs secondary market: - **Primary** — first issuance: IPO, follow-on (FPO), rights issue, QIP, private placement, bond issuance. *The company receives the cash.* - **Secondary** — subsequent trading on an exchange between investors. *The company gets nothing; it provides price discovery and liquidity.*

Functions of capital markets: 1. **Capital formation** — channelling savings into investment. 2. **Price discovery** — continuous pricing of companies via trading. 3. **Liquidity** — the ability to convert holdings to cash. 4. **Risk transfer & sharing** — spreading risk across many investors. 5. **Corporate governance** — market discipline and monitoring by investors.

The ecosystem (previewing later chapters): issuers (companies), investors (retail, mutual funds, insurers, FPIs, pension funds), intermediaries (investment banks, brokers, exchanges, depositories, clearing houses), and the regulator (SEBI in India).

Worked examples

Example 1 — primary vs secondary cash flow. Company Z raises ₹1,000 cr in an IPO (primary) — that ₹1,000 cr goes to the company (or selling shareholders in an OFS). The next day, an investor buys ₹5 cr of Z shares on the NSE (secondary) — that ₹5 cr goes to the *seller*, not to Z. *Z only receives money at issuance.*

Example 2 — why liquidity lowers cost of capital. Two identical companies raise equity. Company A's shares will list on a liquid exchange (easy exit); Company B's won't trade at all. Investors demand a higher return from B for locking up their money, so B's cost of equity is higher and its shares are worth less. Liquidity is valuable.

Example 3 — equity vs debt in a good year and a bad year. A firm earns ₹100 cr. Bondholders get their fixed ₹20 cr interest either way. In a great year (₹200 cr), equity keeps the extra upside; in a terrible year (₹10 cr), equity is wiped out first while bondholders still rank ahead. Equity = uncapped upside, first-loss downside.

Example 4 — a worked cost-of-capital comparison showing why capital-formation efficiency matters at the economy level. Country A has deep, liquid capital markets; a mid-size company there raises equity at a 10% cost of capital. Country B has shallow, illiquid markets with few institutional investors; an identical company there faces a 16% cost of capital purely from the illiquidity/risk premium investors demand. Over a 10-year project requiring steady reinvestment, Country A's company can fund materially more positive-NPV projects (anything returning above 10%) than Country B's company (which can only justify projects returning above 16%) — the same underlying business opportunity set, but a shallower capital market structurally funds less real economic investment. This is the macro-level version of Part 1's "capital formation" function, and a useful answer to "why do governments care about capital-market development, not just individual companies."

Example 5 — tracing money markets and capital markets working together in one transaction. A company needs ₹50 cr for two purposes: ₹10 cr to cover a short-term working-capital gap for the next 60 days, and ₹40 cr to fund a new manufacturing line with a 10-year payback. For the ₹10 cr, it issues commercial paper (a money-market instrument, sub-1-year, matching the short-term need). For the ₹40 cr, it issues long-term bonds or raises equity via a QIP (capital-market instruments, matching the long-term asset being funded). Using a short-term money-market instrument to fund a 10-year asset (or vice versa) creates a maturity mismatch that's a classic, well-documented source of financial fragility (a firm forced to repeatedly refinance short-term debt to fund a long-term asset is exposed to refinancing risk if credit markets tighten at the wrong moment) — matching instrument maturity to the underlying need's time horizon is a first-principles capital-structure discipline, not just a formality.

How it is tested in interviews

- **"Difference between primary and secondary markets?"** — "Primary is where the company issues new securities and actually raises money; secondary is where investors trade those securities among themselves, giving liquidity and price discovery. The company only receives cash in the primary market."
- **"When Infosys shares trade on the NSE, does Infosys get the money?"** — "No — that's the secondary market. Infosys only raised money at its IPO/issuances."
- **"Equity vs debt?"** — "Equity is ownership with uncapped, residual, higher-risk returns; debt is a senior, fixed, capped, lower-risk claim."
- **"What are the functions of capital markets?"** — Capital formation, price discovery, liquidity, risk sharing, governance.

Traps & common mistakes

- Thinking the company receives money from **secondary** trading.
- Confusing **money markets** (short-term) with **capital markets** (long-term).
- Treating equity and debt returns as symmetric — equity's downside is first-loss.
- Underrating the role of **liquidity** in lowering cost of capital.

First-principles recap

- Capital markets connect savers to companies needing long-term funds.
- **Primary** raises capital (company gets cash); **secondary** provides liquidity (investors trade).
- Equity = ownership/residual/uncapped; debt = senior/fixed/capped.
- Functions: capital formation, price discovery, liquidity, risk sharing, governance.
- Secondary-market liquidity lowers the primary-market cost of capital.

Quick-reference

CONCEPT	ONE-LINER
Primary market	New issue; company raises cash
Secondary market	Investors trade; liquidity & price discovery
Money vs capital market	Short-term vs long-term
Equity	Ownership, residual, uncapped
Debt	Creditor, fixed, senior

Q&A — Overview of Equity & Capital Markets

Foundational theory questions and worked scenarios on the primary/secondary distinction and capital-market functions.

Q1. When Infosys shares trade on the NSE today, does Infosys receive any of that money? Explain precisely.

Model answer. No. That is secondary-market trading — an exchange of shares between two investors, with the proceeds going to the selling investor, not the company. Infosys only received cash from the market at the point of an actual issuance (its original IPO, or any subsequent follow-on/QIP/rights issue). This is one of the most common conceptual gaps to close early: the vast majority of daily trading volume on any exchange involves no cash flow to the underlying company at all.

Q2. Why does secondary-market liquidity lower a company's cost of capital, even though the company never touches secondary-market trading proceeds?

Model answer. Investors price in the ability to exit an investment when they need to. If a security is illiquid (hard to sell without moving the price significantly), investors demand a higher expected return to compensate for that liquidity risk — raising the company's cost of equity. A liquid secondary market lowers that required return, which directly lowers the price the company pays (in the form of a higher achievable valuation) to raise capital in the primary market. The two markets are economically linked even though only the primary market involves direct cash flow to the issuer.

Q3. Distinguish money markets from capital markets along two dimensions.

Model answer. Maturity: money-market instruments (T-bills, commercial paper, certificates of deposit, repo) are short-term, generally under one year; capital-market instruments (equities, bonds) are long-term. Purpose: money markets exist primarily for liquidity management (parking or raising cash briefly), while capital markets exist for long-term funding and investment (financing growth, capital projects). A company issuing commercial paper to cover working-capital timing gaps is a money-market transaction; the same company issuing a 10-year bond or new equity is a capital-market transaction.

Q4. Worked — equity vs debt payoff in three scenarios.

A firm has ₹500 cr of debt requiring ₹40 cr annual interest. Compare what debt and equity holders receive in a bad year (₹30 cr operating profit), an average year (₹80 cr), and a great year (₹250 cr).

Model answer. Bad year (₹30 cr): debtholders are owed ₹40 cr but only ₹30 cr is available — this is a default/distress scenario; debtholders receive what's available (₹30 cr, likely triggering restructuring), and equity holders receive **nothing**. Average year (₹80 cr): debtholders receive their fixed ₹40 cr; equity holders receive the residual ₹40 cr. Great year (₹250 cr): debtholders still receive only their fixed ₹40 cr (no upside participation); equity holders capture the full residual ₹210 cr. This illustrates the core asymmetry: debt has a capped, senior claim; equity has an uncapped, residual, first-loss claim — debt's payoff resembles a short put option, equity's resembles a long call.

Q5. List and briefly explain the five core functions of capital markets.

Model answer. (1) Capital formation — channelling household/institutional savings into productive investment. (2) Price discovery — continuous trading aggregates information into a market-clearing price. (3) Liquidity — letting investors convert holdings to cash without large value loss. (4) Risk transfer and sharing — spreading a company's risk across many investors rather than concentrating it with a few. (5) Corporate governance — market discipline (share-price reaction, activist investors, the threat of takeover) monitors and pressures management to act in shareholders' interest.

Q6. A private (unlisted) company and a listed company have identical financials and growth prospects. Would you expect them to trade/be valued at the same multiple? Why or why not?

Model answer. No — the private company would typically warrant a lower valuation, reflecting an illiquidity discount: its equity cannot be readily converted to cash the way a listed company's shares can (no continuous secondary market, no daily price discovery, exit typically requires a full sale process or IPO). Investors demand additional compensation for bearing this illiquidity risk, all else being equal — a direct application of the liquidity-lowers-cost-of-capital logic from Q2, run in reverse.

Q7. What's the difference between "the company raises capital" and "the market provides liquidity," and why do interviewers test this distinction so directly?

Model answer. Raising capital is specifically a primary-market function (new securities issued, cash flows to the issuer). Providing liquidity is specifically a secondary-market function (existing securities change hands between investors, cash flows between investors, not to the issuer). Interviewers test this because it's foundational — a candidate who conflates the two will make downstream errors (e.g. assuming a company benefits financially every time its stock trades, or misunderstanding why a company doesn't "lose money" when its stock price falls on the secondary market post-issuance).

Q8. Why is "risk transfer and sharing" listed as a distinct function from liquidity, when they sound related?

Model answer. Liquidity is about the *ease* of converting a holding to cash. Risk transfer/sharing is about *distributing* a company's business risk across many investors with different risk appetites and diversification needs, rather than concentrating it entirely with a single owner or a small group of founders — a founder who sells 70% of their company via an IPO isn't just gaining liquidity for their own stake, they're also transferring the associated risk of the business's fortunes to a broad, diversified investor base, each of whom bears only a small fraction of the total risk. The two functions are related but conceptually distinct: one is about convertibility, the other about risk distribution across the market.

Primary Markets & the IPO Process

The Problem / Why this matters

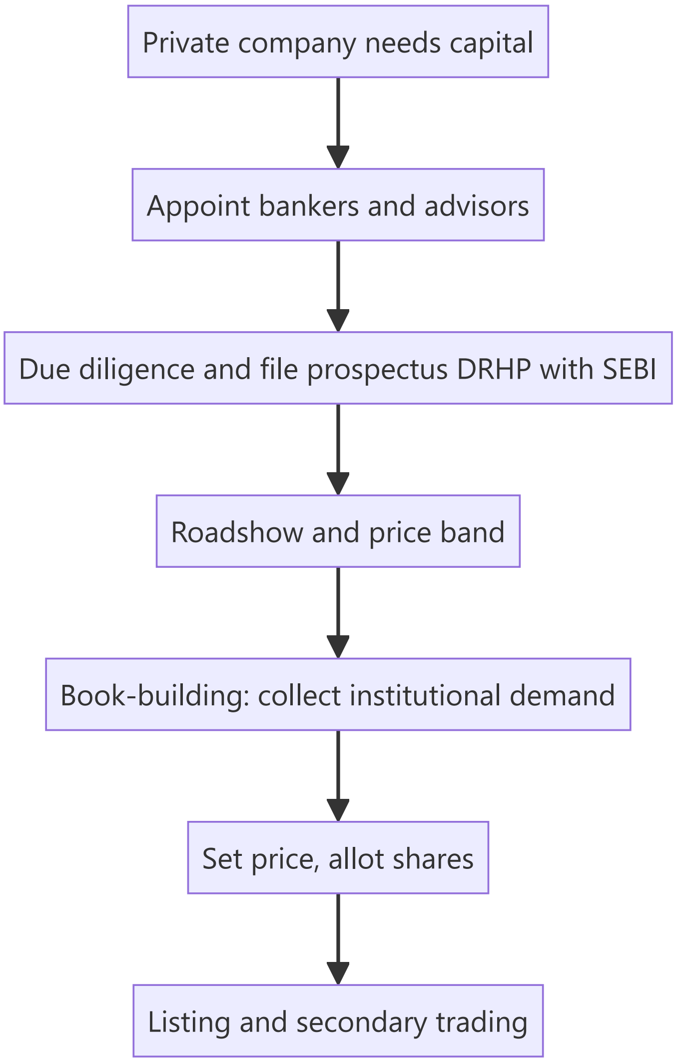
The most visible event in capital markets is a company **going public** — the IPO. It's how a private company raises large-scale equity, gives early investors an exit, and joins the public market. Understanding the IPO process — why companies list, how the price is set (book-building), who does what (the bankers, the regulator), and the risks (underpricing, lock-ups) — is essential for equity capital markets, investment banking, and research roles, and it comes up constantly in interviews.

Core Idea

The **primary market** is where companies issue securities to raise capital for the first time or afresh. An **IPO (Initial Public Offering)** is a private company's first sale of shares to the public, priced via **book-building**, intermediated by investment banks, and regulated (by SEBI in India). Beyond IPOs, primary issuance includes follow-ons, rights issues, QIPs and private placements.

Why it works this way

A company needs a mechanism to (i) discover a fair price when there's no market price yet, and (ii) distribute shares to many investors while managing risk. Book-building solves price discovery by collecting demand from institutions across a price band; underwriters manage distribution and (sometimes) guarantee the raise; the regulator protects investors through disclosure (the prospectus).



Full technical content

Why companies go public: raise growth capital, give founders/PE/VC an exit (offer for sale), acquire currency (listed shares) for M&A, raise profile/credibility, and enable employee stock liquidity. Costs: disclosure, scrutiny, short-term market pressure, listing expense, dilution.

Types of primary issuance:

TYPE	WHAT IT IS
IPO	First public sale of shares
FPO / follow-on	Further public issue by an already-listed firm
Rights issue	Offer of new shares to existing holders, usually at a discount
QIP	Qualified Institutional Placement — quick raise from institutions (India)
Private placement / preferential	Shares sold to select investors
OFS	Offer for Sale — existing holders sell (company gets no cash)

Fresh issue vs offer for sale. A **fresh issue** creates new shares — the *company* receives the money (dilutes existing holders). An **OFS** sells *existing* shares — the selling shareholders receive the money (no dilution, no new company capital). Most IPOs are a mix.

The IPO process (India): 1. Appoint **merchant bankers (BRLMs)**, legal counsel, auditors, registrar. 2. Due diligence; draft the **DRHP** (Draft Red Herring Prospectus) and file with **SEBI**. 3. SEBI review and observations; finalize the RHP. 4. **Roadshow** — market to institutions; set a **price band**. 5. **Book-building** — collect bids across the band from QIBs, non-institutional, and retail categories (India reserves quotas for each). 6. Determine the **cut-off price** from the demand book; allot shares. 7. **Listing** on NSE/BSE; secondary trading begins.

Pricing methods: book-building (price discovered from demand within a band — the norm) vs **fixed price** (set upfront). **Anchor investors** (large institutions) may be allotted a day before to signal quality.

Key phenomena: - **Underpricing** — IPOs often "pop" on day one (priced below where they trade), leaving money on the table but rewarding investors and ensuring a successful listing. - **Lock-up period** — insiders/pre-IPO holders can't sell for a set period after listing, preventing an immediate flood of stock. - **Greenshoe (over-allotment) option** — lets underwriters stabilize the price post-listing by selling up to ~15% extra and buying back if it falls. - **Winner's curse / oversubscription** — hot IPOs are heavily oversubscribed; retail allotment is scaled/lottery-based.

Worked examples

Example 1 — fresh issue vs OFS. An IPO raises ₹2,000 cr: ₹1,200 cr fresh issue (new shares → goes to the *company* for expansion) and ₹800 cr OFS (existing PE investor sells → goes to the *PE fund*, not the company). Only the ₹1,200 cr strengthens the company's balance sheet.

Example 2 — book-building price discovery. Price band ₹95–100. Institutional demand is strong at ₹100 (book covered 8× at the top), so the cut-off is set at **₹100**. Weak demand would have set it lower in the band. The market's bids, not the company's wish, set the price.

Example 3 — underpricing pop. Shares priced at ₹100 open at ₹135 on listing (+35%). Investors who got allotment gain; the company "left money on the table" (could have priced higher) but secured a strong, well-received listing and goodwill for future raises.

Example 4 — a full worked IPO subscription and allotment case study. A company issues 2 crore shares (fresh issue only, no OFS) at a price band of ₹450-₹475, reserved as: 50% QIB (Qualified Institutional Buyers), 15% NII (Non-Institutional Investors, i.e. HNIs), and 35% retail. At close: QIB portion subscribed 18x, NII portion subscribed 42x, retail portion subscribed 3.2x. **Cut-off price**: given the QIB book (the category whose demand carries the most weight in price discovery, since institutions do the deepest diligence) is subscribed nearly 20x even at the top of the band, the cut-off is set at ₹475 — the top of the band. **Retail allotment mechanics**: with the retail portion 3.2x oversubscribed, not every retail applicant gets shares — allotment is via a lottery/proportionate mechanism (for retail, typically each applicant either gets one full lot or nothing, decided by a computerised draw, rather than a proportionate partial allotment, since retail lot sizes are already small) — so a retail investor applying for the minimum lot has roughly a 1-in-3.2 (≈31%) chance of allotment in this scenario, a probability an analyst should be able to explain when asked "how does retail allotment actually work in an oversubscribed IPO." **NII allotment**, by contrast, is typically proportionate (not lottery-based) — an NII applicant subscribed 42x receives roughly 1/42nd of what they applied for, scaled across all NII applicants, reflecting the different allotment methodology SEBI mandates for the NII category versus retail.

Example 5 — a down-round / weak-demand IPO, contrasted with Example 4. A different company, same structure, sees QIB subscription of only 0.8x at the top of the band (i.e. the QIB book is *undersubscribed*) — under SEBI rules, if the QIB portion isn't at least fully subscribed, the issue may not proceed at all (a minimum-subscription threshold), forcing the company to either withdraw the IPO, extend the bidding period, or in some structures reduce the price band and re-open bidding. This scenario — contrasted directly against Example 4's strong 18x QIB demand — is what an analyst means by "the IPO market read this company's valuation as too aggressive": weak institutional demand at the top of the band is the market's most direct, unambiguous pricing signal, well before the stock ever lists and trades.

How it is tested in interviews

- **"Walk me through an IPO."** — Appoint bankers → due diligence and file DRHP with SEBI → roadshow and price band → book-building → set cut-off and allot → list and trade.
- **"Fresh issue vs OFS?"** — "Fresh issue creates new shares and the company gets the cash; OFS sells existing shares and the selling holders get the cash — no new capital for the company."
- **"What is book-building?"** — "Price discovery by collecting institutional demand across a price band and setting the cut-off from the demand book."
- **"Why are IPOs often underpriced?"** — "To ensure a successful, well-subscribed listing and reward investors for the uncertainty; it 'pops' but the issuer leaves some money on the table."
- **"What is a greenshoe / lock-up?"** — Greenshoe stabilizes the price via over-allotment; lock-up stops insiders dumping stock right after listing.

Traps & common mistakes

- Confusing **fresh issue** (company gets cash, dilution) with **OFS** (holders get cash, no dilution).
- Thinking the **company** sets the price — book-building lets **demand** set it.
- Forgetting **SEBI/DRHP** disclosure as the investor-protection backbone.
- Missing the purpose of **lock-ups and greenshoe** (orderly aftermarket).

First-principles recap

- Primary market = companies issue securities to raise capital.
- IPO = first public sale, priced by **book-building**, filed via **DRHP** with SEBI.
- **Fresh issue** funds the company; **OFS** cashes out existing holders.
- Underpricing, lock-ups, greenshoe, and anchor investors shape a successful listing.
- Beyond IPOs: FPO, rights, QIP, private placement.

Quick-reference

TERM	NOTE
IPO	First public share sale
DRHP	Draft prospectus filed with SEBI
Book-building	Demand-based price discovery in a band
Fresh issue vs OFS	Company cash vs selling-holder cash
Greenshoe	Over-allotment price stabilization
Lock-up	Insiders can't sell for a set period

Q&A — Primary Markets & the IPO Process

Theory and worked scenarios on issuance, book-building, and listing mechanics.

Q1. Walk me through the IPO process end to end.

Model answer. Appoint merchant bankers (BRLMs), legal counsel, auditors, and a registrar → due diligence and file the Draft Red Herring Prospectus (DRHP) with SEBI → SEBI review and observations, finalise the RHP → roadshow to market to institutions and set a price band → book-building to collect bids across the band from QIBs, non-institutional, and retail categories → determine the cut-off price from the demand book and allot shares → list on NSE/BSE and begin secondary trading.

Q2. Distinguish a fresh issue from an offer for sale (OFS), and why the distinction matters to a research analyst valuing the deal.

Model answer. A fresh issue creates new shares — the company receives the proceeds, which dilutes existing holders but strengthens the balance sheet (usable for growth capex, debt paydown, etc.). An OFS sells existing shares — the selling shareholder (often a PE/VC investor or promoter) receives the proceeds, with no new capital reaching the company and no incremental dilution beyond the ownership transfer itself. An analyst assessing "what does this IPO do for the company" must check the fresh-issue/OFS split in the prospectus — an IPO that's 90% OFS raises little growth capital for the business despite a large headline deal size.

Q3. Worked — book-building price discovery.

Price band ₹340-360. At ₹360 the book is covered 12x by QIBs; at ₹340 it would be covered 25x. What does this tell you, and roughly where would the cut-off likely land?

Model answer. Strong oversubscription even at the top of the band (12x at ₹360) signals demand comfortably clears the highest price, so the cut-off is very likely to be set at or near the top of the band (₹360) — book-building sets price from where the demand actually clears, not an average; a company with this level of demand has no reason to leave the band's ceiling unclaimed.

Q4. What is the greenshoe (over-allotment) option, and how does it stabilise the post-listing price?

Model answer. Underwriters are permitted to allot up to roughly 15% more shares than the base offer, funded by a share loan from a promoter/existing holder, effectively creating a short position. If the stock falls below the issue price after listing, underwriters buy shares in the open market to cover that short, which supports the price; if the stock trades well above the issue price, the option isn't exercised for buybacks. This is the primary open-market price-stabilisation mechanism during the immediate post-listing window.

Q5. Why do IPOs commonly "pop" on listing day, and is that necessarily a mistake by the company/bankers?

Model answer. Underpricing (setting the offer price modestly below where secondary-market demand will actually clear) is common and partly deliberate — it rewards investors for bearing IPO uncertainty and helps ensure the issue is fully subscribed and lists strongly, building credibility for the issuer's future capital-raising. It is a genuine trade-off, not simply an error: pricing "perfectly" to zero pop risks an undersubscribed, poorly-received listing, which can be more costly to the issuer's reputation and future access to capital than leaving some money on the table once.

Q6. What's the difference between a rights issue and a QIP, and when would a company use each?

Model answer. A rights issue offers new shares to *existing* shareholders (typically at a discount to market price), preserving their proportional ownership if they subscribe — used when a company wants to raise capital while giving existing holders first refusal and avoiding dilution of those who participate. A QIP (Qualified Institutional Placement) is a fast, India-specific route to raise capital directly from institutional investors without the extended regulatory process of a public offer — used when a listed company needs to raise capital quickly and is comfortable diluting existing (especially retail) shareholders who don't get first access, in exchange for speed and lower issuance cost.

Q7. A pre-IPO investor's shares are subject to a lock-up. What happens, mechanically and to the stock price, when the lock-up expires?

Model answer. During the lock-up, those shares cannot be sold on the open market, artificially constraining the freely-tradable float. At expiry, a large block of previously-locked shares becomes sellable at once — if the holder (often a PE/VC fund or promoter group) sells a meaningful portion, the sudden increase in sell-side supply can pressure the price downward even with no change in the company's fundamentals; sophisticated investors and analysts track lock-up expiry dates as a known, mechanical (not fundamentals-driven) source of near-term stock-price risk.

Q8. Worked — anchor investor allotment and its signalling effect.

A company allots 30% of its IPO book to anchor investors (large institutions) the day before the issue opens, at the top of the price band, with strong-name global and domestic funds participating.

Model answer. Anchor allotment serves as a costless signal to the broader market ahead of the public book-building: strong, well-known institutional names committing at the top of the band ahead of retail/other-institutional bidding suggests sophisticated investors, who presumably did deeper diligence, are confident in the pricing — this often improves subscription momentum in the subsequent book-building for the rest of the issue, though an analyst should still independently verify the fundamentals rather than relying on anchor participation alone as a substitute for their own diligence.

Secondary Markets & Trading Mechanics

The Problem / Why this matters

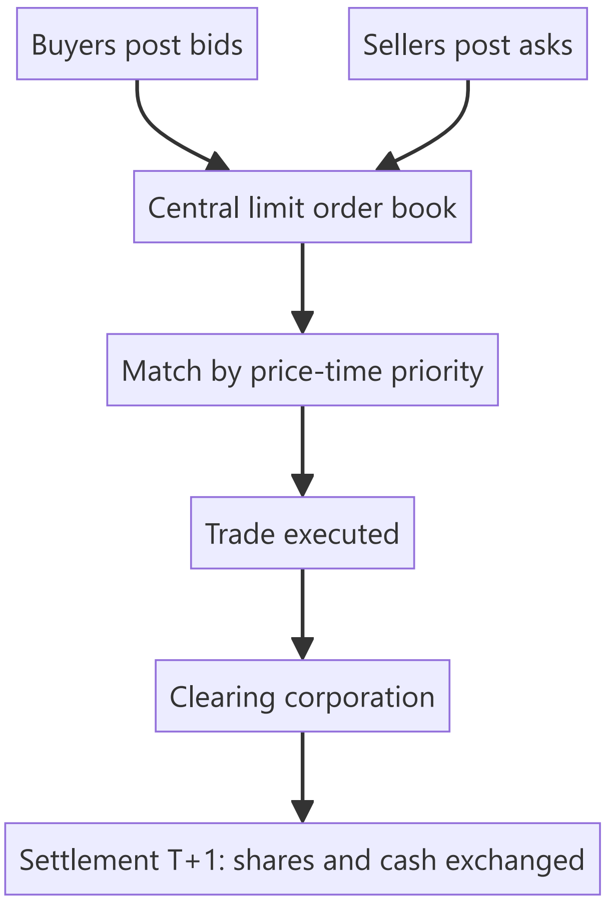
Once shares list, they trade — millions of times a day — on exchanges. How that trading actually works (order types, the order book, bid-ask spreads, matching, settlement) determines the price you get, the cost you pay, and the liquidity you rely on. Anyone touching markets — research, trading, operations, risk — must understand the plumbing. Interviewers test order types, the bid-ask spread, and settlement (T+1) directly.

Core Idea

The secondary market is a **continuous auction**: buyers post bids, sellers post asks, and the exchange's matching engine pairs them by price-time priority. The **bid-ask spread** is the cost of immediacy; **liquidity** is how easily you trade without moving the price; and every trade must **clear and settle** through depositories and clearing corporations.

Why it works this way

Price discovery needs a mechanism that continuously aggregates supply and demand. An **order-driven** market does this via a central limit order book where anyone can post orders and the best-priced ones execute first. Spreads exist because liquidity providers must be compensated for the risk of quoting; settlement infrastructure exists so buyers and sellers who never meet can trust that shares and cash actually change hands.



Full technical content

Market structures: order-driven (a central limit order book, as on NSE/BSE) vs **quote-driven** (dealers/market-makers quote two-way prices). Most modern equity markets are order-driven with electronic matching by **price-time priority** (best price first; at equal price, earliest order first).

The order book & bid-ask spread. The book shows all resting buy orders (bids) and sell orders (asks) at each price. The **best bid** and **best ask** define the spread; a **narrow spread = liquid**, a wide spread = illiquid and expensive to trade. **Depth** (quantity available near the top) shows how much you can trade without moving the price (**market impact**).

Order types:

ORDER	BEHAVIOUR
Market	Execute now at the best available price (certainty of execution, not price)
Limit	Execute only at your price or better (certainty of price, not execution)
Stop-loss	Becomes a market/limit order once a trigger price is hit
Stop-limit	Stop that becomes a limit order
Others	IOC, GTC, iceberg, day orders

Liquidity providers. Market makers and high-frequency traders continuously quote both sides, earning the spread and supplying immediacy; they narrow spreads and add depth (see market microstructure).

Long vs short. Going **long** = buy, profit if price rises (max loss = amount invested). **Short selling** = sell borrowed shares to buy back cheaper; profit if price falls, but loss is theoretically unlimited (price can rise without bound). Shorting requires a stock-borrow and is regulated.

Clearing & settlement. After a trade: the **clearing corporation** (NSE Clearing / Indian Clearing) becomes central counterparty and nets obligations; **depositories** (NSDL, CDSL) hold shares in **demat** form and transfer them; cash and securities settle on **T+1** in India (among the fastest globally; India is moving toward T+0/instant). **Rolling settlement**, margining, and a settlement guarantee fund underpin trust.

Circuit breakers & price bands limit extreme moves (index-level halts and stock-level bands) to curb panic.

Worked examples

Example 1 — market vs limit order. Best bid ₹99.8, best ask ₹100.2 (spread 0.4). A **market buy** executes immediately at ₹100.2 (pays the spread for certainty). A **limit buy at ₹100.0** sits in the book and only fills if a seller drops to ₹100.0 — better price, but it might never execute. Choose by whether you value price or certainty.

Example 2 — spread and liquidity. Stock A (large-cap) shows spread ₹0.05 with 50,000 shares at the top — deep and liquid; you trade large size cheaply. Stock B (small-cap) shows spread ₹2.00 with 200 shares — illiquid; a modest order moves the price sharply (high market impact). Same rupee value order, very different execution cost.

Example 3 — settlement. You buy 100 shares on Monday (T). Under T+1, the shares are credited to your demat and cash debited on Tuesday (T+1). The clearing corporation guarantees the trade so you don't bear your counterparty's default risk.

Example 4 — a circuit breaker halting a stock mid-session. A mid-cap stock with a 10% price band hits its lower circuit after a sudden negative news flash, and trading in that stock halts entirely — no further trades execute at any price below the circuit limit until the next session (or until the exchange revises the band). Contrast this with an index-level circuit breaker (e.g. a 10%/15%/20% Nifty/Sensex decline within a session), which halts trading market-wide for a defined cooling-off period across all stocks, not just one. Both mechanisms serve the same purpose (Part 20's discussion) — preventing a self-reinforcing panic cascade by forcing a pause for information to be digested — but operate at completely different scopes (single-stock vs market-wide), a distinction worth stating precisely rather than conflating the two when asked "what's a circuit breaker."

Example 5 — a worked market-impact cost calculation for an institutional order. A fund needs to buy ₹5 cr worth of a stock currently trading at ₹250, with the order book showing depth of roughly ₹40 lakh at the best few price levels before the price starts moving meaningfully. Executing the full ₹5 cr as a single market order would "walk the book" through many price levels, likely averaging perhaps ₹253-255 versus the ₹250 quoted price — an implicit cost of roughly 1.5-2% purely from market impact, well beyond the visible bid-ask spread alone. This is precisely why institutional desks use algorithmic execution (VWAP/TWAP slicing, covered in the Technical Research material) to spread a large order over time rather than executing it all at once, and why a research analyst recommending a position size for an institutional client should sanity-check the name's actual liquidity depth, not just its market cap, before assuming a position can be built or exited cheaply.

How it is tested in interviews

- **"Market order vs limit order?"** — "Market executes now at the best price (certain execution, uncertain price); limit executes only at your price or better (certain price, uncertain execution)."
- **"What is the bid-ask spread and what does a wide spread mean?"** — "The gap between best bid and best ask — the cost of immediacy. A wide spread means illiquidity and expensive trading."
- **"What's the max loss on a short?"** — "Theoretically unlimited, because the price can rise without bound; a long's max loss is what you invested."
- **"How does settlement work in India?"** — "T+1 rolling settlement: shares (via NSDL/CDSL demat) and cash exchange one day after the trade, guaranteed by the clearing corporation as central counterparty."
- **"What does market depth tell you?"** — "How much you can trade near the top of the book without moving the price — your likely market impact."

Traps & common mistakes

- Confusing **market** (execution certainty) and **limit** (price certainty) orders.
- Forgetting a short's loss is **unlimited**.
- Ignoring **market impact/depth** — a big order in a thin book moves the price against you.
- Not knowing India settles at **T+1** (and is moving faster).
- Overlooking the **clearing corporation** as central counterparty removing settlement risk.

First-principles recap

- Secondary trading is a continuous auction matched by **price-time priority**.
- The **bid-ask spread** is the cost of immediacy; narrow = liquid.
- **Market** = certain execution; **limit** = certain price; **stop-loss** triggers on a level.

- Short selling has **unlimited** loss potential.
- Trades **clear and settle** via clearing corporations and depositories, T+1 in India.

Quick-reference

ITEM	NOTE
Order book	Resting bids/asks by price-time priority
Spread	Best ask − best bid (cost of immediacy)
Market order	Now, best price
Limit order	Your price or better
Short loss	Unlimited
Settlement	T+1 (NSDL/CDSL demat, clearing corp CCP)

Q&A — Secondary Markets & Trading Mechanics

Theory and worked scenarios on order types, spreads, and settlement.

Q1. Market order vs limit order — what does each guarantee, and what does each not guarantee?

Model answer. A market order guarantees execution (it fills immediately against the best available price on the other side of the book) but not the price you'll pay — in a fast-moving or thin market you can be filled well away from the last traded price. A limit order guarantees the price (it will only execute at your specified price or better) but not execution — if the market never trades at your price, the order simply sits unfilled. The choice is a direct trade-off between price certainty and execution certainty.

Q2. Worked — computing market impact from the order book.

Best bid ₹248.50 (2,000 shares), next bid ₹248.20 (5,000 shares); best ask ₹249.00 (1,500 shares), next ask ₹249.40 (4,000 shares). You need to sell 5,000 shares immediately as a market order.

Model answer. Selling into the bid side: the first 2,000 shares execute at ₹248.50, the remaining 3,000 shares execute at ₹248.20 (the next price level down), since the top level's depth is exhausted. Volume-weighted average execution price = (2,000×248.50 + 3,000×248.20) / 5,000 = (497,000 + 744,600) / 5,000 = ₹248.32 — worse than the ₹248.50 best bid, illustrating market impact: a large order "walks the book" and gets a worse average price than the top-of-book quote suggests.

Q3. Why is a short seller's maximum loss theoretically unlimited, while a long position's maximum loss is capped?

Model answer. A long position's downside is capped at the amount invested — the price can only fall to zero. A short position involves selling borrowed shares with the obligation to buy them back later to return them; since a stock's price has no theoretical ceiling, the cost to buy back and close a short position can rise without bound, making the potential loss theoretically unlimited. This asymmetry is why short positions require margin, active risk management, and often stop-loss discipline that a comparable long position may not need as urgently.

Q4. What roles do the clearing corporation and the depository each play in a single trade, and why are they separate entities?

Model answer. The clearing corporation (e.g. NSE Clearing) becomes the central counterparty to every matched trade — it guarantees settlement to both sides, so neither party bears the other's default risk. The depository (NSDL/CDSL) holds securities in dematerialised electronic form and executes the actual transfer of shares between demat accounts on settlement. They're separate because they solve different frictions: the clearing corp manages counterparty/credit risk across the whole market, while the depository manages custody and the mechanical transfer of ownership records — conflating them (a common interview trap) misses that one guarantees the trade and the other physically moves the asset.

Q5. What does "market depth" measure, and why does an analyst or trader care about it beyond the headline bid-ask spread?

Model answer. Depth measures the quantity available at each price level near the top of the order book, not just the best bid/ask. A narrow spread can be misleading if depth is thin — a seemingly liquid-looking stock with a tight spread but only 200 shares at the best bid will show significant market impact (Q2) on any order larger than that, even though the quoted spread alone looks attractive. Depth, not spread alone, determines how much size can actually be traded near the current price.

Q6. Explain price-time priority in order matching with a worked example.

Two limit buy orders both at ₹100: Order A for 500 shares placed at 9:15:02am, Order B for 800 shares placed at 9:15:05am. A sell market order for 600 shares arrives.

Model answer. Price-time priority matches the best price first, and among orders at the same price, the earliest-placed order first. Order A (earlier, same price) fills completely (500 shares), and the remaining 100 shares of the incoming sell order fill against Order B, leaving Order B with 700 shares still resting in the book. Time priority at a given price level rewards being first in the queue, not just posting a competitive price.

Q7. What's the difference between an order-driven and a quote-driven market structure?

Model answer. An order-driven market (NSE/BSE's central limit order book) lets any participant post bids/asks directly, with the exchange's matching engine pairing them by price-time priority — price discovery emerges from aggregated public orders. A quote-driven market relies on designated dealers/market-makers who continuously post two-way (buy and sell) quotes that others trade against — price discovery is mediated through the dealer's quoted spread rather than a fully open book. Most modern equity markets, including Indian exchanges, are primarily order-driven, though market-makers/liquidity providers still operate within that order-driven structure to supply depth.

Q8. Why would a trader use a stop-limit order instead of a plain stop-loss order?

Model answer. A plain stop-loss becomes a *market* order once the trigger price is hit, guaranteeing execution but not the price — in a fast-falling/gapping market, the actual fill can be significantly worse than the trigger level. A stop-limit becomes a *limit* order once triggered, capping the worst acceptable execution price — but risks not executing at all if the price gaps straight through the limit without trading at an acceptable level. The choice reflects the same execution-certainty-vs-price-certainty trade-off as Q1, applied specifically to risk-management/exit orders.

Market Participants & Intermediaries

The Problem / Why this matters

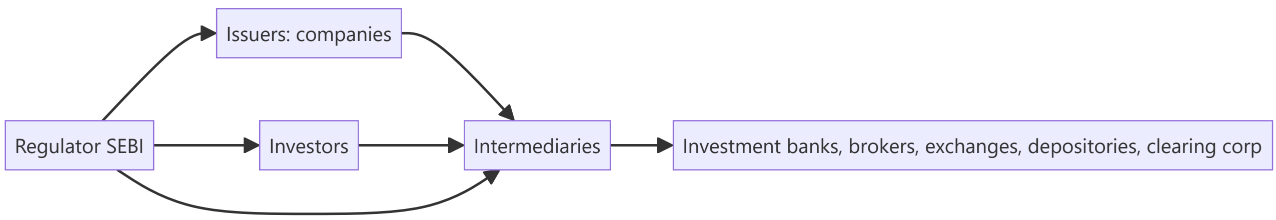
Markets don't run themselves — a web of participants and intermediaries makes issuance, trading, clearing and custody happen safely. Knowing **who does what** — issuers, the different investor types and their motivations, and the intermediaries (banks, brokers, exchanges, depositories, clearing corporations) — tells you how order flow moves, why prices react to certain investors, and where you'd fit in the ecosystem. Interviewers expect you to map the players cleanly.

Core Idea

The capital-market ecosystem has three groups: **issuers** who need capital, **investors** who supply it (each with different horizons and motivations), and **intermediaries** who connect and service them. Overlaying all of it is the **regulator** (SEBI) ensuring fairness, disclosure and investor protection.

Why it works this way

Issuers and investors can't transact directly at scale — they need someone to underwrite issues, match trades, hold securities, guarantee settlement and enforce rules. Each intermediary exists to solve a specific friction (distribution, execution, custody, counterparty risk, oversight), and the mix of investors determines how "smart," patient or reactive the money in a stock is.



Full technical content

Issuers: companies (private and public), governments (bonds), and other entities raising capital.

Investors (by type and motivation):

INVESTOR	HORIZON / MOTIVATION
Retail	Individuals; varied horizons; price-takers
Mutual funds / AMCs	Pooled retail money; benchmark-driven
Insurance & pension funds	Long-term, liability-matching, stable
FPIs / FIIs	Foreign institutional flows; can be large and fast
Domestic institutions (DIIs)	Banks, insurers, MFs domestically
Hedge funds / PMS / AIFs	Absolute-return, flexible, can short/leverage
Promoters / strategic	Control-oriented, long-term holders

Institutional flows (FPI/DII) often drive index moves; "smart money" positioning is watched closely.

Intermediaries:

INTERMEDIARY	ROLE
Investment bank / merchant banker (BRLM)	Advise on and underwrite issues; M&A; capital raising
Brokers / trading members	Execute investor orders on the exchange
Stock exchanges (NSE, BSE)	Provide the trading platform and price discovery
Depositories (NSDL, CDSL)	Hold securities in demat; effect transfer
Clearing corporations	Central counterparty; net and guarantee settlement
Custodians	Safekeep assets for institutions; settlement support
Registrars & Transfer Agents (RTAs)	Maintain shareholder records, process corporate actions
Rating agencies / research	Independent credit/equity opinions
Investment advisers / distributors	Advise and distribute to end investors

The regulator — SEBI. Sets rules for issuers (disclosure), intermediaries (registration, conduct), and markets (surveillance, insider-trading and manipulation enforcement); protects investors and develops the market. RBI oversees banking/monetary aspects; IRDAI insurance.

Sell-side vs buy-side (preview). *Sell-side* (banks, brokers) creates products, executes and publishes research to win client business. *Buy-side* (asset managers, funds) invests client money and consumes that research to make decisions.

Worked examples

Example 1 — the chain of a single trade. A mutual fund (investor) instructs its **broker** (intermediary) to buy 1 lakh shares on the **NSE** (exchange); the trade matches, the **clearing corporation** guarantees it, the **depository (CDSL/NSDL)** moves the shares into the fund's **custodian's** demat account on T+1. Five different intermediaries touched one trade, each solving a distinct friction.

Example 2 — why the investor mix matters. Two similar mid-caps: Stock A is 70% held by long-term insurers and index funds (stable, patient); Stock B is 40% held by fast FPIs and traders (volatile, flow-driven). A negative headline moves B far more, because its holder base reacts and rotates quickly. *Knowing the holder base predicts volatility.*

Example 3 — SEBI's role. A promoter tries to sell shares on undisclosed bad news (insider trading). SEBI's surveillance flags the trades, investigates, and penalizes — protecting other investors and preserving trust that keeps the market liquid.

Example 4 — a hedge fund/PMS using leverage and shorting that a mutual fund can't. A long-only mutual fund identifies an overvalued stock but has no mandate to short it — it can only underweight or avoid holding it, a comparatively weak way to express a negative view. A hedge fund or PMS with a flexible mandate can short the same stock directly, and can additionally use leverage to size a high-conviction long position beyond what its raw capital would allow. This mandate difference — not skill or information — is often the real reason a hedge fund and a mutual fund reach different position sizes or even opposite trades on the same stock, an important nuance when a candidate is asked to compare institutional investor types.

Example 5 — custodian vs broker in an actual settlement failure scenario. An institutional buy order executes on the exchange via the fund's broker, but on settlement date the fund's custodian reports a shortfall — the securities weren't available to deliver from the counterparty side. The clearing corporation's settlement-guarantee mechanism (Part on secondary markets) steps in to make the buyer whole (via the exchange's default/shortfall-handling process, typically an auction to source the shares or a cash equivalent), while the custodian's role throughout is purely to report and reconcile the fund's actual holdings against what was expected — illustrating concretely why an institution needs both a broker (who executed a trade that, from the fund's side, looked completely normal) and an independent custodian (whose job is precisely to catch and report a settlement discrepancy the broker relationship alone wouldn't surface).

How it is tested in interviews

- **"Who are the main market participants?"** — Issuers; investors (retail, mutual funds, insurers/pensions, FPIs, DIIs, hedge funds/PMS, promoters); intermediaries (banks, brokers, exchanges, depositories, clearing corporations, custodians, RTAs); regulator (SEBI).
- **"What does a depository do?"** — "Holds securities in dematerialized form and effects their transfer — NSDL and CDSL in India."
- **"What's the difference between sell-side and buy-side?"** — "Sell-side creates, executes and researches to serve clients; buy-side invests client money and consumes that research."
- **"Why do FPI flows matter?"** — "They're large and can move quickly, so they often drive index-level moves and volatility."
- **"What does SEBI do?"** — "Regulates issuers, intermediaries and markets — disclosure, conduct, surveillance, and investor protection."

Traps & common mistakes

- Confusing **depository** (holds shares) with **clearing corporation** (guarantees settlement) with **exchange** (matches trades) — three different roles.
- Lumping all investors together — horizons and behaviour differ sharply.
- Mixing up **sell-side** and **buy-side**.
- Forgetting the **regulator** as the backbone of trust.

First-principles recap

- Three groups: issuers, investors, intermediaries — plus the regulator.
- Each intermediary solves a friction: distribution, execution, custody, settlement, records.
- The **investor mix** shapes a stock's volatility and behaviour.
- Exchange (match) ≠ clearing corp (guarantee) ≠ depository (hold).
- SEBI ensures disclosure, conduct and investor protection.

Quick-reference

PLAYER	ROLE
Investment bank / BRLM	Underwrite/advise issues
Broker	Execute orders
Exchange (NSE/BSE)	Match trades, price discovery
Depository (NSDL/CDSL)	Hold & transfer demat shares
Clearing corporation	CCP, guarantee settlement
SEBI	Regulate & protect investors

Q&A — Market Participants & Intermediaries

Theory and worked scenarios on the ecosystem of issuers, investors, and intermediaries.

Q1. Map a single institutional trade through every intermediary it touches, and state each one's distinct role.

Model answer. A mutual fund (investor) instructs its broker (execution intermediary) to buy shares on the NSE (exchange, provides the matching venue and price discovery); the trade is guaranteed by the clearing corporation (central counterparty); the depository (NSDL/CDSL) transfers the shares in demat form into the fund's custodian's account (safekeeping/settlement support for institutions) on T+1. Five distinct intermediaries, each solving a different friction: distribution/execution, matching, counterparty guarantee, custody/transfer, and institutional safekeeping.

Q2. Why does the composition of a stock's investor base (retail vs FPI vs DII vs promoter) matter to an analyst forecasting volatility?

Model answer. Different investor types have different horizons and reaction speeds: FPIs can move large sums quickly in response to global risk sentiment or currency moves, amplifying volatility; DIIs (insurers, pension funds) tend to be longer-horizon and more stable, often acting as a stabilising counterweight; promoters are typically long-term, control-oriented holders. A stock heavily held by fast-moving FPIs will react more sharply to negative headlines than an otherwise-similar stock dominated by patient domestic institutional holders — the holder-base composition is itself a volatility signal, independent of the company's fundamentals.

Q3. Distinguish the roles of an exchange, a clearing corporation, and a depository — a classic three-way confusion trap.

Model answer. The exchange (NSE/BSE) provides the trading platform and matches buy/sell orders — price discovery happens here. The clearing corporation becomes the central counterparty to every matched trade, guaranteeing settlement so neither party bears the other's default risk. The depository (NSDL/CDSL) holds securities in dematerialised form and physically executes the transfer of ownership. A common interview trap is treating any two of these as interchangeable — each solves a genuinely distinct friction (matching vs guaranteeing vs custody/transfer).

Q4. What's the difference between sell-side and buy-side, and where would a candidate coming from a trading/derivatives background naturally fit?

Model answer. Sell-side firms (investment banks, brokers) create products, execute trades, and publish research to win and service client business — their research and trading desks are ultimately commercial functions serving external clients. Buy-side firms (asset managers, mutual funds, hedge funds, pension funds) invest their own or client capital directly, consuming sell-side research as one input among many to make investment decisions. A candidate with direct trading/derivatives desk experience (as opposed to client-facing sales or research-publishing experience) often has skills that map most naturally to buy-side execution/trading roles or to sell-side trading desks, as distinct from sell-side equity research roles, which screen more for the research/writing/client-communication skill set.

Q5. What does SEBI actually regulate, concretely, across issuers, intermediaries, and markets?

Model answer. For issuers: disclosure requirements (prospectus content, ongoing reporting, related-party transaction rules). For intermediaries: registration and conduct standards (brokers, merchant bankers, investment advisers must be SEBI-registered and follow conduct codes). For markets: surveillance against insider trading and price manipulation, oversight of exchanges and clearing corporations, and setting rules like circuit breakers and margin requirements. The unifying purpose across all three is investor protection and market integrity — a useful one-line answer is that SEBI's job is to make sure information is disclosed fairly, intermediaries behave honestly, and market mechanics can be trusted.

Q6. Worked scenario — how would surveillance likely detect the insider-trading pattern in this case, and what happens next?

A promoter sells a large block of shares three days before the company announces disappointing earnings, having had no prior pattern of similar sales.

Model answer. SEBI's surveillance systems flag unusual trading patterns ahead of price-sensitive announcements — specifically, a departure from an individual's established trading behaviour (no prior pattern of large sales) combined with timing that precedes a material negative disclosure is a classic red flag pattern. This typically triggers an investigation (examining trading records, communications, and access to the pre-announcement information), and if insider trading is substantiated, results in penalties (fines, trading bans, potential criminal referral) — the broader function this serves is preserving the market's trust that prices reflect fairly available information, which is what keeps outside investors willing to participate.

Q7. What's the difference between a custodian and a broker, and why might an institutional investor use both?

Model answer. A broker executes trading instructions (buy/sell orders) on the investor's behalf. A custodian safekeeps the institution's assets, handles settlement mechanics on the institution's side, and often provides additional services (corporate action processing, reporting, fund administration support). An institutional investor typically uses a broker for execution and a separate custodian for safekeeping and back-office settlement — this separation (rather than one entity doing both) is itself a risk-management practice, reducing the concentration of control over both trade execution and asset custody in a single counterparty.

Q8. Why do RTAs (Registrars & Transfer Agents) matter to a retail shareholder, even though most investors never interact with one directly?

Model answer. RTAs maintain the official record of shareholding and process corporate actions (dividend payments, rights/bonus share credits, name/address changes, dematerialisation requests) on behalf of the issuing company — even though a retail investor's day-to-day interaction is with their broker/demat account, the RTA is the entity that ultimately ensures corporate-action entitlements (a declared dividend, a bonus share) are correctly credited to the right shareholder of record, making it a quietly essential piece of infrastructure behind every corporate action an investor experiences.

Equity Instruments

The Problem / Why this matters

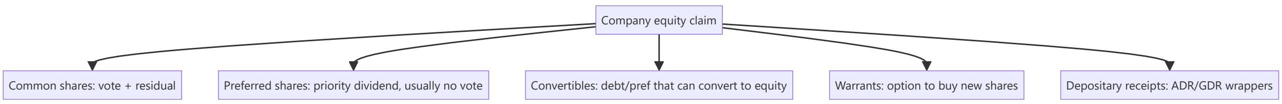
"Equity" isn't a single thing. Companies issue common shares, preferred shares, depositary receipts, warrants, and convertibles — each with different rights to cash flow, control and risk. Knowing the instruments, their rights, and where they sit in the capital structure lets you understand what you actually own, how it's valued, and why one class trades differently from another. It's foundational for equity research and capital markets roles.

Core Idea

Equity instruments are claims on a company's **residual value and control**, ranging from plain common shares (full ownership and residual risk) through preferred shares (a hybrid with priority) to derivative-like claims (warrants, convertibles) and cross-border wrappers (ADRs/GDRs). Each differs in cash-flow priority, voting rights, and payoff.

Why it works this way

Different investors want different risk-return-control packages, and companies want to tailor securities to tap the widest, cheapest capital. So the market slices the equity claim into instruments: some trade voting rights for a fixed dividend (preferred), some bundle upside options (warrants, convertibles), and some repackage foreign shares for local investors (depositary receipts).



Full technical content

Common (equity/ordinary) shares: full ownership — voting rights, residual claim on profits (dividends, discretionary) and on assets in liquidation (last in line). Uncapped upside, first-loss downside. The default meaning of "a share."

Preferred shares: a hybrid. Priority over common for **dividends** (often fixed) and in **liquidation**, but usually **no vote**. Variants: **cumulative** (missed dividends accrue), **participating** (share in extra upside), **convertible** (can convert to common), **redeemable/callable**. Sits between debt and common equity in the capital structure.

Depositary receipts: let investors hold foreign shares locally. - **ADR** (American Depositary Receipt) — a US-listed certificate representing shares of a non-US company (e.g., Infosys ADR on NYSE). - **GDR** (Global Depositary Receipt) — similar, listed outside the home and US markets. - Purpose: access foreign capital and investors without a full foreign listing.

Warrants: a company-issued option giving the holder the right to buy **new** shares at a set price for a period. Exercise creates new shares (dilutive). Often attached to bonds as a sweetener.

Convertible securities: bonds or preferred shares that can convert into common equity at a set ratio. Give the holder debt-like downside protection plus equity upside; give the issuer a lower coupon. Convert when the equity is worth more than the bond.

Rights & entitlements: existing shareholders may receive **rights** (to buy new shares at a discount — see corporate actions) or **bonus** shares.

Key rights of a shareholder: vote (elect directors, major decisions), dividends (if declared), residual assets in liquidation, information/disclosure, pre-emption (rights issues), and to transfer shares.

Where each sits (priority in liquidation): secured debt → unsecured debt → **preferred equity** → **common equity** (last). Higher priority = lower risk = lower expected return.

Worked examples

Example 1 — common vs preferred in a bad year. A firm can pay ₹10 cr of dividends. Preferred holders are owed a fixed ₹6 cr (cumulative) — they're paid first; common gets the remaining ₹4 cr (discretionary). In a terrible year with ₹4 cr available, preferred takes it all and common gets nothing (and cumulative preferred arrears accrue). Preferred = priority but capped; common = residual and variable.

Example 2 — convertible economics. A ₹1,000 convertible bond converts into 8 shares (conversion price ₹125). If the stock is ₹100, the holder keeps the bond (worth more than 8 × ₹100 = ₹800) and enjoys downside protection. If the stock rises to ₹180, converting gives 8 × ₹180 = ₹1,440 — the holder converts and captures the equity upside. The issuer paid a low coupon for embedding that option.

Example 3 — ADR arbitrage link. Infosys trades in Mumbai and as an ADR in New York. If the ADR (adjusted for the ratio and FX) diverges from the Mumbai price, arbitrageurs trade both until they realign — so the ADR essentially tracks the home share.

Example 4 — a worked ADR ratio and arbitrage-band calculation. An Indian company's ADR represents 2 underlying home-market shares (an ADR ratio of 2:1). The home share trades at ₹840, and USD/INR is 83.5. The ADR's "fair" USD price = (840 × 2) / 83.5 ≈ **$20.12**. If the actual ADR price is $20.60, it's trading at a premium of roughly ($20.60 − $20.12)/$20.12 ≈ 2.4% to fair value — small enough that transaction costs (brokerage, FX conversion spread, the operational cost of the deposit/withdrawal mechanism connecting the ADR to the underlying share) likely exceed the arbitrage profit, which is exactly why small ADR premiums/discounts can persist despite the theoretical arbitrage link; a wider gap (say 8-10%) would be a much more actionable, and more surprising, signal worth investigating for a data or execution error before assuming a genuine mispricing.

Example 5 — why a company chooses preferred shares over a straight bond to raise capital. A company wants to raise ₹200 cr without adding to its formal debt covenants (which would trip financial-ratio limits already set with existing lenders) and without immediately diluting common shareholders' voting control. Issuing cumulative redeemable preferred shares achieves both: preferred shares typically aren't counted as "debt" for covenant purposes (they're equity on the balance sheet, even though they behave economically like a hybrid), and standard preferred structures carry no voting rights, leaving common shareholders' control unchanged. The cost: preferred dividends aren't tax-deductible the way interest expense is, making preferred capital more expensive after-tax than equivalent debt — a trade-off (covenant/control flexibility vs after-tax cost) a candidate should be able to state explicitly when asked "why would a company use preferred stock instead of just borrowing."

How it is tested in interviews

- **"Common vs preferred shares?"** — "Common has voting rights and a residual, variable claim (uncapped upside, first-loss). Preferred usually has no vote but priority on a fixed dividend and in liquidation — a hybrid between debt and common."
- **"What is an ADR?"** — "A US-listed certificate representing shares of a foreign company, letting US investors hold it in dollars without a foreign listing."

- **"How does a convertible bond work?"** — "A bond that can convert into equity at a set ratio; the holder gets debt downside protection plus equity upside, and the issuer pays a lower coupon for that option."
- **"Where does preferred sit in the capital structure?"** — "Below debt, above common — priority over common for dividends and liquidation, but behind all creditors."

Traps & common mistakes

- Thinking preferred is "safer equity" without noting it's still **below all debt**.
- Confusing **warrants** (company issues new shares, dilutive) with exchange-traded **options** (no new shares).
- Forgetting convertibles are **dilutive** on conversion.
- Treating ADRs/GDRs as different companies rather than **wrappers** on the same shares.

First-principles recap

- Equity instruments slice the ownership claim by cash-flow priority, control and payoff.
- **Common** = vote + residual (uncapped/first-loss); **preferred** = priority dividend, usually no vote.
- **Convertibles** = debt/pref with an equity option; **warrants** = right to buy new shares (dilutive).
- **ADR/GDR** = wrappers letting foreign investors hold local shares.
- Priority: debt → preferred → common.

Quick-reference

INSTRUMENT	KEY FEATURE
Common share	Vote + residual claim
Preferred share	Priority dividend, usually no vote
Convertible	Converts debt/pref → equity
Warrant	Right to buy new shares (dilutive)
ADR / GDR	Foreign-share wrapper
Priority	Debt > preferred > common

Q&A — Equity Instruments

Theory and worked scenarios on common, preferred, convertible, and depositary-receipt instruments.

Q1. Common vs preferred shares — what's given up and what's gained by holding preferred instead of common?

Model answer. Preferred shareholders give up voting rights (in most standard structures) and uncapped upside participation, in exchange for priority over common holders on dividends (often a fixed rate) and on residual assets in liquidation. Common shareholders retain full voting rights and an uncapped, residual claim on profits and assets, but sit last in the priority stack — first to lose in a downturn, last to be paid even in a good year if the board doesn't declare a dividend. The trade is control-and-upside versus priority-and-a-capped-return.

Q2. Worked — cumulative preferred dividends in a bad year.

A company owes ₹8 cr/year in cumulative preferred dividends. In Year 1 it can only pay ₹3 cr total. In Year 2, profits recover and ₹15 cr is available for distribution. How much does each class receive in each year?

Model answer. Year 1: preferred is owed ₹8 cr but only ₹3 cr is available — preferred receives the full ₹3 cr (still short by ₹5 cr, which accrues as arrears since it's cumulative); common receives ₹0. Year 2: preferred is owed the current year's ₹8 cr *plus* the ₹5 cr arrears from Year 1 = ₹13 cr, paid first from the ₹15 cr available; common receives the remaining ₹15 − 13 = ₹2 cr. This illustrates why "cumulative" matters: a missed preferred dividend doesn't disappear, it compounds as a prior claim against future distributable profit before common sees anything.

Q3. Worked — convertible bond conversion decision at two different stock prices.

A ₹1,000 face-value convertible bond converts into 10 shares (conversion price ₹100). At maturity the bond can be redeemed at face value (₹1,000) or converted.

Model answer. If the stock is at ₹80: conversion value = 10 × 80 = ₹800, less than the ₹1,000 redemption value — the holder redeems for cash, the debt-like floor protects them. If the stock is at ₹150: conversion value = 10 × 150 = ₹1,500, more than the ₹1,000 redemption value — the holder converts to capture the equity upside. The convertible's value is effectively the greater of its bond floor and its as-converted equity value, which is exactly why issuers can offer a lower coupon than a plain bond — investors are compensated with this embedded option instead.

Q4. What is an ADR, and why would the ADR price of an Indian company track its NSE/BSE price closely rather than diverge freely?

Model answer. An ADR (American Depositary Receipt) is a US-listed certificate representing shares of a non-US company held by a depositary bank — it lets US investors hold and trade the company in dollars without a full separate US listing. Because the ADR and the home-market shares represent the same underlying economic claim (adjusted for the ADR ratio and the USD/INR exchange rate), any meaningful price divergence creates a pure arbitrage opportunity: arbitrageurs would buy the cheaper instrument and sell the more expensive one (or convert between them via the depositary mechanism), which pulls the two prices back into alignment — the arbitrage mechanism itself is what keeps the ADR "tracking" the home share, not any inherent property of the ADR.

Q5. Distinguish a warrant from an exchange-traded call option — same payoff shape, different mechanics and consequences.

Model answer. Both give the right to buy shares at a set price by a set date, and both have a similar payoff diagram. The key difference: exercising a warrant creates *new* shares issued by the company (dilutive to existing shareholders, and the exercise proceeds go to the company), while exercising an exchange-traded call option transfers *existing* shares between two market participants (no dilution, no cash to the company — the option writer delivers shares they already own or must acquire in the market). A candidate who conflates the two in an interview signals they haven't thought through where the shares actually come from.

Q6. Where do preferred shares sit in the capital-structure priority stack, and what does that imply about their risk relative to bonds and common equity?

Model answer. Priority order (highest to lowest claim): secured debt → unsecured debt → preferred equity → common equity. Preferred sits below all debt (both secured and unsecured), meaning in a liquidation or severe distress scenario, all debtholders must be paid in full before preferred holders receive anything — so preferred is meaningfully riskier than any debt instrument despite sometimes being marketed as a "safer" or "income" instrument. It is, however, senior to common, so it carries less risk than common equity. This full ordering — not just "preferred is safer than common" — is the complete, correct answer.

Q7. What's the difference between a rights issue entitlement and a bonus share, from the shareholder's perspective?

Model answer. A rights entitlement gives an existing shareholder the option to *buy* additional shares (usually at a discount to market price) — it requires the shareholder to pay in additional capital to exercise it, and if they don't, their proportional ownership is diluted. A bonus share is issued to existing shareholders *for free*, in proportion to their existing holding, funded by capitalising the company's reserves — it requires no payment and, because more shares now represent the same underlying company value, the share price mechanically adjusts downward (similar to a stock split) with no real change in the shareholder's proportional wealth or ownership percentage.

Q8. Why might a company issue convertible preferred shares rather than either straight debt or a straight equity issue?

Model answer. Convertible preferred lets a company raise capital at a lower cost than straight preferred/debt (investors accept a lower fixed return in exchange for the embedded equity-upside option) while deferring dilution — dilution only occurs if and when the instrument converts, typically once the stock has appreciated enough to make conversion attractive to holders. This is a common structure in growth-stage and PE/VC financing rounds specifically because it balances the investor's downside protection (priority, fixed return if it doesn't convert) against the company's desire to avoid selling common equity outright at what may be a depressed current valuation.

Market Indices

The Problem / Why this matters

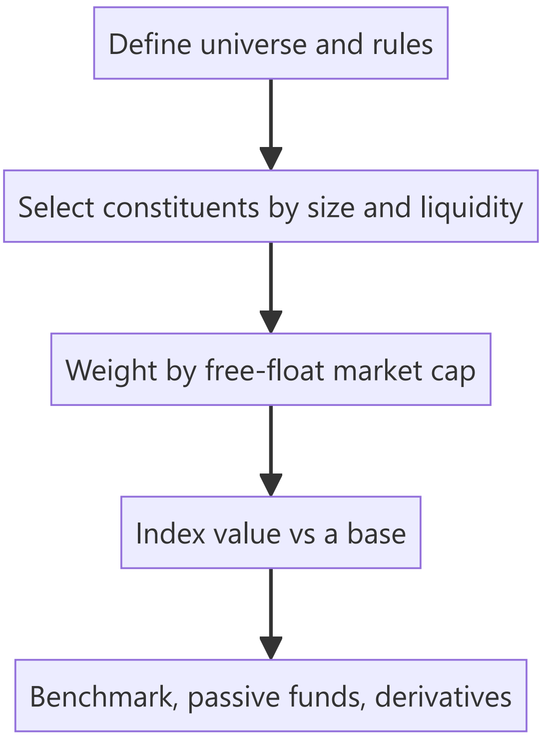
"The market was up 1% today" — but what *is* "the market"? An **index** is the number that represents it. Indices are how performance is measured, how passive funds are built, how benchmarks are set, and how derivatives (index futures/options) are priced. Understanding how they're constructed (and their quirks) matters for research, portfolio, and derivatives roles — and interviewers ask how the Nifty/Sensex are built and what "free-float market-cap weighted" means.

Core Idea

A market index is a **single number tracking the value of a defined basket of stocks**, used as a barometer of the market or a segment. Most modern indices are **free-float market-capitalization weighted**: each stock's weight is proportional to the market value of its publicly tradable shares.

Why it works this way

Investors need a comparable, investable benchmark. Weighting by free-float market cap makes the index reflect where investable money actually sits (bigger, more freely-traded companies matter more) and makes it **replicable** by index funds. It also self-adjusts: as a stock rises, its weight rises, mirroring a real buy-and-hold portfolio.



Full technical content

Construction methods (weighting):

METHOD	WEIGHT BY	EXAMPLES
Free-float market-cap	Value of publicly tradable shares	Nifty 50, Sensex, S&P 500
Full market-cap	Total shares × price	older indices
Price-weighted	Share price (not size)	Dow Jones, Nikkei
Equal-weighted	Same weight each	equal-weight indices

Free-float excludes promoter/strategic/locked holdings — only shares actually available to trade count, so the index reflects investable value.

Indian benchmarks: - **Nifty 50** (NSE) — 50 large-caps, free-float market-cap weighted. - **Sensex** (BSE) — 30 large-caps, free-float weighted. - Broader: Nifty Next 50, Nifty 100/500, Nifty Midcap/Smallcap, and sectoral indices (Bank Nifty, Nifty IT, etc.).

The divisor & continuity. An index uses a **divisor** so that mechanical changes (constituent additions/deletions, corporate actions, new share issuance) don't create artificial jumps — the divisor is adjusted to keep the index continuous. This is why a stock split doesn't move the index.

Rebalancing. Indices are reviewed periodically (e.g., semi-annually for Nifty) to add/drop constituents by size and liquidity rules. Index inclusion/exclusion drives real flows (passive funds must buy the added stock), so it moves prices.

Uses of indices: 1. **Benchmark** — measure a portfolio's relative performance (alpha). 2. **Passive investing** — index funds and ETFs replicate the index cheaply. 3. **Derivatives** — index futures and options (Nifty/Bank Nifty are among the world's most traded). 4. **Market sentiment** — a barometer of the economy/segment.

Price return vs total return index. A **price index** ignores dividends; a **total return index (TRI)** reinvests dividends — TRI is the fair benchmark for a fund that also earns dividends.

Worked examples

Example 1 — free-float weighting. Company X has ₹1,000 cr total market cap but the promoter holds 60% (locked), so free-float = 40% = ₹400 cr. Company Y has ₹600 cr market cap, 100% free-float = ₹600 cr. Despite X being "bigger," Y gets a larger index weight (₹600 cr vs ₹400 cr of investable value). *Free-float, not total size, drives weight.*

Example 2 — why a split doesn't move the index. A constituent does a 1:1 split — price halves, shares double, market cap unchanged. The index divisor absorbs the mechanical change so the index value is unaffected. Corporate actions don't distort the index.

Example 3 — index inclusion flow. A stock is added to the Nifty 50 at the next rebalance. Every passive fund tracking the Nifty must buy it to match the index, creating real buying pressure — the stock often rises into and on inclusion. Index membership has real cash-flow consequences.

Example 4 — sizing the actual passive-flow impact of an index change. Estimated total AUM tracking the Nifty 50 (index funds + ETFs + closet-indexers benchmarked to it) is roughly ₹4,50,000 cr. A stock is added to the index with an assigned weight of 0.6%. The mechanical passive buying this triggers is approximately 0.6% × 4,50,000 cr ≈ **₹2,700 cr** of forced buying across the tracking-fund universe, concentrated around the effective date of the change. If the stock's free-float market cap is, say, ₹40,000 cr and its average daily traded value is ₹150 cr, then ₹2,700 cr of buying represents roughly **18 days' worth of average trading volume** that needs to clear near the rebalance date — explaining why index-inclusion candidates often see meaningfully outsized price moves and volume spikes specifically in the days around the effective date, and why some active managers explicitly trade *ahead* of an anticipated inclusion (once a stock's growing market cap makes inclusion likely at the next scheduled review) to front-run this predictable flow.

Example 5 — why a stock's index weight can change even with no rebalance and no price move. A promoter sells a 10% stake via an open-market block deal, reducing their holding from 55% to 45% — free-float rises from 45% to 55% of total shares. Even with the share price completely unchanged, the stock's free-float market cap (and therefore its index weight) rises proportionally, since free-float weighting (Section 6.1) is a function of *tradable* value, not just price. This is why an index's divisor and constituent weights are reviewed and adjusted not only at scheduled rebalance dates, but also on ad hoc "specified events" like a material shareholding change — a detail that separates a candidate who understands index mechanics from one who only knows the "market-cap weighted" headline.

How it is tested in interviews

- **"How is the Nifty/Sensex constructed?"** — "Free-float market-capitalization weighted baskets of large-caps (Nifty 50 on NSE, Sensex 30 on BSE); weights reflect the value of publicly tradable shares."
- **"What does free-float mean and why use it?"** — "Only publicly tradable shares (excluding promoter/locked holdings). It makes the index reflect investable value and be replicable by index funds."
- **"Why doesn't a stock split move the index?"** — "The divisor adjusts for mechanical changes so the index stays continuous; market cap is unchanged by a split."
- **"Price-weighted vs market-cap-weighted?"** — "Price-weighted (Dow) weights by share price regardless of size; market-cap weighted (Nifty/S&P) weights by company value — the latter reflects the real market."
- **"Price return vs total return index?"** — "TRI reinvests dividends; it's the fair benchmark for a fund that also collects dividends."

Traps & common mistakes

- Saying "market-cap weighted" without the **free-float** nuance for Nifty/Sensex.
- Thinking a **split or bonus** moves the index (the divisor absorbs it).
- Confusing **price-weighted** (Dow) with **market-cap-weighted** (S&P/Nifty).
- Benchmarking a fund against a **price index** instead of the **total-return** index.

First-principles recap

- An index = a single number tracking a defined basket; the market's barometer.
- Nifty/Sensex are **free-float market-cap weighted** — investable value drives weight.
- A **divisor** keeps the index continuous through corporate actions and rebalances.
- Indices power benchmarking, passive funds, and derivatives.
- Use the **total-return** index to fairly benchmark dividend-earning funds.

Quick-reference

ITEM	NOTE
Nifty 50 / Sensex	Free-float market-cap weighted (50 / 30 stocks)
Free-float	Publicly tradable shares only
Divisor	Keeps index continuous through changes
Price-weighted	Dow, Nikkei (by share price)
TRI	Reinvests dividends (fair benchmark)
Inclusion	Forces passive-fund buying (real flow)

Q&A — Market Indices

Theory and worked scenarios on index construction, weighting, and mechanics.

Q1. How are the Nifty 50 and Sensex constructed, precisely — not just "market-cap weighted"?

Model answer. Both are **free-float market-capitalisation weighted** indices of large-cap stocks (50 for Nifty on NSE, 30 for Sensex on BSE) — weight is proportional to the value of each company's *publicly tradable* shares, specifically excluding promoter, government, and other strategic/locked holdings. Saying only "market-cap weighted" without the free-float qualifier is an incomplete answer and a common interview gap, since free-float is the specific design choice that makes the index reflect genuinely investable value rather than raw company size.

Q2. Worked — free-float weight calculation.

Company P: total market cap ₹5,000 cr, promoter holding 55% (locked/strategic). Company Q: total market cap ₹3,200 cr, promoter holding 15%. Which gets a larger index weight, and by how much (in free-float value terms)?

Model answer. Company P free-float = 45% × 5,000 = ₹2,250 cr. Company Q free-float = 85% × 3,200 = ₹2,720 cr. Despite Company P having a larger total market cap (₹5,000 cr vs ₹3,200 cr), Company Q gets the larger index weight because its free-float value (₹2,720 cr) exceeds P's (₹2,250 cr) — a direct illustration that total company size and index weight are not the same thing once free-float is applied.

Q3. Why doesn't a stock split or bonus issue move the index level, even though it changes the share price and share count?

Model answer. The index uses a **divisor** — a normalising constant recalculated whenever a mechanical, non-value change occurs (splits, bonus issues, constituent additions/deletions, buybacks). A split halves the price and doubles the share count with market cap unchanged; the divisor is adjusted so the index's calculated value doesn't move from this purely mechanical change. The divisor is what allows the index to be a continuous, comparable series over time despite constant structural changes to its constituents.

Q4. What's the practical, real-money consequence of a stock being added to (or removed from) a major index like the Nifty 50?

Model answer. Passive funds and ETFs that track the index are mandated to hold constituents in index-matching proportions — when a stock is added at a scheduled rebalance, every fund tracking that index must buy it (and sell any removed stock), creating real, mechanical buying (or selling) pressure independent of the company's fundamentals. This is why stocks often see a price bump running into a confirmed index-inclusion date, and why "will this stock get added to the index" is itself a research question with a trading implication, separate from fundamental valuation.

Q5. Price-weighted vs market-cap-weighted indices — what's the practical flaw of price-weighting that market-cap weighting avoids?

Model answer. A price-weighted index (like the Dow Jones) gives a higher-priced stock more influence on the index level regardless of the company's actual size — a ₹5,000 stock in a small company would move a price-weighted index more than a ₹200 stock in a much larger company, which is economically arbitrary (share price level is just a function of how many shares are outstanding, not company value). Market-cap weighting (Nifty, Sensex, S&P 500) ties influence to actual company/free-float value, which is both more economically meaningful and matches how a real buy-and-hold portfolio would naturally be weighted.

Q6. What is a total return index (TRI), and why does using a price index instead understate a long-term equity fund's true relative performance?

Model answer. A price index tracks only price changes, ignoring dividends paid by constituents. A total return index reinvests those dividends back into the index calculation. Since an actively or passively managed equity fund also earns and (usually) reinvests dividends from its holdings, benchmarking the fund's total return against a price-only index systematically understates the index's true comparable performance — over long periods, the dividend-reinvestment gap between a price index and its TRI equivalent compounds to a meaningful difference, which is why professional performance reporting should always use the TRI, not the more commonly quoted headline price index.

Q7. Worked — divisor mechanics illustrated with numbers.

A simplified 2-stock index: Stock A, 100 shares free-float at ₹50 (₹5,000), Stock B, 200 shares free-float at ₹40 (₹8,000). Total free-float value = ₹13,000. If the divisor is 130, the index value = 13,000/130 = 100. Stock A now does a 2-for-1 split: 200 shares at ₹25 (still ₹5,000 total). What must the new divisor be to keep the index at 100, and why?

Model answer. Post-split total free-float value is unchanged (200 × 25 = ₹5,000 for A, plus B's unchanged ₹8,000 = ₹13,000 total) — since the underlying value hasn't changed, the divisor also doesn't need to change here (it stays at 130, index value = 13,000/130 = 100). The divisor only needs to be *recalculated* when a change alters the *aggregate free-float value* itself without a corresponding real value change — e.g. when a constituent is swapped for a different one, or free-float percentage changes (a promoter selling down increases free-float value without a price change) — a split alone, since it's price × share-count neutral, is actually one of the simpler cases, though it's still routinely cited alongside divisor mechanics because the underlying principle (isolating real value changes from mechanical ones) is the same.

Q8. Beyond Nifty 50 and Sensex, name and briefly describe two other categories of Indian indices and what each is used for.

Model answer. Broader market-cap indices (Nifty 100/500, Nifty Midcap, Nifty Smallcap) — used to benchmark funds and strategies focused outside just the largest 50 companies, giving a fuller picture of mid- and small-cap segment performance. Sectoral/thematic indices (Bank Nifty, Nifty IT, Nifty Auto, Nifty Pharma, etc.) — used to benchmark sector-focused funds, to trade sector-specific derivatives (Bank Nifty options are among the most actively traded derivatives contracts in India), and by analysts to gauge relative sector performance and rotation trends within the broader market.

The Equity Research Process

The Problem / Why this matters

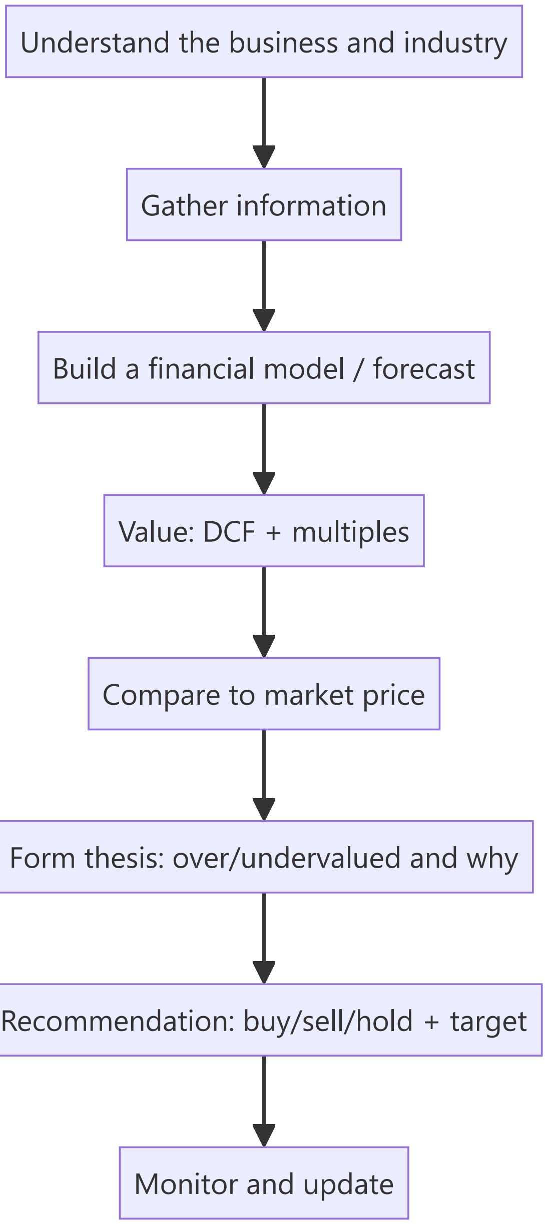
An equity research analyst's job is to form and defend a **view on a stock** — is it worth more or less than the market says, and why? That view drives buy/sell/hold recommendations that clients act on. Knowing the end-to-end process — coverage, information gathering, modelling, valuation, thesis, and the recommendation — is exactly what an equity research or investment-analyst interview tests, often as "walk me through how you'd analyse a company."

Core Idea

Equity research is a repeatable process: **understand the business → model its financials → value it → compare to the market price → form a thesis and recommendation → monitor.** The output is an investment view supported by a model and a written note.

Why it works this way

Markets are mostly efficient, so to add value an analyst must either know something the market underappreciates or interpret known facts better. That requires deeply understanding the business and its drivers, building a defensible forecast, and translating it into a value that can be compared to the price — then articulating *why* the market is wrong.



Full technical content

Step 1 — Understand the business & industry. What does the company do, how does it make money (revenue model, unit economics), what's its competitive position and moat, who are its customers and competitors, and what drives the industry (see fundamental analysis).

Step 2 — Gather information. Annual reports/10-Ks, quarterly results and concalls, management guidance, industry data, channel checks, expert networks, competitor filings, and news. Rigorous, primary-source-driven.

Step 3 — Build the model. A three-statement model with a **driver-based forecast** (revenue drivers → costs → the three linked statements), producing the free cash flows and metrics valuation needs (see the modeling chapter).

Step 4 — Value. Intrinsic (DCF) and relative (comps/multiples), triangulated to a value range and a target price (see applied valuation).

Step 5 — Form the thesis. The core argument: *why* the stock is mispriced. A good thesis is **differentiated** (a view the market doesn't fully hold), **specific**, and **falsifiable** (you can say what would prove you wrong). Identify the **catalysts** that will close the value gap and the **risks**.

Step 6 — Recommendation. Buy / Hold / Sell (or Overweight/Neutral/Underweight) with a **target price** and time horizon, and the expected return vs the current price.

Step 7 — Write and communicate. The research note (initiation or update) and verbal pitch to clients/PMs; then **monitor** results, news and thesis, updating the view as facts change.

The analyst's edge — three sources of alpha:

EDGE	DESCRIPTION
Informational	Knowing a fact the market hasn't priced (rare, and insider info is illegal)
Analytical	Interpreting known facts better (better model/insight)
Behavioural/time-horizon	Exploiting others' biases or short-termism

Most legitimate edge is **analytical** and **behavioural**.

Quality markers of good research: a clear, differentiated thesis; a defensible model with sensible assumptions; explicit catalysts and risks; and intellectual honesty (state what would make you wrong).

Channel checks and primary research in equity analysis

Public filings and models only go so far — the highest-conviction theses are usually backed by **channel checks**: primary research into a company's own ecosystem, run with the same rigour as market research (Part 3-4 of the market-research literature, directly transferable). - **Supplier checks** — calling/meeting key suppliers to gauge order volumes, pricing trends, and capacity utilisation ahead of the company's own disclosure (a leading indicator of revenue). - **Distributor/dealer checks** — for consumer and auto companies, calling a sample of dealers/distributors across regions to triangulate real sell-through (not just sell-in, which a company can inflate near quarter-end by stuffing the channel). - **Customer/end-user surveys** — for B2C names, small structured surveys on brand awareness, repeat-purchase intent, or switching behaviour (identical methodology to Part 5 of market research: sample size, question design, bias avoidance all apply directly). - **Expert networks** — paid platforms connecting analysts to former employees, industry consultants, or specialists for one-off calls; must be used within strict compliance boundaries (no MNPI — material non-public information — can be solicited or used). - **Job-posting and hiring-trend analysis** — a company rapidly hiring in a new geography/function is a public, legal signal of expansion plans, cross-checked against management commentary. - **Store-check/footfall studies** — for retail/QSR names, physically observing footfall and average transaction patterns at a sample of outlets — the equity-research analogue of a market-research "intercept" study (see Market Research, Part 5.6).

Compliance boundary (tested in every legitimate research role's onboarding): any of the above crosses a hard legal line the moment it solicits or knowingly uses **material non-public information** — a supplier confirming "volumes are up" is a legitimate industry read; a supplier disclosing an unreleased quarterly number is insider information, and using it is a criminal offence under India's SEBI (Prohibition of Insider Trading) Regulations. A research analyst must be able to draw this line without hesitation.

A full worked research note — "TechCo India Ltd" (illustrative)

Walking one company through the entire pipeline end to end is the single best interview-prep exercise for this chapter.

Business understanding: TechCo is a mid-cap Indian IT-services company, ₹8,500 crore revenue, historically 14% EBITDA margin, growing 12% p.a., trading at 18x forward P/E vs a peer group average of

22x.

Model (driver-based, simplified): Revenue = existing client base (95% retention) × average contract growth (8%) + new client wins (4% incremental) = ~12% revenue CAGR over the forecast window. Margin: assume 100bps expansion over 3 years from offshore-mix shift and utilisation improvement, taking EBITDA margin from 14% to 15% (this single assumption is the crux of the thesis — see below).

Valuation: DCF using WACC 11.5% (cost of equity via CAPM ~13%, pre-tax cost of debt ~8.5%, D/E modest) and a terminal growth rate of 5% (nominal, in line with long-run Indian GDP-growth assumptions for a mature IT-services business) yields an intrinsic value of ₹1,150/share. Cross-check via relative valuation: applying the peer-average 22x forward P/E (rather than the stock's own 18x) to next-year EPS implies ₹1,190/share — the two methods converge within ~3.5%, a strong triangulation (Part 7.2 of the Market Research handbook's triangulation principle applies identically here).

Thesis: the market is pricing TechCo at a discount to peers (18x vs 22x) because of a perceived margin ceiling; the analyst's differentiated view is that the offshore-mix shift already underway (visible in headcount-location disclosures) will close 100bps of the gap over 3 years, which the market is not yet modelling — this, not revenue growth (already well understood and priced), is the source of edge.

Catalysts: next two quarters' margin prints beating consensus; management commentary on offshore-mix % at the next earnings call; a peer re-rating event that draws attention to the sector's margin trajectory.

Risks: wage inflation offsetting the offshore-mix benefit; a large client loss (client-concentration risk — check top-5-client revenue % in the filings); rupee appreciation compressing margins (a genuine India-specific IT-services risk worth naming explicitly, since revenue is often dollar-denominated while costs are rupee-denominated).

Recommendation: Buy, target ₹1,150-1,190 (DCF/comps range), current price ₹1,000, implying ~15-19% upside over a 12-month horizon; position sized moderately given single-driver (margin) thesis concentration.

What would falsify this thesis: two consecutive quarters of flat or declining offshore-mix % in management disclosures, or a margin miss attributed to wage inflation exceeding the offshore benefit — the analyst commits, in writing, to downgrading on either signal.

Worked examples

Example 1 — the process end to end. Analyst covers an FMCG firm. Understands the business (rural distribution moat); models revenue off volume × price with margin expansion from premiumization; DCFs it and cross-checks EV/EBITDA vs peers → intrinsic ₹1,200 vs market ₹1,000. Thesis: the market underrates margin expansion from premiumization (differentiated view); catalyst: next two quarters of margin beats; risk: rural slowdown. Recommendation: **Buy, target ₹1,200, +20%.**

Example 2 — differentiated vs consensus thesis. Two analysts value the same stock at ₹1,000, exactly the market price, agreeing with consensus. Neither adds value — a research view is only useful when it *differs* from the price and has a reason. The job is to find and defend the gap.

Example 3 — thesis falsification. An analyst's Buy rests on a new product driving 15% volume growth. She states: "If the first two quarters show under 8% volume growth, the thesis is broken and I'll downgrade." That falsifiability is what separates research from cheerleading.

How it is tested in interviews

- **"Walk me through how you'd analyse a company / research a stock."** — Recite: understand the business → gather info → model → value (DCF + comps) → compare to price → form a differentiated thesis with catalysts and risks → recommend with a target → monitor.

- **"What makes a good investment thesis?"** — "Differentiated from consensus, specific, with clear catalysts and risks, and falsifiable — you can state what would prove you wrong."
- **"Where does an analyst's edge come from?"** — "Mostly analytical (better interpretation) and behavioural (exploiting short-termism/biases); informational edge is rare and insider info is illegal."
- **"How do you decide buy vs sell?"** — "Compare intrinsic value/target to the market price; the gap and catalysts drive the rating and expected return."

Traps & common mistakes

- Producing a view that **matches consensus/price** — it adds no value.
- A model with no **thesis** — data without a differentiated argument.
- No **catalysts** (why does the gap close?) or **risks** (what if you're wrong?).
- Not stating what would **falsify** the thesis — a recipe for holding losers.
- Chasing **informational edge** into insider-trading territory.

First-principles recap

- Research = understand → model → value → compare to price → thesis → recommend → monitor.
- Value is only added when your view **differs from the market**, with a reason.
- A good thesis is differentiated, specific, catalyst-driven, and **falsifiable**.
- Edge is mostly analytical and behavioural.
- The output is a rating, a target price, and a defensible note.

Quick-reference

STEP	OUTPUT
Understand business	Revenue model, moat, drivers
Model	Driver-based 3-statement forecast
Value	DCF + multiples → target
Thesis	Why mispriced + catalysts + risks
Recommendation	Buy/Hold/Sell + target + horizon
Edge	Analytical + behavioural

Q&A — The Equity Research Process

A mixed bank of theory and worked scenarios on how an equity analyst actually builds and defends a view.

Q1. Walk me through how you'd research a stock, start to finish.

Model answer. Understand the business and industry (revenue model, moat, competitors) → gather information (filings, concalls, channel checks) → build a driver-based three-statement model → value it via DCF and comps → compare intrinsic value to the market price → form a differentiated, falsifiable thesis with named catalysts and risks → issue a recommendation with a target price and horizon → monitor and update as facts change.

Q2. What are the three sources of an analyst's "edge," and which are legitimate?

Model answer. Informational (knowing a fact the market hasn't priced — rare, and using material non-public information is illegal), analytical (interpreting known, public facts better than consensus), and behavioural/time-horizon (exploiting other investors' biases or short-termism). Legitimate, sustainable edge in a compliant research role is almost entirely analytical and behavioural.

Q3. Two analysts independently value a stock at exactly the current market price. Has either added value?

Model answer. No — a view that matches the price and consensus adds nothing actionable; research only creates value when the analyst's view *differs* from the price, for a stated, defensible reason. Agreement with consensus is not itself wrong, but it isn't a research output worth publishing as a rated call.

Q4. What makes an investment thesis "falsifiable," and why does it matter?

Model answer. A falsifiable thesis states, in advance, the specific evidence that would prove it wrong (e.g. "if offshore-mix % doesn't rise for two consecutive quarters, the margin-expansion thesis is broken and I'll downgrade"). It matters because it forces intellectual honesty and prevents an analyst from silently rationalising every new data point to fit an existing Buy/Sell call — a well-known behavioural trap (confirmation bias) that falsifiability is a direct discipline against.

Q5. What is a channel check, and where is the legal line?

Model answer. A channel check is primary research into a company's own ecosystem — calling suppliers, distributors, or customers to independently gauge demand/pricing/volume trends ahead of official disclosure. The legal line is material non-public information (MNPI): a supplier saying "orders feel stronger this quarter" is a legitimate industry read; a supplier disclosing an actual unreleased number is MNPI, and knowingly trading or advising on it is a criminal offence under SEBI's insider-trading regulations. An analyst must be able to draw this distinction instantly, not as an afterthought.

Q6. Worked scenario — build a one-paragraph thesis from these facts.

Facts: a listed paints company is growing volumes 8% p.a., but raw-material (crude-linked) costs have fallen 12% over two quarters while the company's average selling price has been flat; consensus models flat gross margin for the next year.

Model answer. The market appears to be under-modelling gross-margin expansion: with input costs down 12% and pricing held flat rather than passed through to consumers, gross margin should expand roughly in line with the input-cost decline, which consensus's flat-margin assumption doesn't capture — a differentiated, testable view (falsified if the next two quarters show gross margin flat or down despite the input-cost tailwind, which would suggest the company is instead cutting price to defend volume/share).

Q7. Why is "monitor and update" a formal step in the process, not just an afterthought?

Model answer. A thesis is a hypothesis, not a fact — new quarterly results, management commentary, and competitor moves are continuous tests of it. An analyst who publishes a rating and doesn't systematically re-check it against new data is not doing research, just issuing a one-time opinion; the discipline of monitoring against the pre-stated falsification criteria (Q4) is what makes ongoing coverage credible.

Q8. What's the difference between "informational edge" and insider trading, precisely?

Model answer. Informational edge from *legally obtained, non-material, or already-public-but-underappreciated* information (e.g. noticing a disclosed but overlooked footnote, or aggregating many small legal public data points into an insight nobody else has bothered to compile) is legitimate. The line is crossed the moment the information is both **material** (would move the stock price if disclosed) and **non-public** (not available to the market generally) — using or passing on information meeting both criteria is insider trading regardless of intent or how it was obtained.

Fundamental Analysis: Top-Down & Bottom-Up

The Problem / Why this matters

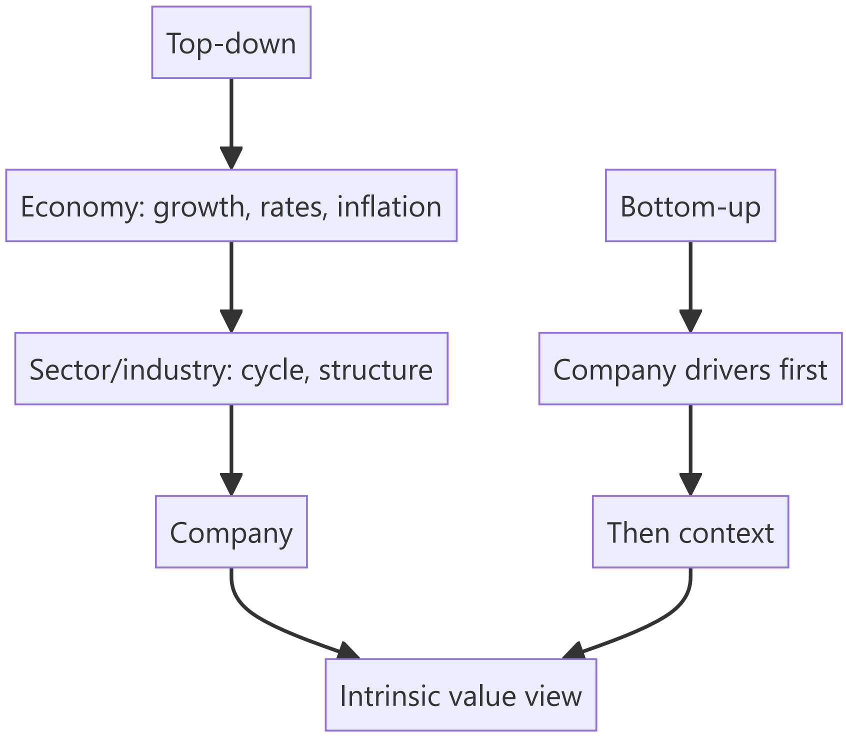
To decide whether a stock is cheap or dear, you must first understand the *fundamentals* — the real economic drivers of the business and its environment. **Fundamental analysis** is the discipline of assessing a company's intrinsic worth from the economy, industry and company up (or down). It's the core skill of equity research and long-term investing, and interviewers probe whether you can structure an analysis rather than just quote a P/E.

Core Idea

Fundamental analysis estimates a company's intrinsic value from its underlying economics. It's done **top-down** (economy → sector → company) and **bottom-up** (company-specific analysis first), usually combining both: understand the macro and industry context, then dig into the specific company's drivers, quality and valuation.

Why it works this way

A company's results are shaped by forces at three levels — the macro economy (growth, rates, inflation), the industry (structure, cycle, competition), and the company itself (strategy, execution, moat). Ignoring any level gives a wrong answer: a great company in a collapsing industry, or a mediocre one riding a boom, will disappoint if you only looked at one layer.



Full technical content

Top-down analysis (economy → sector → stock): 1. **Macro** — GDP growth, interest rates, inflation, currency, policy. Sets the tailwind or headwind (e.g., rate cuts favour rate-sensitives). 2. **Sector/industry**

— pick sectors positioned to benefit; assess industry structure and cycle. 3. **Stock** — pick the best companies within the chosen sectors.

Bottom-up analysis (company first): start with the individual company's fundamentals — competitive position, financials, management, valuation — largely regardless of the macro call, on the belief that a great business at a fair price wins over time. Most research combines both.

Analysing the company — the toolkit:

DIMENSION	WHAT TO ASSESS
Revenue model & unit economics	How it makes money; per-unit profitability
Competitive position / moat	Brand, cost, switching costs, network, scale, IP (Porter's five forces)
Growth drivers	Volume, price, mix, new products, markets
Margins & operating leverage	Profitability trajectory and its drivers
Returns	ROE, ROIC vs cost of capital (value creation)
Balance sheet & cash flow	Leverage, working capital, FCF generation
Management & governance	Track record, capital allocation, alignment
Valuation	Is the price sensible vs intrinsic value?

Qualitative + quantitative. The numbers (financials, ratios) show *what* happened; the qualitative work (moat, management, industry) explains *why* and whether it's sustainable. Great analysis fuses both.

Growth vs value styles. *Value* investors seek cheap stocks relative to fundamentals; *growth* investors pay up for superior growth. Both are fundamental — they weight the drivers differently.

Catalysts. Fundamental value only matters to a trade if something will make the market recognize it — new products, margin inflection, deleveraging, management change, sector re-rating. Identifying catalysts turns a valuation gap into an actionable idea.

Worked examples

Example 1 — top-down chain. Macro: rates are falling and consumption is recovering → favour consumer discretionary. Sector: within it, organized retail is gaining share from unorganized. Stock: pick the best-positioned retailer with a scale/cost moat and clean balance sheet. The macro call, sector selection, and stock pick compound.

Example 2 — bottom-up quality. An analyst ignores the noisy macro and focuses on a company with a durable brand moat, 25% ROIC (well above its ~11% WACC → strong value creation), consistent FCF, and disciplined capital allocation. Even through cycles, a business earning far above its cost of capital compounds value — a classic bottom-up buy.

Example 3 — good company, bad price. A wonderful franchise (high ROIC, wide moat) trades at 60× earnings, pricing in flawless growth for a decade. Fundamentally excellent, but the *price* already reflects it — the fundamental analyst concludes the stock, not the business, is the problem, and passes. Quality ≠ a buy at any price.

Example 4 — a worked Porter's Five Forces on a real sector structure. Applying the framework to Indian organised retail: threat of new entrants is moderate (real estate/capex barriers are high, but e-commerce lowers the barrier for a purely online entrant); bargaining power of buyers is rising (price transparency via apps, easy switching); bargaining power of suppliers is generally low for large retailers (scale gives negotiating leverage over most FMCG suppliers, though it's lower against a must-stock category leader with its own brand power); threat of substitutes is high and growing (quick-commerce, D2C brands bypassing retail entirely); competitive rivalry is intense (multiple well-funded organised players plus unorganised trade still holding majority share). The synthesis an analyst should reach: an industry with intense rivalry and rising buyer power rewards the specific players with genuine structural advantages (private-label margin, supply-chain scale, prime real estate locked in early) over those competing purely on price — exactly the kind of company-specific differentiation a bottom-up analysis (Example 2) should then dig into for the actual stock pick.

Example 5 — separating a cyclical earnings beat from genuine fundamental improvement. A cement company reports 40% YoY EBITDA growth, driving a sharp stock rally. A rigorous fundamental analyst decomposes the beat before extrapolating it: how much came from volume growth (durable, driven by real demand), how much from price/realisation increases (check whether industry-wide, suggesting pricing discipline, or company-specific, suggesting a one-off), and how much from a temporary fuel/input-cost tailwind (coal/pet-coke prices, which are cyclical and can reverse). If the input-cost tailwind explains most of the beat, extrapolating the 40% growth rate forward into a DCF or forward-multiple assumption would materially overstate fair value once costs normalise — the analyst's job is precisely this decomposition, not accepting a reported growth number at face value, directly connecting back to Part 6's "understand the business" discipline applied to a single quarter's result.

How it is tested in interviews

- **"What is fundamental analysis?"** — "Estimating a company's intrinsic value from its underlying economics — the macro, the industry, and the company's own drivers, quality and valuation."
- **"Top-down vs bottom-up?"** — "Top-down starts from the economy and sector and drills to the stock; bottom-up starts with the company's fundamentals. Most research combines both."
- **"How would you analyse a company's competitive position?"** — Porter's five forces / moat sources (brand, cost, switching costs, network, scale, IP) and returns vs cost of capital.
- **"How do you know a company creates value?"** — "ROIC above WACC — it earns more than the cost of the capital it uses."
- **"Why do you need a catalyst?"** — "A valuation gap only pays off if something makes the market recognize it — new products, a margin inflection, deleveraging, a re-rating."

Traps & common mistakes

- Doing only **one level** (company without industry, or macro without the company).
- Confusing a **great business** with a **great stock** — price matters.
- All numbers, no **qualitative** judgment (moat, management, sustainability).
- Ignoring **ROIC vs WACC** — growth without returns above cost of capital destroys value.
- No **catalyst** — a permanently "cheap" stock can stay cheap.

First-principles recap

- Fundamental analysis derives intrinsic value from economy, industry and company.
- **Top-down** (macro → sector → stock) and **bottom-up** (company first) — usually both.
- Fuse quantitative (financials, ROIC) with qualitative (moat, management).
- Value creation = **ROIC > WACC**; a great business isn't a buy at any price.
- Catalysts turn a valuation gap into an actionable idea.

Quick-reference

LEVEL	FOCUS
Macro	Growth, rates, inflation, policy
Sector	Structure, cycle, competition
Company	Moat, drivers, margins, returns, management
Value creation	ROIC > WACC
Catalyst	What makes the market re-rate

Q&A — Fundamental Analysis: Top-Down & Bottom-Up

Theory and worked scenarios on structuring a company/industry/macro analysis.

Q1. Walk through the top-down analysis chain with a concrete example.

Model answer. Start at the macro level (e.g. interest rates are falling and consumption is recovering, favouring rate-sensitive and discretionary sectors), narrow to sector/industry (within consumer discretionary, organised retail is gaining structural share from unorganised players), then to the individual stock (pick the best-positioned retailer within that favoured sector — strongest balance sheet, clearest cost/scale moat). Each layer narrows the opportunity set, and the three calls (macro, sector, stock) compound into the final investment decision.

Q2. How does bottom-up analysis differ philosophically from top-down, and why might a bottom-up investor deliberately ignore a bearish macro call?

Model answer. Bottom-up starts with the individual company's own fundamentals — competitive position, unit economics, management quality, valuation — largely independent of the macro backdrop, on the philosophy that a truly great business bought at a fair price will compound value over time regardless of near-term macro noise, and that macro forecasting itself is notoriously unreliable. A bottom-up investor might deliberately hold a high-quality compounder through a macro downturn they can't predict or time well, betting that business quality matters more over a multi-year holding period than getting the macro call right.

Q3. Worked — is this company creating or destroying value?

Company A has ROIC of 9% and WACC of 11%. Company B has ROIC of 18% and WACC of 12%. Both are growing revenue at 15% per year.

Model answer. Company A is *destroying* value despite growing: it earns 9% on invested capital while that capital costs 11%, so every rupee of growth capital deployed actually erodes value — a classic trap of "growth" that looks impressive on the income statement but destroys shareholder value, since growth in this case simply means investing more capital at a below-cost-of-capital return. Company B is creating substantial value: 18% ROIC against a 12% WACC means each rupee of growth capital earns a 6-point spread above its cost — this is genuine, value-accretive growth. The lesson: revenue growth alone says nothing about value creation; it must be paired with ROIC vs WACC.

Q4. What's the difference between a "great business" and a "great stock," and why does this distinction matter for a recommendation?

Model answer. A great business has strong fundamentals — high and durable ROIC, a real competitive moat, good management. A great stock additionally requires that the *price* doesn't already fully (or more than fully) reflect those strengths. A wonderful business trading at 60x earnings, pricing in a decade of flawless execution, can be a poor investment from the current price even though the underlying business is excellent — the fundamental analyst's job is to separate "is this a good company" from "is this a good price to buy it at," and a recommendation must answer the second question, not just the first.

Q5. Name three sources of competitive moat and give a brief example of each.

Model answer. Cost advantage — a company can produce/deliver at structurally lower cost than competitors (e.g. a scale-driven low-cost manufacturer), letting it either underprice rivals or earn superior margins at the same price. Switching costs — customers face real friction (financial, operational, or data/integration lock-in) in moving to a competitor, common in enterprise software and banking relationships. Network effects — the product/service becomes more valuable as more users join (a payments platform, a marketplace), creating a self-reinforcing advantage that's difficult for a smaller competitor to replicate even with a technically comparable product.

Q6. Why is a "catalyst" necessary even when an analyst has correctly identified that a stock is undervalued?

Model answer. An undervalued stock can, in principle, stay undervalued indefinitely if nothing prompts the broader market to re-examine and re-price it — value alone isn't self-executing. A catalyst (an earnings beat that reveals a hidden margin trend, a management change, a sector re-rating event, a new product launch) is the specific mechanism by which the market's attention is drawn to the mispricing, converting a "this is cheap" observation into an actionable, timed investment thesis. Without an identifiable catalyst, a valuation gap is a real but potentially open-ended and hard-to-time bet.

Q7. A company shows strong revenue growth and margin expansion. What qualitative checks would you still want to run before concluding it's a genuinely strong fundamental story?

Model answer. Check whether growth is broad-based (organic, across products/geographies) or narrowly driven by a single large customer/contract that could be lost; check whether margin expansion is structural (from genuine operating leverage or cost efficiency) or a one-off (a temporary input-cost tailwind, a change in accounting treatment, deferred/cut discretionary spending that will need to resume); assess management's capital-allocation track record (are they reinvesting at good returns, or empire-building); and assess whether the competitive position is actually strengthening or whether the current results reflect a favourable but temporary industry cycle rather than durable company-specific advantage.

Q8. How would you structure an answer to "walk me through how you'd analyse an unfamiliar company in 30 minutes"?

Model answer. State the structure explicitly before diving in: (1) understand the revenue model and unit economics — how does it actually make money; (2) assess the competitive position/moat and industry structure; (3) scan the financial trend — revenue growth, margin trajectory, ROIC vs WACC, balance-sheet health, free-cash-flow generation; (4) form a quick view on management/governance quality from what's available (capital-allocation history, related-party transactions if flagged); (5) sanity-check the current valuation against the quality/growth profile just assessed. Explicitly noting the structure up front, even under time pressure, signals a repeatable process rather than an ad hoc read of whatever numbers happen to be in front of you.

Three-Statement Modeling for Research

The Problem / Why this matters

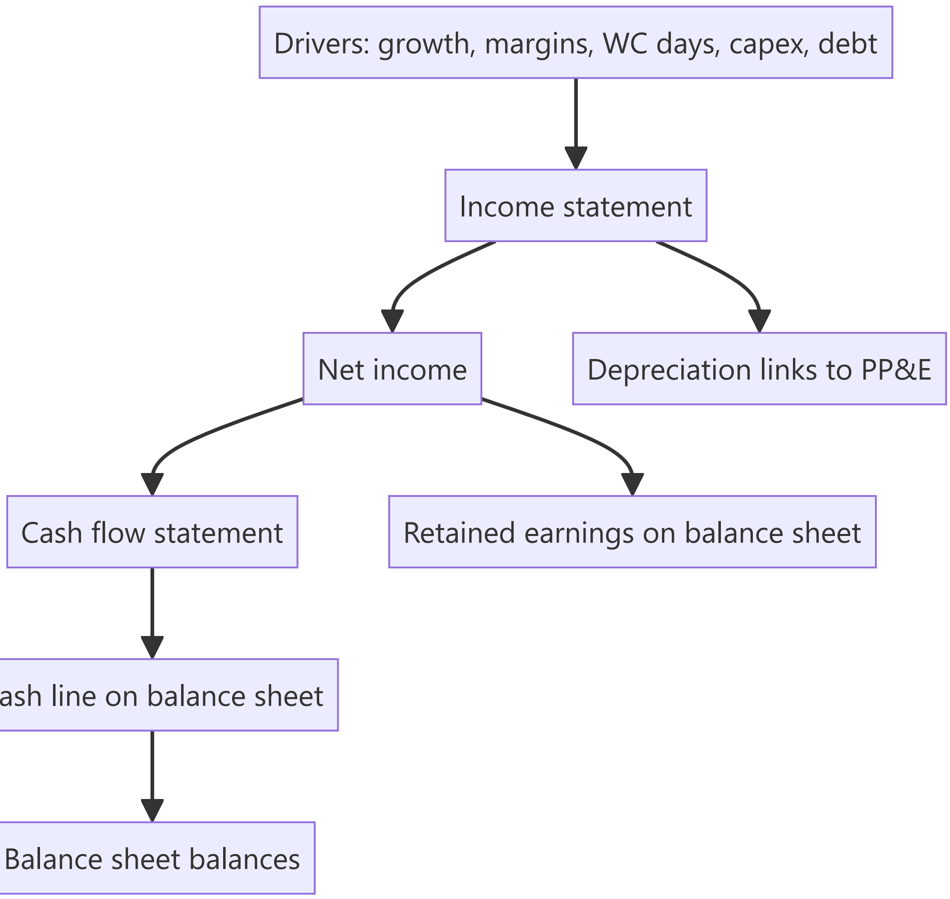
A research view needs numbers: a forecast of what the company will earn and how much cash it will generate. The **three-statement model** — a linked income statement, balance sheet and cash flow driven off a few assumptions — is the workhorse that produces those numbers and feeds valuation. Building and defending one is a core equity-research and IB skill, and "walk me through a 3-statement model" is a standard interview question.

Core Idea

A three-statement model projects the **income statement, balance sheet and cash flow statement**, fully linked so they stay internally consistent, from a set of **operating drivers** (revenue growth, margins, working-capital days, capex, financing). Change an assumption and the whole model — including free cash flow — updates coherently.

Why it works this way

The three statements are connected in reality (net income flows to retained earnings and to cash flow; depreciation links the P&L and the asset base; debt links financing and interest), so a credible forecast must respect those links. Driving everything off a small set of assumptions makes the model transparent, defensible, and easy to sensitize.



Full technical content

Build order (typical): 1. **Revenue** — from drivers (volume × price, or growth rate; segment build). 2. **Costs & margins** — COGS and opex as % of revenue or per-unit; get to EBITDA and EBIT. 3. **Supporting schedules** — depreciation (off PP&E + capex), working capital (off days: DSO, DIO, DPO), debt & interest, equity. 4. **Complete the income statement** — interest (from the debt schedule), taxes, net income. 5. **Build the cash flow** — start from net income, add back D&A, adjust for working-capital changes (from the WC schedule), subtract capex, add/subtract financing. 6. **Complete the balance sheet** — cash (from CF), PP&E (prior + capex − depreciation), working-capital items, debt, retained earnings (prior + NI − dividends). 7. **Check it balances** — assets = liabilities + equity every year; cash ties to the cash-flow statement.

The links that make it consistent: - Net income → retained earnings (BS) and top of cash flow (CF). - Depreciation → reduces PP&E (BS), non-cash add-back (CF), expense (IS). - Capex → increases PP&E (BS), investing outflow (CF). - Working-capital changes → BS balances and CF operating adjustment. - Debt → BS balance, interest on the IS, financing flows on the CF.

Circularity. Interest depends on debt, debt depends on cash, cash depends on interest — a circular reference. Handled with **iterative calculation** (Excel's circular switch) or a **circularity breaker** (average debt / copy-paste). A classic modeling interview question.

Best practices: separate **inputs** (assumptions) from **calculations**; drive off a few clear drivers; keep formulas consistent across columns; build **checks** (balance-sheet balances, cash ties to CF); make scenarios toggle-able; label units. A clean, auditable model beats a clever one.

From model to valuation: the model outputs the **free cash flows** (FCFF/FCFE) that the DCF discounts and the **metrics** (EBITDA, EPS) that multiples apply to — so the model is the engine behind the valuation.

Worked examples

Example 1 — driver-based revenue. A retailer: stores 100 growing to 120, revenue/store ₹5 cr growing 4%/yr. Year-1 revenue = 100 × 5 = ₹500 cr; Year-2 = 110 × 5.2 ≈ ₹572 cr. Costs at 85% of revenue → EBITDA 15% margin. The forecast flows from store count and sales density, not a guessed growth number — defensible and sensitizable.

Example 2 — the balancing check. Net income ₹70 cr, dividends ₹20 cr → retained earnings up ₹50 cr. Cash flow shows net change in cash +₹30 cr. On the balance sheet, cash rises ₹30 cr and retained earnings rise ₹50 cr; other moves (PP&E, debt, working capital) must net so that assets = liabilities + equity. If it doesn't balance, there's a linking error — the check catches it.

Example 3 — the circularity. Adding a revolver whose interest depends on the average debt balance creates a circular reference (interest ↔ debt ↔ cash). Turning on iterative calculation lets Excel converge; alternatively, compute interest on opening (not average) debt to break the loop. Interviewers love this: "where's the circular reference in a 3-statement model?"

Example 4 — full working-capital schedule, driven off days. A company's model needs a working-capital forecast, not a guessed flat number. Given Year-1 revenue ₹800 cr, COGS ₹520 cr, and target days of DSO (Days Sales Outstanding) 45, DIO (Days Inventory Outstanding) 60, DPO (Days Payable Outstanding) 40:

```
Receivables = Revenue × DSO/365 = 800 × 45/365 ≈ ₹98.6 cr
Inventory   = COGS × DIO/365    = 520 × 60/365 ≈ ₹85.5 cr
Payables    = COGS × DPO/365    = 520 × 40/365 ≈ ₹57.0 cr
Net Working Capital = Receivables + Inventory − Payables = 98.6 + 85.5 − 57.0 = ₹127.1 cr
```

If Year-2 revenue grows to ₹880 cr and COGS to ₹560 cr with the *same* days assumptions, NWC becomes 880×45/365 + 560×60/365 − 560×40/365 ≈ 108.5 + 92.1 − 61.4 = ₹139.2 cr — an increase of ₹12.1 cr, which is the **ΔNWC** that flows as a cash *outflow* in the FCFF build (Valuation chapters). This is the concrete mechanism behind the abstract statement "growth costs cash": every rupee of extra revenue tied up in unpaid receivables and unsold inventory (net of the payables the company itself hasn't paid yet) is cash the business doesn't get to keep, and driving it off days — not a flat guess — is what makes the forecast defensible when a PM asks "why did you assume that number."

Example 5 — sanity-checking a model's implied ratios before presenting it. A junior analyst's 5-year forecast shows revenue growing 18% annually while receivable days *fall* from 45 to 30 over the same period, with no stated operational change (no new collections process, no shift in customer mix) driving that improvement. A reviewer should flag this immediately: days ratios don't improve by themselves, and an unexplained, favourable drift in a working-capital assumption is one of the most common ways an overly optimistic forecast quietly inflates projected free cash flow — checking that every year-over-year change in a driver assumption has a stated, defensible reason (not just "the formula extrapolated it") is a core model-review discipline, not a nice-to-have.

How it is tested in interviews

- **"Walk me through a three-statement model."** — Build revenue and costs → schedules (depreciation, working capital, debt) → complete the income statement → cash flow from net income → balance sheet → check it balances. Emphasize the *links*.
- **"How are the three statements linked?"** — Net income → retained earnings and CF; depreciation → PP&E, IS and CF; capex → PP&E and CF; debt → BS, interest and financing.
- **"Where is the circular reference?"** — "Interest ↔ debt ↔ cash. Solved with iterative calculation or by using opening-debt interest."
- **"What checks do you build?"** — "The balance sheet must balance every year, and cash on the balance sheet must equal the cash-flow statement's ending cash."

Traps & common mistakes

- Hard-coding numbers instead of driving off **assumptions**.
- Forgetting a **link** (e.g., depreciation not flowing to PP&E) so the model doesn't balance.
- Not handling **circularity** (or leaving it broken).
- No **checks** — errors go undetected.
- Over-complex models no one (including you) can audit.

First-principles recap

- A 3-statement model links IS, BS and CF, driven off a few operating assumptions.
- Build revenue → costs → schedules → complete IS → CF → BS → **check it balances**.
- Net income, depreciation, capex, working capital and debt are the key links.
- Handle **circularity** (interest ↔ debt ↔ cash) with iteration or a breaker.

- The model outputs the FCF and metrics that feed valuation.

Quick-reference

LINK	FROM → TO
Net income	IS → retained earnings (BS) & CF
Depreciation	IS expense, CF add-back, PP&E ↓
Capex	CF outflow, PP&E ↑
Working capital	CF adjustment, BS items
Debt	BS, interest (IS), financing (CF)
Check	BS balances; CF cash = BS cash

Q&A — Three-Statement Modeling for Research

Theory and worked scenarios on building and troubleshooting a linked financial model.

Q1. Walk me through building a three-statement model from scratch.

Model answer. Start with revenue, driven by explicit operating drivers (volume × price, or a growth rate applied to a segment build) rather than a guessed top-line number. Build costs and margins off revenue to reach EBITDA and EBIT. Build supporting schedules — depreciation (linked to the PP&E and capex build), working capital (driven by DSO/DIO/DPO days), and a debt schedule (linked to financing needs and interest). Complete the income statement with interest (from the debt schedule) and taxes to reach net income. Build the cash flow statement starting from net income, adding back non-cash D&A, adjusting for working-capital changes, subtracting capex, and reflecting financing flows. Complete the balance sheet using the cash flow's ending cash, the PP&E roll-forward, working-capital balances, debt, and retained earnings (prior balance plus net income minus dividends). Finally, check that assets equal liabilities plus equity every single year — if it doesn't balance, there is a linking error somewhere.

Q2. Where exactly does circularity arise in a three-statement model, and name two ways to handle it.

Model answer. Circularity arises because interest expense depends on the debt balance, the debt balance depends on the cash flow (which determines whether the company draws on or pays down a revolver), and cash flow itself depends on net income, which depends on interest expense — a closed loop. Two standard fixes: (1) enable iterative calculation (Excel's circular-reference setting), letting the model converge numerically over repeated recalculation passes; (2) break the circularity structurally by computing interest on the *opening* (prior period) debt balance rather than the average balance, which removes the current period's own cash flow from the interest calculation entirely.

Q3. Worked — trace how a ₹50 cr increase in capex flows through all three statements.

A company increases its planned capex by ₹50 cr in Year 1 versus the prior forecast, funded by drawing on its revolver (debt). Tax rate 25%, no immediate change to depreciation policy in Year 1 itself.

Model answer. Income statement: no immediate Year-1 P&L impact from the capex itself (capex is capitalised, not expensed) — though the higher PP&E base will increase future years' depreciation expense. Balance sheet: PP&E rises by ₹50 cr (gross additions); debt rises by ₹50 cr (the revolver draw funding it) — assuming no cash was used, cash is unaffected by the capex itself in this financing structure. Cash flow statement: investing activities show a ₹50 cr outflow (capex); financing activities show a ₹50 cr inflow (the revolver draw) — the two roughly offset in the net-change-in-cash line, but investing and financing cash flow, viewed separately, both move by ₹50 cr in opposite directions. The check: PP&E is up ₹50 cr and debt is up ₹50 cr on the balance sheet, matching the offsetting ₹50 cr investing outflow and ₹50 cr financing inflow on the cash flow statement — everything ties.

Q4. Why must depreciation be linked to both the PP&E schedule and the income statement, rather than being entered as a flat assumption in each place separately?

Model answer. If depreciation is entered independently in the income statement and in the PP&E roll-forward, the two numbers can drift out of sync (e.g. an analyst updates the capex assumption but forgets to update the resulting depreciation expense to match), silently breaking the model's internal consistency and causing the balance sheet to stop balancing without an obvious error message. Linking depreciation as a single calculated schedule feeding both the income statement (as an expense) and the PP&E roll-forward (as a reduction) is what makes the model a genuinely connected machine rather than three separately-typed documents that happen to sit next to each other.

Q5. What are the two balance-sheet-integrity checks every well-built model should include, and what does a failure of each one tell you?

Model answer. (1) Assets = Liabilities + Equity, checked every forecast year — a failure means there's a broken link somewhere in how an item flows from one statement to another (a classic culprit: net income not fully flowing to retained earnings, or a working-capital change not correctly reflected on both the cash flow and balance sheet). (2) Cash on the balance sheet equals the ending cash balance from the cash flow statement — a failure specifically flags an error in the cash-flow build itself (a financing or investing line not correctly captured) even if the balance sheet appears superficially to balance via some other, unrelated plug.

Q6. Why is "driving off assumptions" (rather than hard-coding forecast numbers) considered a best practice, beyond just being tidier?

Model answer. A driver-based model (e.g. revenue = stores × sales-per-store, rather than a bare "revenue grows 12%" hard-coded number) is transparent about *why* the forecast is what it is, defensible in front of a portfolio manager or interviewer who will ask what's driving the number, and mechanically sensitisable — changing one assumption (e.g. sales-per-store growth from 4% to 6%) automatically ripples correctly through the whole model. A hard-coded forecast number hides the underlying logic, can't be meaningfully stress-tested, and is a strong signal in an interview or work-sample review that the candidate hasn't actually thought through the business drivers.

Q7. A three-statement model outputs FCFF for a DCF. What specific line items from the model does the analyst need to pull, and what must they be careful not to double-count?

Model answer. FCFF = EBIT × (1 − tax rate) + D&A − Capex − ΔNWC, so the analyst pulls EBIT (from the income statement), D&A (from the depreciation schedule), capex (from the investing section of the cash flow build), and the change in net working capital (from the working-capital schedule). The care point: EBIT must be taxed as if unlevered (ignoring the model's actual interest expense) specifically to avoid double-counting the interest tax shield, which is captured separately inside WACC when the DCF discounts these cash flows — pulling the model's actual (post-interest) net income and working backward incorrectly is a common error that inflates or corrupts the FCFF figure.

Q8. Interviewer asks: "Your model isn't balancing — where do you look first?" What's a structured troubleshooting approach?

Model answer. First check the most common culprits in order: (1) does net income correctly flow to both retained earnings on the balance sheet and the top of the cash flow statement — a mismatch here is the single most frequent error; (2) does the depreciation figure match between the income statement expense and the PP&E roll-forward reduction; (3) are all working-capital changes reflected consistently on both the cash flow statement's operating adjustments and the balance sheet's corresponding line items; (4) does the debt schedule's ending balance match what's shown on the balance sheet, and does the associated interest expense match what's in the income statement. Working through these links systematically, rather than randomly changing numbers until the model happens to balance, is both faster and signals real understanding of how the statements connect.

Applied Equity Valuation

The Problem / Why this matters

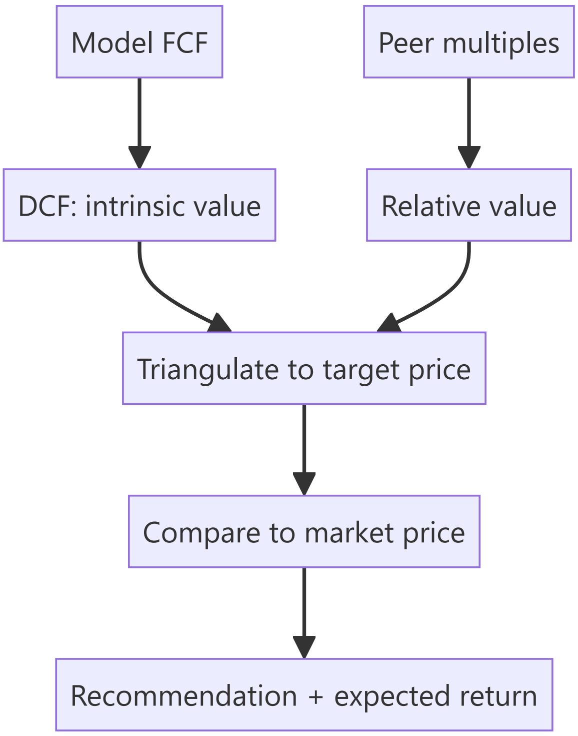
Research turns analysis into a **number** — an estimate of what a share is worth — so it can be compared to the market price. This chapter applies the valuation toolkit (covered deeply in the Valuation book) specifically to **equity research**: how analysts actually use DCF and multiples to set a target price and a recommendation. Interviewers test whether you can value a company end to end and defend the assumptions.

Core Idea

Applied equity valuation triangulates **intrinsic value** (DCF) and **relative value** (multiples vs peers) into a target price. The gap between that target and the market price — plus catalysts — drives the buy/sell/hold recommendation and the expected return.

Why it works this way

No single method is definitive: DCF is theoretically pure but assumption-sensitive; multiples reflect the market's current view but can be collectively wrong. Using both and cross-checking gives a value *range* you can defend, and forces you to understand *why* they differ.



Full technical content

DCF in research: - Project 5–10 years of **free cash flow** (FCFF) from the model. - Discount at **WACC**; add a **terminal value** (Gordon growth or exit multiple). - Sum to **enterprise value**; bridge to **equity value** (subtract net debt, minority interest; add associates); divide by diluted shares → **intrinsic price**. - Run **sensitivity** on WACC and terminal growth (the two swing factors) and present a range.

Relative valuation (multiples): - Pick the right multiple for the sector: **EV/EBITDA** (capital-structure neutral, most sectors), **P/E** (equity-level, mature/stable), **EV/Sales** (loss-making/high-growth), **P/B** (banks/financials), **PEG** (growth-adjusted). - Apply the peer multiple to the company's metric → implied value. - Adjust for differences in growth, margins, returns and risk vs peers.

Setting the target price. Weight the methods (e.g., 60% DCF, 40% comps), or use a football-field range and pick a point. The **12-month target price** implies an expected return vs the current price, which maps to the rating:

EXPECTED RETURN	TYPICAL RATING
> +15%	Buy / Overweight
−10% to +15%	Hold / Neutral
< −10%	Sell / Underweight

Bank/financials valuation uses P/B and ROE (excess-return / residual-income models), and DDM — since FCFF/EV multiples don't work for financials.

High-growth/loss-makers rely on revenue multiples, unit economics, and a path-to-profitability DCF with explicit scenarios.

Sanity checks: does the implied terminal multiple make sense? Is terminal growth below GDP? Does the target imply a reasonable forward P/E? Reconcile DCF and comps — a large gap needs an explanation.

Worked examples

Example 1 — DCF to target. FCFF grows to ₹120 cr in year 5; WACC 11%, terminal growth 4%. Terminal value = 120 × 1.04 / (0.11 − 0.04) = ₹1,783 cr, discounted back; sum of PV of FCFs + PV of TV = enterprise value ₹1,400 cr. Less net debt ₹200 cr = equity value ₹1,200 cr; ÷ 100 mn shares = **₹120/share** intrinsic. Market price ₹100 → ~20% upside → **Buy**.

Example 2 — comps cross-check. Peers trade at 12× EV/EBITDA. The company's EBITDA is ₹130 cr → implied EV ₹1,560 cr; less net debt ₹200 cr = equity ₹1,360 cr ÷ 100 mn = **₹136/share**. The DCF said ₹120, comps say ₹136 — a defensible range of ₹120–136 vs a ₹100 price; the analyst sets a target around ₹128 and rates it Buy.

Example 3 — why DCF and comps differ. DCF (₹120) is below comps (₹136) because the analyst's forecast is more conservative than the growth the market is pricing into peers. Stating *why* they differ — and which you trust — is the analytical value-add. If you think the market's peer growth is too optimistic, lean on the DCF.

Example 4 — a worked football-field range across methods. An analyst triangulates a target price from four methods for the same company: DCF gives ₹115-135/share (a range from WACC/growth sensitivity), EV/EBITDA comps give ₹128-142/share (using the peer group's 25th-75th percentile multiple range, not a single point multiple), P/E comps give ₹120-138/share, and a dividend-discount-model cross-check (for this dividend-paying, mature name) gives ₹110-125/share. Plotting all four ranges as horizontal bars on one chart (the "football field") immediately shows where they overlap — roughly ₹120-135/share across all four methods — which is a far more defensible target range to present to an investment committee than any single method's point estimate, since the overlap itself is evidence that the conclusion isn't an artifact of one method's specific assumptions.

Example 5 — choosing the right multiple when peers have different capital structures. Two peer companies in the same sector look very different on P/E (Peer A trades at 18x, Peer B at 28x) but nearly identical on EV/EBITDA (both around 11x) — the P/E gap is explained almost entirely by Peer B carrying much less debt than Peer A, so more of its enterprise value sits in the equity layer (P/E), inflating the P/E multiple relative to A's more debt-financed structure without reflecting any genuine difference in operating quality. This is exactly why EV/EBITDA (Part 10's "capital-structure neutral" framing) is generally preferred over P/E for comparing companies with different leverage levels — an analyst who compares two differently-levered peers on P/E alone risks drawing a completely wrong conclusion about relative operating performance.

How it is tested in interviews

- **"How would you value this company?"** — "DCF for intrinsic value — project FCFF, discount at WACC, add terminal value, bridge EV to equity, per share — cross-checked with peer multiples (EV/EBITDA, P/E), then triangulate to a target price and compare to the market."
- **"Which multiple would you use for a bank?"** — "P/B with ROE, or a residual-income/DDM model — EV/EBITDA and FCFF don't work for financials."
- **"Your DCF says 120, comps say 136 — what do you do?"** — "Present the range, explain the difference (my forecast is more conservative than the growth peers price in), and lean on the method whose assumptions I trust more."
- **"How does the target price map to a rating?"** — "The implied 12-month return vs the current price: strong upside → Buy, roughly fair → Hold, downside → Sell."

Traps & common mistakes

- Relying on **one method** — always triangulate DCF and comps.
- Using the **wrong multiple** for the sector (EV/EBITDA on a bank).
- Terminal value assumptions that imply **absurd** growth or multiples.
- A target price with no **sensitivity** or range.
- Not explaining **why** DCF and comps differ.

First-principles recap

- Applied valuation triangulates **DCF (intrinsic)** and **multiples (relative)** into a target price.
- Bridge DCF enterprise value to equity value to a per-share number.
- Pick the sector-appropriate multiple; financials use P/B/ROE and DDM.
- The target's implied return vs the price sets the **rating**.
- Explain and reconcile any gap between methods.

Quick-reference

METHOD	USE
DCF (FCFF/WACC/TV)	Intrinsic value
EV/EBITDA	Most sectors, capital-neutral
P/E	Mature/stable, equity-level
P/B + ROE / DDM	Banks & financials
EV/Sales	Loss-making/high-growth
Target → rating	Implied return vs price

Q&A — Applied Equity Valuation

Theory plus fully-solved numerical problems on turning a model into a target price and rating.

Q1. How would you value a company for a research recommendation?

Model answer. Triangulate two independent methods: a DCF (project FCFF, discount at WACC, add terminal value, bridge enterprise value to equity value, divide by diluted shares) for intrinsic value, and relative valuation (apply the appropriate peer multiple — EV/EBITDA, P/E, EV/Sales, or P/B depending on sector) for relative value. Weight or range the two into a target price, compare to the market price, and set a rating from the implied return.

Q2. Worked — DCF to target price.

FCFF in year 5 = ₹150 cr, WACC = 10.5%, terminal growth = 4%. Sum of PV of years 1-5 FCFF = ₹480 cr. Net debt = ₹250 cr. Diluted shares = 80 mn.

Model answer.

```
TV = 150 × 1.04 / (0.105 − 0.04) = 156 / 0.065 = ₹2,400 cr (undiscounted, at year 5)
PV(TV) = 2,400 / (1.105)^5 = 2,400 / 1.6763 ≈ ₹1,431.6 cr
Enterprise Value = 480 + 1,431.6 = ₹1,911.6 cr
Equity Value = 1,911.6 − 250 = ₹1,661.6 cr
Value per share = 1,661.6 / 80 = ₹20.77
```

If the market price is ₹17, upside ≈ (20.77–17)/17 ≈ 22% → a Buy on the standard >15% threshold.

Q3. Worked — reconciling DCF with a comps cross-check.

Continuing Q2: peers trade at 9.5× EV/EBITDA; the company's current-year EBITDA is ₹260 cr.

Model answer.

```
Implied EV = 260 × 9.5 = ₹2,470 cr
Implied Equity Value = 2,470 − 250 = ₹2,220 cr
Implied value per share = 2,220 / 80 = ₹27.75
```

DCF says ₹20.77, comps say ₹27.75 — a real gap. The analyst must explain it (e.g. "peers are pricing in faster medium-term growth than my base-case forecast assumes") and state which figure they trust more, rather than silently averaging the two or picking the more favourable number without justification.

Q4. Why doesn't EV/EBITDA work for valuing a bank?

Model answer. EV/EBITDA assumes debt is a financing choice separate from operations — but for a bank, deposits (debt-like liabilities) *are* the raw material of the business, and "EBITDA" is not economically meaningful when interest income/expense is the core operating activity, not a financing afterthought. Banks are instead valued on **P/B linked to ROE** (a bank earning above its cost of equity should trade above 1x book, and vice versa) or a **dividend-discount/residual-income model**.

Q5. Worked — P/B via ROE for a bank.

Bank's sustainable ROE = 15%, cost of equity = 12%, long-run growth = 8%.

Model answer. Using the Gordon-growth-derived P/B relationship:

```
P/B = (ROE − g) / (Ke − g) = (0.15 − 0.08) / (0.12 − 0.08) = 0.07 / 0.04 = 1.75x
```

If current book value per share is ₹200, the implied fair value is ₹200 × 1.75 = **₹350/share**. The intuition: a bank earning ROE above its cost of equity (15% > 12%) deserves to trade above 1x book, and the multiple compresses as the ROE-Ke spread narrows.

Q6. A high-growth, loss-making company can't be DCF'd on near-term FCFF — how do you value it?

Model answer. Use revenue multiples (EV/Sales) benchmarked to comparable growth-stage peers as the primary cross-check, alongside a longer-horizon DCF with an explicit, well-reasoned path to profitability (modelling the margin trajectory year by year rather than assuming near-term steady-state margins), and always stress-test the terminal-margin assumption specifically, since that single assumption typically drives most of the valuation for a pre-profit name.

Q7. What's wrong with presenting a single-point target price with no range?

Model answer. It implies false precision about inherently uncertain inputs (WACC, terminal growth, peer multiples all carry real estimation error) and hides how sensitive the conclusion is to assumptions at the edge of a reasonable range. A credible target price is always accompanied by a sensitivity table (e.g. WACC × terminal growth) or a football-field chart spanning the methods used, so a reader can see how much of the "Buy" case depends on optimistic inputs.

Q8. Your DCF and comps disagree by more than 10%. Is that a red flag on your work?

Model answer. Not automatically — the two methods answer different questions (DCF reflects the analyst's own forecast, comps reflect what the market is currently paying for similar growth/risk) and can legitimately diverge when the analyst's view differs from what's priced into peers. It becomes a red flag only if the analyst can't articulate *why* they diverge — an unexplained gap suggests an error somewhere (a modelling mistake, a wrong peer set, or an unrealistic terminal assumption) rather than a genuine, defensible difference of view.

The Research Note & Investment Thesis

The Problem / Why this matters

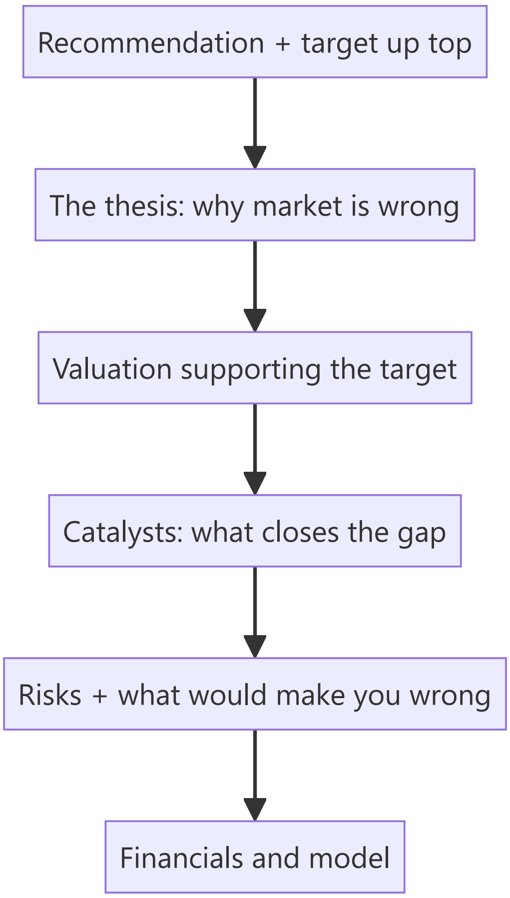
Analysis is worthless if it isn't communicated persuasively. The **research note** is the analyst's product — the written argument that a stock is mispriced, structured so a busy portfolio manager can grasp the thesis, valuation, catalysts and risks in minutes. Writing a tight note and articulating a crisp thesis is exactly what an equity-research interview tests (often via a written case or "pitch me a stock"). This chapter is the writing/communication counterpart to the analysis.

Core Idea

A research note communicates a **differentiated investment thesis** — why the market is wrong about this stock — supported by valuation, backed by catalysts, and honest about risks, ending in a clear **recommendation and target price**. Structure and clarity are as important as the analysis.

Why it works this way

PMs read dozens of notes; they act on the ones that are clear, differentiated, and defensible. A note that buries the thesis, agrees with consensus, or hides the risks gets ignored. So the note leads with the conclusion, states the *variant view*, quantifies the value, and pre-empts objections.



Full technical content

Types of note: initiation (full deep-dive launching coverage), **update/earnings note** (reaction to results or news), **thematic** (sector/idea). All share the core structure.

Anatomy of a research note: 1. **Recommendation & target** — rating (Buy/Hold/Sell), 12-month target price, and expected return — *at the top*. 2. **Investment thesis** — 3–5 crisp points on *why the market is wrong* and you're right (the variant view). 3. **Valuation** — the DCF and/or multiples supporting the target, with the key assumptions. 4. **Catalysts** — the specific events that will close the value gap and their timing. 5. **Risks** — the main downside risks and what would break the thesis. 6. **Business overview & financials** — the model, drivers, and key metrics.

The investment thesis — the heart. A great thesis is: - **Differentiated** — different from consensus (a variant perception); if you agree with the price, there's no trade. - **Specific & supported** — concrete drivers and numbers, not vague optimism. - **Falsifiable** — you can state exactly what would prove you wrong. - **Catalyst-driven** — a reason the market will come around.

The "variant view" test. Ask: *what do I believe that the market doesn't, and why am I right?* If you can't answer, you don't have a thesis — you have a description.

Writing craft: lead with the conclusion (BLUF — bottom line up front); be concise; use specific numbers; separate fact from opinion; make it skimmable (headers, bullets, a chart); and be intellectually honest about risks (it builds credibility).

Position on consensus. Note where you sit vs the Street (above/below consensus estimates and rating) — the value is in the *difference*, and being non-consensus and right is where alpha lives.

Worked examples

Example 1 — a crisp thesis (3 points). *Buy, target ₹1,200 (+20%).* Thesis: (1) The market underrates margin expansion from premiumization — we model 300 bps vs consensus 100 bps. (2) Rural distribution moat is widening, driving volume share gains consensus misses. (3) Valuation at 22× forward P/E is below the 5-year average despite better growth. Catalysts: next two quarters of margin beats. Risks: rural slowdown; raw-material inflation. *Differentiated, specific, catalyst-driven, falsifiable.*

Example 2 — description vs thesis. Weak note: "Company X is a good business with strong brands and steady growth; we rate it Buy." That's a *description* — it says nothing the market doesn't know and gives no reason the stock is mispriced. Strong note: "We are 15% above consensus on FY26 EPS because the Street hasn't modelled the new plant's margin uplift; at the current price that's not in the stock." The second has a variant view.

Example 3 — honest risk section building credibility. A Buy note explicitly says: "If Q1 volume growth comes in below 8% (vs our 15%), our margin thesis breaks and we would downgrade." Stating the falsification point makes the whole note more credible — the PM trusts an analyst who knows what would prove them wrong.

Example 4 — a worked update note reacting to a quarterly result, not a full re-initiation. A company under Buy coverage reports revenue in line but EBITDA margin 80bps below the analyst's estimate, driven by a one-off input-cost spike management flags as temporary. A well-structured update note: states upfront whether the miss changes the rating (here, no — the miss is attributed to a flagged one-off, not a structural issue); quantifies the immediate estimate revision (FY26 EPS cut by 3%, mechanically flowing from the margin miss); explicitly re-confirms or revises the standing falsification criteria from the original thesis (Example 3's kind of statement) in light of the new data; and keeps the note short — a page, not a full re-initiation — since the core investment case is unchanged and a PM reading dozens of update notes that morning needs the delta, not the whole story repeated.

Example 5 — a note that fails the variant-view test, rewritten. Weak: "We reiterate our Buy rating following in-line results, reflecting the company's strong market position and consistent execution." This restates the rating without adding any new information — it doesn't tell a PM anything they didn't already know from the results themselves. Rewritten with an actual variant view: "Results were in line, but we flag that management's commentary on a new export order pipeline (barely mentioned on the call, not yet in consensus models) could add 4-5% to FY27 revenue if it converts — we're not yet baking this into our target given limited disclosure, but it's the specific data point we'll be tracking over the next two quarters and would be a source of upside surprise the Street currently isn't pricing." The rewrite gives the reader something concrete and forward-looking to act on or watch for, exactly the "why publish this note at all" test every update note should pass.

How it is tested in interviews

- **"Pitch me a stock."** — Lead with the recommendation and target, then 2–3 thesis points (your variant view), the valuation, catalysts, and risks — in that order, concisely.
- **"What makes a good research note?"** — "Conclusion up top, a differentiated and falsifiable thesis, valuation that supports the target, specific catalysts, and honest risks."
- **"What's an investment thesis?"** — "A variant view — what you believe that the market doesn't, and why you're right — with catalysts to close the gap."
- **"How is your view different from consensus?"** — Interviewers want to hear you position vs the Street; the value is in the difference.

Traps & common mistakes

- Writing a **description**, not a **thesis** (no variant view).
- Burying the **recommendation** instead of leading with it.
- Omitting **catalysts** (why does the gap close?) or **risks** (looks like cheerleading).
- Agreeing with **consensus/price** — no edge, no trade.
- Not stating what would **falsify** the thesis.

First-principles recap

- The research note is the analyst's product — clarity and structure matter as much as analysis.
- Lead with **recommendation and target**; then thesis, valuation, catalysts, risks.
- A thesis must be **differentiated, specific, falsifiable, and catalyst-driven**.
- Value lives in the **difference from consensus** — the variant view.
- Honest risk disclosure builds credibility.

Quick-reference

SECTION	CONTENT
Top	Rating + target + expected return
Thesis	Variant view (why market is wrong)
Valuation	DCF/multiples supporting target
Catalysts	Events that close the gap + timing
Risks	Downside + falsification point
Thesis test	What do I believe that the market doesn't?

Q&A — The Research Note & Investment Thesis

Theory and worked scenarios on writing a persuasive, defensible research note.

Q1. What does BLUF ("bottom line up front") mean in the context of a research note, and why do PMs specifically demand it?

Model answer. BLUF means the recommendation, target price, and expected return appear at the very top of the note, before the supporting analysis — not as a conclusion the reader has to dig for at the end. Portfolio managers read many notes under real time constraints and need to immediately know what action, if any, is being recommended; a note that builds up to its conclusion risks being abandoned before the PM reaches the actionable part, regardless of how good the underlying analysis is.

Q2. Rewrite this weak thesis statement into a strong one: "The company has good management and a strong brand, so we rate it a Buy."

Model answer. The weak version is a description with no variant view — "good management and a strong brand" are already widely known facts likely reflected in the current price. A strong version names a specific, differentiated view: e.g. "We are 12% above consensus FY26 EBITDA because the Street hasn't yet modelled the margin benefit from the new plant ramping to full utilisation by Q3 — at the current price, that upside isn't reflected." The rewrite adds a concrete, checkable claim about what the market is missing, not just a favourable general description of the company.

Q3. What's the difference between an initiation note and an earnings/update note, and how does the required depth differ?

Model answer. An initiation note is a full deep-dive launching formal coverage — it must establish the complete investment case from scratch (business overview, industry context, full model, valuation, thesis, catalysts, risks) since the reader may have no prior context. An earnings/update note reacts to new information (a results print, management guidance change, a competitive development) against an *existing* thesis — it's shorter and more focused, explicitly stating whether the new information confirms, strengthens, weakens, or breaks the standing thesis, rather than re-deriving the whole investment case each time.

Q4. What is the "variant view" test, and apply it to this scenario: an analyst's DCF and comps both land within 2% of the current market price.

Model answer. The variant view test asks: "what do I believe that the market doesn't, and why am I right?" If an analyst's valuation lands essentially at the market price with no differentiated view on any input, there is no thesis to publish as a rated call — agreeing with the market adds no information the market doesn't already have. In this scenario, the honest output is a Hold/Neutral rating (fairly valued, no edge identified) rather than manufacturing an artificial Buy or Sell case to have something to publish; a credible analyst is comfortable publishing "fairly valued, no strong view" when that's genuinely the finding.

Q5. Why does explicitly stating a falsification point ("if X happens, I'm wrong and will downgrade") make a research note *more* credible rather than less?

Model answer. It demonstrates the analyst has thought rigorously enough about their own thesis to identify what evidence would disprove it, rather than holding a view that can be rationalised indefinitely regardless of new data (a form of confirmation bias that erodes trust over time as a PM watches an analyst explain away every disappointing data point). A PM who sees a clear, specific falsification criterion knows exactly what to watch for and trusts that the analyst will act on that evidence rather than defend a stale call — this is why "what would make you wrong" is one of the most consistently asked follow-up questions after any pitch.

Q6. A research note states the company's own guidance without independent scrutiny, and rates the stock a Buy on that basis. What's the flaw in this approach?

Model answer. Restating management's own guidance and rating the stock accordingly is not independent research — it adds no analytical value beyond what the company itself has already disclosed, and it fails the variant-view test entirely (the market has access to the same guidance). A legitimate research note independently assesses whether that guidance is achievable, conservative, or aggressive (via the analyst's own model, channel checks, or industry comparison) and forms a view that may agree or disagree with management's framing — simply relaying guidance as if it were the analyst's own conclusion is a common junior-analyst failure mode.

Q7. How should a note position itself explicitly relative to Street consensus, and why does this matter for how the thesis is read?

Model answer. A well-written note states directly where its estimates and rating sit versus consensus (e.g. "our FY26 EPS is 8% above consensus; we are the only Buy among 12 covering analysts" or "we are inline with consensus estimates but see a re-rating catalyst the Street hasn't priced"). This matters because it immediately tells the reader where the actual edge (if any) lives — is it a numbers edge (different estimates), a multiple edge (different view on what multiple the market should apply), or a timing edge (agreeing with consensus fundamentals but seeing an unrecognised near-term catalyst) — each requires different supporting evidence and is tested differently by follow-up questions.

Q8. What should a "risks" section actually contain, and what's the difference between a well-written risks section and a legally-defensive boilerplate one?

Model answer. A well-written risks section names the specific, material risks to *this particular thesis* (e.g. "our margin-expansion call breaks if raw-material costs rise faster than the company can pass through in pricing"), ideally with a rough sense of magnitude or probability, and explicitly ties back to the falsification point from the thesis section. A boilerplate risks section lists generic, near-universal risks ("macroeconomic conditions may change," "competition may increase") that apply equally to almost any stock and provide no real information about what's specifically at stake in *this* call — the specificity of the risks section is itself a signal of how rigorously the analyst has actually stress-tested their own thesis.

The Stock Pitch

The Problem / Why this matters

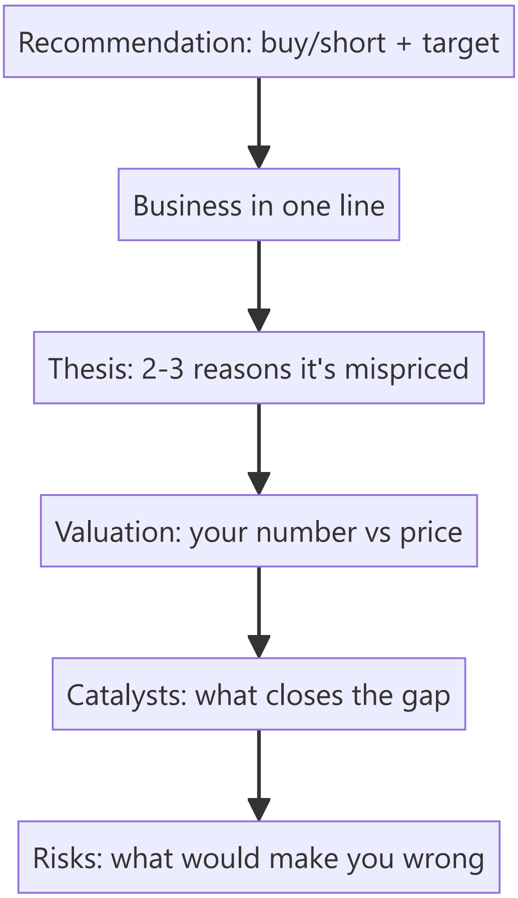
"Pitch me a stock" is the single most common question in equity research, buy-side, and markets interviews — and often the deciding one. It tests everything at once: whether you understand a business, can value it, have a differentiated view, know the catalysts and risks, and can communicate all of it in two minutes. A strong, structured pitch is a skill you can prepare and deliver on demand.

Core Idea

A stock pitch is a **concise, structured argument** for a specific action (buy or short) on a specific stock, delivered in a repeatable format: recommendation → business → thesis → valuation → catalysts → risks. It's the verbal version of the research note, compressed for a live conversation.

Why it works this way

The listener (interviewer or PM) wants to know quickly: *what's the trade, why is it mispriced, what's it worth, what makes it move, and what could go wrong*. Leading with the recommendation and following a clear skeleton signals that you think like an analyst and lets you cover everything without rambling.



Full technical content

The pitch skeleton (in order): 1. **Recommendation** — "I'd buy [X] with a 12-month target of ₹[Y], about [Z]% upside." (Or short.) Lead with it. 2. **Business** — one or two sentences: what the company does and how it makes money. 3. **Thesis** — 2–3 crisp reasons the market is mispricing it (your variant view). This is the core. 4. **Valuation** — your intrinsic value/target and how you got there (DCF and/or a multiple), vs the current price. 5. **Catalysts** — the specific events that will make the market recognize the value, with rough timing. 6. **Risks** — the main risks and what would prove you wrong (and ideally why the risk/reward is still favourable).

Delivery principles: - Lead with the conclusion; don't build up to it. - Be **specific** — real numbers (target, upside, key metric), not vague adjectives. - Have a **differentiated view** — say how you differ from consensus. - Keep it **~2 minutes**; then handle follow-up questions. - Show **balance** — acknowledging risks makes you credible.

Long vs short pitches. A **long** pitch argues undervaluation + positive catalysts. A **short** pitch argues overvaluation, deteriorating fundamentals, or a specific negative catalyst (accounting red flags, structural decline) — and must address the asymmetry (shorts have unlimited downside, borrow cost, and squeeze risk).

Preparation. Have **2–3 pitches ready** (ideally one long, one short), including at least one you genuinely follow. Know the numbers cold (valuation, key drivers) because the follow-ups probe them. Be ready for: "What's the market missing?", "What would make you wrong?", "What's it worth and why?", "What's the catalyst?"

Common follow-ups and how to handle them:

FOLLOW-UP	WHAT THEY'RE TESTING
"Why is it mispriced?"	Your variant view / edge
"What's your target and how?"	Valuation rigour
"What's the catalyst?"	Actionability
"What could go wrong?"	Intellectual honesty
"Why hasn't the market seen this?"	Depth of edge

The written initiation-of-coverage note — full structure

A pitch is the spoken compression of a much longer document: the **initiation of coverage (IoC)** note, typically 15-40 pages, that a sell-side analyst publishes when first taking up formal coverage of a stock. Knowing its structure cold is expected of anyone claiming equity-research experience. 1. **Cover page / summary box** — rating, target price, current price, upside/downside %, market cap, key financial snapshot (revenue, EBITDA margin, EPS, P/E) for the forecast years at a glance. 2. **Investment summary** (1 page) — the thesis compressed to 3-5 bullet points, written to be read standalone by a portfolio manager with 30 seconds. 3. **Business overview** — segments, revenue mix, geographic mix, competitive positioning, moat. 4. **Industry overview** — market size, growth drivers, competitive structure (Porter's Five Forces), regulatory backdrop. 5. **Financial analysis** — historical trend analysis (revenue, margins, returns on capital), quality-of-earnings checks (cash conversion, working-capital trends, one-off items flagged and excluded). 6. **Forecast and assumptions** — the driver-based model's key assumptions stated explicitly and defended (this is where a reader checks whether the analyst's numbers are grounded or arbitrary). 7. **Valuation** — DCF (with WACC/terminal-growth sensitivity table) and relative valuation (peer comp table), reconciled to a target price. 8. **Catalysts and risks** — a dedicated section, not an afterthought; risks should be specific and, where possible, quantified (e.g. "each 1% of rupee appreciation compresses EPS by ~0.6%"). 9. **Appendix** — detailed model outputs, historical financial statements, management background, ESG considerations if material.

Sensitivity table (a near-universal request): a DCF target price is never presented as a single number without a sensitivity grid — typically WACC (rows) × terminal growth rate (columns), showing how the target moves across a realistic range of each assumption (e.g. WACC 10.5%-12.5% in 50bp steps, terminal growth 4%-6% in 50bp steps) — this single table is often the first thing a skeptical PM asks to see, since it reveals how much of the "buy" case depends on optimistic assumptions at the edges of the grid versus holding up across the whole range.

Worked examples

Example 1 — a two-minute long pitch. "I'd buy [FMCG co], target ₹1,200, ~20% upside. It's a branded consumer-staples leader with a rural distribution moat. Thesis: the market underrates margin expansion from premiumization — I model 300 bps versus consensus 100 — and volume share gains from distribution the Street isn't crediting; at 22× forward it's below its historical average despite better growth. On a DCF and comps I get ₹1,200 versus ₹1,000 today. Catalysts: the next two quarters of margin beats. Risks: a rural slowdown or raw-material inflation — but at this valuation, I think the risk/reward is favourable." *Every element, two minutes.*

Example 2 — a short pitch. "I'd short [X], target 30% downside. It's a lender growing loans 40% a year while under-provisioning; my thesis is that credit costs are being deferred, not avoided — restated for realistic provisions, earnings are ~40% lower than reported. Valuation at 4× book assumes durable high ROE that I think is illusory. Catalyst: rising NPAs over the next two quarters and a likely provisioning reset. Risks: it can stay expensive and squeeze shorts, so I'd size it carefully and use a stop." *Addresses the short's asymmetry.*

Example 3 — handling "what would make you wrong?" For the FMCG long: "If the next two quarters show margin *contraction* instead of the expansion I expect — say raw-material inflation the company can't pass through — my core thesis breaks and I'd exit." A crisp falsification point signals a real analyst.

Example 4 — a banking-sector long pitch. "I'd buy [private bank], target ₹950, ~18% upside. It's a private-sector bank with a strong deposit franchise — CASA ratio above 45%, well ahead of peers. Thesis: the market is discounting it for near-term NIM compression from deposit re-pricing, but I think that's already 80% priced in at 2.3x book versus a 5-year average of 2.8x, and the market is underrating the credit-cost normalization as legacy stressed-asset provisions roll off over the next four quarters. On a Gordon-growth-based P/B framework (ROE 16%, cost of equity 13%, growth 12% → implied P/B ~2.8x) applied to next year's book value, I get ₹950. Catalysts: NIM bottoming and credit-cost guidance in the next two quarters' results. Risks: a systemic deposit-cost shock if rate cuts don't materialise as expected, and asset-quality slippage in the unsecured retail book — a segment I'm watching closely given sector-wide unsecured-lending stress." *Shows sector-specific valuation framework (P/B via ROE-CoE-growth, not DCF, which is standard for banks given the difficulty of forecasting bank free cash flow directly) and a sector-specific risk (unsecured retail asset quality).*

Example 5 — a technology-sector short pitch. "I'd short [mid-cap SaaS co], target 25% downside. It trades at 12x forward revenue on the strength of a headline 130% net-revenue-retention figure, but digging into the cohort disclosures, that NRR is concentrated in a handful of large accounts renewing at expanded contract value, while the broader customer base shows flat-to-declining logo retention — a classic 'blended metric hides a worsening core' pattern. My thesis: as the large-account expansion cycle normalises over the next 2-3 quarters, blended NRR converges toward the weaker underlying logo-retention trend, and the multiple re-rates toward peer SaaS names growing similarly but without this concentration risk (6-8x forward revenue). Catalyst: the next NRR print by customer-cohort-size disclosure (if the company provides it) or a deceleration in the headline number itself. Risks: SaaS multiples can stay elevated on momentum regardless of fundamentals for longer than a short position can be held comfortably, and a single large new logo win could reverse the narrative — sizing and stop-discipline matter more here than in the long book." *Demonstrates disaggregating a headline metric — the specific analytical move interviewers look for in a "find the flaw in this metric" style question.*

Sector-specific valuation frameworks — a quick reference

Interviewers routinely ask "how would you value a bank / an insurer / a commodity company differently from a normal company" — a DCF built on standard free cash flow to firm breaks down or requires heavy adaptation for several sectors: - **Banks/NBFCs**: valued primarily on **P/B (price-to-book) linked to ROE**, not DCF — a bank's "free cash flow" is not economically meaningful the way it is for an industrial company, since debt (deposits) is the raw material of the business, not just a financing choice. The Gordon-growth P/B framework (Example 4 above) is the standard tool. - **Insurance**: **Embedded Value (EV)** and **Value of New Business (VNB) multiples** are sector-specific metrics — EV captures the discounted value of in-force policies plus net worth, and the market typically prices life insurers on P/EV rather than P/E or P/B. - **Commodity/cyclical companies** (metals, cement, oil & gas): valued through-the-cycle, often on **normalised EV/EBITDA** (using a mid-cycle commodity-price assumption, not the current spot price, to avoid over/under-valuing at cycle extremes) rather than a single-year DCF that's hostage to where the commodity price happens to sit today. - **Real estate**: **Net Asset Value (NAV)** — summing the discounted value of each project/land parcel individually — is the standard framework, since a blended company-wide DCF obscures project-level economics that actually drive value. - **Early-stage/pre-profit tech**: revenue multiples (EV/Revenue) or, for genuinely forecastable unit economics, a DCF built on a longer explicit forecast period with explicit path-to-profitability assumptions, always paired with a clear statement of how sensitive the valuation is to the terminal-margin assumption specifically.

How it is tested in interviews

- **"Pitch me a stock."** — Deliver the skeleton: recommendation + target → business → 2–3 thesis points → valuation → catalysts → risks, in ~2 minutes.
- **"What's the market missing?"** — Your variant view — the single most important thing; have it sharp.
- **"What's it worth and how did you get there?"** — Give a target with a one-line valuation method; know the numbers.
- **"What would make you wrong?"** — A specific falsification point, delivered confidently.
- **"Long or short something you follow"** — Always have real pitches ready, not textbook examples.

Traps & common mistakes

- **Burying the recommendation** — lead with it.
- No **variant view** — just describing a "good company."
- **Vague** — no target, no numbers.
- Ignoring **risks** — sounds naive; acknowledging them builds credibility.
- Not knowing your own numbers when **probed**.
- For shorts, ignoring **borrow cost, squeeze, and unlimited downside**.

First-principles recap

- A pitch = recommendation → business → thesis → valuation → catalysts → risks, in ~2 minutes.

- **Lead with the conclusion**; be specific with numbers.
- The thesis is a **variant view** — what the market is missing.
- Name **catalysts** (actionability) and **risks/falsification** (credibility).
- Prepare 2–3 real pitches and know the numbers cold.

Quick-reference

ELEMENT	ONE LINE
Recommendation	Buy/short + target + upside
Business	What it does, how it earns
Thesis	2–3 reasons it's mispriced (variant view)
Valuation	Your number vs price + method
Catalysts	What closes the gap + when
Risks	Downside + what proves you wrong

Q&A — The Stock Pitch

Theory and scenario drills for the single most-asked live question in markets interviews.

Q1. "Pitch me a stock" — what's the structure, and how long should it take?

Model answer. Recommendation with target and upside first → one-line business description → 2-3 point thesis (the variant view) → valuation (your number vs the price, and how you got there) → catalysts (what closes the gap, roughly when) → risks (what would prove you wrong). Target roughly two minutes before handling follow-ups; lead with the conclusion, never build up to it.

Q2. What's the single most important element of a pitch, and why?

Model answer. The variant view — what the market is missing that you see. A pitch that just describes a good company with no view on *why the market is wrong* isn't research, it's a book report; the interviewer is specifically screening for whether the candidate can articulate a genuine, differentiated edge.

Q3. How does a short pitch differ structurally from a long pitch?

Model answer. The skeleton is the same, but a short must explicitly address the asymmetry of shorting: unlimited theoretical downside (a stock can rise indefinitely, unlike a long where the max loss is the investment), borrow cost, and squeeze risk — and should generally identify a specific negative catalyst (an accounting red flag, a structural decline, a credit-quality deterioration) rather than a vaguer "it's overvalued" argument, since "overvalued" alone can stay overvalued for a long time and cost a short position dearly while waiting.

Q4. Write a full ~90-second pitch for a hypothetical company from these facts: a listed QSR chain trading at 28x forward P/E, same-store sales growth decelerating from 12% to 6% over the last three quarters, but management guiding aggressive new-store additions (25% unit growth) to offset the deceleration.

Model answer. "I'd short [QSR Co], target 20% downside. It trades at 28x forward earnings — a premium multiple that assumes sustained high growth — but same-store sales growth has decelerated from 12% to 6% over three consecutive quarters, a trend management isn't addressing directly, instead leaning on 25% unit growth to mask the core deceleration in the headline numbers. My thesis: new-store economics take 12-18 months to mature, so aggressive unit growth in the near term will actually depress consolidated margins before it helps, and once same-store-sales deceleration becomes impossible to hide behind unit growth, the multiple should compress toward the 18-20x peer average for decelerating QSR names. Catalyst: same-store sales guidance or print showing continued deceleration in the next one to two quarters. Risks: a same-store-sales re-acceleration (a new product launch or marketing push could reverse the trend), and premium-multiple growth names can stay expensive on momentum longer than expected — I'd size and stop-manage this position carefully given that risk."

Q5. An interviewer asks "why hasn't the market already priced this in?" — what's a strong answer structure?

Model answer. Name a specific, credible reason the market is slow or wrong here — common legitimate reasons include: the signal is buried in a disclosure most analysts don't dig into (e.g. a footnote or a segment breakdown), the market is anchored on a backward-looking headline metric that masks the real trend (see Q4's blended same-store-sales/unit-growth example), the stock is under-covered (few analysts, low institutional ownership) so mispricing persists longer, or the market is systematically short-termist about a benefit that plays out over multiple quarters. A weak answer ("I just think I'm right and everyone else is wrong") signals overconfidence rather than a genuine edge.

Q6. What's the purpose of the sensitivity table in a written pitch/note, and what's the most common grid used?

Model answer. It shows how the target price moves across a realistic range of the most uncertain assumptions, so a reader can judge how much of the "Buy" case depends on optimistic inputs versus holding up broadly. The most common grid is WACC (rows) × terminal growth rate (columns) for a DCF-based target, typically spanning roughly ±1-2% around the base case in each dimension.

Q7. Give the sector-appropriate valuation approach for: (a) a bank, (b) a life insurer, (c) a commodity/cyclical company, (d) a real-estate developer.

Model answer. (a) Bank — P/B linked to ROE (Gordon-growth P/B relationship) or a residual-income/DDM model; standard FCFF/DCF breaks down since deposits are the business's raw material, not just financing. (b) Life insurer — Embedded Value (EV) and Value of New Business (VNB) multiples, typically priced on P/EV rather than P/E or P/B. (c) Commodity/cyclical — normalised EV/EBITDA using a mid-cycle commodity-price assumption rather than current spot, to avoid over/under-valuing at cycle extremes. (d) Real-estate developer — Net Asset Value (NAV), summing the discounted value of each project/land parcel individually rather than a single blended company-wide DCF.

Q8. What does "the risk/reward is still favourable" mean at the end of a pitch, and why include it?

Model answer. After naming the real risks (Q3, Q5), a strong pitch explicitly states why the expected value still favours the trade — e.g. "even if the risk materialises, the downside is roughly X% while the base case upside is Y%, an asymmetric setup" — because simply listing risks without this framing can make the pitch sound tentative; naming risks *and* explaining why they don't change the conclusion is what signals genuine conviction backed by analysis, rather than either naive overconfidence or unearned hedging.

Technical Analysis Essentials

The Problem / Why this matters

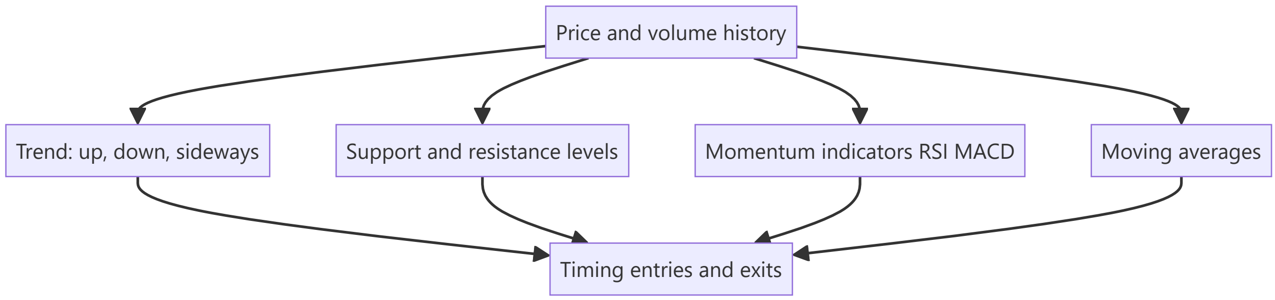
While fundamental analysis asks *what a stock is worth*, **technical analysis** asks *what the price is likely to do next*, from the price and volume history itself. It's the core toolkit for traders and technical research analysts, and it's used alongside fundamentals for timing entries and exits. Even fundamentally-driven roles expect you to understand trends, support/resistance, and the main indicators — and for trading/TRA roles it's central.

Core Idea

Technical analysis studies **price and volume patterns** to forecast future price movement, resting on three assumptions: the price discounts everything, prices move in trends, and history tends to repeat (because human behaviour does). Its tools identify trends, key price levels, momentum, and reversals.

Why it works this way

Price reflects the collective actions of all participants, so patterns in price can reveal the balance of supply and demand and the psychology (fear/greed) driving it. Trends persist because participants react and herd; levels matter because memory and orders cluster at them; patterns repeat because human behaviour under uncertainty is repetitive.



Full technical content

The three assumptions of technical analysis: 1. **The market discounts everything** — all known information is already in the price, so study the price. 2. **Prices move in trends** — once established, a trend is more likely to continue than reverse. 3. **History repeats** — patterns recur because market psychology is consistent over time.

Trend. The primary concept: an **uptrend** = higher highs and higher lows; a **downtrend** = lower highs and lower lows; **sideways/range** = no clear direction. "The trend is your friend" — trade with it until it clearly breaks. **Trendlines** connect the lows (uptrend) or highs (downtrend).

Support & resistance. Support = a price level where buying tends to emerge and halt declines; **resistance** = where selling tends to emerge and cap advances. A broken resistance often becomes new support (and vice versa). These levels are where traders place orders and stops.

Moving averages (MA). Smooth price to reveal the trend. Simple (SMA) or exponential (EMA). A price above its MA is bullish; below is bearish. Crossovers signal trend changes: a **golden cross** (50-day crosses above 200-day) is bullish; a **death cross** (50-day below 200-day) is bearish.

Momentum / oscillators:

INDICATOR	WHAT IT SHOWS
RSI (Relative Strength Index)	Momentum, 0–100; > 70 overbought, < 30 oversold; divergences signal reversals
MACD (Moving Average Convergence Divergence)	Trend + momentum via two EMAs and a signal line; crossovers and histogram
Stochastic	Momentum vs recent range
Bollinger Bands	Volatility bands around an MA

Volume confirms moves — a breakout on high volume is more reliable than on low volume. **Chart patterns** (head-and-shoulders, double top/bottom, triangles, flags) signal continuation or reversal.

Technical vs fundamental — complementary. Fundamentals answer *what* to buy (value); technicals answer *when* (timing). Many analysts use fundamentals for the thesis and technicals for entry/exit and risk (stops).

Worked examples

Example 1 — trend and trendline. A stock makes higher highs and higher lows for months (uptrend); a trendline drawn under the rising lows holds. A trader buys on pullbacks to the trendline with a stop just below it, riding the trend until the line breaks — a break signalling the uptrend may be over.

Example 2 — support turned resistance. A stock repeatedly bounces off ₹100 (support). It finally breaks below to ₹90. On the next rally, ₹100 now acts as *resistance* — sellers who were trapped exit there. The old floor becomes the new ceiling, a classic level flip.

Example 3 — RSI divergence. Price makes a new high but RSI makes a *lower* high (bearish divergence) — momentum is weakening even as price rises, warning of a possible reversal. Combined with resistance, it's a signal to tighten stops or take profit.

Example 4 — golden cross vs a false start. The 50-day MA crosses above the 200-day MA (a golden cross) after a long downtrend, a classically bullish signal. Two weeks later, price stalls and the 50-day MA turns back down without the 200-day MA reversing — a "false" golden cross that didn't lead to a sustained trend. A disciplined technical analyst treats the golden cross as raising the odds of a trend change, not guaranteeing one, and waits for price to also clear a nearby resistance level or hold above the crossed MAs for several sessions before committing full size — precisely the "probabilities, not certainties" discipline this chapter's traps section warns about.

Example 5 — combining fundamentals and technicals on one name. An equity analyst has a fundamental Buy thesis on a stock (undervalued on DCF and comps, per the Applied Equity Valuation chapter) but the stock is in a technical downtrend, trading below its 200-day MA with weak relative strength versus the sector. Rather than buying immediately on the fundamental case alone, the analyst waits for a technical confirmation — a break above the downtrend line with rising volume, or the stock reclaiming its 200-day MA — before initiating the position, using the technical setup purely for entry timing and initial stop placement (just below the reclaimed MA) while the fundamental thesis remains the reason for being long at all. This is the concrete version of "fundamentals for what, technicals for when" that interviewers ask candidates to articulate.

How it is tested in interviews

- **"What is technical analysis and its assumptions?"** — "Studying price and volume to forecast prices, on three assumptions: the price discounts everything, prices move in trends, and history repeats."

- **"Technical vs fundamental analysis?"** — "Fundamentals estimate intrinsic value (what to buy); technicals study price/volume for direction and timing (when). They're complementary."
- **"What is support and resistance?"** — "Levels where buying (support) or selling (resistance) tends to emerge; broken resistance often becomes support."
- **"What do RSI and MACD tell you?"** — "RSI is a 0–100 momentum gauge (>70 overbought, <30 oversold); MACD combines trend and momentum via EMA crossovers. Divergences warn of reversals."
- **"What's a golden cross?"** — "The 50-day MA crossing above the 200-day — a bullish trend signal; the reverse (death cross) is bearish."

Traps & common mistakes

- Treating technicals as **certainty** — they're probabilities, not guarantees.
- Ignoring **volume** confirmation of breakouts.
- Using indicators in **isolation** rather than confluence (trend + level + momentum).
- Fighting the **trend** ("catching a falling knife").
- Forgetting technicals are best **combined** with fundamentals and risk management (stops).

First-principles recap

- Technical analysis forecasts price from price/volume history.
- Three assumptions: price discounts everything, trends persist, history repeats.
- Core tools: **trend**, **support/resistance**, **moving averages**, **RSI/MACD**, **volume**.
- Golden/death crosses and divergences signal trend/momentum shifts.
- Best used **with** fundamentals — value for *what*, technicals for *when*.

Quick-reference

TOOL	SIGNAL
Trend	Higher highs/lows = up; trade with it
Support/resistance	Buy/sell levels; break flips role
Moving average	Above = bullish; golden/death cross
RSI	>70 overbought, <30 oversold, divergence
MACD	EMA crossover, momentum
Volume	Confirms breakouts

Q&A — Technical Analysis Essentials

Theory and worked scenarios on chart-based analysis fundamentals (see the much deeper Technical-Analysis-Complete volume elsewhere in this master for full chart-pattern depth).

Q1. State the three foundational assumptions of technical analysis, and briefly justify each.

Model answer. (1) The market discounts everything — all known information (fundamentals, sentiment, expectations) is already reflected in the current price, so studying price itself captures the net effect of all available information without needing to separately analyse each input. (2) Prices move in trends — once a directional move is established, continuation is statistically more likely than immediate reversal, because participants react to and reinforce established moves (momentum, herding). (3) History repeats — chart patterns recur because they reflect recurring patterns of crowd psychology (fear, greed, FOMO) under uncertainty, which doesn't fundamentally change across market cycles even as the specific assets and eras differ.

Q2. Worked — identify the signal from this scenario.

A stock has been in a clear uptrend for six months. It pulls back to its rising 50-day moving average three separate times, bouncing higher each time. On the fourth pullback, it closes decisively below the 50-day MA on above-average volume.

Model answer. The first three bounces off the rising 50-day MA are consistent with the "trend is your friend" principle — the moving average is acting as dynamic support in an established uptrend, and each successful bounce reinforces that the trend remains intact. The fourth event — a decisive close below the MA on above-average volume — is a meaningfully different signal: volume confirmation matters (a break on light volume is less reliable than one on heavy volume), so this combination suggests the uptrend's support has genuinely failed rather than being another buyable dip, warranting a reassessment of the trend rather than automatically buying the fourth pullback the way the first three were bought.

Q3. What's the difference between what RSI and MACD each measure, and can they disagree with each other?

Model answer. RSI (Relative Strength Index) is a bounded 0-100 momentum oscillator measuring the speed and magnitude of recent price changes, primarily used to flag overbought (>70) or oversold (<30) conditions and divergences. MACD (Moving Average Convergence Divergence) combines trend and momentum information via the relationship between two exponential moving averages and a signal line, primarily used for crossover and histogram signals. Yes, they can disagree — e.g. RSI can show "overbought" while MACD's crossover remains bullish, since RSI is more sensitive to short-term speed of movement while MACD reflects a somewhat longer-term trend-momentum blend; a technical analyst uses multiple indicators in confluence specifically because any single one can give a misleading signal in isolation.

Q4. Explain "support becomes resistance" (or vice versa) with the underlying behavioural logic, not just the rule.

Model answer. When a price level repeatedly holds as support, many market participants place buy orders near it and some who bought elsewhere set mental reference points around it; when the level finally breaks decisively, those who bought at or near the old support level are now underwater and often eager to sell "if I can just get back to even" — creating a cluster of latent sell orders exactly at the old support level, which is now approached from below on any subsequent rally. This behavioural mechanism (trapped buyers becoming eager sellers at their breakeven) is what turns a former support level into new resistance, not some mechanical rule independent of actual participant psychology.

Q5. How should technical and fundamental analysis be combined in a single research process, according to this chapter?

Model answer. Fundamental analysis answers "what to buy" (is the business undervalued, and why) while technical analysis answers "when to buy/sell" (is the current price action and level structure favourable for entry, and where should a stop be placed to manage risk if the thesis is wrong). A common integrated approach: use fundamental research to build the conviction list and set a target price/thesis, then use technical levels (support, trend, moving averages) to time the actual entry and to set objective, rules-based exit/stop points rather than relying purely on discretionary judgment about when to cut a losing position.

Q6. Why is technical analysis theoretically inconsistent with even the weak form of the Efficient Market Hypothesis, and how do practitioners typically respond to that critique?

Model answer. Weak-form EMH holds that all past price and volume information is already reflected in the current price, meaning patterns in historical price data (the entire basis of technical analysis) should carry no predictive power for future prices. Practitioners typically respond that markets are not perfectly efficient in practice — documented behavioural biases (herding, anchoring, disposition effect) create real, if imperfect and hard-to-time, patterns in how information gets incorporated into price, and that technical analysis is better understood as a probabilistic tool reflecting crowd psychology than as a claim that markets are informationally inefficient in a strict academic sense.

Q7. What's the danger of "using indicators in isolation," and how would a disciplined technical analyst avoid it in practice?

Model answer. Any single indicator can and does generate false signals regularly — an RSI reading "oversold" doesn't guarantee a bounce, a single moving-average crossover doesn't guarantee a sustained trend change. A disciplined analyst looks for confluence: multiple independent signals (e.g. a support level, a bullish RSI divergence, and a MACD crossover) aligning at the same point, which meaningfully raises the odds of a signal being genuine versus noise, rather than acting on any single indicator's reading in isolation.

Q8. A stock breaks out above a well-established resistance level, but on noticeably below-average volume. How should this affect confidence in the breakout?

Model answer. Volume is the confirmation mechanism for a breakout's reliability — a genuine breakout driven by real conviction typically attracts increased participation (higher volume) as more buyers step in and short-sellers cover. A breakout on light volume suggests a lack of broad conviction behind the move, raising the probability of a "false breakout" (the price re-entering the prior range shortly after) — a technically disciplined analyst would treat this breakout with reduced confidence and might wait for volume confirmation or a successful retest of the broken level before fully committing, rather than treating the price break alone as sufficient signal.

Market Efficiency & Behavioral Finance

The Problem / Why this matters

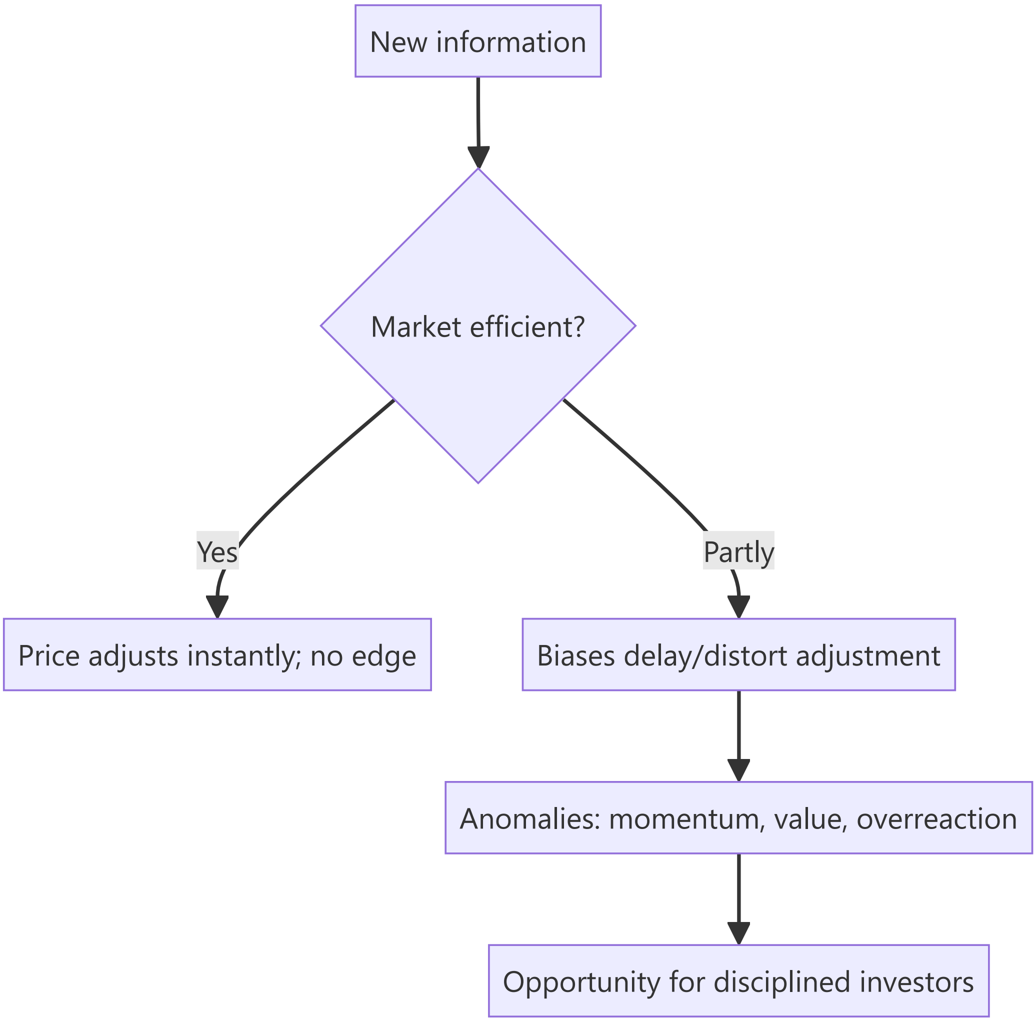
Can you beat the market? The answer shapes everything an investor does. The **Efficient Market Hypothesis (EMH)** says prices already reflect information, so consistently beating the market is nearly impossible — while **behavioral finance** documents the systematic biases that make markets *inefficient* enough to create opportunities. Every research and investing role sits in the tension between these two, and interviewers test whether you understand both.

Core Idea

The **EMH** holds that asset prices reflect available information, so you can't consistently earn risk-adjusted excess returns. **Behavioral finance** counters that investors are systematically irrational — subject to biases that push prices away from fundamentals — creating anomalies a disciplined investor can exploit. Reality lies between: markets are *mostly* efficient but not perfectly.

Why it works this way

If a piece of information predicted higher returns, rational investors would trade on it until the opportunity vanished — that's the efficiency argument. But real investors are human: they overreact, herd, anchor and fear losses, and those biases are *systematic* (not random), so they don't fully cancel out — leaving persistent, if hard-to-capture, mispricings.



Full technical content

The three forms of EMH:

FORM	PRICES REFLECT	IMPLICATION
Weak	All past prices/volume	Technical analysis shouldn't work
Semi-strong	All public information	Fundamental analysis on public data shouldn't give an edge
Strong	Even private/inside information	No one can beat the market, even insiders

Evidence: markets are *broadly* semi-strong efficient (prices react fast to news), but **anomalies** persist that challenge it.

Anomalies (evidence against strict EMH): - **Value effect** — cheap (low P/B, P/E) stocks historically outperform. - **Momentum** — recent winners keep winning over 3–12 months. - **Small-cap effect**, **low-volatility anomaly**, **post-earnings-announcement drift**, calendar effects. These underpin factor investing (Fama-French).

Behavioral biases (why markets misprice):

BIAS	EFFECT
Overconfidence	Excess trading, underestimating risk
Anchoring	Fixating on a reference price
Loss aversion	Losses hurt ~2× gains; holding losers, selling winners (disposition effect)
Herding	Following the crowd → bubbles and crashes
Recency bias	Overweighting recent events
Confirmation bias	Seeking evidence that fits your view
Mental accounting	Treating money differently by source/label
Overreaction / underreaction	To news → reversals and drift

Bubbles and crashes are the macro expression of these biases — herding and overconfidence inflate prices far above value, then fear collapses them.

Implications for investors: - If markets were perfectly efficient → **index passively** (no edge to be had). - Because they're not perfectly efficient → disciplined, unemotional investors can exploit others' biases (the behavioral edge), but it's hard and requires patience. - Know your *own* biases — the first job is not to be the sucker.

Worked examples

Example 1 — semi-strong efficiency in action. A company reports earnings 20% above expectations at 9:00 am; by 9:01 the stock has jumped to reflect it. A retail investor reading the news at noon can't profit from the surprise — it's already priced. That's semi-strong efficiency: public information is incorporated almost instantly.

Example 2 — loss aversion / disposition effect. An investor holds a stock down 30%, refusing to sell "until it comes back," while quickly booking a 10% gain elsewhere. Loss aversion makes them ride losers and cut winners — exactly backwards. Recognizing this bias (and using rules/stops) is the behavioral edge.

Example 3 — herding bubble. During a mania, prices detach from any fundamental value as investors buy because others are buying (herding) and extrapolate recent gains (recency). Fundamentals say "expensive," but the crowd pushes higher — until sentiment turns and it crashes. Bubbles are behavioral finance writ large.

Example 4 — testing weak-form efficiency with a simple rule. A researcher backtests a rule — "buy after two consecutive up-days, sell after two consecutive down-days" — on 10 years of Nifty 50 data and finds it produces a small, statistically insignificant edge after transaction costs, though a raw (pre-cost) backtest shows a modest positive return. This is a textbook weak-form-efficiency result: any exploitable pattern purely in past price sequences is either too small to survive real trading frictions (bid-ask spread, brokerage, slippage) or gets arbitraged away once enough participants exploit it — precisely why weak-form efficiency, though not perfect, holds up well empirically for simple, well-known price-pattern rules on liquid, widely-traded stocks.

Example 5 — anchoring in a real equity-research scenario. An analyst initiated coverage on a stock at ₹500 with a ₹650 target eighteen months ago. New information now suggests fair value is closer to ₹420, but the analyst's revised note sets the target at ₹520 — a number that "splits the difference" between the old target and the new fair-value estimate, rather than moving decisively to ₹420. This is anchoring: the analyst is unconsciously anchoring on their own prior published number rather than updating fully to the new evidence, a well-documented bias among professional forecasters, not just retail investors — one reason many research desks require an explicit, written justification whenever a target price revision is smaller than the underlying model's fair-value change would imply.

How it is tested in interviews

- **"What is the Efficient Market Hypothesis?"** — "Prices reflect available information, so you can't consistently earn risk-adjusted excess returns. Three forms: weak (past prices), semi-strong (public info), strong (private info)."
- **"If markets are semi-strong efficient, what stops working?"** — "Trading on public information/fundamental analysis of public data — it's already in the price. And technical analysis fails under even the weak form."
- **"Do you believe markets are efficient?"** — "Broadly semi-strong efficient — news is priced fast — but not perfectly; behavioral biases and documented anomalies (value, momentum) leave exploitable inefficiencies."
- **"Name a behavioral bias and its effect."** — Loss aversion → holding losers too long (disposition effect); herding → bubbles; anchoring → fixating on a reference price.
- **"If markets were perfectly efficient, how would you invest?"** — "Passively, via low-cost index funds — there'd be no edge to justify active fees."

Traps & common mistakes

- Treating EMH as **all-or-nothing** — it's a spectrum; markets are mostly-but-not-perfectly efficient.
- Confusing the **three forms** (weak/semi-strong/strong).
- Assuming biases **cancel out** — they're systematic, not random.
- Believing anomalies are **free money** — they're real but hard and risky to capture.
- Ignoring your **own** biases.

First-principles recap

- EMH: prices reflect information → hard to beat the market; forms are weak/semi-strong/strong.
- Behavioral finance: systematic biases push prices from value, creating anomalies.
- Reality: mostly efficient, not perfectly — news is priced fast, but inefficiencies persist.
- Anomalies (value, momentum) underpin factor investing.
- The behavioral edge is discipline; the first task is not to be biased yourself.

Quick-reference

CONCEPT	NOTE
Weak EMH	Past prices priced → TA fails
Semi-strong	Public info priced → FA edge hard
Strong	Even private info priced
Anomalies	Value, momentum, small-cap, PEAD
Key biases	Loss aversion, herding, anchoring, overconfidence
If perfectly efficient	Index passively

Q&A — Market Efficiency & Behavioral Finance

Theory and worked scenarios on EMH, anomalies, and behavioural biases.

Q1. State the three forms of the Efficient Market Hypothesis and what each implies for a specific investing strategy.

Model answer. Weak form — prices reflect all past price/volume data, implying technical analysis should not generate consistent excess returns. Semi-strong form — prices reflect all publicly available information, implying fundamental analysis of public data (filings, news, ratios) should not generate a consistent edge, since the information is already priced. Strong form — prices reflect even private/insider information, implying not even insiders could consistently beat the market (empirically the weakest-supported form, and largely why insider trading is illegal and profitable when it occurs — it's real evidence against strong-form efficiency).

Q2. If you believe markets are broadly semi-strong efficient but not perfectly so, how should that shape how you spend research time?

Model answer. Time spent restating or re-deriving already-public, widely-followed information (headline financial ratios, well-known industry trends) adds little value since that's already priced in efficiently. Time is better spent on the sources of genuine edge that survive semi-strong efficiency: deeper, harder-to-replicate analysis of public information (Part 6 of the equity-research-process chapter's "analytical edge"), primary research/channel checks that surface information ahead of wide dissemination, and exploiting behavioural mispricings where other investors' documented biases create persistent, if hard-to-time, opportunities.

Q3. Worked — apply loss aversion and the disposition effect to a portfolio decision.

An investor holds two positions: Stock A is up 25% and Stock B is down 25%, with no new fundamental information on either since purchase. They're considering trimming the portfolio to raise cash for a new idea.

Model answer. The disposition effect predicts the investor will be tempted to sell Stock A (the winner, "locking in a gain feels good") and hold Stock B (the loser, "I'll sell once it comes back to break-even") — even though, absent new information, this is exactly backwards from a purely rational standpoint: there's no fundamental reason the winner is now a worse holding than the loser, and in fact selling winners and holding losers systematically has been shown to hurt realised returns. A disciplined investor would evaluate both positions purely on current expected forward return versus alternatives, ignoring the emotionally-loaded fact of whether each position is currently a paper gain or loss.

Q4. Give a real anomaly that challenges strict EMH, and explain the standard efficient-markets rebuttal to it.

Model answer. Momentum — stocks that have performed well over the past 3-12 months tend to continue outperforming over the subsequent few months, a well-documented empirical anomaly that shouldn't exist under strict EMH (past price data should carry no predictive information). The standard efficient-markets rebuttal is that momentum returns may simply compensate for a real risk factor not being fully understood/priced (a risk-based explanation, consistent with EMH broadly), rather than being unexploited free money — the debate between "momentum is a genuine anomaly from underreaction" (behavioural explanation) and "momentum is compensation for unmeasured risk" (efficient-markets explanation) remains genuinely contested in the academic literature, and a well-prepared candidate should be able to present both sides rather than assert one is obviously correct.

Q5. Distinguish herding from momentum investing — they can look similar from the outside but are conceptually different.

Model answer. Momentum investing is a systematic strategy based on documented historical patterns (recent winners tend to keep winning over a specific time horizon), applied with discipline and typically with defined entry/exit rules. Herding is the behavioural tendency to follow the crowd's actions specifically *because* others are doing it (social proof), often without independent analysis, and is a primary driver of bubbles — herding can push prices far beyond what any momentum-based risk/reward framework would justify, and unlike disciplined momentum investing, herding-driven buying typically has no defined exit discipline, which is exactly what makes bubbles collapse suddenly and violently when sentiment reverses.

Q6. Why does the chapter argue "the first job is not to be biased yourself" before trying to exploit others' biases?

Model answer. Every documented behavioural bias (overconfidence, loss aversion, anchoring, confirmation bias, herding) applies to professional analysts and portfolio managers just as much as to retail investors — an analyst who is unaware of their own susceptibility to, say, confirmation bias (seeking evidence that supports an existing thesis while discounting contrary evidence) will systematically produce worse research regardless of technical skill. Genuine behavioural edge requires first building the discipline and self-awareness to recognise and counteract one's own biases (e.g. deliberately seeking disconfirming evidence, pre-committing to falsification criteria as covered in the research-note chapter) before that discipline can be turned into an edge against less self-aware market participants.

Q7. A stock reports earnings that beat expectations, and the price continues drifting upward gradually over the following weeks rather than jumping immediately to a new level. What phenomenon does this describe, and what does it imply about market efficiency?

Model answer. This describes post-earnings-announcement drift (PEAD), a well-documented anomaly where prices underreact to earnings surprises initially and continue adjusting gradually over subsequent weeks rather than immediately incorporating the full information content of the surprise. It's evidence against strict semi-strong efficiency (which would predict instantaneous, complete adjustment), and is one of the more robust, widely replicated anomalies in the academic literature — though, consistent with the broader theme of this chapter, exploiting it in practice is complicated by transaction costs and the fact that the effect, while statistically real, is not large enough per trade to guarantee reliable profit after costs for every investor who tries to trade it.

Q8. How would you answer "do you believe markets are efficient?" in a way that shows nuanced understanding rather than picking one extreme side?

Model answer. "Markets are broadly semi-strong efficient in the sense that public information gets incorporated into prices quickly — you can't expect to consistently profit just from reading yesterday's news or restating public filings. But they're not perfectly efficient: well-documented behavioural biases are systematic rather than random, meaning they don't fully cancel out in aggregate, and this leaves real, empirically-documented anomalies like value and momentum effects. My view is that genuine edge comes from either deeper analytical work than the market has done on public information, primary research that gets ahead of wide information dissemination, or disciplined exploitation of others' behavioural biases — not from a belief that markets are broadly irrational or easily beaten." This answer explicitly avoids the all-or-nothing trap the chapter warns against.

Corporate Actions

The Problem / Why this matters

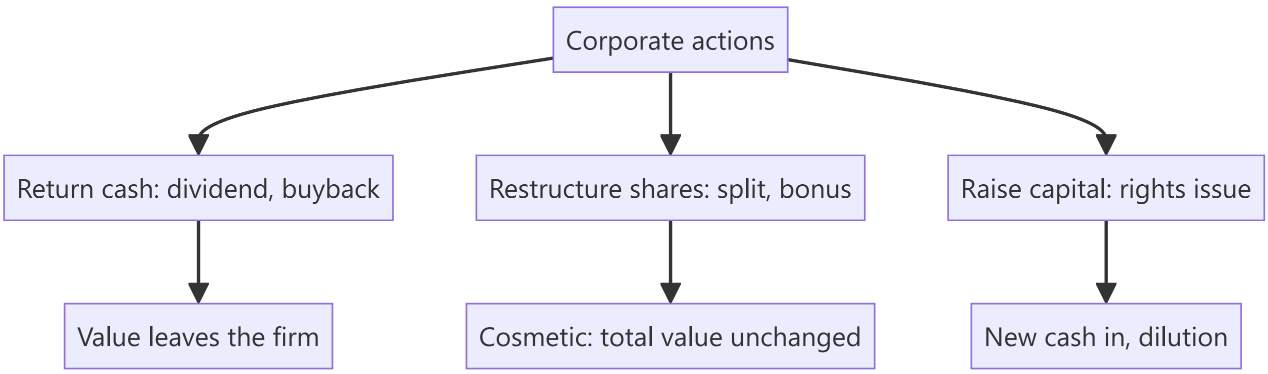
Companies regularly do things that change their shares — pay dividends, split the stock, issue bonus or rights shares, buy back stock. These **corporate actions** change share counts, prices, and per-share metrics, and an analyst must adjust for them correctly (in models, historical price series, and valuation) — and know which ones create value and which merely rearrange it. Interviewers test whether you know that a split or bonus doesn't change value, and how a buyback flows through.

Core Idea

A **corporate action** is an event initiated by a company that affects its securities and shareholders. Some return cash (dividends, buybacks); some change the share structure without changing total value (splits, bonus issues); some raise capital (rights issues). Knowing the mechanics and the value impact of each is essential.

Why it works this way

Per-share numbers depend on the share count, so any action that changes shares changes per-share metrics — but often not total value. A split cuts the price and multiplies shares proportionally (same market cap); a dividend moves cash from the company to shareholders (value leaves the firm). Distinguishing *value-changing* from *cosmetic* actions is the key skill.



Full technical content

Dividends. Cash paid per share from profits/reserves. Reduces cash and retained earnings. Key dates: **declaration, ex-dividend** (buy before ex-date to receive it), **record**, **payment**. The price typically drops by ~the dividend on the ex-date. **Dividend yield** = dividend/price.

Stock split. Divides each share into more shares at a proportionally lower price — e.g., a **1:1 split** (₹100 → two shares of ₹50). Market cap, and your total value, **unchanged**; only the share count and price change. Purpose: improve affordability and liquidity. (A **reverse split** consolidates shares into fewer, higher-priced ones.)

Bonus / scrip issue. Free additional shares to existing holders from reserves (e.g., **1:1 bonus** = one free share per share held). Like a split, it increases shares and cuts the price proportionally — **no change in total value** (reserves just move to share capital). Purely accounting/signaling.

Rights issue. Existing shareholders get the *right* to buy new shares, usually at a **discount**, in proportion to holdings (e.g., 1 new share per 4 held at a discount). Raises capital; those who don't subscribe are **diluted** (though the right itself has value and can be sold). The price adjusts to a **theoretical ex-rights price (TERP)** blending old and new.

Buyback (share repurchase). The company buys its own shares (tender or open-market) and cancels them. Reduces cash and equity, shrinks share count → **boosts EPS**. Returns cash like a dividend but more tax-efficiently/flexibly; signals shares are cheap. Doesn't create value by itself (depends on price paid vs value and alternative uses of cash).

ACTION	SHARES	PRICE/SHARE	TOTAL VALUE	CASH IMPACT
Dividend	Unchanged	Falls ~dividend	Falls (cash out)	Cash out
Split (1:1)	×2	÷2	Unchanged	None
Bonus (1:1)	×2	÷2	Unchanged	None
Rights	Up	TERP (blend)	Up by cash raised	Cash in
Buyback	Down	~Unchanged	Falls (cash out)	Cash out

Analyst adjustments. Adjust **historical prices** for splits/bonus (so charts aren't distorted), adjust **share counts** in models, and use **diluted** shares. Never compare a pre-split price to a post-split price without adjusting.

Worked examples

Example 1 — a split creates no value. A stock at ₹1,000 does a 1:5 split → 5 shares at ₹200 each. An investor who held 10 shares (₹10,000) now holds 50 shares (₹10,000). Nothing changed except affordability. *A 2-for-1 or 5-for-1 split does not make you richer.*

Example 2 — buyback and EPS. A firm has net income ₹100 cr and 100 mn shares → EPS ₹10. It buys back 10 mn shares (cancelled) → 90 mn shares → EPS ₹100 cr / 90 mn = **₹11.1**. EPS rose ~11% with no change in profit — purely from a smaller share count. (Whether it *created value* depends on whether the shares were bought below intrinsic value.)

Example 3 — rights issue TERP. A stock at ₹200; a 1:4 rights issue at ₹100. For every 4 old shares (₹800) you buy 1 new at ₹100 → 5 shares for ₹900 → TERP = ₹180. The price "falls" from ₹200 to ₹180, but that's the dilution of the discounted new shares, not a loss for subscribers.

Example 4 — a full dividend-date sequence, worked with actual dates. A company declares a ₹15/share dividend on 5 June (**declaration date**). It sets 20 June as the **record date** — only shareholders on record that day receive the dividend. Given a T+1 settlement cycle, the **ex-dividend date** is set one business day before the record date, 19 June — an investor must buy the stock by 18 June (so the trade settles by 19 June) to appear on the register by the 20th record date. A trader who buys the stock on 19 June (the ex-date itself) does *not* receive the dividend, and the stock opens on the 19th trading roughly ₹15 lower than the prior close, reflecting the entitlement no longer attaching to a new buyer. **Payment date** — the actual date the ₹15 lands in shareholders' accounts — might be 5 July, several weeks after the ex-date. An analyst modelling total shareholder return must track all four dates correctly: using the wrong date (e.g. the payment date instead of the ex-date) to adjust a historical price series would misalign the mechanical price drop from the actual dividend entitlement.

Example 5 — buyback method comparison: tender offer vs open-market. A company wants to return ₹500 cr to shareholders via buyback and is choosing between two mechanisms. A **tender offer**

buyback sets a fixed price (often at a premium to market, e.g. 20% above the current price) and a fixed quantity, with shareholders tendering shares proportionally if oversubscribed — pros: certainty of price and total outlay, and it signals strong management confidence (paying a visible premium); cons: slower (requires a formal offer process) and the premium paid may not represent good capital allocation if the stock wasn't genuinely undervalued by that much. An **open-market buyback** purchases shares gradually on the exchange over an extended window (up to a regulatory cap, e.g. a maximum % of daily volume) at prevailing market prices — pros: flexible, can be paused if the price rises above what management considers fair value, and avoids overpaying a large explicit premium; cons: slower to complete the full ₹500 cr, and provides a weaker signalling effect than a tender offer's explicit premium commitment. The choice itself is informative to an analyst: a company choosing a tender offer at a large premium is making a stronger public statement about undervaluation than one running a measured open-market programme.

How it is tested in interviews

- **"Does a 2-for-1 stock split create value?"** — "No — twice the shares at half the price; total value and market cap are unchanged. It improves affordability and liquidity."
- **"How does a buyback affect EPS and the statements?"** — "Cash and equity fall; share count shrinks, so EPS rises. It returns cash like a dividend but more flexibly/tax-efficiently. It only creates value if shares are bought below intrinsic value."
- **"Split vs bonus issue?"** — "Both increase shares and cut the price proportionally with no value change; a split changes the face value, a bonus capitalizes reserves into share capital."
- **"What is a rights issue and TERP?"** — "Existing holders can buy new shares at a discount; the price settles at a theoretical ex-rights price blending old and new. Non-subscribers are diluted."
- **"Why adjust historical prices for a split?"** — "So the chart and returns aren't distorted by the mechanical price change."

Traps & common mistakes

- Thinking a **split/bonus** makes shareholders richer — it's cosmetic.
- Confusing a **buyback's EPS boost** with value creation (depends on price paid).
- Forgetting **rights issues dilute** non-subscribers.
- Comparing pre- and post-split prices **without adjusting**.
- Missing the **ex-date** logic for dividends/entitlements.

First-principles recap

- Corporate actions change shares, price, and per-share metrics — often not total value.
- **Splits and bonus issues are cosmetic** (value unchanged).
- **Dividends and buybacks** return cash (value leaves the firm); buybacks lift EPS.
- **Rights issues** raise capital and dilute non-subscribers (price → TERP).
- Always adjust historical prices and share counts for splits/bonus.

Quick-reference

ACTION	VALUE EFFECT
Dividend	Cash out; price drops ex-date
Split / bonus	Cosmetic; total value unchanged
Buyback	Cash out; EPS up; value if bought cheap
Rights issue	Capital in; dilutes non-subscribers; TERP
Adjust for	Splits/bonus in prices & share counts

Q&A — Corporate Actions

Theory and worked numerical scenarios on dividends, splits, bonuses, rights issues, and buybacks.

Q1. Does a 2-for-1 stock split make a shareholder wealthier? Walk through why or why not with numbers.

Model answer. No. If an investor holds 100 shares at ₹500 (₹50,000 total value) and the company executes a 2-for-1 split, they now hold 200 shares at ₹250 (still ₹50,000 total value) — the market capitalisation and the investor's total holding value are mechanically unchanged; only the share count and per-share price have changed. The purpose of a split is to improve affordability (a lower per-share price may attract more retail participation) and liquidity (more shares outstanding can mean tighter spreads and more trading volume), not to create value.

Q2. Worked — buyback impact on EPS.

A company has net income of ₹240 cr and 60 mn shares outstanding (EPS = ₹4.00). It executes a buyback, repurchasing and cancelling 6 mn shares.

Model answer. New share count = 60 − 6 = 54 mn. New EPS = 240 / 54 = ₹4.44. EPS rose from ₹4.00 to ₹4.44 (an 11.1% increase) purely from the reduced share count, with no change in the underlying net income or business performance. Whether this buyback *created value* for remaining shareholders depends entirely on whether the repurchase price was below the shares' intrinsic value — buying back overvalued shares actually destroys value even while mechanically boosting EPS, a distinction the chapter emphasises as a common analytical trap.

Q3. Worked — theoretical ex-rights price (TERP) calculation.

A stock trades at ₹450 before a 1:3 rights issue (one new share for every three held) priced at ₹300.

Model answer. For every 3 existing shares (value 3 × 450 = ₹1,350), the holder can buy 1 new share at ₹300, giving 4 shares for a total cost of 1,350 + 300 = ₹1,650. TERP = 1,650 / 4 = ₹412.50. The price is expected to settle around ₹412.50 post-rights — a fall from ₹450, but this is not a loss for a shareholder who subscribes to their full entitlement: they now hold more shares at a lower average cost, and total portfolio value is preserved. A shareholder who does *not* subscribe is diluted and their proportional ownership falls, though they can potentially recoup some value by selling the rights entitlement itself if it's tradeable.

Q4. Distinguish which corporate actions are "cosmetic" (no total value change) from which are "value-changing," and explain the underlying reason for the distinction.

Model answer. Cosmetic actions — stock splits and bonus issues — merely restructure the existing equity claim into a different number of shares/different face value, drawing from the company's own reserves (bonus) or simply redefining share units (split), with no cash or assets actually leaving or entering the company; total value is mechanically preserved. Value-changing actions — dividends and buybacks — involve actual cash leaving the company to shareholders, directly reducing the company's asset base (and, in the case of a dividend, typically causing an ex-date price adjustment roughly equal to the dividend). Rights issues are value-changing in the other direction — real new cash enters the company, funding growth or debt reduction, at the cost of dilution for non-subscribers.

Q5. Why does a stock's price typically fall by roughly the dividend amount on the ex-dividend date, and is this a "loss" for existing shareholders?

Model answer. On the ex-dividend date, buyers of the stock are no longer entitled to the declared dividend (only holders as of the record date receive it) — so the stock is mechanically worth roughly the dividend amount less than it was the day before, reflecting that upcoming cash payment no longer being part of what a new buyer receives. This is not a loss for the existing shareholder who is entitled to the dividend: they hold a slightly lower-priced stock plus a dividend payment equal (approximately) to the price drop — their total value (stock plus declared dividend receivable) is essentially unchanged by the mechanical ex-date adjustment itself.

Q6. What's the difference between a bonus issue and a rights issue, and why would a company prefer one over the other?

Model answer. A bonus issue gives existing shareholders additional shares for free, funded by capitalising the company's own reserves — no new cash enters the company, and it's purely a cosmetic restructuring (Q4) often used as a signal of management confidence or to improve share liquidity/affordability. A rights issue requires shareholders to pay for the new shares (at a discount to market) — it genuinely raises new capital for the company. A company needing actual growth capital or debt reduction uses a rights issue; a company wanting to signal confidence or improve trading liquidity without needing new cash uses a bonus issue.

Q7. An analyst is building a 10-year historical price chart for a stock that underwent a 1:1 bonus issue five years ago. What adjustment must they make, and what happens if they skip it?

Model answer. They must adjust all historical prices *before* the bonus issue date by the bonus ratio (halving pre-bonus prices, in a 1:1 case) so the entire price series is on a consistent, comparable per-share basis. If they skip this adjustment, the chart will show an artificial, dramatic price "crash" exactly at the bonus-issue date that has nothing to do with the company's actual performance — a classic and easily avoidable analytical error that would badly distort any technical analysis, historical-return calculation, or valuation multiple trend built on the unadjusted series.

Q8. A company announces a large buyback funded entirely by increasing debt (leverage), rather than using existing cash reserves. What should an analyst think about beyond the simple EPS-accretion math?

Model answer. Beyond the mechanical EPS boost (Q2), a debt-funded buyback increases financial leverage and interest expense, raising the company's financial risk — the analyst should assess whether the resulting capital structure remains prudent (debt/EBITDA, interest coverage) and whether management is buying back shares because they're genuinely undervalued (a legitimate capital-allocation decision) or simply to engineer a short-term EPS increase without a corresponding improvement in the underlying business, sometimes to hit management compensation targets tied to EPS — a distinction directly relevant to assessing management quality and capital-allocation discipline (a recurring theme from the fundamental-analysis chapter).

Capital Raising: Follow-ons, Rights, QIPs & Block Deals

The Problem / Why this matters

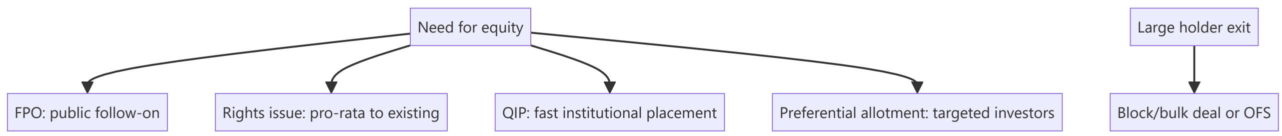
An IPO is a company's first equity raise, but listed companies raise equity again and again — to fund growth, cut debt, or let large holders exit. **Equity capital markets (ECM)** covers all these follow-on ways to issue or place equity: FPOs, rights issues, QIPs, preferential allotments, and block/bulk deals. Knowing each mechanism, when it's used, and its effect on existing shareholders is core to ECM, IB, and research roles.

Core Idea

Beyond the IPO, listed firms raise or place equity through several routes, differing by **who can buy, how fast, and at what dilution**: broad public offers (FPO), pro-rata offers to existing holders (rights), fast institutional placements (QIP), targeted placements (preferential), and secondary sales of existing shares (blocks/OFS). Each balances speed, price, investor base, and dilution.

Why it works this way

Different situations call for different tools: a company wanting a quick institutional raise uses a QIP (fast, no lengthy prospectus); one wanting to treat all shareholders fairly uses a rights issue; a large holder wanting to exit uses a block deal or OFS. The menu exists so issuers can optimize for speed, price, and the desired investor base.



Full technical content

Primary raises (new shares → company gets cash, dilution):

ROUTE	WHO BUYS	SPEED	NOTES
FPO	Public	Slow (prospectus)	A full follow-on public offer
Rights issue	Existing holders, pro-rata	Moderate	Usually at a discount; non-subscribers diluted; price → TERP
QIP	Qualified institutional buyers	**Fast**	India-specific; no lengthy prospectus; institutions only
Preferential allotment	Selected investors	Fast	Targeted (e.g., a strategic/PE investor); priced per SEBI formula

Secondary sales (existing shares → selling holder gets cash, no new capital, no dilution):

ROUTE	WHAT IT IS
OFS (Offer for Sale)	Promoters/large holders sell via an exchange mechanism
Block deal	A large negotiated trade (min size) in a special window at a pre-agreed price
Bulk deal	Large trades executed in the normal market (disclosed)

Key distinctions: - Fresh issue vs secondary sale — fresh issue raises capital for the company and dilutes; secondary sale just transfers existing shares (holder gets cash, no dilution). - **Dilution** — new shares reduce existing holders' ownership % and EPS unless the capital earns above the cost of equity. - **Pricing** — placements are usually at a small discount to market to attract buyers; rights at a larger discount; SEBI sets floor-price formulas for QIPs/preferential. - **Signaling** — equity issuance can signal management thinks shares are fairly/over-valued (pecking-order); a raise to cut debt vs to fund high-return growth is read very differently.

ECM's job. Investment banks (ECM desks) advise on the route, timing, size, and pricing, and place the shares with investors — balancing the issuer's proceeds against a successful, well-received deal.

Worked examples

Example 1 — QIP for speed. A company needs ₹1,500 cr quickly to fund expansion. Rather than a months-long FPO, it does a **QIP**: places new shares with institutions at a ~3% discount to the market over a couple of days. Fast, institutional, and dilutive — the company gets the cash without a full prospectus process.

Example 2 — rights issue to deleverage. A leveraged firm does a **1:2 rights issue** at a 30% discount to cut debt. Existing holders can maintain their stake by subscribing; those who don't are diluted but can sell their rights. The stock re-rates to TERP; the balance sheet strengthens.

Example 3 — block deal exit (no dilution). A PE fund holding 8% wants to exit. It sells the entire stake in a single **block deal** to institutions at a small discount in the block window. The *fund* receives the cash; the company issues no new shares, so there's **no dilution** — just a change of ownership. (Contrast with a QIP, which is new shares and dilutive.)

Example 4 — a worked dilution and EPS-impact calculation for a QIP. A company with 20 crore shares outstanding and current-year net profit of ₹400 cr (EPS = ₹20) raises ₹600 cr via QIP at ₹300/share, issuing 2 crore new shares. Post-QIP share count = 22 crore. If the newly raised capital is simply held as cash or used to pay down debt (not yet deployed into earnings-generating assets), diluted EPS falls to 400/22 ≈ **₹18.18** — an immediate ~9% EPS dilution. If instead the ₹600 cr is deployed into a project expected to earn a 15% post-tax return (₹90 cr incremental profit once ramped), pro-forma EPS becomes (400+90)/22 ≈ **₹22.27** — accretive. This is the calculation an analyst must run before mechanically treating any capital raise as automatically negative for EPS: the answer depends entirely on the return the new capital earns relative to what a simple share-count increase alone would imply, precisely the same ROIC-vs-WACC logic from the fundamental-analysis chapter applied to a specific financing event.

Example 5 — a preferential allotment used for a strategic partnership, not just capital. A mid-cap technology company issues a preferential allotment of 5% of its equity to a large global technology firm at a modest premium to market price, as part of a broader commercial partnership (the global firm becomes a channel partner and technology licensor). An analyst evaluating this shouldn't only check the SEBI floor-price compliance and dilution math (Part 16's core content) — the more important question is whether the strategic relationship itself creates value beyond the capital raised (access to new markets, technology, or credibility that wouldn't otherwise be available), which is exactly the kind of qualitative judgment call (similar to Part 16's earlier discussion of assessing a preferential allottee) that separates a mechanical dilution calculation from genuine investment analysis of whether the deal is good for existing shareholders.

How it is tested in interviews

- **"Ways a listed company can raise equity?"** — FPO, rights issue, QIP, preferential allotment (primary/dilutive); plus OFS/block deals for existing holders to sell (secondary, non-dilutive).
- **"What is a QIP and why use it?"** — "A Qualified Institutional Placement — a fast raise from institutions without a full prospectus; used when a company wants speed and an institutional base."
- **"Rights issue vs QIP?"** — "Rights is pro-rata to all existing holders (fair, at a discount, non-subscribers diluted); QIP is a quick placement to institutions only."
- **"Block deal vs QIP — is it dilutive?"** — "A block deal is a secondary sale of existing shares — the seller gets the cash, no dilution. A QIP is new shares — the company gets the cash, dilutive."
- **"Why might a raise move the stock down?"** — "Dilution, plus the signal that management may think shares are fully valued; the use of proceeds (debt reduction vs high-return growth) matters."

Traps & common mistakes

- Confusing **primary** (new shares, company cash, dilution) with **secondary** (existing shares, holder cash, no dilution).
- Thinking a **block deal/OFS** raises capital for the company — it doesn't.
- Forgetting **rights issues dilute** non-subscribers and reset the price to TERP.
- Ignoring the **signaling** and use-of-proceeds read of a raise.

First-principles recap

- Listed firms raise equity via FPO, rights, QIP, preferential (primary, dilutive) and sell via OFS/blocks (secondary, non-dilutive).
- **QIP** = fast institutional raise; **rights** = pro-rata to existing holders at a discount.
- Fresh issue → company cash + dilution; secondary sale → holder cash, no dilution.
- Placements price at a discount; SEBI sets floors.
- Read every raise for **dilution, signaling, and use of proceeds**.

Quick-reference

ROUTE	TYPE	CASH TO	DILUTION
FPO	Primary	Company	Yes
Rights	Primary, pro-rata	Company	Yes (non-subscribers)
QIP	Primary, institutions	Company	Yes
Preferential	Primary, targeted	Company	Yes
OFS / block	Secondary	Selling holder	No

Q&A — Capital Raising: Follow-ons, Rights, QIPs & Block Deals

Theory and worked scenarios on ECM mechanisms beyond the IPO.

Q1. List the primary (dilutive, company receives cash) capital-raising routes and the secondary (non-dilutive, existing holder receives cash) routes, and state the key distinguishing question to classify any deal you encounter.

Model answer. Primary routes: FPO, rights issue, QIP, preferential allotment — all involve newly issued shares, with proceeds going to the company, and all dilute existing non-participating shareholders. Secondary routes: OFS, block deals, bulk deals — all involve a transfer of *existing* shares, with proceeds going to the selling shareholder, and no dilution or new capital for the company. The key distinguishing question for any deal: "are new shares being created (primary), or are existing shares just changing hands (secondary)?" — this single question resolves the classification and its capital-structure/dilution implications immediately.

Q2. Why would a company choose a QIP over a rights issue when both raise capital from investors, and what does the company give up by choosing QIP?

Model answer. A QIP is dramatically faster (days, versus the weeks-to-months a rights issue with its formal offer document and subscription period requires) and doesn't require the extended prospectus/disclosure process, since it's placed only with qualified institutional buyers. What the company gives up: fairness/inclusivity to retail and smaller existing shareholders (who get no opportunity to participate and maintain their proportional stake, unlike in a rights issue), and potentially a broader/more diversified new-investor base — a QIP concentrates the new shares among a smaller number of institutional holders. Companies needing capital urgently, or comfortable diluting retail without offering them participation, favour QIP; companies wanting to treat all existing shareholders equitably favour rights issues despite the slower timeline.

Q3. Worked — is this transaction dilutive? A private equity fund holding 12% of a listed company sells its entire stake to a group of mutual funds via a block deal at a 2% discount to the prevailing market price.

Model answer. This is a secondary transaction — the PE fund is selling *existing* shares it already owns to other investors; no new shares are created, so there is zero dilution to any other shareholder, and the company itself receives no cash from this transaction (only the exiting PE fund receives proceeds). The only effect on the company's shareholder register is a change in *who* holds that 12% stake (institutional buyers replacing the PE fund), not a change in total shares outstanding or the company's capital position. A common interview trap is assuming any large equity transaction involving a listed company is dilutive — always check whether shares are newly issued (primary) or merely transferred (secondary) first.

Q4. What does it mean that "SEBI sets floor-price formulas for QIPs and preferential allotments," and why does this regulation exist?

Model answer. Rather than letting a company negotiate an arbitrarily low placement price with select institutional or preferential investors, SEBI mandates a minimum ("floor") price, typically calculated from a formula based on recent average trading prices over a specified look-back window. This exists to protect existing (particularly retail) shareholders from being diluted at an unfairly cheap price that would transfer value from existing holders to the new placement investors — without a floor-price rule, a company (or its promoters, in a preferential-allotment scenario benefiting an insider) could place new shares to favoured parties at a steep, unjustified discount, effectively expropriating value from other shareholders.

Q5. How should an analyst interpret the signal when a company announces a large equity raise specifically to fund debt repayment, versus one raising equity to fund high-return growth capex?

Model answer. Equity raised to repay debt is often read as a defensive, deleveraging move — potentially signalling the company (or its bankers) judge the current leverage level as risky, and it dilutes existing shareholders without adding new earning assets, generally a less positive signal for near-term EPS/growth. Equity raised to fund high-return growth capex can be read more positively if the analyst believes the incremental capital will earn well above its cost (ROIC > WACC, per the fundamental-analysis chapter) — dilution is a real cost, but it's justified if it funds genuinely value-accretive investment. The "why" behind a raise (stated use of proceeds) is therefore as important to the analyst's read as the raise's size and pricing.

Q6. What is the "pecking order" signalling concern raised in this chapter regarding equity issuance, and how does it apply to interpreting a surprise equity raise?

Model answer. Pecking-order theory suggests companies prefer to fund needs with internal cash first, then debt, and equity issuance last — because managers, who typically have better information about the company's prospects than outside investors, are more likely to issue equity when they believe the stock is fairly valued or overvalued (since issuing undervalued equity would unnecessarily give away value to new shareholders at existing holders' expense). This means a surprise equity raise can itself be read by the market as a mild negative signal about management's view of the current valuation — a nuance an analyst should be alert to beyond just the raise's stated purpose, since the *choice* to raise equity at all (versus debt) carries information.

Q7. Explain the difference between a block deal and a bulk deal, and why the distinction (special window vs normal market) matters for how each affects the stock's observed price.

Model answer. A block deal is executed in a special, separate trading window with a minimum transaction size, at a pre-agreed price negotiated directly between buyer and seller — because it happens outside the regular continuous order book, a large block deal doesn't directly "walk the book" and cause the market-impact price movement that an equivalent-sized order would in normal trading (see the secondary-markets chapter's market-impact discussion). A bulk deal is a large trade executed within the normal market during regular trading hours (just disclosed due to its size) — it does interact with and can move the regular order book, similar to any other large market order. The special block window exists specifically to let large holders transact size without unduly disrupting the continuous market price.

Q8. A company does a preferential allotment to a single strategic investor at a price close to the SEBI floor price, shortly after a period of share-price weakness. What should an analyst check before concluding this is either clearly positive or clearly negative for existing shareholders?

Model answer. Check: (1) who the strategic investor is and whether their involvement brings genuine strategic value (technology, market access, credibility) beyond just capital — a well-regarded strategic investor committing capital can be a positive signal even at a discount; (2) the actual discount to the pre-announcement price relative to the SEBI-mandated floor — a price right at the regulatory floor (the minimum legally allowed) versus a smaller discount matters for how much value existing shareholders are giving up; (3) the stated use of proceeds and whether it addresses a genuine, previously-flagged need (e.g. deleveraging, funding a specific growth initiative) versus appearing opportunistic; (4) whether the preferential allottee has any pre-existing relationship with promoters/management that could raise related-party or governance concerns. No single factor alone determines whether the deal is good or bad for existing shareholders — it requires weighing all of these together.

Sell-Side vs Buy-Side

The Problem / Why this matters

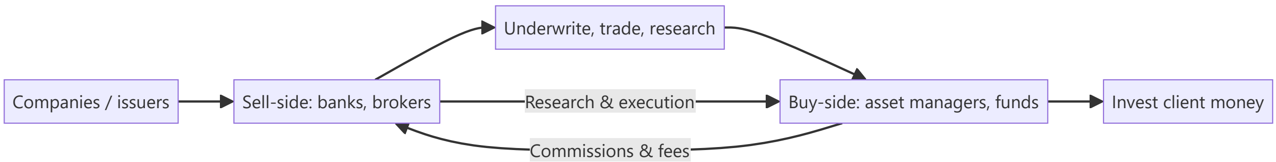
The finance industry splits into two worlds — the **sell-side** (banks and brokers who create, sell, and research products) and the **buy-side** (asset managers who invest client money). Every markets role sits on one side, they interact constantly, and interviewers frequently ask which you want and why. Understanding the distinction — who does what, how each makes money, and how research flows between them — is basic industry literacy.

Core Idea

The **sell-side** creates and sells financial products and services (underwriting, trading, research) to clients and earns fees/commissions. The **buy-side** buys and holds investments on behalf of clients (funds) and earns management/performance fees on assets. Sell-side research is *published* to win business; buy-side research is *private* and drives the fund's own decisions.

Why it works this way

The two exist because issuers and investors need different services. The sell-side intermediates — helping companies raise capital and helping investors execute and understand — monetized through transactions. The buy-side is the end investor of capital — monetized through investment performance on assets under management. Research flows from sell-side (marketing/service) to buy-side (decision input).



Full technical content

Sell-side (investment banks, brokerages):

FUNCTION	WHAT IT DOES
Investment banking (ECM/DCM/M&A)	Raise capital and advise on deals for companies
Sales & trading	Execute trades, make markets, provide liquidity
Sell-side research	Publish analysis and ratings on stocks/sectors to clients
Earns: **underwriting fees, advisory fees, trading commissions, spreads.** Sell-side research is *distributed* to institutional clients to win their trading/banking business — it's a service, not a profit centre itself.	

Buy-side (asset managers):

TYPE	DESCRIPTION
Mutual funds / AMCs	Pooled long-only funds, benchmark-relative
Hedge funds	Flexible, absolute-return, can short/leverage
Pension & insurance funds	Long-term, liability-driven
PE / VC	Private investing
PMS / AIFs	Portfolio management for HNIs
Earns: **management fees (% of AUM)** and often **performance fees** (e.g., 2-and-20 for hedge funds). Buy-side research is *internal* and confidential — it directly drives what the fund buys.	

How they interact: - Sell-side provides **execution** (trading), **capital-raising access** (IPO allocations), and **research** to the buy-side. - Buy-side pays via **commissions** and consumes sell-side research as one input (increasingly unbundled post-MiFID II). - The buy-side ultimately makes the investment decision; the sell-side facilitates.

Research differences:

	SELL-SIDE RESEARCH	BUY-SIDE RESEARCH
Audience	Published to many clients	Internal to the fund
Purpose	Win business/commissions	Make the fund's own decisions
Coverage	Broad, many names	Focused on holdings/ideas
Incentive	Volume, access, relationships	Being *right* (performance)

Career framing. Sell-side research/banking: broad coverage, client-facing, deal/idea flow. Buy-side: fewer names, deeper conviction, direct accountability for returns. Many move sell-side → buy-side over a career.

Worked examples

Example 1 — a research idea's journey. A sell-side analyst publishes a Buy on Stock X with a note. It's distributed to hundreds of institutional clients. A buy-side PM reads it, does her *own* work, disagrees on one assumption, but likes the idea — and buys through the same bank's trading desk (paying commission). The sell-side earned the commission and strengthened the relationship; the buy-side made the actual decision.

Example 2 — how each makes money. A brokerage (sell-side) earns ₹5 cr in commissions from a fund's trading this year. The fund (buy-side) manages ₹5,000 cr and earns a 1% management fee = ₹50 cr, plus performance fees if it beats its benchmark. Different economics: transaction fees vs fees on assets and performance.

Example 3 — incentive difference. A sell-side analyst is rewarded for coverage, client access, and generating trading interest; a buy-side analyst is rewarded purely for whether the calls make money. This is why buy-side research is more focused and conviction-driven — being *right* is the only thing that pays.

Example 4 — a worked management-fee-plus-performance-fee calculation. A hedge fund manages ₹2,000 cr, charges a 1.5% management fee and a 20% performance fee above a hurdle rate of 8%. In a year the fund returns 22%. Management fee = 1.5% × 2,000 cr = ₹30 cr, charged regardless of performance. Performance fee applies only to the return *above* the 8% hurdle: excess return = 22% − 8% = 14%, on which the fund earns 20% = 14% × 20% = 2.8% of AUM = ₹56 cr. Total fund revenue this year: ₹30 cr + ₹56 cr = **₹86 cr**, versus an investor's net return after fees of roughly 22% − 1.5% − 2.8% ≈ **17.7%**. This concrete "2-and-20-with-a-hurdle" calculation is a standard technical question for anyone claiming buy-side/hedge-fund interest, and the hurdle-rate mechanic specifically (performance fees only above a minimum threshold, not on the whole return) is the detail most candidates get wrong by omitting it.

Example 5 — how MiFID-II-style research unbundling changed sell-side economics in practice. Before unbundling, a buy-side fund's trading commissions implicitly paid for both execution *and* the sell-side research bundled alongside it — a fund couldn't easily tell how much it was really paying for research specifically. Post-unbundling, funds must pay for research explicitly and separately from execution costs, forcing them to actively budget and justify research spend against its measured value. The practical effect on sell-side desks: consolidation toward fewer, higher-conviction analysts covering fewer names with more differentiated views, since broad, generic coverage that used to be "free" (bundled into commissions) no longer automatically earns its keep once a fund has to consciously decide whether a specific research relationship is worth paying for directly.

How it is tested in interviews

- **"What's the difference between sell-side and buy-side?"** — "Sell-side (banks/brokers) creates and sells products — underwriting, trading, research — for fees and commissions. Buy-side (asset managers) invests client money for management and performance fees. Sell-side research is published to win business; buy-side research is private and drives the fund's decisions."
- **"How does each make money?"** — Sell-side: underwriting/advisory fees, trading commissions/spreads. Buy-side: % of AUM management fees plus performance fees.
- **"Do you want sell-side or buy-side, and why?"** — Have a genuine answer: sell-side for broad coverage and client/deal exposure; buy-side for deep conviction and direct accountability for returns.
- **"How do sell-side and buy-side research differ?"** — Audience (published vs internal), purpose (win business vs make money), and incentive (volume/access vs being right).

Traps & common mistakes

- Mixing up which side **invests** (buy-side) vs **intermediates** (sell-side).
- Thinking sell-side research is the sell-side's **profit centre** — it's a service to win trading/banking.
- Not having a **genuine preference** with reasons when asked.
- Forgetting the buy-side does its **own** work and makes the final call.

First-principles recap

- **Sell-side** intermediates (underwrite, trade, research) for fees/commissions.
- **Buy-side** invests client money for fees on assets and performance.
- Research flows sell-side → buy-side; the buy-side decides.
- Incentives differ: sell-side rewards access/volume; buy-side rewards being right.
- Careers often move sell-side → buy-side.

Quick-reference

	SELL-SIDE	BUY-SIDE
Role	Create/sell/execute/research	Invest client money
Examples	IBs, brokerages	Mutual/hedge/pension funds, PMS
Revenue	Fees, commissions, spreads	% AUM + performance fees
Research	Published, wins business	Internal, drives decisions
Reward	Access/volume	Performance (being right)

Q&A — Sell-Side vs Buy-Side

Theory and scenario drills on industry structure, economics, and career framing.

Q1. Explain precisely why sell-side research is described as "not a profit centre itself" — if it doesn't directly make money, why do banks maintain large research departments?

Model answer. Sell-side research is typically distributed to institutional clients without a direct per-report fee (especially pre-unbundling regulatory regimes) — its function is to win and retain the trading commissions, IPO allocation access, and banking relationships that *are* the actual revenue sources. A highly-regarded analyst who consistently produces valuable research attracts institutional client relationships and trading flow to the bank's desks; research is best understood as a marketing and relationship function that indirectly drives commission and banking revenue, not a standalone profit line measured against research-department costs alone.

Q2. How do sell-side and buy-side analysts' incentives differ, and how does that difference shape the character of their research output?

Model answer. A sell-side analyst is rewarded substantially for breadth of coverage, client access, and generating trading interest/relationships — incentives that favour maintaining ratings on many names, being responsive to client questions, and sometimes avoiding overly controversial calls that could damage banking relationships with covered companies. A buy-side analyst is rewarded purely on whether their calls make money for the fund — incentives that favour deep conviction on a smaller number of ideas and genuine intellectual honesty about being wrong, since there's no relationship-management layer diluting the accountability. This difference is why buy-side research is typically described as more focused and conviction-driven.

Q3. Worked scenario — a sell-side analyst's Buy rating and a buy-side fund's actual position disagree. What does this reveal, and is it a problem?

Model answer. A sell-side Buy rating represents one analyst's published view, distributed as one input among many to potentially hundreds of buy-side clients, each of whom does their own independent analysis. It is entirely normal and healthy for a buy-side fund to read the sell-side note, disagree with a specific assumption, and reach a different conclusion (or even the opposite conclusion) — the sell-side analyst's job is to provide a well-reasoned, differentiated view and supporting analysis, not to dictate the buy-side's actual investment decision. If every buy-side fund simply followed every sell-side rating without independent work, there would be no genuine price discovery or alpha generation on the buy-side at all.

Q4. Describe the economics difference between a sell-side brokerage and a buy-side asset manager with a worked comparison.

A brokerage earns ₹8 cr in annual commissions from a specific fund's trading activity. That same fund manages ₹4,000 cr in assets, charging a 1.25% management fee plus a 15% performance fee on returns above its benchmark, and beats its benchmark by 3% this year.

Model answer. Brokerage revenue from this relationship = ₹8 cr (pure commission on transaction volume, independent of whether the fund's trades were ultimately profitable). Fund's management fee = 1.25% × 4,000 cr = ₹50 cr (charged regardless of performance). If outperformance is measured on, say, a ₹4,000 cr base, a 3% outperformance generates ₹120 cr of excess return, of which the fund captures 15% = ₹18 cr in performance fees. Total buy-side economics (₹50 cr + ₹18 cr = ₹68 cr) dwarfs the sell-side's ₹8 cr commission from this single relationship — illustrating why buy-side economics scale with assets under management and performance, while sell-side economics scale with transaction volume, a structurally different (and, at meaningful AUM, typically larger) revenue model.

Q5. What genuine reasons might a candidate give for preferring buy-side over sell-side, and vice versa, beyond generic statements like "I want to make investment decisions"?

Model answer. For buy-side: direct accountability (the P&L reflects your own calls, not a published rating diluted across many clients' independent decisions), typically fewer, higher-conviction names to cover allowing real depth, and a career path more directly tied to demonstrated investing skill. For sell-side: broader exposure across many companies/sectors early in a career (useful for building pattern recognition before specialising), more client-facing and deal-flow exposure (useful for those interested in the transactional/advisory side or building a wide professional network), and a well-trodden training-ground role that many buy-side firms explicitly like to hire from. A strong interview answer states a genuine, specific reason tied to the candidate's actual background and goals rather than a generic preference.

Q6. Why might a company or its investment bank prefer that sell-side research remains largely aligned with, rather than sharply critical of, that company — and what tension does this create for research objectivity?

Model answer. A bank's research on a company it also does (or hopes to do) investment banking business with (underwriting, M&A advisory) creates a structural conflict of interest — overly negative research could jeopardise that banking relationship, while research too closely aligned with management's own narrative undermines the research's actual value and credibility to buy-side clients. This is precisely why regulatory "Chinese wall" separations between research and banking departments exist, and why analysts are formally required to disclose banking relationships alongside their ratings — a well-prepared candidate should be able to name this conflict-of-interest tension directly rather than pretend sell-side research is fully independent of commercial pressures.

Q7. What does it mean that sell-side research economics have shifted "post-MiFID II" toward unbundling, and why does this matter to how research value is now assessed?

Model answer. MiFID II (a European regulatory regime with global ripple effects on institutional research practices) required buy-side firms to pay explicitly for research rather than have it bundled implicitly into trading commissions — this forced buy-side firms to directly justify and budget research spend against its actual, measurable value, rather than treating it as a "free" byproduct of trading relationships. The practical effect has been increased scrutiny on which sell-side research genuinely adds value (leading some firms to consolidate research spend on fewer, higher-conviction providers) — a structural shift a candidate discussing the sell-side research business model should be aware of.

Q8. Describe a realistic career path that moves from sell-side to buy-side, and what specifically makes an analyst attractive for that transition.

Model answer. A common path: several years as a sell-side research associate/analyst building deep sector expertise and a track record of well-reasoned, differentiated calls (not just accurate but well-argued and risk-aware, since buy-side hiring managers scrutinise the *reasoning* behind calls, not just whether they happened to be right), followed by a move to a buy-side fund as a research analyst or associate portfolio manager. What makes a sell-side analyst attractive for this move: demonstrated ability to form genuinely differentiated (not consensus-hugging) views, a track record where calls can be checked against actual subsequent stock performance, deep sector/company knowledge built over the coverage years, and often personal relationships built with buy-side clients during the sell-side tenure who can vouch for the analyst's judgment.

Portfolio Construction & Risk

The Problem / Why this matters

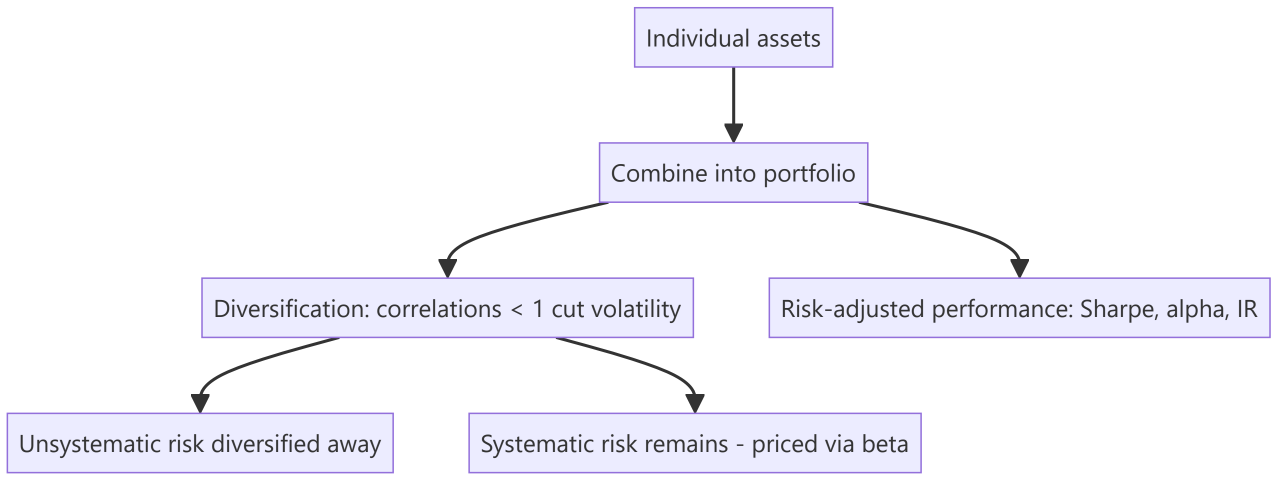
Owning a single great stock is risky; investing is really about building a **portfolio** — combining positions so the whole is better (higher return per unit of risk) than the parts. How you size positions, diversify, benchmark, and measure risk-adjusted performance is the core of asset management and any investing role. Interviewers test diversification, systematic vs unsystematic risk, beta, and the performance ratios.

Core Idea

Portfolio construction combines assets to maximize expected return for a given level of risk, exploiting **diversification** (imperfectly correlated assets reduce portfolio volatility). Risk splits into **systematic** (market, undiversifiable) and **unsystematic** (specific, diversifiable), and performance is judged on a **risk-adjusted** basis (Sharpe, alpha, information ratio), not raw return.

Why it works this way

Because asset returns aren't perfectly correlated, combining them means their ups and downs partly offset, lowering portfolio volatility without proportionally lowering return — the "only free lunch in finance." But the shared exposure to the economy (systematic risk) can't be diversified away, so it's what investors are ultimately compensated for (via beta and the risk premium).

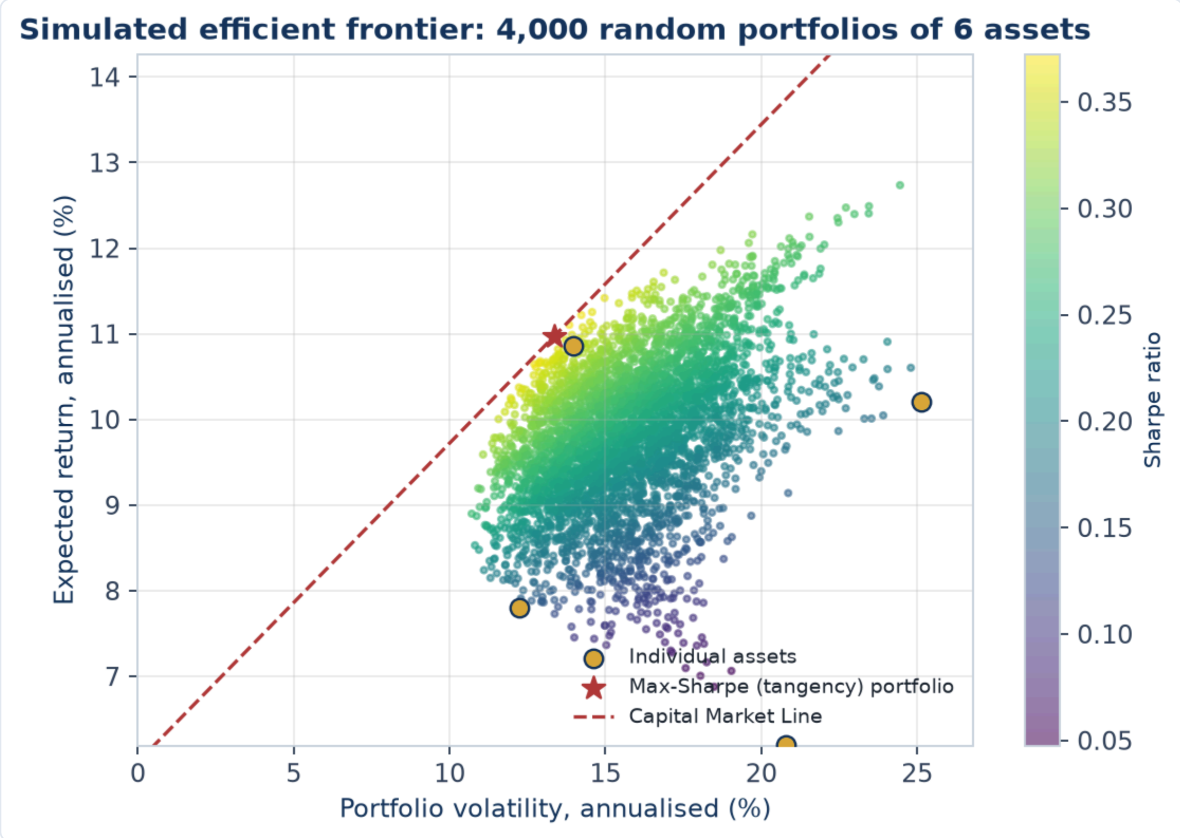


Full technical content

Risk decomposition: - Unsystematic (specific/idiosyncratic) risk — company/sector-specific; **diversifiable** by holding many uncorrelated names. - **Systematic (market) risk** — common exposure to the economy; **not diversifiable**; measured by **beta**; it's what earns the risk premium.

Diversification. Combining assets with correlation < 1 reduces portfolio standard deviation for a given return. The benefit is largest when correlations are low/negative; most idiosyncratic risk is removed with ~20–30 well-chosen names, leaving mainly systematic risk.

Modern Portfolio Theory (Markowitz). For a given return, there's a minimum-variance portfolio; the set of best risk-return combinations is the **efficient frontier**. Adding a risk-free asset gives the **Capital Market Line**; the best risky portfolio is the market portfolio (→ CAPM).



The chart above simulates 4,000 random-weight portfolios of 6 assets, coloured by Sharpe ratio. The individual assets (gold dots) all sit *inside* the cloud of possible portfolios — no single asset is as efficient as a well-diversified combination, the visual proof of Example 1's diversification math. The star marks the max-Sharpe (tangency) portfolio, and the dashed Capital Market Line shows the best achievable risk-return trade-off once a risk-free asset is available to blend with it — exactly the CAPM intuition this section describes.

Position sizing & construction approaches: - **Concentration vs diversification** — high-conviction concentrated books (fewer, larger positions) vs broadly diversified books; a trade-off between idiosyncratic risk and edge. - **Active vs passive** — active tries to beat a benchmark (higher fees, tracking error); passive replicates it cheaply. - **Benchmark-relative** — most funds are measured vs an index; **active weights** (overweight/underweight vs the benchmark) express views; **tracking error** measures deviation. - **Constraints** — position limits, sector caps, liquidity, mandate.

Risk-adjusted performance metrics:

METRIC	FORMULA	MEASURES
Sharpe ratio	(Rp − Rf) / σp	Excess return per unit of total risk
Treynor ratio	(Rp − Rf) / βp	Excess return per unit of market risk
Alpha (Jensen's)	Rp − [Rf + β(Rm − Rf)]	Return above CAPM (skill)
Information ratio	Active return / tracking error	Consistency of outperformance vs benchmark
Beta	Cov(Rp,Rm)/Var(Rm)	Sensitivity to the market

Sharpe uses **total** risk (σ), Treynor uses **systematic** risk (β) — use Treynor when the portfolio is one part of a diversified whole. **Alpha** is the holy grail: return not explained by market exposure.

Rebalancing. Periodically returning to target weights (as prices drift) enforces "sell high, buy low" and controls risk.

Worked examples

Example 1 — diversification cuts risk. Two stocks each with 20% volatility and a correlation of 0.3. A 50/50 portfolio has volatility = $\sqrt{(0.5^2 \cdot 0.2^2 + 0.5^2 \cdot 0.2^2 + 2 \cdot 0.5 \cdot 0.5 \cdot 0.3 \cdot 0.2 \cdot 0.2)} = \sqrt{(0.01 + 0.006)} \approx \sqrt{0.016}$ ≈ **12.6%** — well below the 20% of either stock alone, for the average return. That's the diversification benefit from correlation < 1.

Example 2 — Sharpe ratio comparison. Portfolio A returns 15% with 10% volatility; Portfolio B returns 20% with 25% volatility; risk-free 5%. Sharpe A = (15−5)/10 = **1.0**; Sharpe B = (20−5)/25 = **0.6**. B has the higher return but A is the better *risk-adjusted* performer — you'd prefer A (and could lever it to match B's return at lower risk).

Example 3 — alpha vs beta. A fund returned 18% when the market (beta 1.2 exposure) returned 12%, risk-free 5%. CAPM-expected = 5 + 1.2(12−5) = 13.4%. **Alpha = 18 − 13.4 = +4.6%** — genuine outperformance beyond what its market exposure explains. Distinguishing this skill (alpha) from just taking market risk (beta) is the whole game.

Example 4 — building a 4-stock portfolio from scratch, with a concentration/diversification trade-off. A PM has ₹10 cr to deploy across a high-conviction book. Four candidate holdings have expected returns and volatilities of: Stock A (18%, 22%), Stock B (14%, 15%), Stock C (22%, 30%), Stock D (11%, 12%), with an average pairwise correlation of 0.35 among them (a realistic same-market, imperfectly correlated basket). An equal-weight (25% each) portfolio's expected return is simply the weighted average: 0.25×(18+14+22+11) = **16.25%**. Portfolio volatility, using the full correlation structure, comes out meaningfully below the weighted-average volatility of 19.75% — illustrating the same diversification math as Example 1, but now across four holdings instead of two, where the benefit compounds as more imperfectly-correlated positions are added. **The concentration trade-off**: if the PM instead conviction-weights toward the highest-expected-return, highest-volatility names (say 40% A, 10% B, 40% C, 10% D), expected return rises to 0.4×18+0.1×14+0.4×22+0.1×11 = **18.5%**, but so does portfolio volatility, since the weight is now concentrated in the two highest-volatility, still-correlated names — a direct, numeric illustration of Part 18's "concentration vs diversification" trade-off (7.1's position-sizing discussion generalised to a full portfolio): more conviction-weighting raises both expected return *and* risk, and the PM must judge whether the extra ~2.25 points of expected return justifies the higher realised volatility and larger single-name drawdown exposure, given the fund's actual risk mandate and client risk tolerance.

Example 5 — factor investing in practice. A fund constructs a "value + momentum" combined strategy: it ranks its investable universe by a value score (low P/B, low P/E) and a momentum score (12-month trailing return, per Part 14's anomaly discussion), buying stocks that rank favourably on *both* factors simultaneously rather than either alone. The logic: value and momentum have historically shown low or even negative correlation as standalone factors (value tends to do best when momentum struggles, and vice versa, since they capture different market inefficiencies — value bets on mean-reversion in cheap stocks, momentum bets on trend continuation in recent winners) — combining them in a single portfolio can produce a smoother overall return stream than either factor run alone, the same diversification logic from Example 1 and Example 4, applied across *factors* rather than across individual stock-specific risk. This is why many quantitative equity strategies deliberately blend multiple, historically-uncorrelated factors (value, momentum, quality, low-volatility) rather than betting on a single factor.

How it is tested in interviews

- **"Why diversify?"** — "Because assets aren't perfectly correlated, combining them lowers portfolio volatility for a given return — it removes diversifiable, idiosyncratic risk, leaving mostly systematic risk."
- **"Systematic vs unsystematic risk?"** — "Unsystematic is company-specific and diversifiable; systematic is market-wide, undiversifiable, and measured by beta — it's what earns the risk premium."
- **"What does the Sharpe ratio measure?"** — "Excess return per unit of total risk (volatility) — a risk-adjusted performance measure; higher is better."
- **"Sharpe vs Treynor?"** — "Sharpe uses total risk (σ); Treynor uses systematic risk (β). Use Treynor when the portfolio is part of a broader diversified portfolio."
- **"What is alpha?"** — "Return above what CAPM/beta predicts — the manager's skill, not explained by market exposure."

Traps & common mistakes

- Judging on **raw return** instead of **risk-adjusted** return.
- Thinking diversification removes **all** risk — systematic risk remains.
- Confusing **Sharpe (total risk)** and **Treynor (systematic risk)**.
- Confusing **alpha** (skill) with **beta** (market exposure) — high returns from leverage/beta aren't alpha.
- Over-diversifying to closet-indexing (paying active fees for index-like returns).

First-principles recap

- Portfolios exploit **diversification** (correlations < 1) to raise return per unit of risk.
- Risk = **systematic** (market, priced via beta) + **unsystematic** (specific, diversifiable).
- Judge performance **risk-adjusted**: Sharpe, Treynor, alpha, information ratio.
- **Alpha** = skill beyond market exposure; **beta** = market exposure.
- Rebalance to targets; manage tracking error vs the benchmark.

Quick-reference

METRIC	FORMULA
Beta	Cov(Rp,Rm)/Var(Rm)
Sharpe	(Rp − Rf)/σp
Treynor	(Rp − Rf)/βp
Alpha	Rp − [Rf + β(Rm − Rf)]
Information ratio	Active return / tracking error
Diversifiable	Unsystematic only

Q&A — Portfolio Construction & Risk

Theory and worked numerical problems on diversification, risk decomposition, and risk-adjusted performance.

Q1. Worked — two-asset portfolio volatility.

Stock A: 25% volatility. Stock B: 18% volatility. Correlation between A and B: 0.2. Portfolio is 60% A, 40% B.

Model answer.

```
σp² = w_A² σ_A² + w_B² σ_B² + 2 w_A w_B ρ σ_A σ_B
    = 0.6² × 0.25² + 0.4² × 0.18² + 2 × 0.6 × 0.4 × 0.2 × 0.25 × 0.18
    = 0.36 × 0.0625 + 0.16 × 0.0324 + 0.48 × 0.2 × 0.045
    = 0.0225 + 0.005184 + 0.00432
    = 0.032004
σp = √0.032004 ≈ 17.89%
```

The portfolio's volatility (≈17.9%) is below the simple weighted-average volatility of the two holdings (0.6×25 + 0.4×18 = 22.2%) — the gap between 22.2% and 17.9% is the diversification benefit, arising directly from the correlation being well below 1.

Q2. Why can't systematic risk be diversified away, no matter how many additional uncorrelated stocks are added to a portfolio?

Model answer. Systematic risk represents exposure to broad market/economic factors (interest rates, GDP growth, systemic shocks) that affect essentially all stocks simultaneously to some degree — since virtually every stock has some positive correlation to the overall market, adding more holdings can eliminate the *company-specific* (unsystematic) portion of risk that's genuinely uncorrelated across names, but it cannot eliminate the shared, market-wide component that every stock carries. This is precisely why systematic risk (measured by beta) is the risk that's compensated with a risk premium — it's the risk investors are stuck bearing collectively no matter how they diversify within the equity asset class.

Q3. Worked — Sharpe ratio decision.

Fund X: 22% return, 18% volatility. Fund Y: 16% return, 9% volatility. Risk-free rate: 6%.

Model answer.

```
Sharpe X = (22 − 6) / 18 = 16/18 ≈ 0.89
Sharpe Y = (16 − 6) / 9 = 10/9 ≈ 1.11
```

Fund Y has the higher Sharpe ratio despite the lower raw return — it delivers more excess return per unit of total risk taken. An investor could, in principle, lever Fund Y (e.g. via borrowing) to match Fund X's volatility level and would expect to end up with a higher return than Fund X at that same risk level, which is the practical implication of comparing Sharpe ratios rather than raw returns alone.

Q4. Distinguish alpha from beta with a worked numerical example, and explain why conflating the two is a serious analytical error.

A fund returns 19% in a year when the market returned 10%, risk-free rate is 5%, and the fund's beta is 1.5.

Model answer.

```
CAPM-expected return = Rf + β(Rm − Rf) = 5% + 1.5 × (10% − 5%) = 5% + 7.5% = 12.5%
Alpha = Actual return − CAPM-expected return = 19% − 12.5% = +6.5%
```

The fund's 19% return looks impressive, but 12.5 percentage points of it are simply explained by taking on 1.5x the market's systematic risk (beta exposure) — anyone could have replicated most of that 19% by leveraging a passive index fund to a 1.5 beta, at much lower fees. Only the 6.5% alpha represents genuine skill (return not explained by market exposure). Conflating a high raw return achieved mostly through high beta with genuine investment skill is a serious error — it leads to overpaying active-management fees for what is, economically, just leveraged market exposure available far more cheaply.

Q5. When would an analyst use the Treynor ratio instead of the Sharpe ratio to evaluate a fund's risk-adjusted performance, and why does the choice matter?

Model answer. Sharpe ratio uses total risk (standard deviation, σ), making it the right choice when evaluating a portfolio as a standalone investment (all of an investor's risk exposure). Treynor ratio uses only systematic risk (beta), making it the right choice when evaluating a portfolio as *one component* of a broader, diversified overall portfolio — since in that context, the fund's unsystematic/idiosyncratic risk may already be diversified away by the investor's other holdings, and what matters is how efficiently the fund generates excess return per unit of the market risk it does contribute. Using Sharpe when Treynor is appropriate (or vice versa) can lead to incorrectly favouring a fund with low idiosyncratic risk but poor systematic-risk-adjusted returns, or the reverse.

Q6. Worked — information ratio and what it signals about consistency.

Fund A: active return (vs benchmark) averaging 4% annually, with a tracking error of 8%. Fund B: active return averaging 2.5% annually, with a tracking error of 2%.

Model answer.

```
IR_A = 4 / 8 = 0.50
IR_B = 2.5 / 2 = 1.25
```

Despite Fund A's higher average active return (4% vs 2.5%), Fund B has the much higher information ratio — its outperformance is far more *consistent* relative to how much it deviates from the benchmark. Fund A's higher average return comes with much more volatile/inconsistent relative performance (a high tracking error), meaning an investor experiences a bumpier ride of over- and under-performance to earn that average, whereas Fund B delivers steadier, more reliable outperformance per unit of active risk taken — a meaningfully different (and often more prized by institutional allocators) quality than Fund A's higher but less consistent alpha.

Q7. What's the trade-off in choosing a concentrated, high-conviction portfolio (e.g. 15-20 positions) versus a broadly diversified one (e.g. 100+ positions)?

Model answer. A concentrated portfolio allows each position to genuinely reflect the manager's highest-conviction ideas and lets real skill (analytical edge) show up meaningfully in returns if the manager's calls are good — but it also carries more idiosyncratic risk per position, since fewer holdings mean any single stock-specific negative surprise has a larger portfolio-level impact, and the overall portfolio's realised volatility can deviate more sharply from the benchmark. A broadly diversified portfolio diversifies away most idiosyncratic risk (per the diversification math in Q1-Q2), producing smoother, more benchmark-like returns — but if a manager is genuinely skilled at stock selection, over-diversifying dilutes that skill's impact toward benchmark-like ("closet indexing") returns while still charging active-management fees, itself a well-known problem the chapter flags.

Q8. Why does rebalancing a portfolio back to target weights function as a disciplined "sell high, buy low" mechanism, even without any explicit market-timing view?

Model answer. As asset prices move, a position that has appreciated becomes an overweight relative to its original target allocation, and a position that has declined becomes an underweight. Rebalancing back to target weights mechanically requires trimming the now-overweight (appreciated, "expensive" relative to target) position and adding to the now-underweight (declined, "cheap" relative to target) position — enforcing a disciplined sell-high/buy-low pattern purely through the rebalancing rule itself, without requiring the manager to have any specific forecasting view on which asset will outperform next. This also controls risk by preventing any single position's appreciation from silently growing the portfolio's concentration and risk profile beyond its intended target.

ESG in Equity Analysis

The Problem / Why this matters

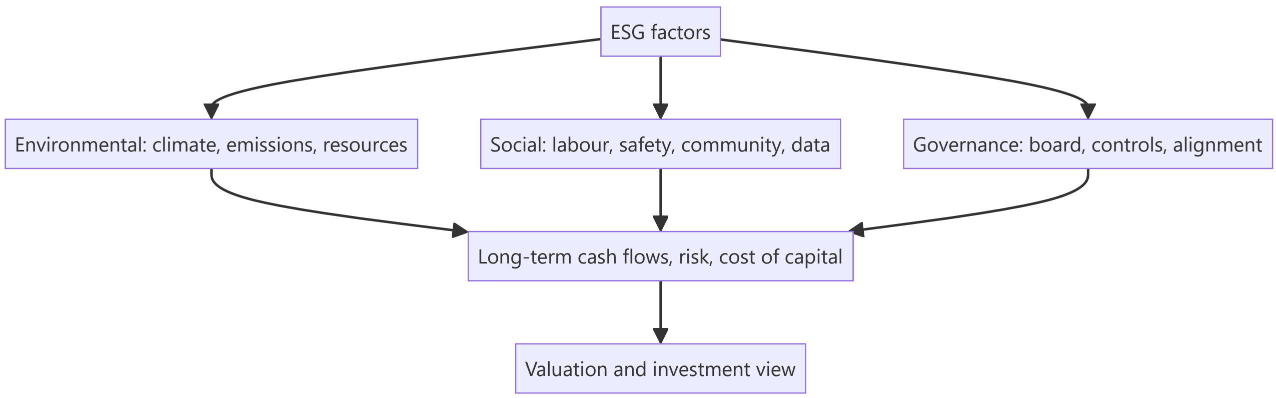
Investors increasingly judge companies not just on financials but on **Environmental, Social and Governance (ESG)** factors — because these non-financial issues create real financial risks and opportunities (regulation, reputation, stranded assets, talent). ESG has moved from niche to mainstream, is increasingly mandated in disclosure (BRSR in India), and comes up in interviews as "why do investors care about ESG?" Understanding how to integrate it into equity analysis is now part of the job.

Core Idea

ESG analysis assesses a company's exposure to and management of **environmental, social, and governance** risks and opportunities, integrating them into the investment view. The premise: material ESG factors affect long-term cash flows, risk, and therefore valuation — so ignoring them means missing risks that standard financials don't capture.

Why it works this way

Many ESG issues are **financially material but slow-moving** — a carbon-intensive business faces future regulation and stranded-asset risk; poor governance invites fraud and value destruction; weak labour or safety practices bring litigation and reputational damage. These show up in cash flows and cost of capital eventually, so a forward-looking analyst prices them in before they hit the financials.



Full technical content

The three pillars:

PILLAR	EXAMPLE FACTORS
Environmental	Carbon emissions, energy/water use, waste, climate transition risk, biodiversity
Social	Labour practices, health & safety, diversity, product safety, data privacy, community relations
Governance	Board independence/quality, executive pay alignment, shareholder rights, audit quality, related-party dealings, anti-corruption

Governance is often seen as the most immediately financially material — weak governance (dominant promoters, poor controls) is a leading cause of fraud and value destruction, especially in emerging markets.

How investors use ESG: - Integration — embed material ESG factors into fundamental analysis and valuation (adjust growth, margins, cost of capital, or apply a risk discount). - **Screening** — exclude (negative screening: tobacco, weapons) or tilt toward leaders (positive/best-in-class). - **Thematic / impact** — invest in a theme (clean energy) or for measurable impact. - **Stewardship** — engage and vote to improve company behaviour.

Materiality. The key discipline: focus on ESG factors that are **financially material** to *that* industry (e.g., emissions for a utility, data privacy for a tech firm, safety for a miner) — not a generic checklist. Frameworks like SASB map materiality by sector.

Instruments & disclosure: green bonds, **sustainability-linked loans**, ESG funds/ETFs; disclosure via GRI, TCFD, and in India the **BRSR (Business Responsibility and Sustainability Report)** mandated for top listed firms by SEBI.

Debates & challenges: - Greenwashing — overstating ESG credentials. - **Inconsistent ratings** — ESG scores from different providers often disagree (different methodologies). - **Performance debate** — whether ESG helps or hurts returns is contested; the strongest case is **risk management** (avoiding blow-ups) rather than guaranteed outperformance.

Effect on cost of capital. Strong ESG can lower cost of capital (broader investor base, lower risk perception); poor ESG can raise it (exclusion, higher risk), feeding directly into valuation.

Worked examples

Example 1 — governance red flag. An analyst finds a company with a promoter-dominated board, frequent related-party transactions, and an auditor change after a qualification. Governance risk is high — historically a precursor to value destruction. The analyst applies a valuation discount (or avoids the stock), a call standard financials alone wouldn't trigger.

Example 2 — environmental transition risk. A thermal-coal power producer looks cheap on current earnings, but tightening emissions regulation and the energy transition threaten **stranded assets** and rising compliance costs over the next decade. Integrating this, the analyst lowers long-term cash flows and raises the discount rate — the stock is less cheap than it looks.

Example 3 — materiality focus. For a software company, carbon emissions are minor but **data privacy and talent retention** are highly material (a breach or attrition could hit revenue and margins). A good ESG analyst weights the *material* factors for that sector, not a generic environmental score.

Example 4 — quantifying a governance discount in a valuation, not just flagging it qualitatively. An analyst values a company at ₹450/share on a clean DCF, but the company has two governance red flags: 35% of revenue runs through related-party transactions with promoter-owned entities (raising round-tripping/value-leakage concerns), and the board has no independent directors on the audit committee. Rather than simply writing "governance risk — be cautious" in the note, a rigorous approach quantifies it: apply a discount rate premium (e.g. +150 basis points to the cost of equity, reflecting the additional risk premium investors should demand for weaker minority-shareholder protection) and re-run the DCF. If the +150bp premium takes WACC from 11.0% to 12.5%, the same cash flows might value the stock at ₹390/share instead of ₹450 — a concrete, defensible ₹60/share governance discount an investment committee can actually act on, rather than a vague qualitative caution that doesn't change the number anyone actually trades on.

Example 5 — BRSR disclosure as a research input, not just a compliance checkbox. India's SEBI-mandated Business Responsibility and Sustainability Report (BRSR, required for the top listed companies by market cap) discloses standardised metrics — GHG emissions (Scope 1 and 2), water/waste intensity, employee attrition and diversity data, and related-party transaction details. An equity analyst covering a manufacturing company can use the BRSR's disclosed emissions-per-unit-of-revenue trend across several years as a genuine research input: a rising emissions-intensity trend at a company operating in a sector facing tightening carbon-pricing regulation is a quantifiable, forward-looking risk signal — directly usable in the Example 2 stranded-asset-style analysis — rather than treating BRSR filing as a purely regulatory box-ticking exercise disconnected from the investment thesis.

How it is tested in interviews

- **"Why do investors care about ESG?"** — "Because material ESG factors create real financial risks and opportunities — regulation, stranded assets, reputation, governance failures — that affect long-term cash flows and cost of capital but aren't captured in standard financials."
- **"What are the three pillars?"** — Environmental (climate, emissions, resources), Social (labour, safety, privacy, community), Governance (board, controls, alignment).
- **"Which is most financially material?"** — "Often governance — weak governance is a leading cause of fraud and value destruction, especially in emerging markets — but materiality is industry-specific."
- **"What are the criticisms of ESG?"** — Greenwashing, inconsistent ratings across providers, and a contested performance record (the strongest case is risk management).
- **"How would you integrate ESG into a valuation?"** — "Focus on material factors for the sector; adjust growth, margins, or the discount rate, or apply a governance/risk discount."

Traps & common mistakes

- Treating ESG as a **generic checklist** rather than **industry-material** factors.
- Ignoring **governance** — the most immediately material pillar.
- Taking a single ESG **rating** at face value (providers disagree).
- Assuming ESG **guarantees** outperformance — the robust case is risk management.
- Missing the **cost-of-capital** channel through which ESG hits valuation.

First-principles recap

- ESG assesses environmental, social and governance risks/opportunities that are financially material.
- **Materiality is industry-specific** — focus on what matters for that sector.
- Governance is often the most immediately material pillar.
- Integrate via adjusted cash flows/discount rate, screening, or stewardship.
- Watch greenwashing and rating inconsistency; the strongest case is risk management and cost of capital.

Quick-reference

PILLAR	MATERIAL EXAMPLES
Environmental	Emissions, transition risk, resources
Social	Labour, safety, data privacy
Governance	Board, controls, related-party, alignment
Uses	Integration, screening, thematic, stewardship
India disclosure	BRSR (SEBI)
Key discipline	Focus on financially material factors

Q&A — ESG in Equity Analysis

Theory and worked scenarios on integrating ESG factors into fundamental equity research.

Q1. Why is "materiality" described as the key discipline in ESG analysis, and what's the error a generic ESG checklist commits?

Model answer. Materiality means focusing specifically on the ESG factors that are genuinely financially relevant to a *particular* company's industry and business model — carbon emissions are highly material for a utility or heavy manufacturer but nearly irrelevant to a software company, while data privacy and talent retention are highly material for a tech firm but not for a cement producer. A generic ESG checklist that scores every company against the same broad list of environmental, social, and governance factors regardless of industry dilutes the signal with irrelevant noise and can produce misleading comparisons (e.g. penalising a software company for a low environmental score on factors that don't actually bear on its business risk) — frameworks like SASB exist specifically to map which factors are material by sector.

Q2. Why does the chapter identify governance as "often the most immediately financially material" ESG pillar, particularly in emerging markets?

Model answer. Weak governance — a dominant, unchecked promoter, frequent undisclosed related-party transactions, weak board independence, poor audit quality — is a well-documented, historically recurring precursor to fraud, capital misallocation, and outright value destruction, and these risks can materialise suddenly and severely (a governance blow-up can wipe out a large fraction of equity value in a short period), unlike environmental or social risks which more often play out gradually over years. In emerging markets specifically, weaker regulatory enforcement and more concentrated ownership structures make governance risk both more prevalent and harder to detect from standard financial disclosures alone, making independent governance assessment a genuinely high-value, differentiated piece of analysis.

Q3. Worked scenario — apply ESG integration (not exclusion) to a valuation. A thermal-power generation company trades at 6x forward earnings, well below the sector average of 9x, with strong current cash flows. What ESG-related analysis should inform whether this "cheap" valuation is a genuine opportunity or a value trap?

Model answer. The low current multiple may already partially reflect market awareness of transition risk, but a rigorous ESG-integrated analysis goes further: model the specific regulatory trajectory for emissions/carbon pricing in the relevant jurisdiction and its likely impact on the plant's economics over the next 10-15 years; assess stranded-asset risk — whether the plant may need to be retired before the end of its natural economic life due to policy or financing-access changes (increasingly, lenders and insurers are reducing exposure to thermal assets, raising the cost and availability of future capital); and explicitly model these effects into extended-horizon cash flows and/or a higher discount rate rather than treating current-year earnings as representative of the long-run value. The conclusion may still be a legitimate contrarian buy if the market is overstating transition risk, or a genuine value trap if the market hasn't yet fully priced a realistic regulatory trajectory — the ESG-integration approach earns that conclusion through explicit analysis rather than either ignoring the risk or reflexively avoiding the sector.

Q4. Distinguish ESG integration from negative screening, and explain why a fund might use one, the other, or both.

Model answer. Integration embeds material ESG factors directly into fundamental analysis and valuation (adjusting growth, margin, or discount-rate assumptions based on ESG-related risk/opportunity) while still potentially owning any company if the valuation is attractive after that adjustment. Negative screening excludes entire categories of companies outright (e.g. tobacco, weapons manufacturers) regardless of valuation, typically driven by a mandate's ethical/values constraints rather than a purely financial materiality judgment. A fund might use integration alone (financially-driven ESG-aware investing with no exclusions), screening alone (values-driven exclusions with otherwise standard financial analysis), or both together (a values-constrained universe, analysed with ESG-integrated fundamental work within that universe) — the choice reflects the fund's mandate and client base, not a single "correct" approach.

Q5. What is "greenwashing," and what should an equity analyst specifically check to avoid being misled by it?

Model answer. Greenwashing is a company overstating or misrepresenting its actual ESG credentials/performance, often through selective disclosure, vague sustainability marketing claims, or ESG-labelled initiatives that are small relative to the company's overall footprint. An analyst should check: whether sustainability claims are backed by third-party-verified, quantified data (not just marketing language) with clear year-over-year metrics; whether disclosed ESG initiatives are material relative to the company's overall scale of operations (a large polluter's small solar-power pilot project doesn't offset its core emissions profile); and whether the company's governance track record (Q2) itself is credible enough to trust its self-reported ESG data in the first place — governance quality is itself a useful filter for how much to trust a company's other ESG disclosures.

Q6. Why do ESG ratings from different data providers often disagree significantly on the same company, and what does this imply for how an analyst should use third-party ESG scores?

Model answer. Different ESG rating providers use different methodologies — different weightings across the E/S/G pillars, different underlying data sources, and different judgments about which specific factors matter for a given industry — meaning it's genuinely common for the same company to receive meaningfully different ESG scores from different providers, sometimes even contradictory qualitative assessments. This implies an analyst should treat any single third-party ESG score as one data point rather than ground truth, and should understand the specific methodology behind a score before relying on it for an investment decision — the chapter's broader point about materiality (Q1) applies here too: an analyst's own judgment about what's genuinely material to *this* company, rather than a black-box aggregate score, is the more defensible basis for investment analysis.

Q7. What is the "cost of capital channel" through which ESG factors affect valuation, distinct from the direct cash-flow channel?

Model answer. Beyond directly affecting projected cash flows (e.g. lower future revenue from reputational damage, higher costs from regulatory compliance), a company's ESG profile can affect the discount rate applied to those cash flows: strong ESG performance can broaden the pool of investors willing to hold the stock (some large institutional mandates now explicitly exclude poor-ESG names or require minimum ESG scores) and can be perceived as lowering certain tail risks, both of which can modestly lower the required return (cost of equity) investors demand. Conversely, poor ESG performance can shrink the addressable investor base (exclusion from ESG-mandated funds) and raise perceived risk, pushing the required return up — meaning two companies with identical projected cash flows could reasonably receive different valuations purely through this cost-of-capital channel if their ESG profiles differ meaningfully.

Q8. How would you respond to the claim "ESG investing is proven to boost returns" in an interview, given the chapter's framing of the performance debate?

Model answer. "The empirical evidence on whether ESG investing boosts returns is genuinely contested and methodology-dependent — I wouldn't claim it's proven to boost returns. The more defensible and robust case for ESG integration is risk management: material ESG factors, when ignored, have been repeatedly shown to precede real financial blow-ups (governance failures leading to fraud, environmental liabilities becoming stranded-asset write-downs, social/labour issues triggering costly litigation or reputational damage) — so ESG-aware analysis is better framed as helping avoid left-tail risk and value-destroying surprises, rather than as a reliable source of additional upside. That's a more defensible, evidence-consistent position than claiming guaranteed outperformance." This response demonstrates nuanced understanding rather than either dismissing ESG or overselling it.

Indian Equity Markets (Capstone)

The Problem / Why this matters

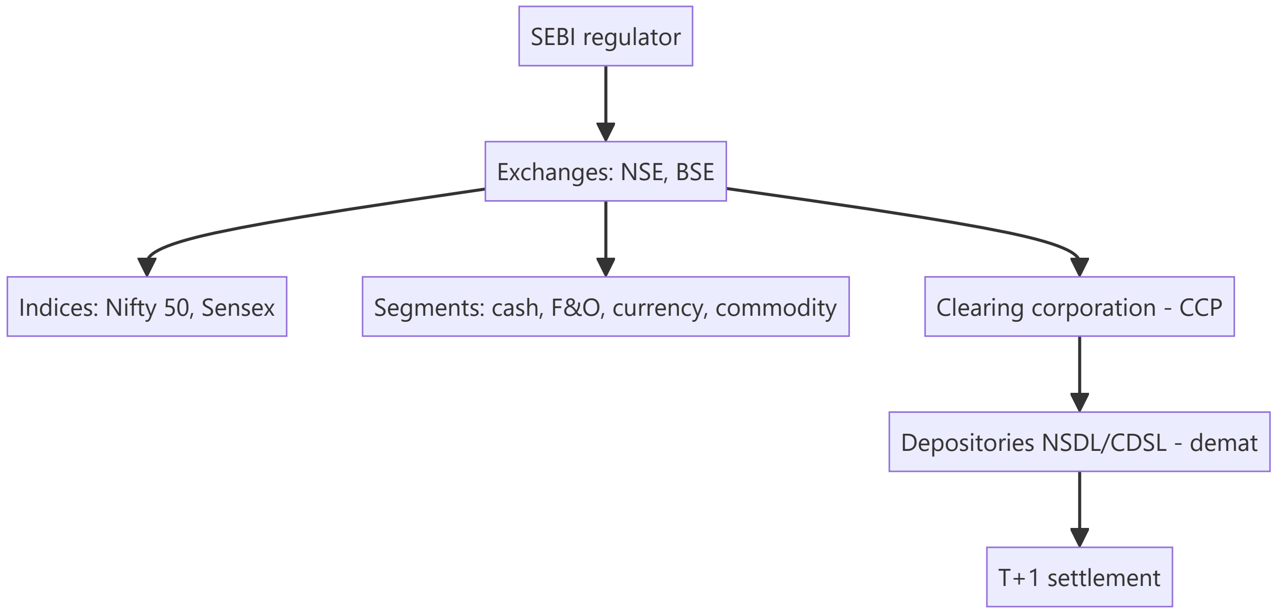
For any finance role in India, you must know your home market cold — the regulator, the exchanges, the indices, how shares are held and settled, the segments, and the participants. It's the most common "home turf" interview check, and it ties together everything in this book in the specific institutional context you'll actually work in. This capstone maps the Indian equity ecosystem end to end.

Core Idea

The Indian equity market is a **modern, electronic, well-regulated** market: **SEBI** regulates it; **NSE and BSE** are the exchanges; **Nifty 50 and Sensex** are the benchmarks; shares are held in **demat** via NSDL/CDSL and settle at **T+1**; and it spans cash, F&O, currency and commodity segments with deep participation from retail, domestic institutions, and foreign investors.

Why it works this way

India built a fully dematerialized, exchange-traded, centrally-cleared market with strong disclosure and one of the world's fastest settlement cycles precisely to ensure trust, liquidity and investor protection — the conditions that let a large, diverse investor base participate and companies raise capital efficiently.



Full technical content

Regulator: SEBI (Securities and Exchange Board of India) — regulates issuers (disclosure), intermediaries (registration/conduct), and markets (surveillance, insider-trading and manipulation enforcement); protects investors. **RBI** oversees monetary policy and banking; **IRDAI** insurance.

Exchanges: NSE (National Stock Exchange — the larger by volume, especially derivatives) and **BSE** (Bombay Stock Exchange — Asia's oldest). Both are fully electronic, order-driven markets.

Benchmark indices: Nifty 50 (NSE, 50 large-caps) and **Sensex** (BSE, 30 large-caps), both **free-float market-cap weighted**. Broader: Nifty Next 50, Nifty 100/500, Midcap/Smallcap, and sectoral indices

(**Bank Nifty**, Nifty IT, etc.).

Holding & settlement: shares held in **dematerialized (demat)** form via depositories **NSDL** and **CDSL** (through Depository Participants); trades cleared by the **clearing corporation** (central counterparty) and settled on **T+1** (India moved to T+1 fully in 2023 and is piloting T+0/instant) — among the fastest globally.

Market segments:

SEGMENT	WHAT TRADES
Cash / equity	Delivery-based and intraday shares
F&O (derivatives)	Index and stock futures & options (Nifty, Bank Nifty are among the world's most traded)
Currency derivatives	USD/INR and other pairs
Commodity	Via MCX/NCDEX (and NSE/BSE segments)

Participants: retail (large and growing, via demat/UPI-based investing), **domestic institutions (DIIs)** — mutual funds, insurers (LIC), pension (EPFO/NPS) — and **foreign portfolio investors (FPIs)**, whose flows often drive index moves. Promoters hold large stakes in many firms.

Trading mechanics: order-driven, price-time priority; **circuit breakers** (index-level halts) and **price bands** (stock-level) curb extreme moves; **UPI/ASBA** streamline IPO applications and payments.

Recent evolution: surging retail participation (record demat accounts), the rise of discount brokers and index/SIP investing, T+1 (moving to instant) settlement, and world-leading F&O volumes (prompting SEBI curbs on retail options speculation).

Worked examples

Example 1 — the full trade lifecycle. A retail investor places a market buy for 100 shares of an NSE-listed company via a broker app. The order matches on the NSE by price-time priority; the **clearing corporation** guarantees it; on **T+1**, shares are credited to the investor's **CDSL/NSDL demat** account and cash debited. Regulated end-to-end by SEBI. Every institution in this chapter touched one small trade.

Example 2 — FPI flows move the Nifty. Foreign investors turn net sellers of ₹15,000 cr in a week on global risk-off. Because FPIs hold large weights in Nifty heavyweights and trade fast, the index falls sharply even without India-specific news — illustrating why FPI flows are watched as a key driver.

Example 3 — F&O depth. Bank Nifty and Nifty options are among the most actively traded derivatives in the world, giving Indian markets deep hedging and speculation venues — but also prompting SEBI to tighten rules to protect retail traders from outsized options losses. India's derivatives market is globally significant.

Example 4 — DII flows offsetting FPI selling. Continuing Example 2: the same week FPIs sell ₹15,000 cr, DIIs (domestic mutual funds, insurers) net-buy ₹11,000 cr, cushioning roughly three-quarters of the FPI outflow's index impact. This DII-as-stabiliser pattern has become a structural feature of Indian markets over the past decade, driven substantially by rising domestic SIP (Systematic Investment Plan) mutual-fund inflows — a monthly, relatively flow-insensitive-to-sentiment source of buying that didn't exist at the same scale two decades ago. An analyst explaining "why didn't the market fall as much as the FPI selling would suggest" should be able to name this DII/SIP-flow offset explicitly, not just cite "buying support" vaguely.

Example 5 — a circuit breaker in action. A stock-specific circuit filter (e.g. 10% for a mid-cap) halts trading in a single name after unexpected, severely negative news (a fraud allegation, a regulatory action) causes a rush of sell orders. The halt gives the market a cooling-off period — allowing information to be digested and genuine two-sided price discovery to resume — rather than letting a single wave of panic-selling execute against a thin, one-sided order book at an arbitrarily bad price. Index-level circuit breakers (triggered by a large move in Nifty/Sensex itself) work the same way at the market-wide level during systemic shocks. Knowing that circuit filters differ by stock category (larger, more liquid stocks typically have wider bands than smaller, more volatile ones) is the kind of granular, "home turf" detail this capstone chapter flags as a common interview differentiator.

How it is tested in interviews

- **"Walk me through the Indian equity market structure."** — SEBI regulates; NSE/BSE are the exchanges; Nifty 50/Sensex are the benchmarks (free-float market-cap weighted); shares held in demat via NSDL/CDSL; cleared by the clearing corporation; settled T+1; segments cash/F&O/currency/commodity.
- **"Who regulates the Indian securities market?"** — "SEBI — issuers, intermediaries, and markets, with investor protection; RBI covers banking/monetary, IRDAI insurance."
- **"What are the main indices and how are they built?"** — "Nifty 50 (NSE) and Sensex (BSE), free-float market-cap weighted large-cap baskets."
- **"How are trades settled in India?"** — "T+1 rolling settlement, shares in demat via NSDL/CDSL, guaranteed by the clearing corporation as central counterparty — among the fastest cycles globally."
- **"What drives Indian market moves?"** — "FPI and DII flows, global cues, rates/inflation and RBI policy, earnings, and sentiment; FPI flows especially move the index."

Traps & common mistakes

- Confusing **SEBI** (securities) with **RBI** (banking/monetary) roles.
- Not knowing India settles at **T+1** (and is going faster).
- Forgetting Nifty/Sensex are **free-float** market-cap weighted.
- Mixing up the **exchange** (NSE/BSE), **depository** (NSDL/CDSL), and **clearing corporation** roles.
- Underrating **FPI flows** and India's globally-large **F&O** volumes.

First-principles recap

- **SEBI** regulates; **NSE/BSE** are the exchanges; **Nifty 50/Sensex** the free-float benchmarks.
- Shares are held in **demat** (NSDL/CDSL) and settle **T+1** (moving to instant).
- Segments: cash, F&O (globally large), currency, commodity.
- Participants: retail (surging), DIIs, FPIs (flow-drivers), promoters.
- A modern, electronic, centrally-cleared, well-regulated market — the context for every Indian finance role.

Quick-reference

ELEMENT	INDIA
Regulator	SEBI (RBI banking, IRDAI insurance)
Exchanges	NSE, BSE
Indices	Nifty 50, Sensex (free-float mcap)
Depositories	NSDL, CDSL (demat)
Settlement	T+1 (→ instant), CCP-guaranteed
Segments	Cash, F&O, currency, commodity
Flow drivers	FPIs, DIIs

Q&A — Indian Equity Markets (Capstone)

The "home turf" questions every India-based markets interview checks for.

Q1. Who regulates Indian equity markets, and what are the two main exchanges?

Model answer. SEBI (Securities and Exchange Board of India) is the primary regulator, covering market conduct, disclosure, intermediary registration, and investor protection. The two main exchanges are NSE (National Stock Exchange) and BSE (Bombay Stock Exchange); NSE is the larger by traded volume in cash equities and derivatives, BSE is India's oldest exchange (founded 1875).

Q2. What are India's two main benchmark indices, and how do they differ in construction?

Model answer. Nifty 50 (NSE, 50 large-cap stocks) and Sensex (BSE, 30 large-cap stocks) — both free-float market-cap-weighted, meaning weights reflect only the shares actually available to trade (excluding promoter/locked-in holdings), not total shares outstanding. This free-float methodology matters because it means an index's composition reflects genuinely tradable liquidity, not raw company size.

Q3. Explain India's settlement cycle and why it matters for an analyst.

Model answer. Indian equities settle on a T+1 basis (trade date plus one business day) — among the fastest cycles globally, reduced from T+2 in a phased rollout completed in 2023. For an analyst, faster settlement reduces counterparty risk in the system and means trading strategies involving quick turnaround face less operational lag than in slower-settling markets.

Q4. What is "demat," and what are the two depositories?

Model answer. Dematerialisation (demat) is holding securities in electronic form rather than physical share certificates — mandatory for trading in India. The two depositories are NSDL (National Securities Depository Limited) and CDSL (Central Depository Services Limited), which maintain electronic records of ownership; an investor's shares sit in a demat account held via a registered depository participant (typically a broker).

Q5. What are the main market segments on Indian exchanges?

Model answer. Cash/equity segment (spot trading in listed shares), F&O (futures & options — index and stock derivatives, one of the largest derivatives markets globally by contract volume), currency derivatives (USD/INR and other pairs), and commodity derivatives (via MCX/NCDEX, and NSE for select commodities) — each with its own participant base, margin rules, and settlement mechanics.

Q6. Who are the main categories of participants in Indian equity markets, and how does their behaviour differ?

Model answer. Retail investors (a rapidly growing base via discount brokers and direct market access), Domestic Institutional Investors — DIIs (mutual funds, insurance companies, pension funds — generally longer-horizon, often a stabilising counterweight to foreign flows), and Foreign Portfolio Investors — FPIs (foreign institutional money, registered and regulated by SEBI, historically a major swing factor in short-term market direction, especially around global risk-sentiment shifts and currency moves). Analysts routinely track monthly FPI/DII net flow data as a sentiment indicator, since large, sustained FPI selling offset by DII buying (or vice versa) is itself informative about market positioning.

Q7. What's the STT (Securities Transaction Tax), and why does it matter to an options trader/analyst specifically?

Model answer. STT is a tax levied on transactions in listed securities and derivatives on Indian exchanges, with different rates for cash-market trades versus options versus futures. It matters specifically at options expiry, where STT on the notional value of a physically/cash-settled in-the-money option that isn't closed out before expiry can be disproportionately large relative to the option's premium — a well-known trap ("the STT trap") that active options traders and anyone modelling realistic F&O trading costs must account for explicitly.

Q8. Why is "knowing your home market cold" specifically emphasised as an interview screen in India?

Model answer. It's a fast, low-effort way for an interviewer to separate candidates who have genuinely engaged with markets (reading regulatory changes, tracking flow data, understanding settlement mechanics) from those with only textbook/global theoretical knowledge — since every India-based markets role, regardless of specific coverage sector, operates within this specific regulatory and market-structure context daily, a candidate who can't answer basic structural questions about SEBI, NSE/BSE, or settlement raises doubts about how seriously they've engaged with the actual job, independent of their technical modelling skill.

Sector-Specific Analysis Frameworks

The Problem / Why this matters

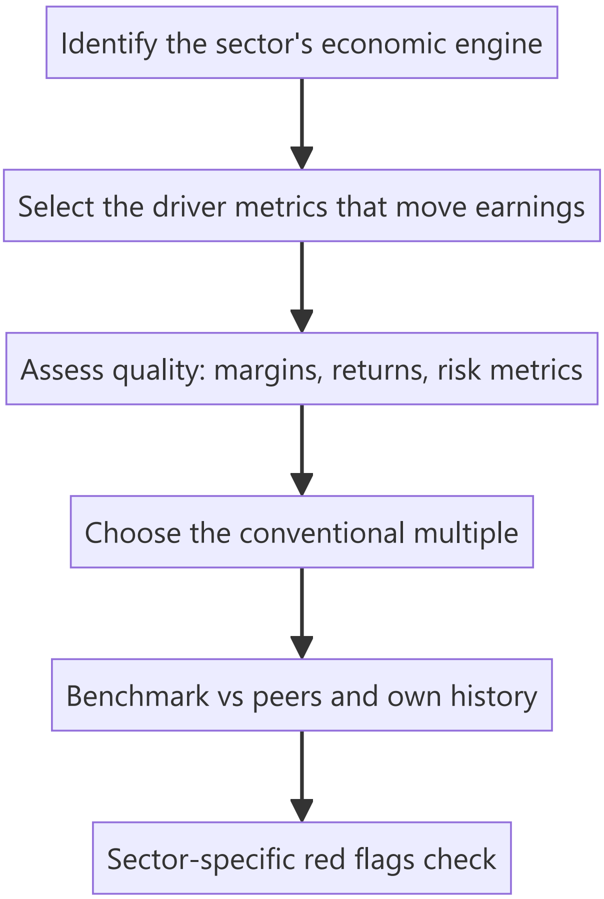
The generic research process (coverage → model → value → thesis) is the same for every stock, but the **drivers, metrics and red flags are not**. An analyst who values a bank the way they value an FMCG company will produce a confidently wrong number. Equity research is organised by sector precisely because each sector has its own economics, its own vocabulary, and its own valuation convention — and "walk me through how you'd analyse a bank" is one of the most common senior equity-research interview questions.

Core Idea

Every sector has a **primary value driver**, a **sector-specific metric set**, and a **conventional valuation multiple** that reflects its economics. Master the framework for the sectors you cover: what actually moves earnings, which metrics reveal quality, and which multiple the market actually uses.

Why it works this way

Valuation multiples are shorthand for underlying economics. Banks trade on P/B because their earnings power is a return *on the balance sheet* (equity), so book value is the anchor. Asset-light IT services trade on P/E because earnings are the asset. Cyclicals trade on EV/EBITDA through the cycle because leverage and depreciation distort net income at cycle extremes. The multiple follows the economics — it is not an arbitrary convention.



Full technical content

Banks and NBFCs — a balance-sheet business

A lender's product *is* its balance sheet. Earnings = income earned on assets minus cost of funds minus credit losses minus operating cost.

The core P&L bridge:

LINE	MEANING
Interest income − Interest expense	**Net Interest Income (NII)** — the core revenue
NII ÷ average earning assets	**Net Interest Margin (NIM)** — pricing power/spread
+ Fee & other income	Non-interest income (cards, distribution, treasury)
− Operating expense	Measured as **Cost-to-Income ratio**
− Provisions	Credit cost, the swing factor
= Profit	Measured as **RoA** and **RoE**

Asset quality — the metrics that matter most: - **GNPA / NNPA (%)** — Gross and Net Non-Performing Assets as a share of advances. NNPA is after provisions already taken. - **Provision Coverage Ratio (PCR)** — provisions held against GNPAs. A low PCR means future P&L pain is still to come. - **Slippage ratio** — fresh NPAs created in the period, the *forward-looking* stress signal. Rising slippages precede rising GNPA. - **Credit cost (%)** — provisions ÷ average advances. The single most important swing variable in a bank's earnings. - **Restructured book / SMA-1 / SMA-2** — loans showing early stress but not yet NPA. A leading indicator.

Funding and capital: - **CASA ratio** — Current Account Savings Account deposits as a share of total. High CASA = cheap, sticky funding = structurally better NIM. The single biggest quality differentiator between Indian banks. - **Credit-Deposit (CD) ratio** — how much of deposits are lent out; very high CD ratio constrains further growth. - **CAR / Tier-1 capital** — regulatory capital adequacy. Low Tier-1 means dilution risk (an equity raise) is coming.

How to value: P/B (Price to Book), cross-checked against RoE. The relationship is fundamental: a bank earning RoE above its cost of equity deserves P/B > 1, and the higher the sustainable RoE, the higher the justified P/B. Use **P/ABV** (Adjusted Book Value, net of unprovided NPAs) for stressed lenders, because reported book value overstates real equity when provisioning is inadequate.

NBFC-specific differences: no deposit franchise (mostly), so funding is wholesale (bank borrowings, NCDs, CPs) and therefore more expensive and less sticky. Watch **ALM (Asset-Liability Management) mismatch** — borrowing short to lend long is the structural NBFC risk, and it is exactly what turns a liquidity event into a solvency event. Also watch **spread** (yield minus cost of funds) rather than NIM, and **leverage** (debt/equity), which is far higher than in most sectors by design.

Red flags: falling PCR while GNPA rises; sharp loan growth in a single risky segment; rising restructured book; repeated capital raises; NIM held up only by rising risk (lending down the credit ladder); divergence between RBI-assessed and company-reported NPAs.

IT services — a people-and-contracts business

Drivers: headcount, utilisation, billing rate, and currency.

METRIC	WHY IT MATTERS
Revenue growth in constant currency (cc)	Strips out FX so you see real business momentum
Utilisation (%)	Billable hours ÷ available hours — the operating leverage lever
Attrition (%)	High attrition raises replacement and training cost, hurts margin and delivery quality
Deal TCV (Total Contract Value)	Forward-looking order book; the leading indicator of revenue
Revenue per employee	Productivity / value-add of the mix
Offshore-onsite mix	Offshore work is far higher margin
Client concentration	Top-5/top-10 client revenue share = risk

Margin bridge: wage inflation and rupee appreciation hurt; utilisation, offshoring, automation and pricing help. In an Indian IT model, a **1% INR depreciation vs USD typically adds roughly 20–30bps to EBIT margin** — currency is a genuine earnings driver, not noise.

How to value: P/E, benchmarked to growth (a PEG-style view) and to the company's own historical band. Tier-1 names command a premium for scale, client relationships and delivery consistency.

Red flags: growth held up by low-margin deals; rising attrition with flat margins (cost pressure not yet visible in P&L); declining deal TCV while revenue still grows (the pipeline is emptying); heavy dependence on one vertical (e.g. BFSI) heading into that vertical's downturn.

Pharma — a pipeline-and-regulation business

Two very different business models under one sector label: - Generics / API — commodity-like, competes on cost and regulatory compliance. Price erosion in the US generics market is the structural headwind. - **Branded / domestic formulations** — brand-led, better pricing power, closer to an FMCG model.

Metrics: R&D as % of sales, **ANDA filings and approvals** (US pipeline), **Para-IV / FTF (First-to-File)** opportunities (180-day exclusivity = outsized but temporary profit), US price erosion rate, and the domestic formulation growth rate versus IPM (Indian Pharmaceutical Market) growth.

The regulatory factor that dominates everything: a **USFDA inspection outcome** — specifically a Form 483 observation, a Warning Letter, or an Import Alert on a plant — can wipe out a facility's US revenue overnight. Plant-level regulatory status is a first-order equity risk, not a compliance footnote. Always know which plants serve which revenue.

How to value: P/E for stable domestic-led businesses; **EV/EBITDA** where leverage or heavy depreciation distorts; sum-of-the-parts where a company has genuinely different segments (domestic branded vs US generics vs API vs CDMO).

Red flags: revenue concentrated in one product's exclusivity period (a cliff is coming); repeated regulatory observations across plants; capitalising R&D aggressively; receivables stretching in the US channel.

FMCG / consumer — a brand-and-distribution business

Drivers: volume growth versus value growth — the single most important decomposition. Value growth without volume growth means price hikes are carrying the quarter, which is not durable; volume growth is real demand.

Metrics: volume growth, gross margin (raw-material sensitive), **A&P (advertising & promotion) spend as % of sales**, distribution reach (outlets covered, direct vs indirect), rural-urban mix, and new-product contribution.

How to value: P/E, typically at a structural premium to the market, justified by high RoCE, low capital intensity, and predictable cash generation. Also cross-check **EV/EBITDA**.

Red flags: value growth persistently outrunning volume growth; A&P cut to protect reported margin (borrowing from future brand equity); channel stuffing (primary sales to distributors outpacing secondary sales to consumers — watch distributor inventory days).

Autos and cyclicals — a volume-and-operating-leverage business

Drivers: monthly **volume data** (a rare high-frequency, publicly disclosed operating metric — auto companies report monthly dispatches), realisation per unit, product mix, and raw-material cost (steel, aluminium, rubber).

Key structural distinction: dispatches (to dealers) versus retail sales (to customers). Divergence signals dealer inventory build — a warning that future dispatches must fall.

How to value: EV/EBITDA through the cycle, because net income is distorted by leverage and depreciation at cycle extremes, and P/E is famously misleading for cyclicals — a cyclical looks *cheapest* on P/E at the peak (peak earnings, low multiple) and *most expensive* at the trough. This is the classic cyclical trap.

Cross-sector summary

SECTOR	PRIMARY DRIVER	SIGNATURE METRICS	CONVENTIONAL MULTIPLE
Banks / NBFC	Balance-sheet growth × spread − credit cost	NIM, CASA, GNPA/NNPA, PCR, slippage, RoA, CAR	**P/B** vs RoE
IT services	Headcount × utilisation × rate	cc growth, utilisation, attrition, deal TCV	**P/E**
Pharma	Pipeline + regulatory status	ANDA filings, US price erosion, plant status	**P/E**, EV/EBITDA, SOTP
FMCG	Volume growth × pricing	Volume vs value growth, A&P%, distribution	**P/E** (premium)
Autos/cyclicals	Volumes × operating leverage	Monthly volumes, realisation, RM cost	**EV/EBITDA** through cycle

Common mistakes

- Applying **P/E to a bank** instead of P/B, or to a cyclical at peak earnings.
- Reading a bank's falling GNPA as improving asset quality without checking whether **write-offs** (not recoveries) drove the fall.
- Treating an IT company's reported USD revenue growth as business momentum without stripping out **currency**.
- Assuming FMCG value growth equals demand strength when it is entirely **price-led**.
- Ignoring **plant-level USFDA status** in pharma because it looks like a compliance rather than a financial issue.

Interview angle

"Walk me through how you'd analyse a bank" is a standard question. Structure the answer: (1) the balance sheet is the product — growth in advances and deposits; (2) spread — NIM, driven by CASA and the lending mix; (3) asset quality — GNPA/NNPA, PCR, and *especially* slippages as the forward indicator; (4) efficiency — cost-to-income; (5) capital — CAR/Tier-1 and therefore dilution risk; (6) output — RoA and RoE; (7) valuation — P/B justified by sustainable RoE versus cost of equity. Naming slippages and CASA specifically is what separates a prepared candidate from a generic one.

Earnings Season and Concall Analysis

The Problem / Why this matters

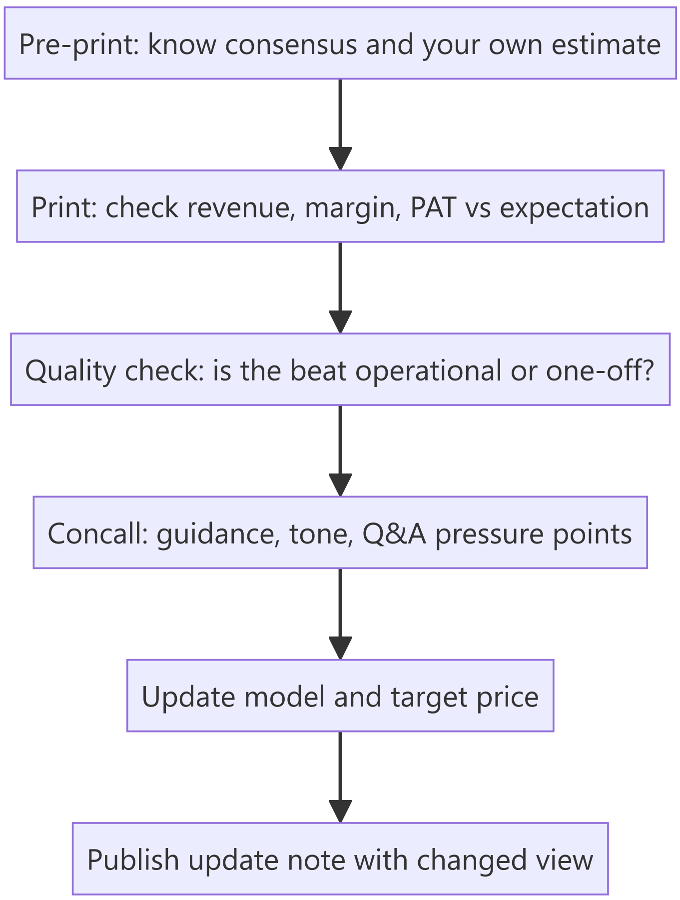
Four times a year, every company you cover reports — and an analyst's credibility is made or lost in the 48 hours around each print. The job is not to read the press release; it is to know **before** the print what the market expects, judge **within minutes** whether the result beat or missed on the lines that matter, extract what management said that wasn't in the numbers, and update the model and the view. This is the single most repetitive, highest-frequency task in an equity research seat.

Core Idea

An earnings event has three separable information layers: the **reported numbers** (versus consensus, versus your own estimate), the **quality** of those numbers (how they were achieved), and the **forward guidance and tone** from the concall. Price reacts to all three, and the third is often the largest mover.

Why it works this way

Markets price expectations, not levels. A company can grow profit 30% and fall 8% because consensus expected 40%, or report a loss and rally because management guided to a turnaround. The reported number only matters relative to what was already discounted — which is why knowing consensus *before* the print is a prerequisite, not a nicety.



Full technical content

Stage 1 — Pre-print preparation

Before the result, you must have on one page: - **Consensus estimates** for revenue, EBITDA, margin and PAT (Bloomberg/Refinitiv, or a poll of broker estimates). - **Your own estimate**, and explicitly *where you differ from consensus and why* — this is where research value is created. - The **key swing variables** for this specific quarter (a raw-material move, a price hike taken mid-quarter, a plant shutdown, a large order execution). - **What the stock has already done** into the print. A stock up 20% into results has a high bar; the same result produces a different price reaction depending on positioning. - The **questions you want answered** on the concall.

Stage 2 — Reading the print, in order

Read in this sequence, because it is the order of information value:

1. **Revenue** — versus consensus, and decomposed into **volume versus price/realisation** where disclosed. Volume-led beats are higher quality than price-led ones.
2. **Gross margin** — the rawest read on pricing power and input costs, before management discretion enters via opex.
3. **EBITDA and EBITDA margin** — the operating result. Compare margin YoY *and* QoQ; QoQ catches the recent inflection, YoY catches the trend.
4. **Below EBITDA** — depreciation, interest, other income, tax rate. A PAT beat driven by a **low tax rate** or a spike in **other income** is a low-quality beat and should not be extrapolated.
5. **Exceptional / one-off items** — always identify and strip these. Reported PAT and *adjusted* PAT can differ enormously.
6. **Balance sheet and cash flow**, if disclosed (in India, half-yearly for most) — receivables, inventory, and debt movement. Profit without cash conversion is the classic warning.

The beat/miss quality matrix

BEAT DRIVEN BY	QUALITY	EXTRAPOLATE?
Volume growth	High	Yes
Sustainable price increase holding	High	Yes
Cost efficiency / operating leverage	Medium-high	Partly
Favourable raw-material swing	Medium	No — mean-reverts
Lower tax rate	Low	No
Other income / treasury gains	Low	No
One-off asset sale	Very low	No
Lower provisions/depreciation charge	Low — check why	Investigate

Stage 3 — The concall

The earnings call is where the **forward-looking** information lives. Structure your listening:

Management's prepared remarks — usually the narrative they want you to leave with. Note what they lead with, and note carefully **what they do not mention** that they discussed last quarter. A segment that was a highlight last quarter and goes unmentioned this quarter is usually a problem.

Guidance — the most price-sensitive content on the call. Track: - Was guidance **raised, maintained, or cut**? - Is guidance for revenue, margin, or both? - Compare the *language* to last quarter's exact language. A shift from "we are confident of" to "we are hopeful of" is a real downgrade even with the same number attached.

The Q&A section — the most valuable part, because it is not scripted. Watch for: - Which questions get **specific numeric answers** versus deflection ("we don't guide on that", "let's take this offline"). - **Repeated questions** from multiple analysts on the same issue — the buy side has collectively identified a concern. - Whether the CFO or CEO answers a difficult question — deflection to a junior executive can signal discomfort. - Hedging language density (see the tone-tracking discipline: "broadly", "largely", "we believe", "should normalise") — rising hedging around a specific topic across successive calls is a genuine, trackable signal.

Stage 4 — Model update and the response note

Update the model for: the actual quarter (replacing your estimate with reported), any guidance change, and any structural change to your forecast drivers. Then recompute the target price.

The output is a **results update note**, typically published within hours, containing: the beat/miss summary, what drove it, what changed in the concall, your revised estimates (with the % change to EPS clearly shown), and whether your rating and target price change — and if they do not change, *why not*, which is equally a view.

Estimate revision as a signal in itself

The **direction and breadth of consensus estimate revisions** after a print is itself one of the more robust equity signals: stocks with broad upward EPS revisions tend to be re-rated, and the market often under-reacts initially to a genuine change in the earnings trajectory (post-earnings-announcement drift). Track whether *your* revision is with or against the consensus direction, and be explicit when you are the outlier.

Common mistakes

- Reacting to **reported PAT** without adjusting for one-offs and tax-rate effects.
- Not knowing consensus before the print, so being unable to judge beat/miss at all.
- Treating a **raw-material-driven margin beat** as structural.
- Listening only to prepared remarks and skipping Q&A — where the real information is.
- Failing to note **language changes** in guidance because the headline number was unchanged.
- Publishing an update that restates the numbers without stating a **view change or explicit reaffirmation**.

Interview angle

Expect: "A company reports 25% PAT growth and the stock falls 6%. Explain." A strong answer covers: consensus expected more (the beat/miss frame); the *quality* of the growth (was it other income or a low tax rate rather than operations?); guidance was cut or the tone on the call deteriorated; and the stock had already run into the print so expectations were elevated. Naming all four dimensions — expectations, quality, guidance, positioning — is a complete answer.

Forensic Accounting and Accounting-Quality Red Flags

The Problem / Why this matters

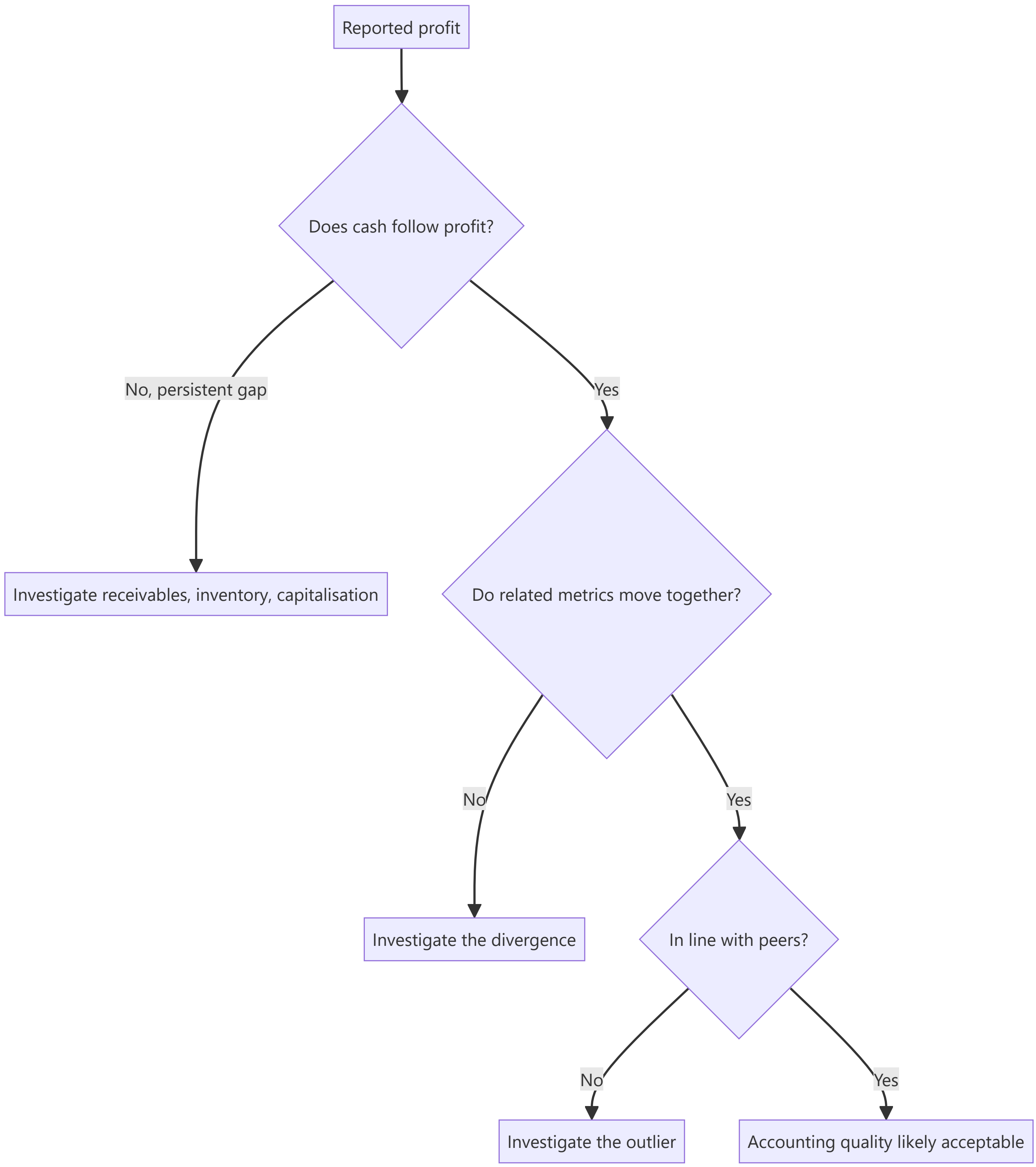
A model built on misstated financials produces a precise, confident, worthless valuation. The most damaging outcomes in equity research are not valuation errors of 15% — they are companies where the reported earnings were not real. Detecting deteriorating accounting quality *before* it becomes a headline is among the highest-value skills a senior analyst has, and "what red flags would you look for in a company's accounts?" is a standard senior-level interview question.

Core Idea

Accounting quality problems almost always show up as a **divergence between reported profit and cash**, between **related figures that should move together**, or between **a company and its peers** on the same metric. You find them by systematically checking those relationships rather than reading the P&L alone.

Why it works this way

Profit involves judgement — revenue recognition timing, provisioning, depreciation rates, capitalisation choices. Cash does not. Management can influence reported profit far more easily than it can conjure cash. Therefore any sustained, unexplained gap between profit and operating cash flow is the single most reliable place to start.



Full technical content

Category 1 — Cash flow versus profit

The core test: cumulative CFO versus cumulative PAT over 3–5 years. For a healthy, non-growing-working-capital business these should track reasonably closely. A company reporting consistent profit while operating cash flow lags persistently is the highest-priority red flag in this entire chapter.

Cash conversion ratio = CFO ÷ EBITDA. Persistently below ~60–70% without a structural explanation (a genuinely working-capital-intensive business model) warrants investigation.

Where the gap hides: - **Receivables growing faster than revenue** — revenue may be recognised on sales that will not collect. Check **Days Sales Outstanding (DSO)** trend, and against peers. - **Inventory**

growing faster than sales — either demand is weakening or inventory is not being written down. Check **inventory days**. - **Unusual "other current assets"** growth — a common parking place for items that should be expensed.

Category 2 — Revenue recognition

- **Revenue recognised well ahead of cash** — especially in project/EPC businesses using percentage-of-completion, where the completion estimate itself is a management judgement.
- **Unbilled revenue** rising sharply as a share of revenue — work claimed as earned but not yet invoiced.
- **Channel stuffing** — primary sales (to distributors) outpacing secondary sales (to end consumers). Compare the company's reported growth to distributor inventory commentary and to retail-audit/industry data.
- **Quarter-end concentration** — a large share of quarterly revenue landing in the final days.
- **Related-party revenue** — sales to entities connected to the promoter. Always read the related-party transactions note in full.

Category 3 — Expense and capitalisation choices

- **Capitalising costs that peers expense** — R&D, software development, or interest. Capitalisation moves cost off the P&L onto the balance sheet, flattering current profit and inflating assets. Compare the capitalisation policy to peers directly.
- **Depreciation rate lower than peers** or a sudden change in useful-life assumptions — a change in estimate that conveniently boosts profit.
- **Under-provisioning** — for doubtful debts, inventory obsolescence, or warranty. Check provision as a % of the relevant base, trended and versus peers.
- **Falling employee cost per employee** or A&P cut sharply while revenue grows — borrowing from the future.

Category 4 — Balance sheet and structure

- **Rising debt alongside rising reported profit** — if the company is profitable, why is it borrowing more?
- **Contingent liabilities** large relative to net worth — read the note; disputed tax demands and guarantees given to group companies both matter.
- **Loans and advances to related parties / subsidiaries** — a classic route for funds to leave the listed entity.
- **Complex subsidiary structures** with material unconsolidated entities, or frequent restructuring of the group.
- **Goodwill that is never impaired** despite the acquired business underperforming.
- **Cash on the balance sheet alongside high-cost debt** — genuinely odd; ask why the cash is not repaying the debt. Sometimes the "cash" is not freely available.

Category 5 — Governance and behavioural signals

These often precede the accounting problems becoming visible: - **Auditor resignation mid-term**, or a change to a materially smaller audit firm. Read the stated reason in the resignation letter — a reason citing inability to obtain satisfactory explanations is severe. - **Qualified audit opinion** or an emphasis-of-matter paragraph. - **CFO churn** — repeated CFO exits are among the most reliable soft signals. - **Resignation of independent directors**, especially with a stated reason. - **Rising promoter pledge** — the promoter's own funding stress, which creates incentive pressure on reported results. - **Frequent changes of**

accounting policy or restatements of prior periods. - Aggressive, promotional investor communication disproportionate to the business's actual scale.

Ratio-based screening tools

The Beneish M-Score combines eight ratios (days-sales-in-receivables index, gross margin index, asset quality index, sales growth index, depreciation index, SG&A index, leverage index, and total accruals to total assets) into a single score designed to flag possible earnings manipulation. **The Altman Z-Score** screens for financial distress rather than manipulation. Both are *screens, not verdicts* — they generate a shortlist for investigation, and both produce false positives (fast-growing companies frequently trip the M-Score legitimately).

The accruals ratio — (Net income − CFO) ÷ average total assets — is a simpler, robust version of the same idea: high accruals mean profit is largely non-cash.

How to use this in practice

Run the checks in order of information value: **cash versus profit first**, then working-capital trends, then policy comparison versus peers, then the governance signals. A single flag is a question, not a conclusion. A **cluster** of flags — rising receivables *and* falling PCR-equivalent provisioning *and* a CFO exit *and* a rising promoter pledge — is a pattern, and patterns are what you act on.

Common mistakes

- Reading the P&L and ignoring the **cash flow statement** entirely.
- Treating a single red flag as proof of fraud, or dismissing a cluster because no single item is conclusive.
- Comparing a company's ratios only to its own history and not to **peers** — an industry-wide change is different from a company-specific one.
- Skipping the **notes to accounts**, which is where related-party transactions, contingent liabilities, and policy changes actually live.
- Assuming a Big-4 auditor guarantees quality.

Interview angle

"What red flags would make you distrust a company's reported earnings?" Structure it: (1) profit not converting to cash — the cumulative CFO vs PAT test, and where the gap hides (receivables, inventory); (2) revenue-recognition aggressiveness — unbilled revenue, channel stuffing, related-party sales; (3) expense choices — capitalisation versus peers, under-provisioning, depreciation assumptions; (4) balance-sheet structure — related-party loans, contingent liabilities, debt rising alongside profit; (5) governance signals — auditor resignation, CFO churn, rising promoter pledge. Then close with the key judgement point: one flag is a question, a cluster is a pattern.

Working Capital and Cash Conversion Analysis

The Problem / Why this matters

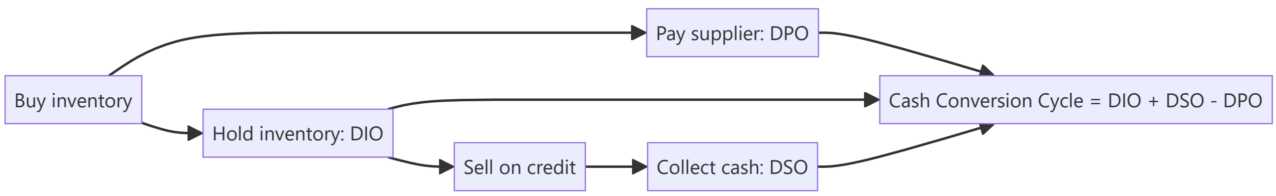
Two companies can report identical revenue growth and identical EBITDA margins, and one can be generating cash while the other is consuming it. The difference is working capital — and it determines whether growth creates value or destroys it. For any business selling on credit or holding inventory, working capital is where growth is funded, where quality shows up, and where the earliest signs of demand weakness or channel stress appear.

Core Idea

Working capital is the cash tied up in running the business day to day: **receivables + inventory − payables**. The **cash conversion cycle** measures how many days of cash are locked up in that loop. A shortening cycle releases cash; a lengthening cycle absorbs it — and a lengthening cycle during revenue growth is one of the most reliable early warnings in fundamental analysis.

Why it works this way

Revenue is recognised at sale; cash arrives at collection. Between those two points the company has funded production and given the customer credit. The faster that loop turns, the less capital the business needs to grow, the higher its return on capital, and the less it depends on external funding — which is precisely why working-capital-light businesses command premium multiples.



Full technical content

The three components

METRIC	FORMULA	MEANING	DIRECTION THAT HELPS
DIO — Days Inventory Outstanding	(Inventory ÷ COGS) × 365	Days of stock held	Lower
DSO — Days Sales Outstanding	(Receivables ÷ Revenue) × 365	Days to collect from customers	Lower
DPO — Days Payable Outstanding	(Payables ÷ COGS) × 365	Days taken to pay suppliers	Higher

Cash Conversion Cycle (CCC) = DIO + DSO − DPO

A CCC of 60 means roughly 60 days of operating cash is permanently tied up in the business; grow revenue and that tied-up amount grows proportionally.

Negative working capital — where DPO exceeds DIO + DSO — means suppliers and customers are funding the business. This is the structural advantage of quick-service restaurants, some retail, and subscription businesses collecting in advance: **growth generates cash rather than consuming it**. It is a genuine, durable competitive and valuation advantage.

The growth-funding arithmetic

The cash a company must fund for each rupee of incremental revenue is roughly **CCC ÷ 365**. A business with a 90-day CCC growing revenue by ₹100 crore must fund about ₹25 crore of additional working capital to do so. This is why:

- A high-CCC business growing fast will show **profit without cash flow**, and will need debt or equity to fund the gap.
- **Return on capital employed** is mechanically depressed by a long CCC, because working capital sits in the capital-employed denominator.
- A company that improves CCC while growing releases a one-time slug of cash — genuinely valuable, but **non-recurring**, and should not be extrapolated into the forecast as sustainable cash generation.

Reading the trend — what each movement signals

Rising DSO (collections slowing): - Demand weakening, so the company is extending credit to hold volumes — an early demand-stress signal, visible before revenue falls. - A shift in customer mix toward weaker or larger, more powerful buyers. - Aggressive revenue recognition on sales unlikely to collect (see the accounting-quality chapter). - Genuine one-off: a large customer's payment cycle straddling the period end.

Rising DIO (inventory building): - Demand slowing while production continued — the classic pre-downturn signal in manufacturing. - Deliberate build ahead of a launch, a price increase, or a seasonal peak (benign — check management commentary). - Obsolescence risk not yet written down.

Rising DPO (paying suppliers slower): - Improved negotiating power (benign, and genuinely value-creating). - **Or liquidity stress** — stretching creditors because cash is tight. Distinguish these by checking whether cash balances and debt are simultaneously deteriorating. Stretching payables to flatter reported CFO is a real and common tactic.

The critical discipline: decompose before concluding

CCC moving is not itself informative — *which component moved* is. A CCC improvement driven entirely by stretching payables (rising DPO) while DSO and DIO also deteriorate is not an improvement at all; it is deterioration masked by supplier financing. Always report the three components separately.

Sector context

Working-capital norms vary enormously, so **always benchmark against sector peers, not an absolute standard**:

BUSINESS TYPE	TYPICAL CCC CHARACTER
FMCG / quick-service	Low or negative — fast turns, cash sales, supplier credit
IT services	Moderate — no inventory, but receivables from enterprise clients
Pharma (generics/US)	Long — extended channel receivables, regulatory inventory
Capital goods / EPC	Very long — project cycles, retention money, unbilled revenue
Retail	Varies — inventory-heavy but often strong payables terms
Banks / NBFC	Not applicable — the balance sheet *is* the product

Linking to valuation

Working capital enters valuation directly: in a DCF, **change in working capital** is subtracted in deriving free cash flow. A forecast that grows revenue aggressively while holding working capital days flat is implicitly assuming an efficiency improvement — make that assumption explicit and defensible, because it is a common way DCFs are quietly inflated.

Also note the **RoCE linkage**: RoCE = EBIT ÷ (Fixed assets + Working capital). Two companies with identical EBIT margins will show materially different RoCE if their working-capital intensity differs — and RoCE, not margin, is what drives sustainable multiple differences between them.

Common mistakes

- Reporting CCC as a single number without decomposing into DIO/DSO/DPO, hiding offsetting moves.
- Reading rising DPO as improved bargaining power when it is liquidity stress.
- Comparing CCC across sectors rather than against relevant peers.
- Extrapolating a one-time working-capital release as recurring cash generation.
- Forecasting revenue growth in a DCF without funding the corresponding working-capital build.
- Ignoring seasonality — comparing a peak-season quarter-end to an off-season one and calling the difference a trend.

Interview angle

"A company's profit is growing but its operating cash flow is falling. What's happening?" The structured answer: working capital is absorbing the cash, so decompose it — receivables rising faster than revenue (collections slowing, possibly demand stress or aggressive revenue recognition), inventory building (demand slowing or obsolescence unprovided), or payables normalising after having been stretched. Then check whether it is seasonal, one-off, or a genuine trend by looking at the multi-period series and peer comparison. Being able to name the three components and what each movement implies is the whole answer.

Management Quality and Corporate Governance Assessment

The Problem / Why this matters

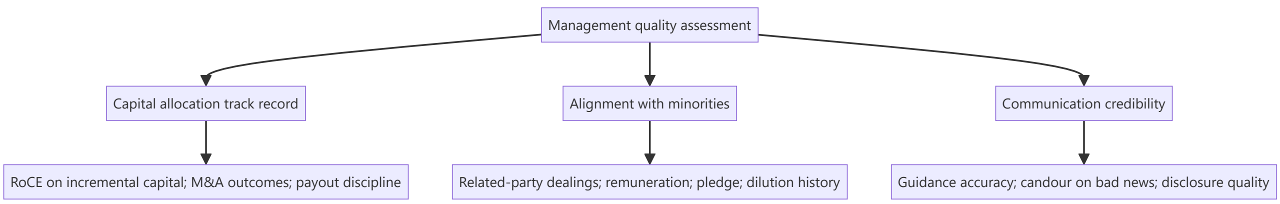
Over a long holding period, the quality of the people allocating a company's capital matters as much as the business economics they started with. Two companies with identical assets and identical market positions will diverge enormously depending on whether management reinvests well, treats minority shareholders fairly, and tells the truth about setbacks. Governance failure is also the fastest destroyer of equity value in Indian markets — and unlike a valuation error, it is rarely recoverable.

Core Idea

Management quality is assessable, not merely intuitive. Judge it on three dimensions with evidence: **capital allocation track record**, **alignment with minority shareholders**, and **credibility of communication** — each measurable against what management previously said and what they actually did.

Why it works this way

Talk is free and universally optimistic; every management team says they are focused on shareholder value. The evidence is in the historical record: what returns did their past investments earn, who benefited when value was created, and did their previous guidance prove accurate? Because that record is public and cumulative, management quality is one of the few "soft" factors that can be assessed rigorously.



Full technical content

Dimension 1 — Capital allocation

This is the CEO's primary job: deciding what to do with the cash the business generates. There are only five options — reinvest in the core business, acquire, pay down debt, pay dividends, or buy back stock — and the historical record of those choices is fully visible.

Assessment tools:

- **Incremental RoCE** — the return earned on capital deployed over the last 5–10 years, calculated as the change in EBIT divided by the change in capital employed over the period. This is far more informative than the reported RoCE level, which is dominated by legacy assets. Management deploying fresh capital at returns below the cost of capital is destroying value however profitable the legacy business looks.
- **Acquisition track record** — for each material acquisition: what was paid, what was promised, and what the acquired business actually delivered. Then check whether goodwill from those deals has ever been impaired. Serial acquirers who never impair goodwill despite underperforming acquisitions are not being straight with you.

- **Capex discipline through the cycle** — did they expand capacity at the top of the cycle (value-destructive, and extremely common in commodities and cyclicals) or counter-cyclically?
- **Returning cash** — is the dividend/buyback policy consistent and rational, or does the company hoard cash on the balance sheet earning treasury returns while claiming reinvestment opportunities it never funds?

Dimension 2 — Alignment with minority shareholders

In promoter-driven markets, the central governance question is whether value created accrues to *all* shareholders or is diverted to the promoter group.

What to examine, all of it publicly disclosed:

- **Related-party transactions** — read the full note every year. Sales to, purchases from, loans to, and guarantees given to promoter-group entities. Look at scale relative to revenue and net worth, and at whether the terms appear arm's-length.
- **Promoter remuneration** — as a share of profit, and its trend versus profit. Remuneration rising while profit falls is a direct alignment failure.
- **Promoter pledge** — a high or rising pledge means the promoter's personal financial position is levered to the share price, which creates pressure on reported results and forced-selling risk (covered in the technical-signal material).
- **Dilution history** — repeated equity raises at prices unfavourable to existing minorities, or preferential allotments/warrants to promoter entities at advantageous prices.
- **Royalty and brand-fee payments** to promoter entities — a legitimate structure, but check quantum and trend; escalating royalty as a share of profit is a classic value-transfer route.
- **Group-company entanglement** — loans, cross-guarantees, and receivables from group entities that sit outside the listed vehicle.

Dimension 3 — Communication credibility

The most efficiently assessable dimension, because it self-scores over time.

- **Guidance accuracy** — build a simple historical table: what did management guide to, and what did they deliver? Chronic over-promising is a durable trait, not a series of unlucky quarters.
- **Candour on bad news** — does management flag a problem early and explain it, or does the problem only surface once it is undeniable? Managements that disclose early are systematically more reliable on everything else.
- **Consistency of narrative** — compare the strategy described three years ago to today's. Frequent strategic pivots described each time as a long-held plan is a credibility signal.
- **Disclosure quality** — segmental detail, volume/realisation splits, and clear reconciliation of exceptional items. Reducing disclosure granularity (e.g. ceasing to report a segment separately just as it weakens) is a meaningful negative signal.
- **Concall behaviour** — willingness to take hard questions directly, versus deflection (covered in the earnings-call material).

Board and structural governance

- **Board independence** — genuinely independent directors, or long-tenured associates of the promoter? Check tenure and other directorships.
- **Audit committee composition and auditor tenure**; any auditor resignation or qualification (a severe signal, per the accounting-quality material).
- **Separation of Chairman and CEO roles.**

- **Independent director resignations**, especially with a stated reason.
- **Succession planning** in promoter-led businesses — a genuine long-horizon risk that is rarely disclosed proactively.

Turning it into a research output

Do not write "management is good." Write an evidence-based assessment: *"Incremental RoCE of 18% over FY19–24 versus a ~12% cost of capital; two acquisitions in the period, both delivering above the stated revenue case; related-party transactions immaterial at under 1% of revenue; promoter pledge nil; guidance met or exceeded in 9 of the last 12 quarters."* That is a defensible assessment. Its opposite is equally defensible and far more valuable to a client.

Management quality legitimately affects the **multiple** you apply and the **discount rate**, and it belongs explicitly in the risk section of a note — but it must be argued from the record, not asserted from impression.

Common mistakes

- Confusing a charismatic, articulate CEO with a good capital allocator — they are uncorrelated.
- Assessing management on the reported RoCE level rather than **incremental** RoCE on capital they actually deployed.
- Skipping the related-party transactions note because it is dense.
- Treating governance as a binary compliance checkbox rather than a spectrum affecting valuation.
- Being reluctant to write a negative governance assessment for a company with a strong stock price — the market repricing that gap is precisely where research adds value.
- Assuming large size or index membership implies governance quality.

Interview angle

"How would you assess management quality?" Give the three-dimension structure with the specific evidence under each: capital allocation (incremental RoCE, M&A track record versus what was promised, capex timing through the cycle, cash-return discipline); alignment (related-party transactions, remuneration versus profit, pledge, dilution history, royalty payments); credibility (guidance accuracy tracked over time, candour on bad news, disclosure consistency). Close with the key point that all of it is assessable from public disclosure — this is analysis, not impression.

Consensus Estimates, Surprise and Revision Analysis

The Problem / Why this matters

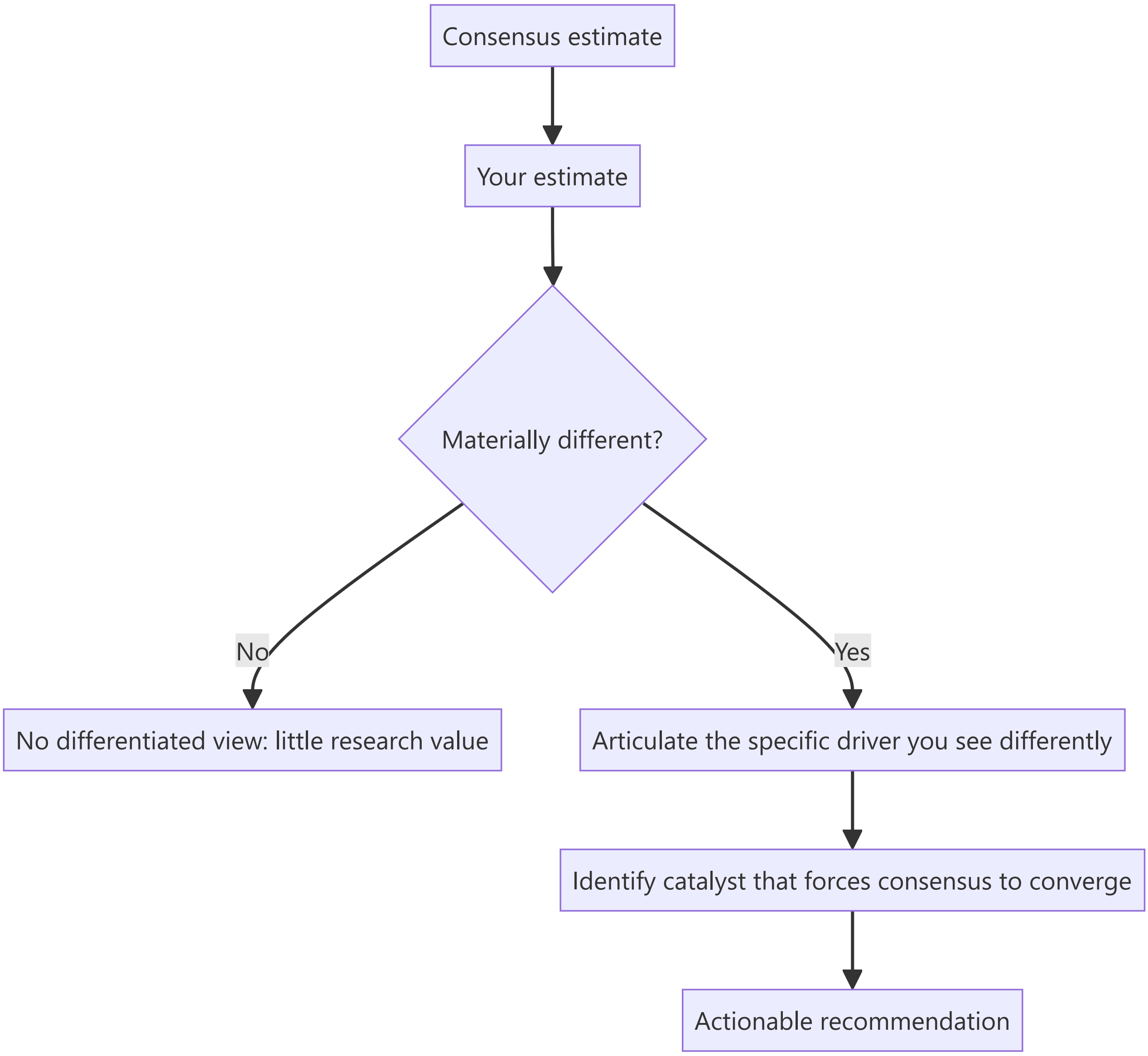
A stock's price already contains the market's collective forecast. An analyst who does not know what that forecast is cannot know whether their own view is differentiated — and a view that merely agrees with consensus, however well-researched, generates no alpha and no reason for a client to read the note. Understanding consensus, how it forms, how it moves, and where it is likely wrong is the mechanism by which research becomes actionable.

Core Idea

Consensus is the aggregated estimate of covering analysts. Value is created by being **differentiated and right** — holding a forecast materially away from consensus for an articulable reason, and being correct. Estimate *revisions* are themselves one of the more durable equity signals, because the market systematically under-reacts to genuine changes in the earnings trajectory.

Why it works this way

Price ≈ earnings expectation × multiple. If your earnings forecast matches consensus and your multiple matches the market's, your target price will match the current price — you have produced a Hold with no informational content. Alpha requires disagreeing with the market on one of those two inputs, with evidence.



Full technical content

What consensus actually is

Consensus is compiled by data providers (Bloomberg, Refinitiv, FactSet, and in India also broker polls) from the published estimates of covering sell-side analysts. Key characteristics an analyst must understand:

- It is typically a **mean or median** of contributing analysts, so it hides dispersion. Always look at the **high/low range and the standard deviation**, not just the point estimate. Wide dispersion means the market itself is unsure — genuine uncertainty and therefore genuine opportunity.
- It is **stale to varying degrees** — some contributors update within hours of a result, others weeks later. Immediately after an event, the "consensus" may not reflect the event at all.
- **Coverage count matters.** A stock with 30 analysts has an efficiently-formed consensus; a stock with 3 has a fragile one, and under-covered mid-caps are where differentiated work is most likely to pay.
- Providers vary in whether estimates are **adjusted or reported** earnings, which can create apparent surprises that are purely definitional. Know which basis you are comparing against.

The whisper number

Beyond published consensus, an informal **"whisper number"** often circulates — the buy side's actual working expectation, which can differ from published consensus, especially when the print is anticipated to be strong or weak. This is why a company can "beat consensus" and still fall: it missed the number the market was actually positioned for. Gauge this from recent price action into the print, positioning data, and buy-side conversations.

Earnings surprise

Surprise % = (Actual − Consensus) ÷ |Consensus|

The academically robust finding is **post-earnings-announcement drift (PEAD)**: stocks that deliver a large positive surprise tend to continue drifting in that direction for weeks afterward, and negative-surprise stocks likewise — implying the market under-reacts initially to genuinely new earnings information. This is one of the most replicated anomalies in equity markets, and it is why the *quality* assessment of a surprise (from the earnings-season material) matters: drift follows surprises reflecting genuine operational change, not one-off tax or other-income effects.

Estimate revisions — the more powerful signal

More durable than a single surprise is the **trend and breadth of estimate revisions**:

SIGNAL	CONSTRUCTION	INTERPRETATION
Revision direction	Change in mean EPS estimate over 1/3/6 months	Rising = improving earnings trajectory
Revision breadth	(# analysts raising − # cutting) ÷ total	Broad-based revisions are more reliable than one outlier
Revision magnitude	% change in consensus EPS	Larger changes carry more information
Dispersion trend	Narrowing or widening spread of estimates	Narrowing = converging conviction

Upward-revision momentum tends to persist, because analysts revise incrementally and conservatively rather than jumping straight to a new view — so a first upgrade is often followed by more. Being **early** in that revision cycle is where sell-side research most directly creates client value.

Building and defending a differentiated estimate

The practical workflow:

1. **Decompose consensus.** Do not just note that consensus EPS is ₹42; work out what revenue growth, margin and tax rate that implies. Consensus is a set of assumptions, and you can only disagree usefully with a specific assumption.
2. **Identify your point of difference.** Typically one or two drivers — you think volume growth is 4% not 8%, or that gross margin holds where the market expects compression.
3. **Evidence it.** Channel checks, industry data, capacity data, management commentary, competitor disclosures.
4. **Quantify the earnings gap.** "We are 12% below FY26 consensus EPS, driven entirely by our lower realisation assumption."
5. **Identify the catalyst.** A differentiated view only becomes a return when the market converges to it. What forces convergence, and when? A quarterly print, a capacity announcement, a price-hike attempt failing?

6. **State the falsification condition.** What would prove you wrong — because the analysts who state this in advance are the ones clients trust.

Where consensus is most often wrong

- **Turning points** — consensus extrapolates recent trends and is systematically late at cycle inflections, both up and down.
- **Under-covered mid- and small-caps** — fewer analysts, weaker consensus formation.
- **Structural change** versus cyclical change — the market frequently misreads a permanent margin shift as a temporary one, or vice versa.
- **Second-order effects** — a raw-material move's impact on a customer's customer.
- **Post-event staleness** — the days immediately after a major announcement, before contributors update.

Common mistakes

- Comparing your **adjusted** EPS to a **reported**-basis consensus and declaring a surprise that is purely definitional.
- Quoting consensus as a single number without checking dispersion or the number of contributors.
- Building a model that lands on consensus and calling it research.
- Having a differentiated estimate but no **catalyst** — being right eventually is not an investable recommendation.
- Ignoring that consensus immediately post-event may be stale and not yet reflect the event.
- Treating any beat as bullish without assessing whether the beat was operational or one-off.

Interview angle

"How do you add value if the market already knows everything you know?" The answer is the three sources of edge, applied through consensus: **informational** (you did work others did not — channel checks, primary data), **analytical** (you interpret the same public facts differently, usually by decomposing consensus assumptions and finding one that is wrong), and **behavioural/time-horizon** (you are willing to hold a view through a period the market is discounting). Then make it concrete: state where you differ from consensus, on which specific driver, by how much on EPS, and what catalyst forces convergence.

Scenario Analysis, Sensitivity and Probability-Weighted Targets

The Problem / Why this matters

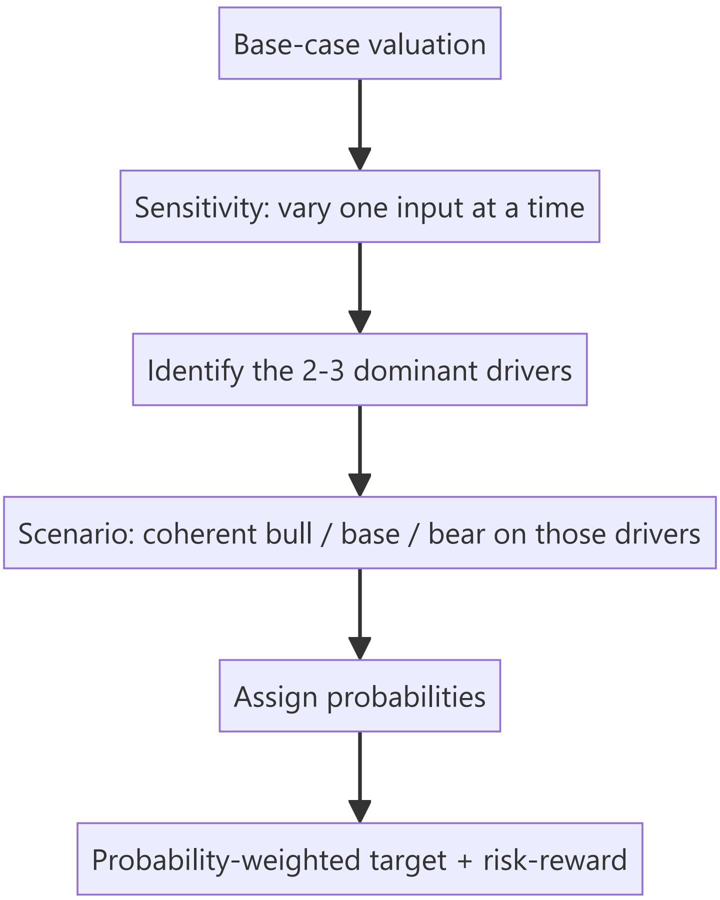
A single-point target price implies a precision no forecast possesses. Every valuation rests on assumptions that could reasonably be otherwise, and a client's real question is not "what is it worth?" but "what do I make if I'm right, what do I lose if I'm wrong, and how likely is each?" Presenting a range with explicit drivers, rather than one number, is what distinguishes a senior analyst's output from a junior's.

Core Idea

Test the valuation against the assumptions that actually move it. **Sensitivity analysis** varies one input at a time to find which matter; **scenario analysis** varies a coherent set of inputs together to describe genuinely different futures; **probability weighting** combines those into an expected value and, more usefully, into an explicit risk-reward.

Why it works this way

Not all assumptions matter equally. In most DCFs, a handful of inputs — revenue growth, terminal margin, WACC, terminal growth — dominate the output, while a dozen others barely move it. Finding which few dominate tells you where to concentrate your research effort, and tells the client which specific things to watch.



Full technical content

Sensitivity analysis — finding what matters

Vary one input, hold everything else constant, record the change in value. The standard presentation is a **two-way data table** — the classic being WACC against terminal growth rate in a DCF:

VALUE PER SHARE	G = 3.0%	G = 3.5%	G = 4.0%	G = 4.5%
WACC 10.5%	892	968	1,062	1,182
WACC 11.0%	812	874	949	1,042
WACC 11.5%	743	794	855	929
WACC 12.0%	683	726	776	836

This table is doing real analytical work: it shows immediately that a 150bp WACC range and a 150bp terminal-growth range together span roughly ₹683 to ₹1,182 — a 73% spread on inputs that are all individually defensible. That is an honest statement about the precision of a DCF, and presenting it prevents the false confidence a single number creates.

Which inputs to test (in typical order of impact): 1. Revenue growth rate (near-term and terminal) 2. Terminal/steady-state EBIT margin 3. WACC (via cost of equity, beta, or equity risk premium) 4. Terminal growth rate 5. Capex and working-capital intensity 6. Tax rate

The elasticity view: express sensitivity as "a 100bp change in terminal margin moves value by 9%" — more useful to a client than a raw table, because it directly tells them how much a given piece of news matters.

Scenario analysis — coherent alternative futures

Sensitivity's weakness is that it varies inputs independently, which is often unrealistic — in a genuine downturn, volume growth, margin and multiple all deteriorate *together*. Scenario analysis fixes this by defining internally consistent states of the world.

Constructing scenarios properly:

	BEAR	BASE	BULL
Narrative	Demand slows, competitive pricing	Guidance broadly met	Capacity ramps, mix improves
Revenue CAGR (3yr)	4%	9%	14%
Terminal EBIT margin	13%	16%	18.5%
Exit multiple / WACC	11.5% WACC	11.0%	10.75%
Value per share	**620**	**950**	**1,290**
Probability	25%	55%	20%

Three disciplines make this genuinely useful rather than decorative:

1. **Each scenario must have a narrative**, not just different numbers. "Bear" should describe a specific, plausible world — not simply "base minus 20%."
2. **Assumptions must move coherently.** A bear case with lower volumes but *unchanged* margins is usually incoherent, because operating leverage works in both directions.

3. **The multiple should move too.** In a genuine bear case the market applies a lower multiple to lower earnings — the double effect is precisely why drawdowns exceed earnings declines. A scenario analysis that flexes earnings but holds the multiple constant systematically understates downside.

Probability weighting and the expected value

Expected value = Σ (probability × scenario value)

Using the table above: (0.25 × 620) + (0.55 × 950) + (0.20 × 1,290) = 155 + 522.5 + 258 = **₹935.5**

Note that the probability-weighted value (₹935) sits below the base case (₹950) — because the bear case is closer to base than the bull case is, and carries higher probability. That asymmetry is itself the finding, and it is invisible without doing this arithmetic.

Where probabilities come from: they are judgements and should be stated as such. Anchor them to something — historical frequency of similar outcomes, the implied probability in options markets, or the dispersion of consensus estimates. State them explicitly so a client who disagrees can substitute their own and recompute.

Risk-reward — usually the most useful output

More decision-relevant than the expected value is the **risk-reward ratio**:

- Upside to bull = (1,290 − 800) ÷ 800 = **+61%** (assuming current price ₹800)
- Downside to bear = (620 − 800) ÷ 800 = **−22.5%**
- **Risk-reward ≈ 2.7 : 1**

A common professional convention is that a Buy requires a risk-reward of at least roughly 2:1 or 3:1 — a threshold that forces discipline, because it means a stock with 20% upside and 20% downside is not a Buy regardless of how attractive the base case narrative sounds. It also explains why a recommendation can change without the base case changing at all: if the stock rallies to ₹1,100, the base case may be unchanged but the risk-reward has collapsed to roughly 0.9:1, and the correct rating is now Hold or Sell.

Presenting it in a note

The professional format states: base-case target and its key assumptions; the bull and bear values with their one-line narratives; the resulting risk-reward; and the **two or three variables** the whole thesis actually hinges on, each with its sensitivity quantified. That last element is what a portfolio manager actually uses — it tells them what to monitor.

Common mistakes

- Publishing a **single target price** with no range, implying false precision.
- Building scenarios by mechanically applying ±20% rather than from coherent narratives.
- Flexing earnings across scenarios while holding the **multiple constant**, understating true downside.
- Assigning probabilities without stating them, or assigning them to make the expected value support a predetermined recommendation.
- Running sensitivity on inputs that barely move value while ignoring the dominant ones.
- Confusing sensitivity (one input at a time) with scenario analysis (coherent joint moves) and presenting one as the other.

Interview angle

"How confident are you in your target price?" The strong answer never claims precision: explain that the base case is ₹950 but the DCF spans roughly ₹680–1,180 across a defensible WACC and terminal-growth range; that the thesis hinges primarily on terminal margin and volume growth, where a 100bp margin change moves value ~9%; that the bear/bull scenarios give ₹620/₹1,290 for specific narrative reasons; and that at the current price the risk-reward is 2.7:1, which is what actually supports the Buy. Showing you know *which* assumptions carry the valuation is the point of the question.

Initiating Coverage — The Full Initiation Note

The Problem / Why this matters

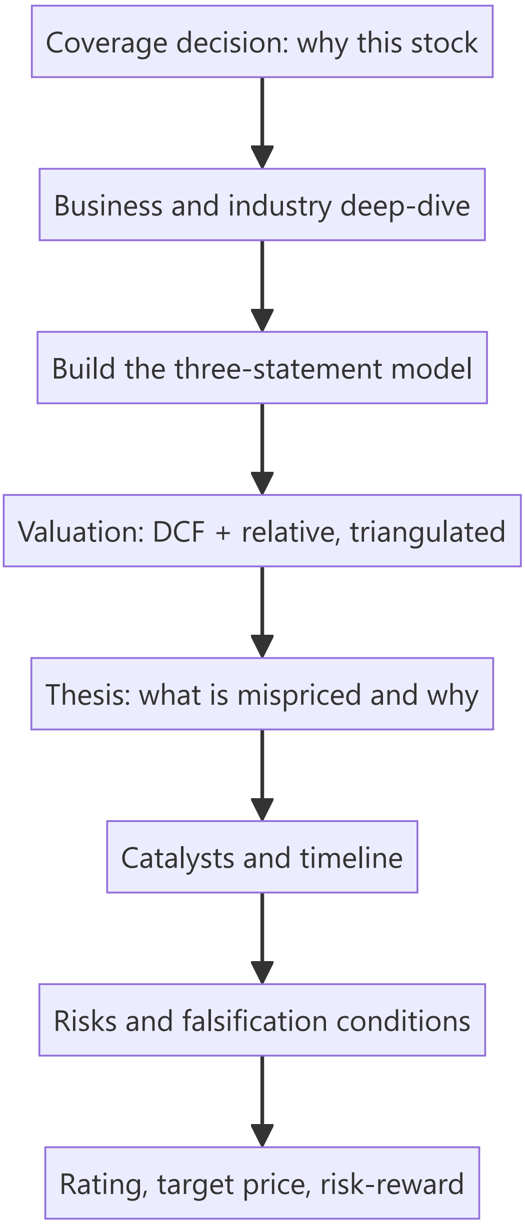
An initiation note is the single largest deliverable in a sell-side analyst's year — typically 40–100 pages, establishing a company as part of your covered universe and, more importantly, establishing *you* as the analyst clients call about that stock. Unlike a results update, it must build the entire investment case from zero: the business, the industry, the model, the valuation, the thesis, and the risks. Being asked "how would you initiate on a company you've never covered?" is a standard senior interview question precisely because it tests whether a candidate can structure a complete analytical project rather than react to a print.

Core Idea

An initiation is a **complete, self-contained investment case** that a reader with no prior knowledge of the company can follow from business model to recommendation. Its distinguishing feature versus every other note type is that it must justify the *coverage decision* itself — why this stock is worth a client's attention at all.

Why it works this way

Clients allocate scarce attention. An initiation competes against every other name in a portfolio manager's universe, so it must answer not just "is it cheap?" but "why should I care about this company, and what do you know that the market doesn't?" That is why initiations lead with a differentiated thesis rather than a company description — the description supports the thesis, not the reverse.



Full technical content

The standard structure of an initiation note

1. Front page / summary (the most-read page by far) - Rating, target price, current price, implied upside - A three-to-five-line **investment summary** stating the thesis in plain language - Key financial table: revenue, EBITDA, EBITDA margin, PAT, EPS, growth, and valuation multiples for the last two actuals and next two-to-three forecast years - The two or three bullet points that carry the entire argument

Most institutional readers read only this page. If the thesis is not clear and differentiated here, the remaining 60 pages will not be read.

2. Investment thesis (3–5 pages) The core argument, typically structured as three or four distinct pillars, each with evidence. A strong thesis is: - **Differentiated** — states explicitly where you differ from consensus and by how much on EPS - **Specific** — names the driver, not a generality ("capacity utilisation rises from 68% to 84% as the new line ramps," not "operational improvements") - **Falsifiable** — you can state what would prove it wrong

3. Company and business model (5–10 pages) What it does, how it earns, segment-wise revenue and margin split, geographic mix, customer concentration, manufacturing/delivery footprint, and its position in the value chain. Include the history that matters — past capital allocation, prior cycles survived, promoter background.

4. Industry analysis (8–15 pages) Market size and growth, structure and concentration, competitive dynamics, regulation, and the sector's own economics (drawing on the sector-framework material). Where does industry profit pool sit, and is it shifting? A frequent weakness in junior initiations is treating this as a data dump rather than answering "what does this industry structure mean for *this* company's returns?"

5. Competitive positioning (3–6 pages) Market share and its trend, the specific source of any moat (cost, brand, distribution, switching costs, regulatory), and evidence for it — sustained above-peer RoCE and margins are the empirical test of a claimed moat.

6. Financial analysis and forecast (8–15 pages) Historical performance decomposed (volume/price, mix, margin bridge), then the forecast with **every material driver stated explicitly**. The reader must be able to see and disagree with your assumptions. Include RoCE/RoE decomposition and working-capital analysis.

7. Valuation (4–8 pages) DCF with full assumptions disclosed, relative valuation versus domestic and global peers, a triangulated fair-value range, and the sensitivity/scenario tables. State clearly which method you weight most and why.

8. Risks (2–4 pages) Specific and quantified, not boilerplate. "A 5% adverse move in the rupee reduces our FY26 EPS by 4%" beats "currency risk." Include the governance assessment.

9. Appendices — financial statements in full, management profiles, shareholding pattern, glossary.

The workflow — roughly 4–8 weeks

PHASE	WORK
1. Scoping	Read 3–5 years of annual reports, all recent concall transcripts, existing broker notes, industry reports. Build the question list.
2. Primary work	Management meeting, plant/store visits, channel checks with distributors and customers, competitor and supplier conversations, expert calls
3. Modelling	Build the three-statement model from historicals; construct the driver-based forecast
4. Valuation	DCF, comps, scenarios, sensitivities
5. Thesis formation	Reconcile the work into a differentiated view; test it against the bear case
6. Writing and review	Draft, internal review, compliance and legal review, publication

The management meeting deserves specific attention: it is not for extracting guidance (which is public) but for understanding how management thinks about capital allocation, competition and risk — the qualitative inputs behind the governance assessment. Prepare specific questions from the model's uncertainties; generic questions get generic answers.

What makes an initiation good rather than merely complete

- It has a **view**, not just information. A comprehensive, well-researched note with no differentiated conclusion has no value to a client.
- It states **where consensus is wrong** and quantifies the gap.

- It identifies **catalysts with timing** — a mispricing with no mechanism for correction is not actionable.
- It is **honest about the bear case**, presenting it strongly enough that a reader who disagrees still respects the work.
- It provides **monitorables** — the specific two or three data points a client should watch to know whether the thesis is tracking.

Initiating with a Sell or Hold

A common misconception is that initiations are always Buys. Initiating with a Sell is the highest-conviction, highest-risk act in sell-side research — it invites management hostility and loses corporate access — but a well-argued Sell initiation builds enormous credibility with the buy side, who are structurally short of genuine negative research. Initiating with a Hold is legitimate when the work concludes the stock is fairly valued, though it must still contain a view: what would make it a Buy, and at what price.

Common mistakes

- **Description without a thesis** — 60 pages that tell the reader what the company does but not what to do about it.
- Burying the differentiated point on page 30 instead of the front page.
- Industry sections that are data dumps disconnected from the company's returns.
- Forecast assumptions that are not visible, so the reader cannot disagree with them.
- **Boilerplate risk sections** that mention "competition" and "regulation" without quantifying anything.
- No catalyst, so the thesis has no timeline.
- Anchoring the target price to the current price rather than to the analysis.

Interview angle

"How would you initiate coverage on a company you don't know?" Structure the answer as the workflow: scoping (annual reports, transcripts, existing research, building a question list) → primary research (management, channel checks, competitors) → three-statement model with explicit drivers → valuation via DCF and comps, triangulated with scenarios → thesis formation stating where you differ from consensus and why → catalysts and risks → rating with risk-reward. Then add the point that distinguishes a senior answer: the note's job is to state a differentiated, falsifiable view with monitorables — not to be comprehensive.

Channel Checks and Primary Research for Equity Analysts

The Problem / Why this matters

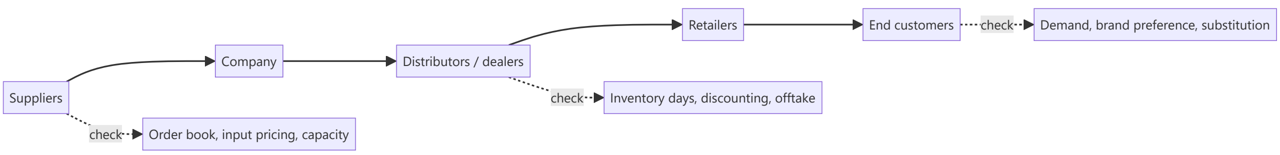
Every analyst covering a stock reads the same annual report, hears the same concall, and sees the same consensus. Public information is, by definition, already in the price. The one reliable source of **informational edge** in equity research is primary work — talking to distributors, customers, suppliers, former employees and industry experts to learn something before it appears in a filing. It is also the area where compliance risk is highest, which is why the methodology and its boundaries must both be understood precisely.

Core Idea

A **channel check** is structured primary research along a company's value chain — upstream (suppliers), midstream (distributors, dealers, retailers) and downstream (customers) — designed to verify or challenge what management has said and to detect changes before they appear in reported numbers.

Why it works this way

Reported financials are a lagging, quarterly, aggregated summary. The underlying business generates signals continuously and locally: a distributor's inventory building, a dealer offering unusual discounts, a supplier's order book thinning. These are observable weeks or months before they show up in a P&L — and they are observable to anyone willing to do the work.



Full technical content

Who to talk to, and what each reveals

SOURCE	WHAT THEY KNOW	TYPICAL QUESTION
Distributors / dealers	Primary vs secondary sales, inventory, scheme intensity	"How many days of stock are you carrying versus normal?"
Retailers	Consumer offtake, shelf movement, competitor activity	"Which brand is moving fastest in this category now?"
Suppliers	Order volumes, payment behaviour, capacity plans	"Have order volumes changed in the last two months?"
Customers (B2B)	Vendor share, pricing, satisfaction, switching intent	"Has your allocation to this vendor changed?"
Former employees	Process, culture, historic issues	Structural and historical, never current confidential data
Industry experts / consultants	Technology shifts, regulation, structural change	Context and framing

The discipline that makes checks reliable

1. Sample properly. Three distributors in one city is an anecdote. Checks should span geography (metro/tier-2/rural), channel type (modern trade/general trade/e-commerce), and size of counterparty. A finding that holds across a diverse sample is a signal; one that appears in a single location is local noise.

2. Ask about behaviour, not opinion. "How many days of inventory are you holding?" produces a fact. "Do you think the company is doing well?" produces a pleasantry. This is the same stated-versus-revealed principle that governs consumer research.

3. Establish a baseline. A datapoint without a comparison is uninterpretable. Always anchor: "versus three months ago," "versus this time last year," "versus the competing brand."

4. Check consistency across the chain. If the company reports strong primary sales but distributors report inventory building and retailers report flat offtake, you have detected channel stuffing — a finding no single source could have given you.

5. Track the same sources over time. A panel of the same distributors contacted each quarter produces far more reliable trend data than a fresh sample each time, because it removes sample-composition change from the signal.

6. Triangulate against hard data. Where available, cross-check qualitative findings against monthly volume disclosures, GST/e-way bill data, import-export data, satellite/footfall data, app-download rankings, or job postings.

Interpreting what you hear

Channel participants have their own incentives, and a professional analyst adjusts for them: - **Distributors** may understate stock to protect their relationship with the company, or overstate pressure to justify seeking better terms. - **Competitors** will characterise the subject company unfavourably. - **Former employees** may carry grievance or nostalgia. - **Management-arranged** channel visits are a curated sample by construction — useful, but never treat them as an independent check.

The correction is not to distrust everything but to weight sources by incentive and to require corroboration across independent parties before acting on a finding.

The compliance boundary — this matters more than the technique

Primary research is legitimate and central to the profession. It becomes illegal when it involves **unpublished price-sensitive information (UPSI)** obtained from someone with a duty of confidentiality.

The bright lines: - Never seek or accept **pre-release financial results, guidance, or specific order wins** from anyone inside the company or its counterparties. - The **mosaic theory** is the governing principle: assembling many pieces of individually non-material public and non-confidential information into a material conclusion is legitimate research. Receiving one piece of material non-public information is not, regardless of how it was obtained. - **Expert networks** must be used through compliance-approved channels, with the expert confirming they are not sharing confidential employer information, and typically with call chaperoning or recording per firm policy. - Never speak to a **current employee of a company you cover** about that company's current performance outside official investor-relations channels. - If a source begins to disclose something material and non-public, **stop the conversation** and report it to compliance. The obligation is affirmative, not passive.

For a SEBI-registered Research Analyst in India, this sits alongside the disclosure obligations covered elsewhere: holdings, conflicts, and the firm's relationship with the covered company must be disclosed in the note itself.

Documenting and using the work

Record for each check: date, source type (never the individual's name in the published note), geography, the specific questions asked, and the answers. In the published research, findings appear as *"our checks across 14 distributors in five states indicate inventory of 25–30 days versus a normal 18–20"* — specific about method and scale, anonymous about individuals, and clearly labelled as the analyst's own primary work rather than company-provided data.

This labelling matters commercially as well as ethically: primary work is the part of a note a client cannot get elsewhere, and it should be visibly identified as such.

Common mistakes

- Generalising from a **tiny or geographically concentrated sample**.
- Asking opinion questions and receiving polite, uninformative answers.
- Taking a **management-arranged** channel visit as an independent check.
- Failing to establish a baseline, so a datapoint cannot be interpreted.
- Not adjusting for the **incentive** of the person speaking.
- Continuing a conversation after a source begins disclosing material non-public information.
- Presenting channel findings without disclosing the method and sample size, which makes them unverifiable and therefore less credible.

Interview angle

"How would you check whether a consumer company's reported growth is real?" A strong answer moves down the chain: compare **primary sales** (company to distributor, which is what gets reported) against **secondary sales** (distributor to retailer) and retail offtake; check distributor inventory days versus normal; ask about scheme/discount intensity, since growth bought with promotions is different from growth from demand; sample across geographies and channel types; and corroborate against hard data such as GST/e-way-bill volumes or retail-audit data. Then close on the compliance frame — mosaic theory, and never soliciting UPSI.

Sum-of-the-Parts and Holding-Company Valuation

The Problem / Why this matters

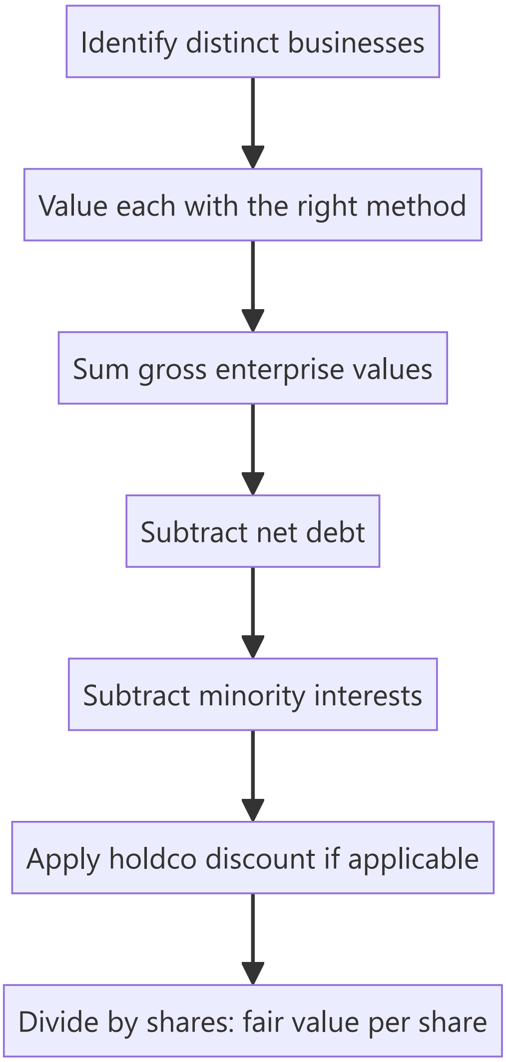
A single multiple applied to a conglomerate's consolidated earnings is almost always wrong. A company running a high-growth digital business, a mature manufacturing business and a stake in a listed subsidiary has three different sets of economics that the market will never value at one multiple. Getting this right matters enormously in Indian markets specifically, where conglomerate and holding-company structures are common and where holding-company discounts of 30–70% are routine and frequently misunderstood.

Core Idea

Sum-of-the-parts (SOTP) values each business separately using the method appropriate to *that* business, then aggregates and adjusts for net debt, minority interests and any holding-company discount. The output is a fair value that a single blended multiple cannot produce.

Why it works this way

Multiples encode growth, risk and capital intensity. Combining a 25% growth business and a 4% growth business into one earnings number and applying an average multiple systematically misvalues both — it understates the growth business and overstates the mature one. Because the market prices the parts separately when it can observe them, the analyst must too.



Full technical content

When to use SOTP

- **Conglomerates** with genuinely unrelated segments (different growth, margins, capital intensity, risk).
- **Holding companies** whose main assets are stakes in other listed or unlisted companies.
- Companies with a **large, separately-valuable asset** — surplus land, a treasury portfolio, a stake in an associate.
- Businesses containing a **loss-making but valuable** segment (a scaling digital arm) that drags consolidated earnings and makes P/E meaningless.
- Ahead of an anticipated **demerger**, where the market will shortly value the parts separately anyway.

Step 1 — Segment properly

Use the company's reported segments as a starting point, but interrogate them. Two tests: do the segments have genuinely different economics, and is there enough disclosure (segment revenue, EBIT, and ideally capital employed) to value them separately? Where disclosure is inadequate, you must estimate — and disclose that you have.

Step 2 — Choose the right method per segment

SEGMENT TYPE	APPROPRIATE METHOD	REASONING
Mature manufacturing	EV/EBITDA on peer multiple	Capital-intensive, depreciation-heavy
High-growth consumer	EV/Sales or DCF	Margins not yet at steady state
Financial services arm	P/B on RoE	Balance-sheet business
Listed subsidiary stake	**Market value** of the stake	Directly observable
Unlisted associate	Peer multiple or last transaction price	No market price
Surplus land / real estate	Independent valuation or circle rate	Non-operating asset
Treasury / investment book	Marked-to-market value	Directly observable
Loss-making scaling business	EV/Sales, or DCF, or last funding round	Earnings are meaningless

The critical discipline: **value listed stakes at their observable market price**, not at your own estimate of their fair value. Mixing "what I think the subsidiary is worth" into a SOTP double-counts your view and makes the output uninterpretable. If you disagree with the subsidiary's market price, say so separately.

Step 3 — Aggregate and adjust

```
Σ Segment enterprise values
+ Value of listed stakes (at market)
+ Surplus non-operating assets (land, treasury)
− Net debt (consolidated, at parent level)
− Minority interest (value of what you don't own in consolidated subsidiaries)
− Present value of unallocated corporate costs
= Equity value
÷ Diluted shares outstanding
= SOTP fair value per share
```

Unallocated corporate cost is frequently forgotten and materially matters: a conglomerate's head-office cost is real, recurring, and belongs nowhere in the segments. Capitalise it (divide by an appropriate rate, or run it as a perpetuity) and subtract it.

Minority interest must be handled consistently: if you have valued a subsidiary's full enterprise value but own only 60% of it, subtract the 40% you do not own.

The holding-company discount

Holding companies — whose principal assets are stakes in other companies — routinely trade well below the market value of those stakes. Discounts of **30–70%** are common in India and persist for structural reasons:

1. **Tax on realisation** — monetising a stake triggers capital gains tax, so the after-tax value to shareholders is genuinely below the gross market value.
2. **No control over cash flows** — the holdco's shareholders cannot compel the operating companies to distribute cash.
3. **Double taxation of dividends** flowing through the structure in some cases.
4. **Governance and capital-allocation risk** — the holdco may reinvest realisations rather than distribute them, and minority shareholders cannot force otherwise.

5. **Low liquidity** in the holdco stock itself.
6. **No catalyst** — absent a demerger, buyback, or distribution policy, there is no mechanism by which the gap closes.

How to handle the discount analytically: do not apply a round number by convention. Estimate the company's **own historical discount range** over several years and use that as the base, then adjust for anything that has changed — a newly announced distribution policy, a demerger proposal, an improvement in governance, or a change in liquidity. A note that says "we apply a 45% holdco discount, versus this company's five-year average of 52%, reflecting the newly announced dividend policy" is doing analysis; one that says "we apply a standard 50% discount" is not.

The trap: a holdco trading at a wide discount looks perpetually cheap, and inexperienced analysts repeatedly recommend it on that basis. The discount is only an opportunity if there is a specific, identifiable catalyst for it to narrow. Without one, the discount is a permanent feature and the stock is fairly valued *with* it.

Worked illustration

A conglomerate with three parts:

COMPONENT	BASIS	VALUE (₹ CR)
Manufacturing EBITDA ₹800cr	9× EV/EBITDA	7,200
Consumer EBITDA ₹300cr	22× EV/EBITDA	6,600
55% stake in listed subsidiary (subsidiary m-cap ₹12,000cr)	Market value × 55%	6,600
Surplus land	Independent valuation	900
Gross value		**21,300**
Less: net debt		(3,400)
Less: capitalised corporate cost (₹120cr ÷ 11%)		(1,090)
Equity value before discount		**16,810**
Less: holdco discount @ 35% applied to the stake component only		(2,310)
SOTP equity value		**14,500**
Shares outstanding	50 cr	
Fair value per share		**₹290**

Note the judgement embedded here: the discount is applied **only to the listed-stake component**, not to the wholly-owned operating businesses — because the discount's rationale (tax on realisation, no control over cash flows) applies to holdings in other companies, not to businesses the company operates directly. Applying a blanket discount to the entire SOTP is a common and material error.

Common mistakes

- Applying **one multiple** to a genuinely diversified consolidated business.
- Valuing a listed subsidiary stake at your own fair value rather than **market value**, double-counting your view.
- Forgetting **unallocated corporate costs**, overstating value by a material margin.
- Mishandling **minority interest**, valuing 100% of a subsidiary the parent owns 60% of.

- Applying a **conventional round-number holdco discount** rather than deriving it from the company's own history and current catalysts.
- Applying the holdco discount to **wholly-owned operating businesses** as well as to stakes.
- Recommending a holdco purely because the discount is wide, with **no catalyst** for it to narrow.
- Double-counting: including a subsidiary's earnings in consolidated EBITDA *and* adding the stake's market value separately.

Interview angle

"How would you value a conglomerate?" Structure it: identify genuinely distinct businesses; value each with the method matching its economics (EV/EBITDA for mature manufacturing, EV/Sales or DCF for a scaling business, P/B for a financial arm, market value for listed stakes); sum; subtract net debt, minority interest and capitalised unallocated corporate costs; apply a holding-company discount **only where warranted and derived from the company's own historical range**; divide by diluted shares. Then add the senior point: a wide holdco discount is only an opportunity with a specific catalyst — otherwise it is a permanent structural feature, and the stock is fairly priced with it.

Quality of Earnings and Adjusted Metrics

The Problem / Why this matters

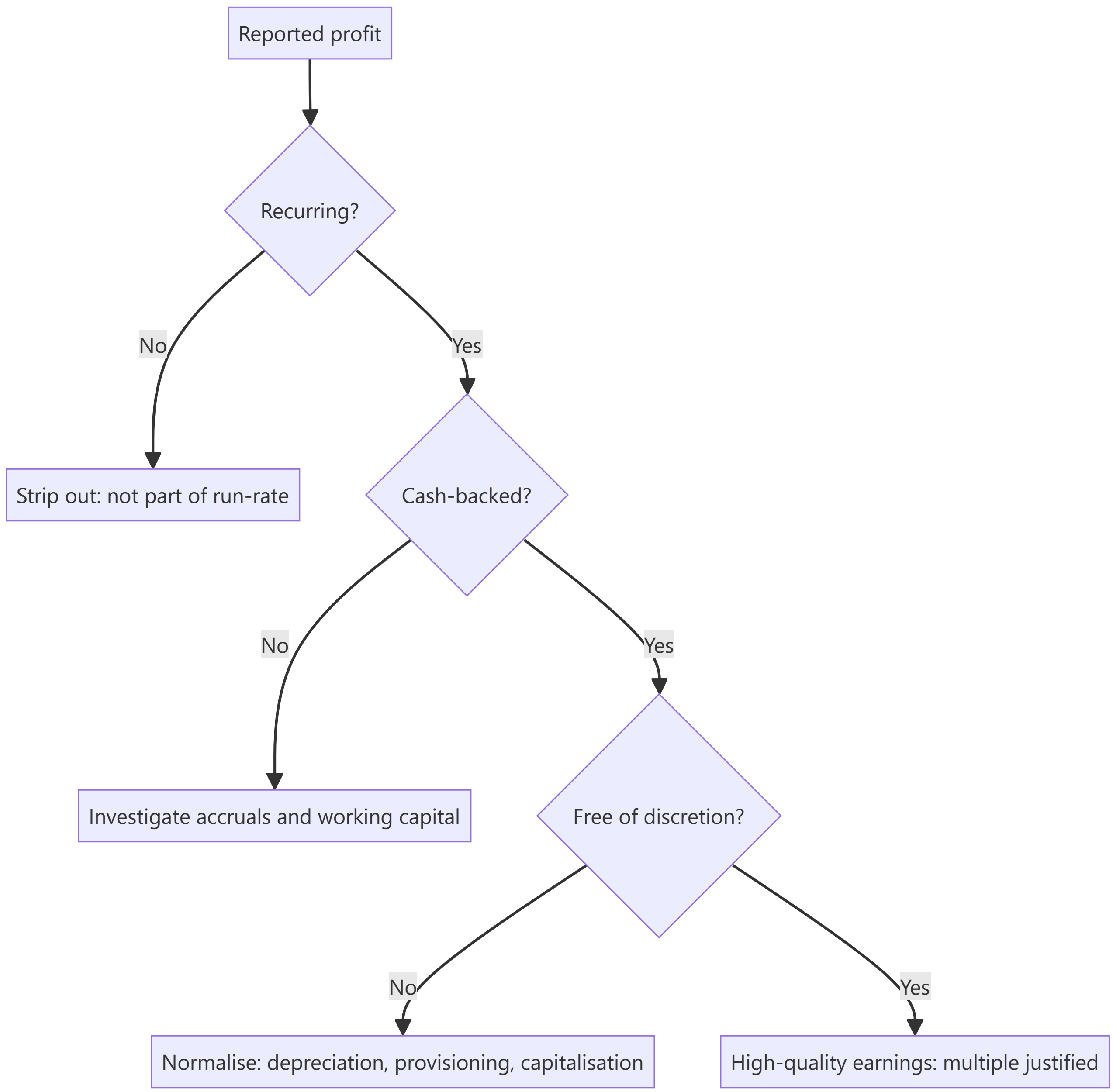
Companies increasingly report two sets of numbers: statutory results and "adjusted", "normalised", "underlying" or "pro-forma" figures excluding items management considers non-representative. Sometimes those adjustments genuinely clarify the run-rate; sometimes they systematically remove every bad item and retain every good one. An analyst who accepts adjusted numbers uncritically is being handed management's preferred narrative; one who rejects them entirely misses real economic signal. The skill is knowing which adjustments to accept.

Core Idea

Quality of earnings asks whether reported profit reflects **repeatable, cash-backed operating performance**. Assess it by testing three properties: is the profit **recurring**, is it **cash-converting**, and is it **free of accounting discretion**?

Why it works this way

Valuation multiplies an earnings figure by a multiple. That multiple implicitly assumes the earnings persist. If a substantial part of reported profit will not recur — a one-time land sale, an unusually low tax rate, a favourable raw-material window — then applying a persistence-assuming multiple to it overvalues the company by exactly that amount, compounded.



Full technical content

Test 1 — Is it recurring?

Identify and separate items that will not repeat:

Genuinely non-recurring (strip out for run-rate purposes): - Asset or land sale gains - One-time litigation settlements (received or paid) - Restructuring charges from a discrete, completed programme - Insurance claim receipts - Gain or loss on sale of a business - Impairment of a specific asset

Frequently mislabelled as non-recurring (usually should NOT be stripped): - **"Restructuring charges" that appear every single year** — if a company restructures annually, restructuring is an operating cost of that business, not an exception. - **Share-based compensation** — a real economic cost of employing people, and excluding it (common in "adjusted EBITDA") systematically overstates profitability. It is non-cash but it is not non-economic; it dilutes shareholders. - **Recurring impairments** — a company impairing goodwill every other year has a capital-allocation problem, not a series of unrelated exceptions. - **Forex losses** in a company with structural foreign-currency exposure — that exposure is part of the business model. - **"One-time" marketing spend** for a launch, in a company that launches continuously.

The diagnostic test: **plot "exceptional items" over 5–7 years**. If exceptionals appear in most years, they are not exceptional, and the adjusted figure is systematically flattering.

Test 2 — Is it cash-backed?

The accrual-based checks (developed in the working-capital and forensic-accounting material) applied to earnings quality:

- **Cumulative CFO vs cumulative PAT** over 3–5 years — the single most robust test.
- **Cash conversion** = CFO ÷ EBITDA — persistently low without a structural reason is a warning.
- **Accruals ratio** = (Net income − CFO) ÷ average total assets — high accruals mean profit is largely non-cash and, empirically, mean-reverts.
- Check whether **CFO itself has been flattered** by stretching payables or by classifying items favourably between operating, investing and financing.

Test 3 — Is it free of accounting discretion?

Compare against peers and against the company's own history: - **Depreciation rate / useful-life assumptions** — a change that boosts profit - **Capitalisation policy** for R&D, software, or interest - **Provisioning levels** for doubtful debts, inventory, warranties - **Revenue recognition timing**, especially percentage-of-completion - **Tax rate** — an unusually low effective rate needs explanation and is usually temporary

Building your own adjusted number

The professional approach is to construct an **analyst-adjusted earnings figure** rather than accepting either the statutory or the company-adjusted number:

1. Start with statutory PAT.
2. **Add back** genuinely non-recurring charges.
3. **Strip out** genuinely non-recurring gains.
4. **Do not strip** share-based compensation, recurring restructuring, or structural forex.
5. **Normalise the tax rate** to the sustainable rate.
6. **Normalise for accounting policy** differences versus peers where material (e.g. adjust for a peer-divergent capitalisation policy).
7. Disclose every adjustment made, so a reader can reverse any they disagree with.

That final point matters: an analyst's adjustments are also judgements. Presenting them transparently — a visible bridge from statutory to adjusted — is what makes the number credible rather than merely another narrative.

The EBITDA warning

Adjusted EBITDA is the most manipulated metric in public markets. It excludes interest (a real cost of the capital structure), tax (a real cash outflow), depreciation (the consumption of assets that must eventually be replaced), *and* whatever else management designates as adjustable. For a capital-intensive business, EBITDA can look robust while the company never generates a rupee of free cash flow, because maintenance capex consumes it all.

The corrective: always pair EBITDA with **free cash flow** and with **EBITDA less maintenance capex**, and be especially sceptical where "adjusted EBITDA" differs materially from statutory EBITDA.

Quality-of-earnings scoring in practice

A practical summary an analyst can include in a note:

DIMENSION	STRONG	WEAK
Exceptionals frequency	Rare, genuinely discrete	Present most years
CFO/PAT (3-yr cumulative)	~1.0 or above	Persistently below 0.7
Cash conversion (CFO/EBITDA)	Above peer median	Below, with no structural reason
Tax rate	At or near statutory	Persistently low, unexplained
Capitalisation vs peers	In line	More aggressive
Company-adjusted vs statutory gap	Small, well-explained	Large, one-directional

Common mistakes

- Accepting **company-adjusted** figures uncritically because they are the ones in the press release headline.
- Excluding **share-based compensation** from earnings — it is a genuine cost to shareholders.
- Treating a recurring "exceptional" as exceptional because that is what it is labelled.
- Valuing on **EBITDA alone** for a capital-intensive business without checking free cash flow.
- Extrapolating a **low tax rate** that resulted from a one-off credit.
- Making adjustments without disclosing them, so the reader cannot evaluate the judgement.
- Assuming all adjustments are illegitimate — some genuinely do clarify the underlying run-rate, and rejecting them wholesale is as unanalytical as accepting them wholesale.

Interview angle

"A company reports adjusted EPS well above statutory EPS. How do you respond?" Work through it: identify what the adjustments are; test each for whether it is genuinely non-recurring (checking whether similar items appear in prior years); flag adjustments that should not be made — share-based compensation, recurring restructuring, structural forex; check whether the earnings are cash-backed via cumulative CFO versus PAT; normalise the tax rate; then build your own adjusted figure with a transparent bridge from statutory. Close on the principle: the multiple you apply assumes persistence, so the earnings you apply it to must be the persistent ones.

Relative Valuation and Peer-Set Construction

The Problem / Why this matters

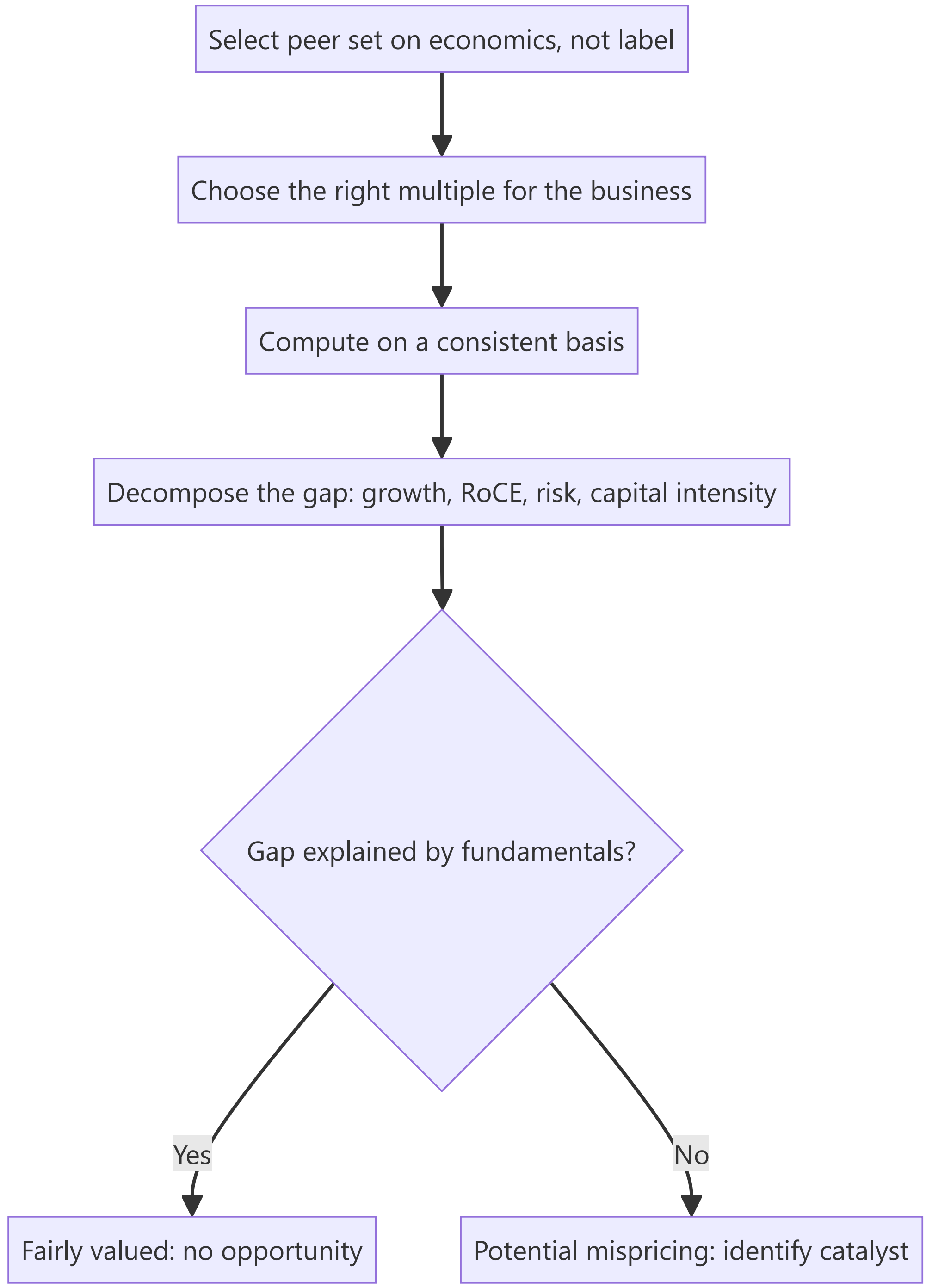
"It trades at 18x versus the sector at 24x, so it's cheap" is the most common sentence in bad equity research. A multiple is meaningless without a defensible peer set and without understanding *why* the gap exists. Most apparent valuation gaps are not mispricings at all — they are the market correctly pricing differences in growth, returns, risk or capital intensity. The analytical work is separating the gaps that are justified from the ones that are not.

Core Idea

Relative valuation compares a company to peers on a standardised multiple. Its validity rests entirely on two things: a **genuinely comparable peer set**, and an explicit account of **what drives any multiple differential** between the company and that set.

Why it works this way

A multiple is a compressed summary of a DCF. P/E embeds growth, risk and payout; EV/EBITDA embeds growth, capital intensity and risk. Two companies deserve the same multiple only if those underlying variables match. When they don't, the multiple gap is *information*, not opportunity — and the analyst's job is to decompose it.



Full technical content

Constructing a defensible peer set

Peers should share **economics**, not merely an industry label. Screen on:

CRITERION	WHY IT MATTERS
Business model	A branded pharma company is not comparable to an API manufacturer
Growth profile	A 20% grower and a 5% grower deserve different multiples
Margin and RoCE	Return profile is the single biggest multiple driver
Capital intensity	Determines how much growth costs
Scale	Larger companies typically command a liquidity/stability premium
Geography and regulation	Different risk premia and tax regimes
Customer concentration / cyclicality	Risk affects the multiple directly

Include global peers when the domestic set is thin, but adjust for country risk premium, tax rate differences, and accounting-standard differences (IND-AS vs IFRS vs US GAAP). A common error is comparing an Indian IT company's P/E directly to a US-listed peer without noting the different growth, currency and tax context.

Peer-set size: 5–10 is usually right. Too few and one outlier dominates; too many and you have diluted the set with companies that aren't truly comparable, which pulls the "sector average" toward meaninglessness.

Always show median alongside mean — a single extreme multiple (often a company with near-zero earnings producing a huge P/E) distorts the mean severely.

Choosing the right multiple

MULTIPLE	USE WHEN	AVOID WHEN
P/E	Stable, profitable, comparable capital structures	Cyclicals at extremes; loss-makers; different leverage
EV/EBITDA	Different leverage; capital-intensive; cyclicals	Very different capex intensity within the set
EV/EBIT	Depreciation policies differ materially	—
EV/Sales	Loss-making or early-stage; margins not normalised	Mature businesses where margins are comparable
P/B	Banks, financials, asset-heavy businesses	Asset-light service businesses
P/B vs RoE	Financials — the correct paired view	—
EV/Capacity (per tonne, per MW, per room)	Commodity/infrastructure with homogeneous assets	Differentiated products
PEG	Comparing across different growth rates	Low-growth or negative-growth companies

Consistency discipline: an equity multiple (P/E, P/B) uses an equity numerator and equity denominator; an enterprise multiple (EV/EBITDA) uses enterprise value over a pre-interest metric. Mixing them — EV/PAT, or P/EBITDA — is a definitional error that produces a meaningless number.

Computing multiples correctly

Enterprise Value = Market cap + Total debt – Cash and cash equivalents + Minority interest + Preference capital. Common errors: forgetting minority interest (which overstates value for companies with large partly-owned subsidiaries), and netting off cash that isn't genuinely surplus.

Which period: trailing (last 12 months, factual but backward-looking) or forward (next 12 months or FY+1/FY+2, more relevant but estimate-dependent). Markets price forward. **The critical discipline is comparing like with like** — a forward P/E for your company against a trailing P/E for peers is a systematic, and very common, error that makes a growing company look artificially cheap.

Diluted share count, including outstanding ESOPs, warrants and convertibles — not the basic count.

Decomposing the multiple gap — the actual analysis

This is what separates real relative valuation from multiple-quoting. Suppose your company trades at 18x forward P/E versus a peer median of 24x. Work through:

1. **Growth.** Is your company's forecast EPS CAGR below the peer median? A company growing 8% versus peers at 16% *should* trade at a discount. Compute PEG for both to normalise.
2. **Returns.** Compare RoCE and RoE. Sustainably lower returns justify a lower multiple — this is the most fundamental driver of multiple differences and the most frequently ignored.
3. **Capital intensity.** Higher capex/sales means less of the earnings converts to distributable cash, justifying a lower multiple on the same earnings.
4. **Risk.** Customer concentration, cyclicality, leverage, regulatory exposure, governance quality — each justifies a discount.
5. **Cash conversion.** Lower CFO/EBITDA means lower-quality earnings, justifying a lower multiple.
6. **Free float and liquidity.** Genuinely affects the multiple, especially for smaller companies.

Only after accounting for these can you claim a residual, unexplained gap — and *that* residual is the potential opportunity.

The historical-band cross-check

A second, independent reference: where does the multiple sit versus the **company's own history**? Compute the 5-year and 10-year average, and the percentile of the current multiple within that range. Then ask the essential question: **has anything structurally changed** to justify a permanent re-rating or de-rating? A stock at the bottom decile of its own historical band is cheap only if its fundamentals are unchanged; if growth has structurally slowed or returns have permanently fallen, the low multiple is correct and the "cheapness" is a value trap.

Common structural traps

- **The cyclical trap** — a cyclical looks cheapest on P/E at the earnings peak and dearest at the trough. Use mid-cycle earnings or EV/EBITDA through the cycle.
- **The value trap** — persistently cheap because the business is in structural decline. The multiple is not a mistake.
- **Averaging away the outliers** — a "sector P/E" including one 200x company is not a benchmark.
- **The conglomerate error** — applying one multiple to a diversified business rather than SOTP.
- **Ignoring accounting differences** — capitalisation policies, lease treatment, and share-based-comp treatment all distort cross-company multiples.

Common mistakes

- Building a peer set from an **industry classification code** rather than from economics.
- Quoting a sector mean without the median, letting one outlier set the benchmark.
- Comparing **forward** multiples for one company against **trailing** for peers.
- Using basic rather than **diluted** share count.

- Forgetting **minority interest** in enterprise value.
- Concluding "cheap versus peers" without decomposing growth, returns and risk.
- Treating a discount to the historical band as opportunity without asking what has structurally changed.

Interview angle

"This stock trades at 18x versus its sector at 24x. Is it cheap?" The expected answer refuses the premise until the work is done: first, is the peer set genuinely comparable on business model, growth, returns and capital intensity? Second, decompose the gap — lower growth, lower RoCE, higher cyclicality, weaker cash conversion, governance discount, lower liquidity all *justify* a discount. Third, compare against the company's own historical band and ask whether anything has structurally changed. Only a residual gap unexplained by fundamentals is an opportunity, and even then it needs a catalyst to close.

Risk Assessment and Position Sizing for Research Analysts

The Problem / Why this matters

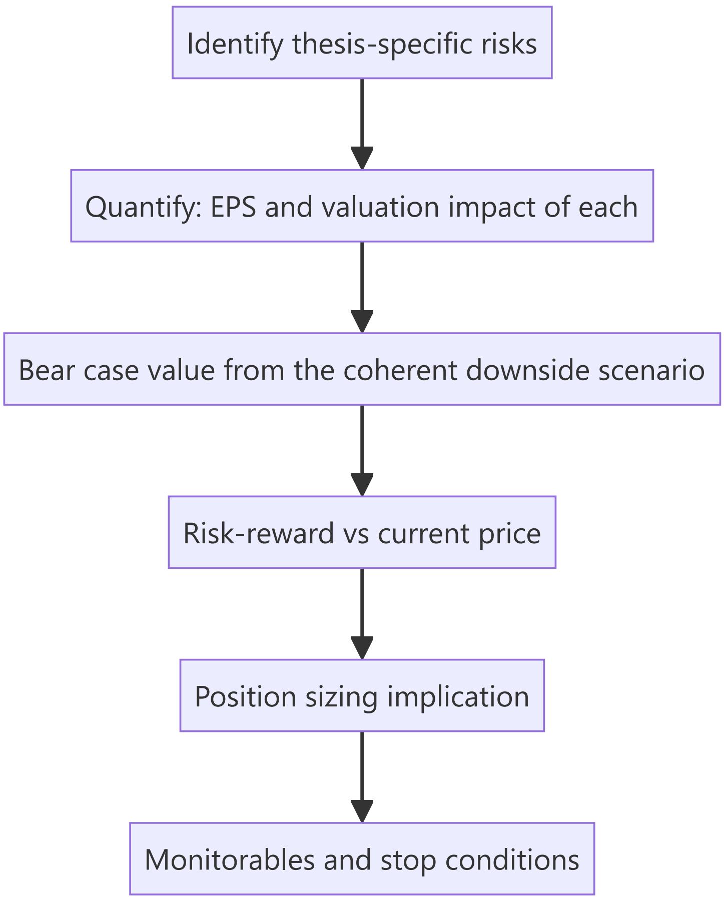
A recommendation without a risk assessment is a half-finished piece of work. Clients do not act on conviction alone — a portfolio manager needs to know what can go wrong, how much they can lose, and therefore how much of the portfolio the idea deserves. An analyst who says "Buy, target ₹1,200" without quantifying downside has given the PM a number but not a decision. This is also the dimension where sell-side and buy-side analysts differ most: the buy side must size, and thinking like they do makes sell-side research far more useful.

Core Idea

Risk work has three layers: **identify** the specific risks to *this* thesis, **quantify** their earnings and valuation impact, and translate that into **risk-reward** that supports a sizing decision. Boilerplate risk sections fail all three.

Why it works this way

Investment outcomes are distributions, not points. The expected return of a position is meaningful only alongside its dispersion — a 20% expected return with a possible 50% drawdown is a different proposition from a 15% expected return with a possible 10% drawdown, and a portfolio manager will size them very differently.



Full technical content

Layer 1 — Identifying the right risks

Generic risk sections list "competition, regulation, currency." Useful risk sections identify what would specifically break **this thesis**. Start from the thesis pillars and invert each one:

If the thesis is *"margin expands from 14% to 18% as the new plant ramps and mix improves"*, the risks are: the ramp is delayed; utilisation disappoints; the mix shift doesn't happen because the premium segment slows; a competitor adds capacity simultaneously and pricing breaks.

A taxonomy to work through systematically:

CATEGORY	EXAMPLES
Business	Demand slowdown, market-share loss, customer concentration, key-person dependence
Operational	Project delay, plant shutdown, supply-chain disruption, quality/recall
Financial	Refinancing risk, covenant breach, forex exposure, receivable default
Regulatory	Price control, licence/approval risk, tax change, environmental compliance
Governance	Related-party dealings, promoter pledge, auditor issues, minority-shareholder treatment
Valuation	Multiple de-rating independent of earnings; sector-wide re-rating reversal
Thesis-specific	Whatever your particular argument requires to be true

Layer 2 — Quantifying

Every material risk should carry a number. The professional format:

> *"A 200bp shortfall versus our 18% terminal EBITDA margin assumption reduces our FY27 EPS by 11% and our DCF value by 14%, taking fair value to ₹820."*

Build a **risk sensitivity table** in the note:

RISK	TRIGGER	EPS IMPACT	VALUE IMPACT
Plant ramp delayed 2 quarters	Commissioning slips	−7% FY26	−5%
Terminal margin 16% not 18%	Mix shift fails	−11% FY27	−14%
Key customer (18% of revenue) loss	Contract not renewed	−15%	−17%
INR appreciates 5%	Currency move	−6%	−6%

This is directly usable by a PM in a way that prose is not.

Layer 3 — Risk-reward and sizing

From the scenario work: bull, base and bear values, and therefore upside/downside from the current price.

Risk-reward = upside to bull ÷ downside to bear.

Conventional thresholds: a Buy generally requires **at least 2:1, often 3:1**. This discipline forces a rating change when the price moves even if the fundamental view has not — a stock that rallies into the target has a deteriorating risk-reward and eventually stops being a Buy regardless of how much you like the business.

Sizing frameworks a PM applies (worth understanding even on the sell side):

- **Conviction-weighted** — larger positions in higher-conviction, better risk-reward ideas.
- **Volatility-adjusted** — size inversely to expected volatility so each position contributes comparable risk to the portfolio.
- **Fixed fractional risk** — risk a fixed percentage of the portfolio per idea, so position size = (portfolio risk budget) ÷ (distance to the stop/bear case). A wider bear-case gap mechanically means a smaller position.
- **Correlation-aware** — an idea highly correlated with existing holdings adds less diversification and deserves less size even at equal conviction.

The link that matters: **the wider your bear case, the smaller the position that same conviction justifies.** This is why quantifying downside is not pessimism — it directly determines how much capital the idea should receive.

Monitorables and falsification

The most valuable closing element of a note. Specify:

- **What to watch** — the two or three data points that will confirm or break the thesis (monthly volumes, quarterly margin, the plant commissioning announcement, a specific regulatory decision).
- **What would prove you wrong** — stated in advance. An analyst who has pre-committed to a falsification condition is dramatically more credible than one who rationalises after the fact.
- **When to revisit** — the specific event or date at which the thesis gets re-tested.

Portfolio-level risks the analyst should flag

Even a single-stock note should note where the idea sits in a portfolio context: sector concentration if the PM already owns peers, factor exposure (is this simply a leveraged bet on a commodity price or the rupee?), liquidity (days-to-exit at a reasonable share of ADV — critical for mid- and small-caps), and event risk (an upcoming binary regulatory or legal decision).

Liquidity deserves specific mention: a mid-cap with ₹8 crore average daily volume cannot absorb a ₹200 crore position without material impact, and a genuinely correct recommendation that cannot be implemented at size is of limited use to a large fund. Stating days-to-build and days-to-exit at, say, 20% of ADV is a professional courtesy that many notes omit.

Common mistakes

- **Boilerplate risk sections** that would apply to any company in any sector.
- Risks listed without **quantification** — no EPS or valuation impact.
- Bear cases constructed by mechanically cutting the base case rather than from a coherent narrative, and holding the **multiple constant** while cutting earnings (which understates real downside).
- No stated **falsification condition**, so the thesis can never be wrong.
- Maintaining a Buy after the stock has rallied into the target and risk-reward has collapsed.
- Ignoring **liquidity**, recommending a position size the stock cannot absorb.
- Treating risk analysis as a compliance section at the back rather than as part of the investment case.

Interview angle

"You're bullish on a stock. What could go wrong?" A weak answer lists generic risks. A strong answer inverts the thesis pillars specifically — for each thing that must be true for the call to work, state what happens if it isn't, and quantify the EPS and value impact. Then give the bear-case value, the resulting risk-reward from the current price, the specific monitorables that would tell you early that the thesis is breaking, and what would make you change the rating. Naming the falsification condition unprompted is what marks a senior candidate.

Special Situations and Event-Driven Analysis

The Problem / Why this matters

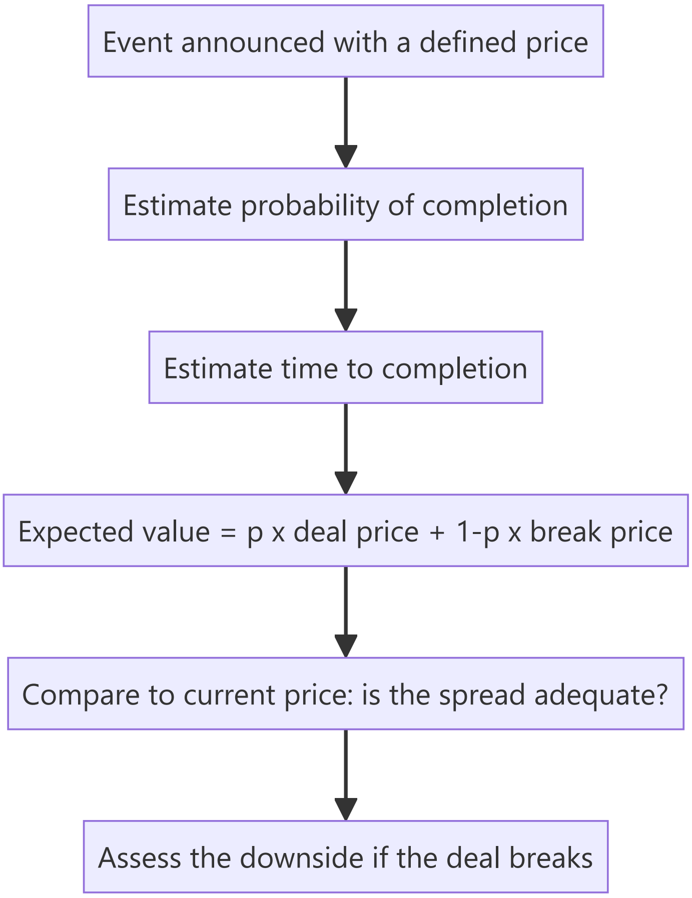
Most equity research values a company as an ongoing business. But a meaningful share of returns comes from **corporate events** — mergers, demergers, open offers, delistings, buybacks, rights issues, insolvency resolutions — where the value driver is not next year's earnings but the mechanics, probability and timing of a specific transaction. This requires a genuinely different analytical toolkit, and it is where an analyst who understands corporate-action mechanics precisely can add value that a pure fundamental modeller cannot.

Core Idea

In a special situation, price is driven by **the expected value of a defined outcome** — the deal price, weighted by the probability of completion, discounted for the time to completion — rather than by an ongoing DCF. The analysis is therefore about probability, timeline, and downside if the event fails.

Why it works this way

Once a transaction is announced with a defined price, the stock's future is largely determined by whether that transaction completes. The market prices the probability-weighted outcome, and the residual spread between the current price and the deal price compensates for completion risk and time. Analysing the situation means analysing that risk, not the company's long-term prospects.



Full technical content

The core arithmetic — merger arbitrage

For an announced acquisition at a fixed cash price:

Gross spread = (Deal price − Current price) ÷ Current price **Annualised return** = Gross spread × (365 ÷ days to expected close)

But the honest calculation must include the break scenario:

Expected value = p × Deal price + (1 − p) × Estimated break price

The **break price** is typically the undisturbed price before the announcement, often adjusted downward — a failed deal frequently leaves the stock below where it started, because the failure itself signals something (a regulator's objection, due-diligence findings, or a deteriorating business).

Worked example: stock at ₹470, cash offer at ₹500, expected close in 5 months, pre-announcement price ₹400. - Gross spread = 6.4%; annualised ≈ 15.3% - If p(completion) = 85%: EV = 0.85 × 500 + 0.15 × 400 = 425 + 60 = **₹485** - Current price ₹470 is below EV ₹485 — a positive, but the downside on a break is −15%, so risk-reward is roughly 1:5 against on the raw numbers. This spread is only attractive if you assess completion probability meaningfully above 85%.

That inversion is the entire point of the arithmetic: merger arb is a **negative-skew** strategy — many small gains, occasional large losses — so the probability estimate has to be genuinely rigorous, not a comfortable assumption.

Assessing completion probability

FACTOR	RAISES PROBABILITY	LOWERS PROBABILITY
Financing	Cash on hand, committed facility	Contingent on raising finance
Regulatory	No overlap, small share	Antitrust/CCI concerns, sectoral caps, foreign-investment approval
Shareholder approval	Acquirer already holds a large stake; board unanimous	Contested; large dissenting holders
Due diligence	Completed	Still open, or MAC clause broad
Strategic logic	Clear synergy, adjacent business	Diversification with weak rationale
Acquirer's position	Strong balance sheet, prior deal record	Stretched leverage, hostile approach
Board stance	Recommended	Hostile / rejected

Watch also for **competing bids**, which change the calculation entirely — the downside becomes much shallower and the upside opens up.

Open offers under the takeover framework

When an acquirer crosses the substantial-acquisition threshold, a **mandatory open offer** follows at a regulator-prescribed minimum price. The analytical work:

- Compute the **open-offer price** from the prescribed formula (highest of: negotiated price, volume-weighted average of the acquirer's own purchases over a defined lookback, and market VWAP over a defined period). This is a calculable, regulator-anchored floor.

- Estimate the **acceptance ratio** — the offer is for a fixed quantity, so if tendering exceeds it, shareholders receive only a proportion. The stock typically trades below the offer price by roughly the amount reflecting the expected acceptance ratio.
- The residual (unaccepted) shares return to the shareholder at the post-offer market price — so the blended outcome is (accepted portion × offer price) + (unaccepted portion × expected post-offer price).

Demerger and spin-off situations

Value here comes from the market re-rating the parts separately (drawing on the SOTP material). The analysis: - SOTP value of the parts versus the current consolidated market price. - The **when-issued** market, if operating, as real-time price discovery of the implied split. - Post-listing technical dynamics: forced selling by index funds that cannot hold the demerged entity, and by shareholders who wanted only the parent business — which often creates temporary, mechanical pressure unrelated to fundamentals and is a recurring source of opportunity.

Delisting situations

Under reverse book-building, the exit price is discovered from shareholder bids and the acquirer may **accept or reject** the discovered price. The stock trading below the floor price signals genuine market scepticism about completion. The asymmetry to note: a successful delisting caps the upside at the accepted price, while a failed one returns the stock to fundamental value, which may be materially lower after the speculative bid premium unwinds.

Insolvency and restructuring

Once a company enters insolvency proceedings, ordinary technical and fundamental frameworks largely stop applying. The critical point for equity: in the resolution waterfall, **equity holders rank last**, and in most resolved cases existing equity is heavily diluted or extinguished entirely. A stock trading at a small fraction of its pre-admission price is not necessarily cheap — the residual equity claim may genuinely be near zero, and the trading price often reflects speculation rather than any recoverable value.

Index inclusion and exclusion events

Covered mechanically elsewhere; from an event-driven view the key points are that the flow is **predictable and dated** (index funds must trade at the effective date, concentrated in the closing window), and therefore substantially **anticipated and pre-priced** — meaning the return is usually captured before the announcement, not after.

What makes event-driven analysis distinctive

- **Defined outcomes and timelines**, unlike open-ended fundamental theses.
- **Probability is the main variable**, not growth or margin.
- **Negative skew** in most deal situations — size accordingly.
- **Legal and regulatory documents are primary sources** — the scheme of arrangement, the offer document, the merger agreement's MAC and break-fee clauses. Reading the actual document is the work; most of the edge is there.
- **Time decay matters** — a spread that doesn't close erodes annualised returns even if the deal eventually completes.

Common mistakes

- Quoting the **gross spread** without the probability-weighted expected value or the break-price downside.

- Assuming a deal announced is a deal completed.
- Ignoring the **acceptance ratio** in an open offer and modelling full tendering at the offer price.
- Treating a stock in insolvency as cheap because it has fallen 95% — equity ranks last.
- Not reading the actual transaction documents, where the conditions precedent and MAC clauses live.
- Sizing a merger-arb position as if it were symmetric when the payoff is sharply negative-skewed.

Interview angle

"A company announces an all-cash acquisition of a target at ₹500. The target trades at ₹470. What do you do?" Structure: compute the gross spread and annualise it over the expected close; estimate completion probability from financing certainty, regulatory overlap, board stance and shareholder approval; estimate the break price (usually near or below the undisturbed pre-announcement price); compute the probability-weighted expected value and compare to ₹470; then assess the asymmetry — the downside on a break is typically several times the upside on completion, so the position only makes sense with high completion confidence and modest size. Mentioning that the payoff is negative-skewed is what signals genuine familiarity.

The Regulatory Framework for Research Analysts

The Problem / Why this matters

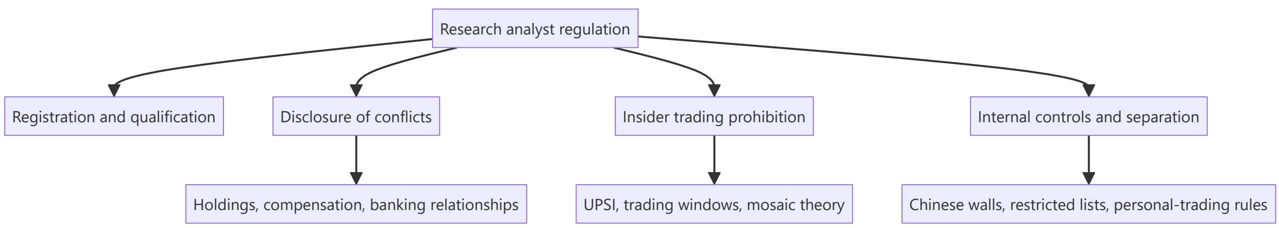
Equity research is a regulated activity. An analyst publishing recommendations to clients operates under a specific legal framework governing registration, disclosure, conflicts of interest, and the handling of price-sensitive information. Breaching it carries personal consequences — penalties, debarment, and in serious cases criminal liability — quite apart from the reputational damage to the firm. Interviewers ask about this because a candidate who cannot articulate the disclosure obligations is a compliance liability regardless of their analytical skill.

Core Idea

The regulatory architecture exists to address one structural problem: research is produced by firms with **many other commercial relationships** with the companies being researched, and consumed by clients who cannot see those relationships. Regulation forces disclosure of conflicts and prohibits trading on information the market does not have.

Why it works this way

Research recommendations move prices, so they can be used to profit at clients' expense — by front-running the recommendation, by publishing favourably to win investment-banking business, or by selectively disclosing to preferred clients first. Each of these is addressed by a specific rule, and understanding *why* each rule exists makes the rules themselves easy to remember.



Full technical content

Registration and qualification

In India, an individual or entity providing research recommendations to clients must be registered with SEBI as a **Research Analyst** under the Research Analyst Regulations. Requirements include prescribed **qualifications** (a relevant postgraduate degree or professional qualification), mandatory **certification** — in practice the **NISM Series-XV: Research Analyst** certification — minimum net worth, and compliance infrastructure.

The registration requirement applies to the act of making a **buy/sell/hold recommendation or target price to clients**, which is why the boundary between "research" and "general market commentary" matters, and why unregistered stock tipping on social media falls foul of the framework.

Mandatory disclosures in a research report

Every published report must disclose:

DISCLOSURE	PURPOSE
Analyst's own holdings in the subject company	The analyst may benefit from the recommendation
Holdings of the analyst's relatives / associates	Indirect benefit
The firm's holdings (usually above a threshold, e.g. 1%)	Firm-level interest
Investment-banking relationship — past 12 months and anticipated	The strongest conflict: the firm may be paid by the company
Compensation received from the subject company for any service	Any commercial relationship
Whether the analyst or firm acts as market maker	Trading interest
Whether the analyst served as an officer/director/employee of the subject	Personal entanglement
The rating distribution across the firm's coverage	Reveals systemic optimism — if 95% are Buys, ratings are uninformative
Definitions of rating terms and the target-price horizon	Prevents ambiguity
Whether the report was shared with the company before publication	Should be for factual accuracy only, never for the view

The insider-trading prohibition

Governed in India by the **Prohibition of Insider Trading (PIT) Regulations**. The central concept is **Unpublished Price-Sensitive Information (UPSI)** — information relating to a company, not generally available, which upon becoming available would materially affect the price.

Typical UPSI: financial results before publication, dividend declarations, M&A, capital restructuring, changes in key management, and material regulatory or legal outcomes.

Core prohibitions: trading while in possession of UPSI; communicating UPSI to anyone except for legitimate purposes; and procuring UPSI.

The mosaic theory — the analyst's protection and the professional standard: assembling many pieces of individually non-material public and non-confidential information into a material conclusion is **legitimate research**, even where the conclusion itself is price-sensitive. What is prohibited is receiving one piece of material non-public information from someone with a duty of confidentiality. This is the principle that makes channel checks and expert calls lawful — and it defines exactly where they stop.

Practical obligations for an analyst: never solicit pre-release results or guidance; stop a conversation immediately if a source begins disclosing UPSI and escalate to compliance; use expert networks only through approved channels with the appropriate attestations; and maintain records demonstrating the mosaic — that is, evidence of how a conclusion was assembled from permissible sources.

Internal controls at the firm level

- **Chinese walls / information barriers** — physical and systems separation between research and investment banking, so the deal side cannot influence or preview the research side.
- **Restricted and watch lists** — securities on which the firm has material non-public information, where research publication and personal trading are curtailed.
- **Quiet periods** — restrictions on publishing research around a transaction the firm is involved in (e.g. an IPO the firm is underwriting).

- **Personal-trading rules** — pre-clearance requirements, minimum holding periods, and a prohibition on trading ahead of one's own published recommendation.
- **Distribution fairness** — research must be distributed to entitled clients simultaneously; selectively giving a favoured client an advance look is a serious breach.
- **Supervisory review** — reports reviewed by a supervisory analyst and by compliance before release.

Conflicts of interest specific to sell-side research

The structural tension: research is a cost centre funded by trading commissions and, historically, by proximity to investment banking. This creates pressure toward optimism — negative research risks losing corporate access, banking mandates, and management goodwill. The regulatory response is disclosure plus separation, but analysts should understand that **the rating distribution disclosure is the clearest evidence of whether a firm has resisted that pressure**. A research house whose ratings are 95% Buy has, in effect, informed clients that its ratings carry little information.

MiFID II in Europe changed this economics by requiring research to be **unbundled** — paid for explicitly rather than through trading commissions — which reduced research budgets industry-wide but strengthened the independence of what remains. Analysts working with global clients should understand the concept.

Analyst behaviour and professional standards

Beyond the regulations, professional bodies (notably the CFA Institute's Code and Standards) impose obligations that frequently arise in interviews: independence and objectivity, reasonable basis for recommendations, fair dealing among clients, preservation of confidentiality, and — importantly — a duty to distinguish clearly between **fact, estimate and opinion** in published work.

Common mistakes

- Assuming disclosure obligations apply only to the firm and not to the **individual analyst's** own holdings.
- Believing that if information came from a "third party" it cannot be UPSI — the test is the nature of the information and the source's duty of confidentiality, not the number of hops.
- Continuing an expert call after the expert begins describing their current employer's unpublished performance.
- Sharing a draft report's **conclusions or target price** with company management before publication (factual verification only is acceptable; the view is not).
- Giving a favoured client advance notice of a rating change.
- Trading personally ahead of publishing one's own recommendation.
- Treating compliance as an obstacle to research rather than as the framework that makes published research credible.

Interview angle

"What must a research analyst disclose, and why does it matter?" Cover the specific items — personal and firm holdings, investment-banking relationships in the past 12 months and anticipated, any compensation from the company, market-making status, and the firm's rating distribution — then give the reason that shows genuine understanding: disclosure exists so clients can correctly *weight* the recommendation knowing the incentives behind it. Credibility is the only asset research has, and undisclosed conflicts destroy it. Adding the mosaic-theory distinction — how legitimate primary research differs from receiving UPSI — signals that you understand where the line actually sits in daily practice.

DCF Deep Dive — WACC, Terminal Value and the Assumptions That Matter

The Problem / Why this matters

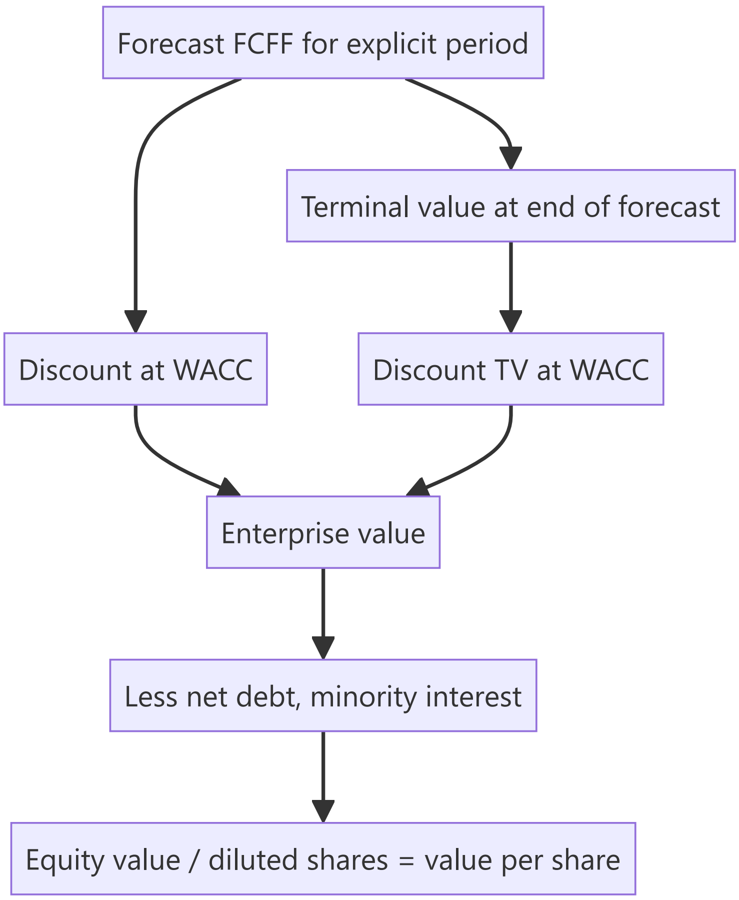
Most of a DCF's output is determined by two inputs that receive the least scrutiny: the discount rate and the terminal value. In a typical model, **60–80% of the computed value sits in the terminal value** — meaning the explicit five-year forecast an analyst spends weeks building often drives less than a third of the answer. Understanding where the value actually comes from, and how to defend those inputs, is what separates a DCF that survives challenge from one that collapses under the first question.

Core Idea

A DCF discounts projected free cash flows at a rate reflecting their risk. Its integrity depends on three things being internally consistent: the **cash flow definition**, the **discount rate applied to it**, and the **terminal value method** — and on the terminal assumptions being economically defensible rather than reverse-engineered.

Why it works this way

Value is the present value of cash available to capital providers. Because a going concern generates cash indefinitely, and no analyst can forecast indefinitely, the model splits into an explicit forecast period and a terminal value capturing everything after. The discount rate converts future rupees into present ones at a rate reflecting the risk of not receiving them.



Full technical content

Getting the consistency right

The single most common structural error is mismatching cash flow and discount rate:

CASH FLOW	DISCOUNT AT	GIVES	THEN
FCFF (free cash flow to firm — pre-financing)	**WACC**	Enterprise value	Subtract net debt to get equity
FCFE (free cash flow to equity — post-interest, post-debt-flows)	**Cost of equity**	Equity value directly	No further subtraction

Discounting FCFF at the cost of equity, or FCFE at WACC, is a definitional error that produces a meaningless number. Most equity research uses the FCFF/WACC route.

FCFF build:

```
EBIT
× (1 − tax rate)              = NOPAT
+ Depreciation & amortisation
− Capital expenditure
− Increase in working capital
= FCFF
```

The discount rate — WACC

WACC = (E/V × Cost of equity) + (D/V × Cost of debt × (1 − tax rate))

Cost of equity via CAPM: **Rf + β × ERP** (+ any size or specific-risk premium).

Each input carries judgement:

- **Risk-free rate (Rf)** — the 10-year government bond yield, matched to the currency of the cash flows. Use a **current** yield, but be aware that using a cyclically depressed yield embeds an aggressive assumption into a perpetual valuation; some practitioners normalise.
- **Equity risk premium (ERP)** — the excess return demanded over the risk-free rate. For India typically in the region of 6–8%, drawn from published country-risk-premium work. State your source; this is a genuinely contested number and a 100bp change moves value materially.
- **Beta (β)** — sensitivity to the market. Practical guidance: raw regression betas are noisy and period-dependent, so use an **adjusted beta** (the common Blume adjustment shrinks toward 1: `0.67 × raw + 0.33 × 1`) or, better, a **peer-median unlevered beta re-levered to the target capital structure**. That approach removes single-stock estimation noise and is more defensible under challenge.

Unlever: $\beta u = \beta L \div [1 + (1 - t) \times D/E]$ → take the peer median → *re-lever* at the subject's capital structure.

- **Cost of debt** — the company's actual marginal borrowing cost (from the interest-rate disclosure or its credit rating's spread over the risk-free rate), not the historical average rate on legacy debt. Applied after tax, because interest is deductible.
- **Capital-structure weights** — use **market values**, not book, and use a **target/sustainable** structure rather than a temporarily distorted current one.

Terminal value — where most of the value lives

Two accepted methods, and best practice is to compute both and cross-check.

1. Perpetuity growth (Gordon) method:

TV = $FCFF_{(n+1)}$ ÷ (WACC – g)

The critical discipline on **g**: the terminal growth rate is a *perpetual* rate. A company cannot grow faster than the economy forever, so **g must not exceed long-run nominal GDP growth** — for India, realistically in the 4–6% nominal range, and for a global business closer to 2–3%. A model using g = 8% is asserting the company eventually becomes the entire economy. This is the single most abused input in equity research.

Note the extreme sensitivity: with WACC 11% and g 4%, the denominator is 7%. Moving g to 5% makes it 6% — increasing TV by roughly 17% from a 100bp change in one assumption. This is precisely why the WACC-versus-g sensitivity table is mandatory rather than decorative.

2. Exit multiple method:

TV = Terminal-year EBITDA × Exit EV/EBITDA multiple

The exit multiple should reflect a **mature-business** multiple — typically at or below the current sector multiple, because a company growing more slowly at the end of the forecast should command a lower multiple than it does today. Using today's multiple for a business that will be mature and slower-growing is a common way of quietly inflating value.

The cross-check that professionalises the model: compute TV both ways and back out the implied other variable. If your exit multiple implies a perpetual growth rate of 9%, the multiple is too high. If your perpetuity growth implies an exit multiple of 4x for a quality business, it is too low. Disclosing both, and the implied cross-check, is standard in good research.

Terminal-year normalisation

The terminal year must represent a **steady state**, not the last year of an unusual forecast: - **Capex should approximate depreciation** in perpetuity — a business cannot grow forever while spending less on assets than it consumes. A terminal year with capex well below depreciation overstates FCFF permanently. - **Working capital change** should reflect only the growth rate, not a one-off release. - **Margins** should be at a sustainable, competitively-plausible level — not a cyclical peak. - **Tax rate** at the statutory/sustainable rate, not a temporarily concessional one.

Mid-year convention

Cash flows arrive through the year, not on the final day. Discounting at t = 0.5, 1.5, 2.5 rather than 1, 2, 3 raises value by roughly (1 + WACC)^0.5 — typically 4–6%. Either convention is acceptable; applying it inconsistently between the explicit period and the terminal value is not.

From enterprise value to value per share

```
Enterprise value
− Net debt (debt − surplus cash)
− Minority interest (market or book value of what you don't own)
− Underfunded pension / other debt-like items
+ Value of non-operating assets (surplus land, investments, associates)
= Equity value
÷ Diluted shares (incl. ESOPs, warrants, convertibles)
= Value per share
```

Honest use of the output

A DCF produces a *range*, not a point. Best practice is to present the sensitivity table (WACC × g), state the two or three assumptions carrying the value, and triangulate the DCF output against relative valuation rather than presenting it as an independently precise answer. Where a DCF and comps diverge materially, that divergence is itself the finding worth investigating — usually it means the market is assuming something structurally different about growth or returns than your model is.

Common mistakes

- **Mismatching** FCFF with cost of equity or FCFE with WACC.
- Terminal **g above long-run nominal GDP growth** — economically impossible in perpetuity.
- Terminal-year **capex below depreciation**, permanently overstating cash flow.
- Using an exit multiple equal to today's multiple for a business that will then be mature.
- Using **book-value** capital-structure weights instead of market values.
- Using a **raw regression beta** without adjustment or peer cross-check.
- Forgetting **minority interest** or using basic rather than diluted shares.
- Reverse-engineering WACC or g to reach a target price that was decided first — the most damaging error, because it makes the entire model an exercise in justification rather than analysis.

Interview angle

"Walk me through a DCF, and tell me which assumption matters most." Cover the mechanics — forecast FCFF, discount at WACC, terminal value, bridge to equity value per share — then demonstrate seniority by going to where the value actually sits: 60–80% is typically in the terminal value, so terminal growth and the WACC spread over it dominate, and terminal g must be capped at long-run nominal GDP growth. Add that you'd compute terminal value both ways and check the implied cross-consistency, and that you'd present a WACC-versus-g sensitivity table rather than a single number, because a defensible range of inputs can easily span ±30% of value.

Writing for Impact — Research Communication Craft

The Problem / Why this matters

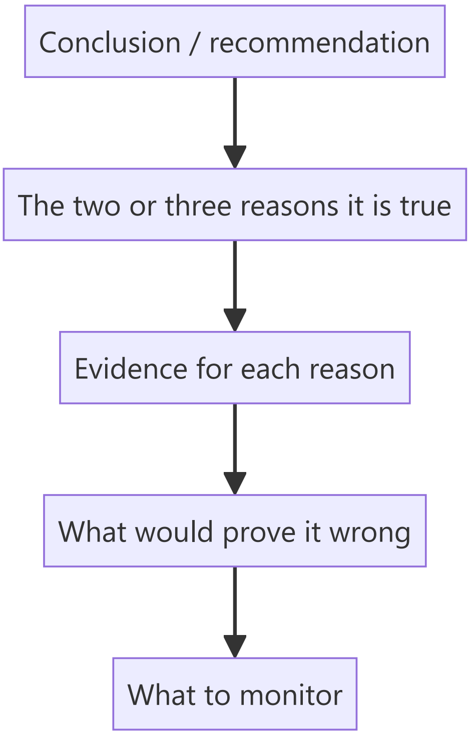
Research that is not read has no value, and research that is read but misunderstood has negative value. Institutional clients receive hundreds of notes a week and read a fraction of them. The analyst's analytical work is only half the job; converting it into something a portfolio manager can absorb in ninety seconds and act on is the other half — and it is the half most junior analysts underestimate.

Core Idea

Professional research writing is **conclusion-first, evidence-second**, quantified wherever possible, and explicit about the distinction between fact, estimate and opinion. Its structure serves a reader who may stop after the first paragraph.

Why it works this way

A portfolio manager's scarcest resource is attention. Academic writing builds to a conclusion because the reader is committed to finishing; research writing leads with the conclusion because the reader is not. Every structural convention in research writing follows from that single asymmetry.



Full technical content

The BLUF principle — bottom line up front

Open with the conclusion and the action, not with background. Compare:

> ❌ *"The Indian speciality chemicals sector has seen significant capacity additions over the past three years, driven by the China+1 supply-chain diversification theme. Against this backdrop, Company X has been*

> *expanding its Dahej facility..."*
>
> ✅ *"We initiate on Company X with a Buy and a ₹1,150 target (34% upside). We are 14% above FY27 consensus EPS because we believe the market is under-modelling the Dahej ramp: our checks indicate commissioning is running a quarter ahead of guidance."*

The second version tells a PM in one sentence what the call is, what makes it differentiated, and what evidence supports it. Everything after that is elaboration for those who want it.

The inverted pyramid

Structure information in descending order of importance, because readers stop at different depths:

1. **Recommendation, target, upside, and the differentiated point** (everyone reads)
2. **The two or three thesis pillars** with the key number under each (most read)
3. **Evidence and analysis** supporting each pillar (interested readers)
4. **Model detail, methodology, appendices** (the few who dig)

A note that buries its differentiated insight on page 30 has structurally failed regardless of how good that insight is.

Quantify everything

Vague language is the most common weakness in junior research. Every qualitative claim should carry a number:

WEAK	STRONG
"Margins should improve"	"We model EBITDA margin at 18.2% in FY27 vs 14.6% in FY25, driven 250bp by mix and 110bp by operating leverage"
"The company faces competitive pressure"	"Two competitors are adding 400kt of capacity by FY27, ~18% of current industry supply; we model realisations down 6%"
"Valuations are attractive"	"At 14x FY27E EPS versus a 5-year average of 21x and peers at 19x, with unchanged RoCE"
"Management is confident"	"Management guided to 15–17% revenue growth, up from 12–14% last quarter"

Distinguish fact, estimate and opinion

A professional standard and, in most codes of conduct, an explicit obligation. The reader must always know which is which:

- **Fact:** "Revenue grew 18.4% YoY in Q2FY26." (reported)
- **Estimate:** "We forecast FY27 revenue of ₹4,820cr, 14% above consensus."
- **Opinion:** "We believe the market is underestimating the durability of the pricing gain."

Blurring these — presenting an estimate in the grammar of a fact — is the most damaging habit in research writing, because it destroys the reader's ability to calibrate how much of what they are reading is verified.

Note types and their conventions

TYPE	LENGTH	PURPOSE	TURNAROUND
Initiation	40–100 pp	Establish the full investment case	Weeks
Results update	2–6 pp	Beat/miss, what changed, revised estimates	Same day
Event note	1–3 pp	Reaction to a specific announcement	Hours
Thematic / sector	15–40 pp	A cross-company idea or structural shift	Weeks
Model update	1–2 pp	Estimate revision without a view change	Same day
Flash / first take	<1 pp	Immediate reaction, pre-full-analysis	Minutes

Each has an expected shape, and violating it wastes the reader's time — a six-page results update that buries the EPS revision is a failure of format, not just of style.

Charts and tables

- **One message per chart.** If a chart needs a paragraph to explain, it is doing too much.
- **Title the chart with the conclusion**, not the contents: "Margin has expanded 400bp as mix shifted to premium" beats "EBITDA margin trend."
- Always label **axes, units and source**, and state clearly whether figures are actuals or estimates.
- Tables should have the **comparison built in** — YoY, versus consensus, versus your prior estimate — because a reader should not have to compute a delta themselves.

Language discipline

- **Active voice, short sentences.** "We cut FY27 EPS 8%" not "FY27 EPS estimates have been revised downwards by 8%."
- **Avoid hedging stacks.** "We believe it is possible that margins may potentially improve somewhat" communicates nothing. Take a position or say you are uncertain and why.
- **Define jargon on first use**, especially sector-specific terms — your reader may be a generalist PM.
- **Consistent units and periods** throughout. Mixing ₹cr and ₹mn, or FY and CY, within one note is a common and confusing error.
- **Say what changed.** In any update, the reader's first question is "what's different from last time?" Answer it in the first line.

The rating-change note

The highest-stakes writing an analyst does. It must explicitly address: what changed (fundamentals, price, or your view), why you were previously wrong if you were, the revised numbers, and what would need to happen for you to change back. Analysts who acknowledge an error plainly build far more credibility than those who present a downgrade as though it were always foreseen — and buy-side readers notice which kind of analyst they are dealing with.

Common mistakes

- Burying the conclusion under background context.
- Qualitative claims with **no numbers attached**.
- Failing to separate **fact from estimate from opinion**.
- Charts titled with contents rather than conclusions.

- Not answering "**what changed?**" in an update note's opening line.
- Hedging so heavily that no view is discernible.
- Writing for another analyst rather than for a **generalist portfolio manager**.
- Length as a proxy for effort — a 60-page note with no differentiated view is worse than a 2-page note with one.

Interview angle

"How would you structure a research note?" Answer with the reader in mind: conclusion first — rating, target, upside, and the one differentiated point in the opening lines; then the two or three thesis pillars each with a supporting number; then evidence; then risks with quantified impact; then monitorables and what would falsify the view. Add the professional standards: quantify every claim, keep fact/estimate/opinion clearly separated, and in an update lead with what changed. The underlying principle to state explicitly is that a PM may only read the first paragraph, so the first paragraph must contain the entire investable idea.

Macro Linkages for Equity Analysts

The Problem / Why this matters

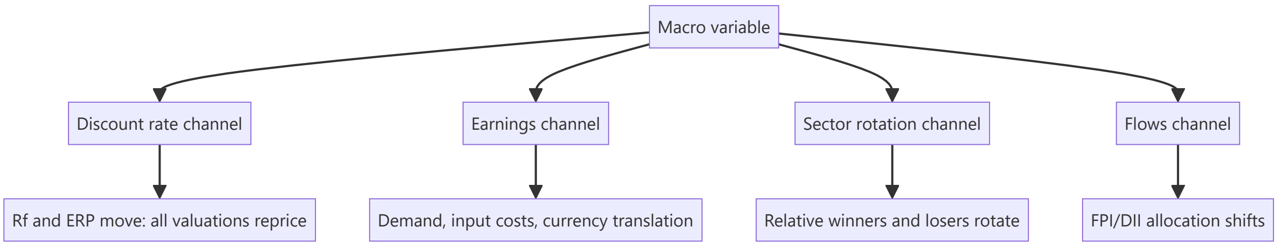
A bottom-up analyst can build a flawless company model and still be wrong because rates rose, the rupee moved, or commodity prices reversed. Macro variables enter equity valuation through specific, traceable channels — the discount rate, input costs, demand, and translation effects — and an analyst who cannot articulate those channels will be blindsided by moves that were, in principle, analysable. "How do rising interest rates affect your stock?" is a routine interview question that separates candidates who understand transmission from those who have memorised a directional rule.

Core Idea

Macro affects equities through **four transmission channels**: the discount rate, corporate earnings, sector rotation, and flows. Every macro variable acts through one or more of these, and the analyst's job is to identify which channel dominates for *their* specific company.

Why it works this way

A stock's value is discounted future cash flows. Macro can move the discount rate (rates, risk premium), the cash flows (demand, costs, currency), or the relative attractiveness of equities versus other assets (flows). Because different companies have different sensitivities to each channel, the same macro event produces genuinely different outcomes across a coverage universe.



Full technical content

Interest rates

Channel 1 — the discount rate. A higher risk-free rate raises WACC, mechanically reducing present value. Critically, this hits **long-duration equities hardest** — companies whose value sits mostly in distant cash flows (high-growth, low current earnings) fall more than mature cash-generative businesses for the same rate move. This is the equity analogue of bond duration and explains why growth stocks de-rate sharply in rate-rising cycles while value/cyclical stocks hold up better.

Channel 2 — earnings. Higher rates raise interest cost for leveraged companies (direct P&L hit), and suppress demand in rate-sensitive sectors — housing, autos, consumer durables, anything bought on credit.

Channel 3 — sector rotation. Rate rises are generally positive for **banks** (asset repricing typically faster than liability repricing, so NIM expands in the near term) and negative for rate-sensitive consumption and for high-duration growth names.

The analyst's task: compute your company's actual sensitivity. Quantify: what does a 100bp WACC increase do to your DCF value? What does 100bp on the borrowing cost do to EPS given the floating-rate debt balance? Those two numbers are the answer to the interview question.

Currency (USD/INR)

Direction of impact depends entirely on where the company sits in the trade flow:

COMPANY TYPE	RUPEE DEPRECIATES	MECHANISM
IT services, pharma exporters	**Positive**	Revenue in USD, costs in INR — margin expands
Oil marketing, importers of raw materials	**Negative**	Input costs rise in INR terms
Companies with USD debt	**Negative**	Translation loss and higher servicing cost in INR
Domestic-only businesses	**Mostly neutral directly**	Indirect via inflation and rates

Quantify with an explicit sensitivity: *"every 1% INR depreciation adds ~25bp to EBIT margin"* for a typical Indian IT services company. Note also the distinction between **transaction exposure** (cash flows), **translation exposure** (reporting of foreign subsidiaries), and **hedging** — many exporters hedge 6–12 months forward, which delays rather than eliminates the impact and means a currency move shows up in earnings with a lag.

Inflation and commodity prices

- **Input-cost inflation** compresses gross margin unless passed through. The key analytical question is **pricing power** — can the company raise prices without losing volume? Test it empirically: look at what happened in the last input-cost spike.
- **Pass-through lag** — even companies with pricing power raise prices with a delay, so margins compress temporarily then recover. Distinguishing a timing effect from a structural one is often the whole call.
- **Commodity producers** benefit from the same move that hurts commodity consumers — within one coverage universe, a steel price rise is positive for steel producers and negative for autos and appliances.
- **Inflation also raises the risk-free rate**, so it hits through the discount-rate channel simultaneously.

GDP growth and the demand channel

Map your company's revenue to its actual macro driver rather than headline GDP: consumer staples track nominal consumption and rural income; autos track disposable income and credit availability; capital goods track the private capex cycle and government infrastructure spend; banks track credit growth, itself roughly a multiple of nominal GDP growth.

Beta to the cycle varies enormously — staples might grow at 0.7× nominal GDP with low variance; capital goods might grow at 2× in an upcycle and contract outright in a downturn. Knowing your company's historical elasticity to its driver is a genuinely useful, computable number.

Flows — FPI and DII

Indian equities are meaningfully driven by **foreign portfolio investor** flows, which respond to global risk appetite, the US dollar, US rates, and relative emerging-market valuations — factors entirely outside any Indian company's control. **Domestic institutional investor** flows (mutual funds via SIPs, insurance) have grown into a substantial stabilising counterweight, which is why sustained FPI selling has, in recent cycles, produced smaller index drawdowns than it once did.

For a stock-level analyst the practical relevance is: high-FPI-ownership stocks are more exposed to global risk-off episodes regardless of fundamentals, and this is checkable from the quarterly shareholding pattern.

Building a macro sensitivity table for your coverage

The professional output is a table making your macro exposures explicit and quantified:

VARIABLE	MOVE	EPS IMPACT	VALUE IMPACT
USD/INR	+1% depreciation	+2.1%	+2.4%
Interest rates	+100bp	−1.8% (interest cost)	−9% (WACC)
Key raw material	+10%	−4.5%	−5%
Volume growth	−200bp	−6%	−8%

This is far more useful to a client than prose, and it is what allows a PM to overlay their own macro view on your bottom-up work — which is precisely how sell-side research gets used in practice.

The honest boundary

An equity analyst is not a macro forecaster and should not pretend to be. The professional position is: **do not forecast macro; quantify sensitivity to it.** State your base-case assumptions explicitly (the rate path, the currency level, the commodity price you have assumed), make them easy to find, and show what changes if the client disagrees. A note that hides its macro assumptions inside the model is unusable for anyone with a different macro view — which is most of your readership.

Common mistakes

- Directional rules without transmission logic — "rate hikes are bad for stocks" without knowing which channel and how much.
- Forgetting that rate moves hit **long-duration growth stocks** disproportionately.
- Ignoring **hedging**, so currency impact is modelled immediately when it will actually arrive with a lag.
- Treating raw-material-driven margin expansion as **structural** when it will mean-revert.
- Using headline GDP as the demand driver when the company's actual driver is rural income or the capex cycle.
- Hiding macro assumptions in the model rather than stating them prominently.
- Attempting to forecast macro rather than quantifying sensitivity to it.

Interview angle

"Rates are rising 200bp. What happens to your coverage?" Answer through the channels rather than directionally: discount rate — WACC rises, hitting long-duration growth names hardest, quantified as X% of DCF value per 100bp; earnings — interest cost rises for leveraged names (quantify from floating-rate debt), demand falls in credit-sensitive sectors; rotation — banks typically benefit near-term via faster asset repricing, rate-sensitive consumption suffers; flows — higher global rates can pressure FPI allocation to emerging markets. Then close with the professional boundary: you don't forecast the rate path, you publish the sensitivity so the client can apply their own.

Building and Auditing the Financial Model

The Problem / Why this matters

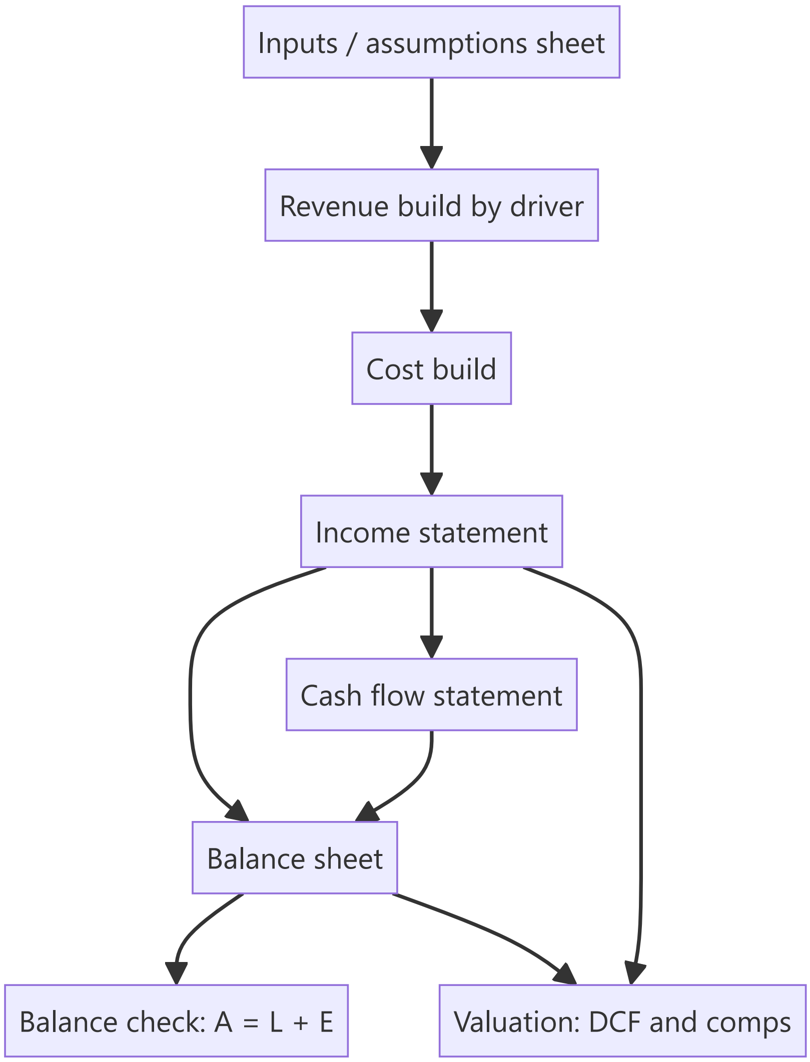
The model is the analyst's core working tool, and a model that is wrong, opaque, or impossible to update is worse than useless — it produces confident, precise, incorrect numbers and it cannot be handed to anyone else. Equity research interviews frequently include a modelling test, and the assessment is rarely about whether the candidate knows the formulas; it is about structure, transparency, and whether the model can be audited by someone who did not build it.

Core Idea

A good model is **driver-based, transparent, internally consistent, and auditable**. Every number is either a clearly-flagged input or a formula traceable to inputs — never a hardcoded number buried inside a calculation.

Why it works this way

A model's purpose is not to produce one answer but to let you and your reader ask "what if?" That requires assumptions to be visible and changeable in one place. The moment a growth rate is typed inside a revenue formula, the model has stopped being an analytical tool and become a calculator with hidden settings.



Full technical content

Structural conventions

Separate sheets by function: Assumptions → Historicals → Revenue build → P&L → Balance sheet → Cash flow → Valuation → Output/summary. Never mix inputs and calculations on the same sheet.

Colour convention (near-universal in finance, and interviewers look for it): - **Blue** — hardcoded inputs and assumptions - **Black** — formulas within the same sheet - **Green** — links from another sheet - **Red** — links to an external file (to be minimised)

One row, one formula. A formula should copy consistently across the entire forecast period. If a row needs different logic in different years, that is a signal to restructure rather than to write a bespoke formula in one cell — inconsistent rows are the single most common source of undetected model error.

No hardcodes inside formulas. `=B12*1.08` is a defect; `=B12*(1+Assumptions!$C$5)` is correct. Every driver lives on the assumptions sheet where it can be seen and flexed.

Consistent time axis across all sheets — same columns, same periods, clearly labelled actual (A) versus estimate (E).

The driver-based revenue build

The distinguishing feature of a research-quality model. Rather than growing revenue at an assumed percentage, decompose it into the quantities that actually drive it:

BUSINESS	REVENUE BUILD
Manufacturing	Capacity × utilisation × realisation per unit
Retail	Stores × sales per store (or per sq ft) × footfall × conversion
Bank	Advances × yield, and deposits × cost
Subscription	Subscribers × ARPU × (1 − churn)
IT services	Billable headcount × utilisation × realisation

This matters because it makes the forecast **challengeable and updatable** — when the company announces a new plant, you change capacity, not a growth percentage; when a competitor cuts price, you change realisation. It also forces the analyst to understand the business rather than extrapolate.

Linking the three statements

The integrity test of any model. The essential links:

- **Net income** flows from P&L to retained earnings on the balance sheet and to the top of the cash flow statement.
- **Depreciation** is added back in cash flow and accumulates against fixed assets on the balance sheet.
- **Capex** reduces cash and increases gross fixed assets.
- **Working capital changes** flow from balance-sheet movements into operating cash flow.
- **Debt movements** flow through financing cash flow and change the balance-sheet debt balance; interest expense links to the average debt balance.
- **Closing cash** from the cash flow statement is the balance-sheet cash figure.

The balance check — Assets = Liabilities + Equity — must be a live formula displayed prominently, ideally as a single cell showing zero. A model without a visible balance check cannot be trusted, and interviewers ask to see it.

Circularity: interest expense depends on average debt, which depends on cash flow, which depends on interest expense. Handle it either with iterative calculation enabled plus a circuit-breaker switch, or by using **opening-balance** debt for the interest calculation — the cleaner approach for a research model, since the precision loss is trivial and the model becomes far more robust.

The revolver / cash sweep

A model should not produce negative cash. Build a simple funding mechanism: if the cash balance falls below a minimum threshold, a revolver draws to cover it; if there is surplus cash above the threshold, it repays the revolver. This makes the model behave sensibly under stress scenarios rather than showing an impossible negative cash line.

Auditing a model — yours or someone else's

A structured review process:

1. **Trace the balance check** — is it live, is it zero across all periods?
2. **Scan for hardcodes** in formula cells (Excel: Find & Select → Formulas, or a colour-convention scan).
3. **Check formula consistency across rows** — inconsistent formulas within a row are the most common defect.
4. **Test the extremes** — set growth to zero, set it very high, set margin to zero. Does the model break, produce negative cash, or show implausible results?

5. **Verify the links** — change one input and confirm the change propagates correctly all the way to the output.
6. **Sanity-check outputs** against history: are forecast margins, RoCE and working-capital days plausible relative to what the company has actually achieved?
7. **Check terminal-year normalisation** — capex versus depreciation, sustainable margin, sustainable tax rate.
8. **Look for the classics** — sign errors (adding an outflow), off-by-one period references, and SUM ranges that miss the last row or double-count.

The plausibility check that catches most errors

Before trusting any model output, compare the forecast to the company's own history and to peers:

CHECK	QUESTION
Revenue CAGR	Has the company ever grown at this rate? Has anyone in the sector?
Terminal margin	Is it above the best margin the company has ever achieved?
RoCE trajectory	Does it rise indefinitely without the capital base growing?
Working-capital days	Have they been assumed to improve with no stated reason?
Capex/sales	Sufficient to support the assumed growth?
Implied market share	Does the revenue forecast imply an implausible share of the total market?

That last check is the most powerful and most neglected: forecast revenue divided by forecast industry size gives implied market share. A model implying a company reaches 60% share of a fragmented market has an arithmetic error or an unrealistic assumption, and this test finds it immediately.

Common mistakes

- **Hardcoded numbers inside formulas**, making assumptions invisible.
- **No balance check**, or a balance check that has been broken and ignored.
- Inconsistent formulas within a row.
- Growing revenue by an assumed percentage instead of by **drivers**.
- Forecast margins or returns that exceed anything the company or sector has achieved.
- **Terminal year not normalised** — capex below depreciation, peak-cycle margin.
- Circular references left uncontrolled, so the model breaks unpredictably.
- Building a model so complex only its author can use it — research models are collaborative documents.
- Not checking **implied market share**.

Interview angle

"How would you build a model for this company?" Structure: separate assumptions from calculations with a strict colour convention; build revenue from operational drivers rather than a growth rate; forecast costs by nature with explicit margin assumptions; link the three statements fully with a live balance check; handle circularity via opening-balance interest; add a revolver so cash never goes negative; then run sanity checks — forecast margins versus history, RoCE trajectory, working-capital days, and implied market share. Mentioning the balance check and the implied-market-share test unprompted is what signals you have actually built and audited models rather than only read about them.

Competitive Strategy and Moat Analysis

The Problem / Why this matters

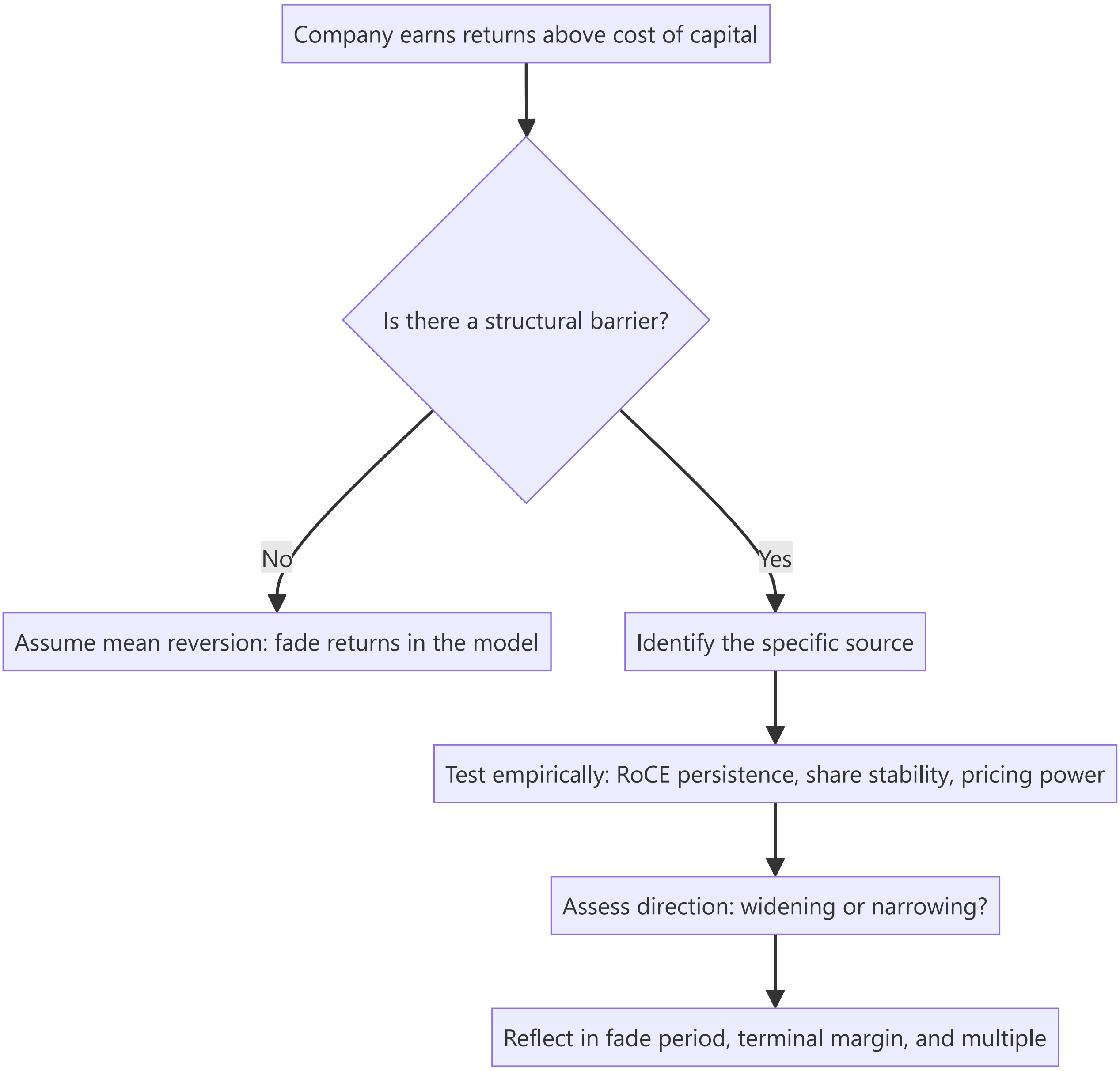
The single most important question in long-horizon equity analysis is whether a company's current profitability will persist. Capitalism's default is that high returns attract competition and get competed away — so a company earning 30% RoCE will, absent something protecting it, drift toward the cost of capital. Whether that erosion happens over three years or thirty is the difference between a stock worth 12x earnings and one worth 40x. "Moat" is the name for whatever prevents that erosion, and assessing it rigorously is the core of fundamental research.

Core Idea

A **moat** is a structural characteristic that allows a company to sustain returns above its cost of capital against competitive attack. It must be identified specifically (which of the recognised sources?), tested empirically (does the financial record actually show persistence?), and monitored (is it widening or narrowing?).

Why it works this way

Excess returns are, in a competitive market, temporary by default. Persistent excess returns therefore require an explanation — some barrier preventing competitors from replicating the economics. If you cannot name that barrier specifically, you should assume the returns will mean-revert, and value the company accordingly.


Company earns returns above cost of capital
Is there a structural barrier?
No
Yes
Assume mean reversion: fade returns in the model
Identify the specific source
Test empirically: RoCE persistence, share stability, pricing power
Assess direction: widening or narrowing?
Reflect in fade period, terminal margin, and multiple

Full technical content

The recognised sources of competitive advantage

SOURCE	MECHANISM	SECTOR EXAMPLES	HOW TO VERIFY
Intangibles — brand	Customers pay more for the same functional product	FMCG, luxury, paints	Sustained price premium vs unbranded; gross margin vs peers
Intangibles — patents/regulatory	Legal exclusion of competitors	Pharma, licensed businesses	Patent expiry schedule; licence conditions
Switching costs	Changing supplier is costly, risky or disruptive	Enterprise software, banking, medical devices	Customer retention rate; contract length; share of wallet
Network effects	Product becomes more valuable as more people use it	Exchanges, marketplaces, payment networks	User growth vs competitors; take rate stability
Cost advantage — scale	Fixed costs spread over more volume	Cement, telecom, retail	Cost per unit vs peers at different scales
Cost advantage — process/location	Structurally cheaper production or distribution	Commodities with captive resources	Cost-curve position
Efficient scale	Market only supports one or two players profitably	Regional utilities, pipelines, airports	Market structure; history of entry attempts failing
Distribution reach	Physical presence competitors cannot economically replicate	Indian FMCG, paints, building materials	Outlets covered; direct reach vs peers

Distribution deserves emphasis in the Indian context: reaching millions of small retail outlets across a fragmented market takes decades and enormous capital, and it is why incumbents in Indian consumer categories have been so durable against well-funded challengers.

Testing a claimed moat empirically

A moat claim without evidence is a story. Three tests:

1. Return persistence. Plot RoCE (or RoE for financials) over 10+ years, and against peers. A genuine moat shows returns *sustained* above the cost of capital across a full cycle — including the downturn. Returns that were high only during a boom are cyclical, not structural.

2. Market-share stability. A moat should show up as share that is stable or rising over time, particularly during periods of aggressive competitive entry. Share lost during a competitive attack, later recovered by discounting, indicates a weak moat.

3. Pricing power. The cleanest single test: can the company raise prices without losing volume? Look at historical episodes of input-cost inflation and check whether gross margin was preserved. A company that fully passed through a cost spike within two quarters has pricing power; one whose margins compressed and never recovered does not.

A useful supplementary check is **the incumbent's response to attack**: when a well-funded competitor entered, what happened to the incumbent's margins and share? That natural experiment is more informative than any qualitative argument.

Porter's Five Forces — as a diagnostic, not a checklist

The framework's value is in identifying *where the profit pool sits and why*:

- **Rivalry among existing competitors** — concentration, capacity utilisation, exit barriers. High fixed costs plus high exit barriers produce chronic overcapacity and poor returns (the structural problem in steel, airlines, and telecom in many markets).
- **Threat of new entry** — capital intensity, regulation, distribution access, brand.
- **Supplier power** — concentration of suppliers, switching costs, forward-integration threat.
- **Buyer power** — concentration of customers, price sensitivity, backward-integration threat. This is the key structural risk for auto-component makers and IT services with concentrated clients.
- **Threat of substitutes** — often the most under-analysed, because substitutes usually come from *outside* the industry definition.

The analytical output is not "rivalry is high" but a conclusion about **who captures the industry's profit** and whether that is shifting.

Moat direction — the more valuable question

Most analysts assess whether a moat exists. The higher-value question is whether it is **widening or narrowing**, because that determines re-rating or de-rating:

Widening signals: rising market share with stable or rising margins; increasing switching costs as products embed deeper; scale advantages compounding; distribution reach extending; regulatory position strengthening.

Narrowing signals: technology change altering the basis of competition; a new distribution channel bypassing the incumbent's advantage (e-commerce versus physical distribution reach); deregulation; patent cliff approaching; customer consolidation increasing buyer power; a well-capitalised entrant willing to sustain losses.

The disruption question: the most dangerous moats are those that are strong against *existing* competition but irrelevant against a new mode of competing. A distribution moat is formidable against another physical-distribution competitor and much weaker against a direct-to-consumer digital model.

Translating moat analysis into the model and valuation

This is where the analysis becomes financially consequential, and where most notes stop short:

- **Fade period.** In a DCF, how long before returns converge toward the cost of capital? A wide, verified moat justifies a long fade (15–20 years or effectively none); no identifiable moat justifies a short one (5–7 years). This single choice moves DCF value enormously.
- **Terminal margin and RoCE.** A moat justifies a terminal margin above the industry average; its absence does not.
- **Terminal growth.** Only a genuinely durable franchise can justify growth at the upper end of the plausible range in perpetuity.
- **The multiple.** Sustainably higher RoCE mathematically justifies a higher multiple — this is the rigorous version of "quality deserves a premium."
- **Risk assessment.** The moat-narrowing signals become the specific, monitorable risks in the note.

A note that asserts a moat but then models returns fading to the cost of capital in five years has not connected its qualitative and quantitative work — a very common internal inconsistency.

Common mistakes

- Calling a company's strong current margins a "moat" — high profitability is the *evidence to be explained*, not the explanation.
- Naming a moat without specifying **which source** it comes from.
- Confusing **operational excellence** (replicable) with structural advantage (not replicable). Good management is not a moat.
- Assessing existence but never **direction**.
- Assuming a moat that was durable against past competition is durable against a new mode of competition.
- Failing to connect the moat conclusion to the **fade period and terminal assumptions** in the model.
- Treating first-mover advantage or scale as automatic moats — both are frequently competed away.

Interview angle

"What makes this a good business?" Do not answer with margins. Name the specific moat source — brand, switching costs, network effects, cost advantage, efficient scale, distribution — then give the empirical evidence: RoCE sustained above cost of capital through a full cycle, stable-to-rising share, and demonstrated pricing power through an input-cost spike. Then add the two things that mark a senior answer: whether the moat is widening or narrowing and why, and how that conclusion shows up in your model as the fade period and terminal return assumption.

Small- and Mid-Cap Research

The Problem / Why this matters

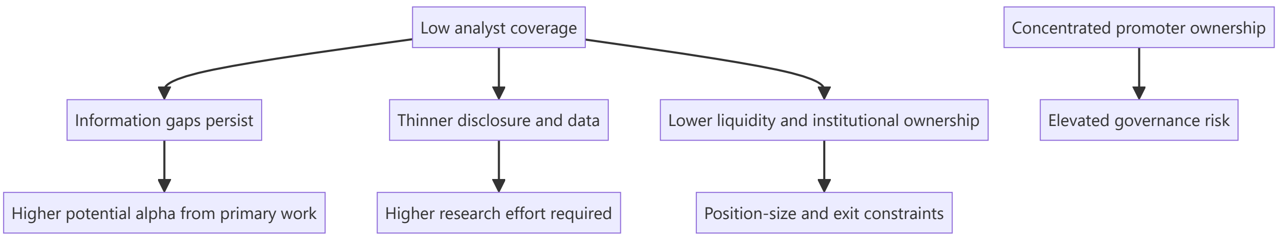
Large-cap research is crowded — thirty analysts cover the same company, all reading the same filings, and genuine informational edge is scarce. Small- and mid-caps are where the structural inefficiency lives: fewer analysts, thinner disclosure, less institutional ownership, and therefore more mispricing. But the same conditions that create the opportunity also create the risks — liquidity constraints, governance concerns, and data scarcity — and an analyst who applies large-cap methods unmodified to a small-cap will get hurt.

Core Idea

Small- and mid-cap research offers **higher potential edge** in exchange for **higher research effort, higher governance risk, and hard liquidity constraints**. The methodology must adapt on all three dimensions, not just be scaled down.

Why it works this way

Market efficiency is a function of the number of informed participants. Where twenty well-resourced analysts compete, prices reflect available information quickly. Where two do, and where institutional mandates prevent many funds from participating at all, prices can diverge from value for extended periods — which is precisely the condition an analyst is looking for.



Full technical content

Where the edge actually comes from

SOURCE	WHY IT EXISTS IN SMALL/MID CAPS
Coverage gap	0–3 analysts versus 25–40 for large caps; often no consensus at all
Disclosure gap	Less granular segment reporting, shorter concalls, sometimes no concall at all — rewarding primary work
Institutional-mandate exclusion	Many funds cannot buy below a market-cap or liquidity threshold, so a genuine buyer base only appears later
Index exclusion	No passive ownership, so no automatic bid
Under-modelled inflections	A single new plant or product can change the earnings base materially in a way that is easy to model but few have modelled
Re-rating potential	Moving from unknown to covered, or from small-cap to mid-cap index inclusion, brings a valuation re-rating on top of earnings growth

That last point is the specific small-cap return engine: returns compound from **earnings growth × multiple re-rating** simultaneously, which is why successful small-cap calls produce outsized outcomes.

The modified research process

Primary research matters far more. With thin disclosure, channel checks, supplier and customer conversations, plant visits and dealer conversations move from "helpful supplement" to "the core of the work." A small-cap thesis built only on published filings is usually not differentiated, because those filings are all anyone has and they are thin.

Management access is usually easier and more valuable. Small-cap managements are typically more accessible than large-cap ones — often the promoter directly. This is a genuine advantage for understanding strategy and capital allocation, but it carries a specific hazard: proximity creates the risk of becoming an advocate rather than an analyst. Maintain the same evidentiary standards you would apply to a company whose management never returns your calls.

Longer time horizons. Small-cap mispricings can persist for years because there is no institutional pressure forcing correction. The catalyst discipline matters more here, not less — but the realistic catalyst timeline is longer.

Balance-sheet scrutiny is disproportionately important. Small companies fail. Large caps rarely go to zero; small caps do. Every small-cap thesis needs an explicit solvency assessment: debt maturity schedule, refinancing requirement, covenant headroom, promoter pledge, and whether the business survives a demand downturn.

The governance premium on due diligence

Small- and mid-caps are typically promoter-controlled with high concentration, which raises the stakes on everything in the governance framework:

- **Related-party transactions** — proportionately more common and more material relative to company size.
- **Promoter pledge** — a small-cap promoter under personal financial stress has both stronger incentive and greater ability to affect reported results.
- **Auditor quality** — smaller audit firms, less scrutiny. An auditor change or resignation is a severe signal.
- **Disclosure quality** — inconsistent segment reporting, sparse notes, sometimes no earnings call.
- **Minority-shareholder track record** — has the promoter previously diluted minorities on unfavourable terms, or transferred value to unlisted group entities?

The practical rule: **governance failure is the dominant loss mechanism in small caps, not valuation error.** Weight the governance work accordingly, and be willing to pass on a fundamentally attractive business where the governance evidence is poor — "cheap enough to compensate for governance risk" is a proposition that fails far more often than it works.

Liquidity — the constraint that shapes everything

Liquidity is not a footnote in small-cap work; it determines whether a correct call is implementable.

Metrics to compute and state explicitly: - **Average daily traded value (ADTV)** over 3 and 6 months. - **Days to build** a target position at, say, 20–25% of ADTV. - **Days to exit** — critically, assessed under *stressed* conditions, where liquidity typically falls sharply exactly when you need it. - **Free float** — promoter holding of 70%+ leaves a small tradeable float regardless of headline market cap. - **Impact cost** — the price move caused by transacting meaningful size.

The asymmetry that matters: **liquidity evaporates precisely when you most want to sell.** A stock with adequate normal-conditions liquidity can become effectively untradeable during a drawdown, which is why position sizing in small caps must be set against stressed liquidity, not average.

Valuation adjustments

- **Illiquidity discount** — a genuine, defensible adjustment, though it should be estimated rather than applied by convention.
- **Higher cost of equity** — small-cap risk premium in CAPM, reflecting genuine higher fundamental and liquidity risk.
- **Governance discount** where warranted, argued from evidence.
- **Peer-set difficulty** — often no directly comparable listed peer, so relative valuation may require larger peers with an explicit size adjustment, or cross-market comparables.
- **Higher scenario dispersion** — small-cap bull and bear cases should be genuinely wider than large-cap ones, because the business outcomes genuinely are.

The re-rating thesis structure

A well-constructed small-cap thesis usually has this shape:

1. **The business is better than its multiple implies** — evidence from returns, moat, growth.
2. **Why the market hasn't noticed** — no coverage, recent listing, a past problem now fixed, an under-modelled capacity addition, index exclusion.
3. **The catalyst that forces attention** — earnings inflection, index inclusion, coverage initiation, a large institutional investor entering, promoter pledge release.
4. **The dual return driver** — earnings growth of X% plus re-rating from Ay to By.
5. **Honest liquidity and governance assessment** — including what would make you exit.

Common mistakes

- Applying large-cap methods unmodified — relying on filings alone where primary work is required.
- Ignoring **liquidity**, recommending a position that cannot be built or exited at size.
- Under-weighting **governance**, which is the dominant loss mechanism in this segment.
- Assuming a low multiple is opportunity when it reflects genuine risk — small caps are often cheap for reasons.
- Becoming an **advocate** through close management access.
- Failing to assess **solvency** explicitly — small companies do go to zero.
- Assuming the mispricing will correct on a large-cap timeline.
- Sizing against average liquidity rather than stressed liquidity.

Interview angle

"Why would you look at small caps rather than large caps?" Frame it as a structural inefficiency: coverage of 0–3 analysts versus 30, thinner disclosure, and institutional-mandate exclusion mean information gaps persist, so primary research earns a genuine return — and successful calls compound earnings growth *and* multiple re-rating together. Then show you understand the cost of that edge: governance is the dominant loss mechanism so due diligence must be disproportionate; balance-sheet solvency needs explicit assessment because small caps do fail; and liquidity must be assessed under stressed rather than average conditions, because it disappears exactly when you need to exit. That balanced answer is what distinguishes a considered candidate from one who has only heard that small caps outperform.

Capital Allocation Analysis

The Problem / Why this matters

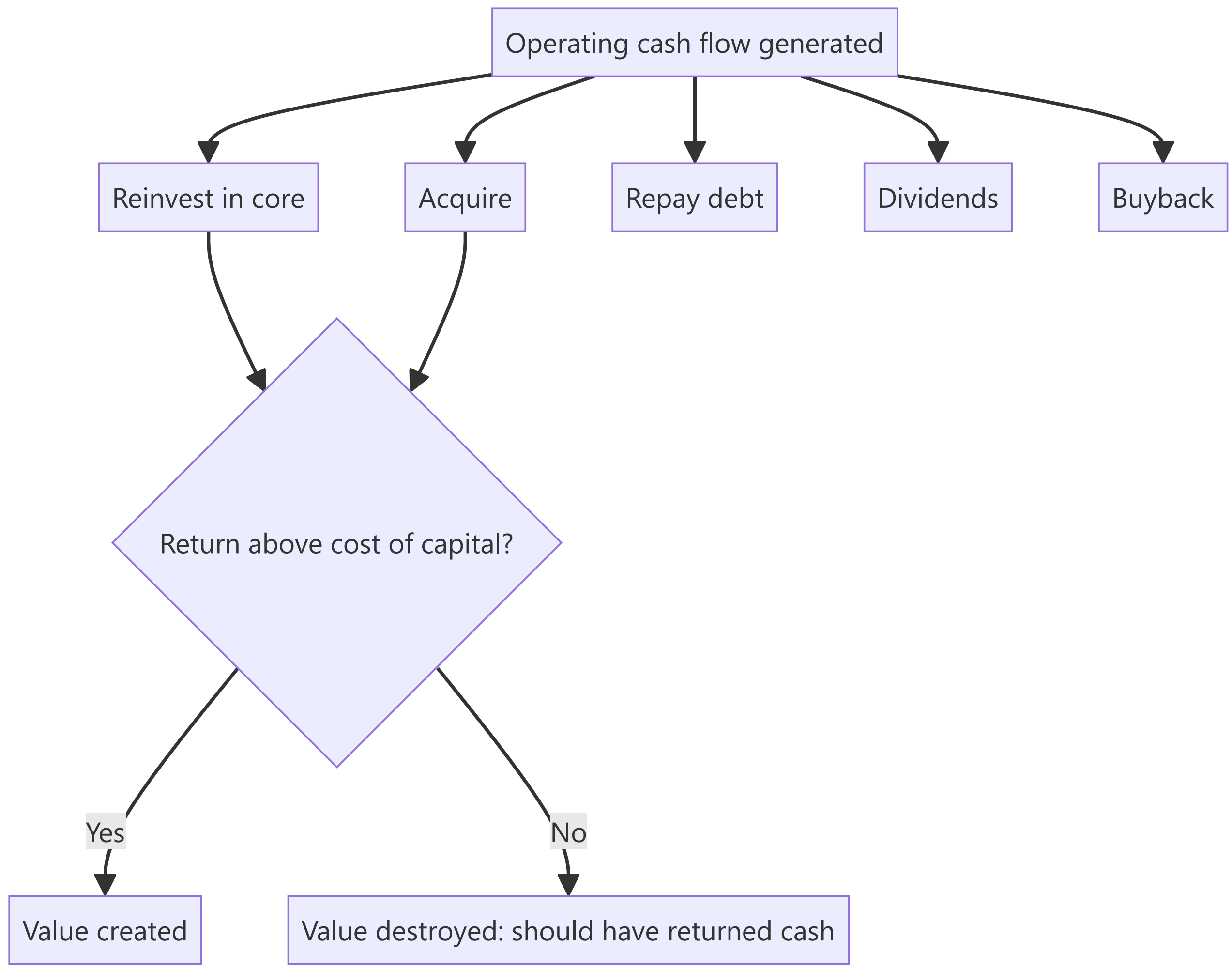
Over a decade, the compounding effect of capital-allocation decisions frequently exceeds the effect of operating performance. A company earning a 20% RoCE that reinvests all its cash at 8% will destroy the value its operations create; a company earning 15% that returns excess cash rather than reinvesting badly will compound shareholder value steadily. Yet capital allocation is the least-analysed dimension in most equity research, because it requires assembling a decade of history rather than reading one quarter.

Core Idea

Management has exactly **five uses** for cash generated: reinvest in the core business, acquire, repay debt, pay dividends, or buy back stock. Capital-allocation analysis evaluates the historical record of those choices against the returns they generated, and assesses whether current policy is value-creating.

Why it works this way

Every rupee retained is a rupee not returned to shareholders, who could have invested it themselves. Retention is therefore only justified if the company can earn more on that rupee than the shareholder's alternative — the cost of equity. This is the single test that governs the whole subject.



Full technical content

The central test — incremental return on invested capital

Reported RoCE reflects the whole asset base, most of which was deployed long ago by possibly different management. What matters for judging *current* management is the return on capital **they** have deployed:

Incremental RoCE = Δ EBIT(1−t) over period ÷ Δ Capital employed over period

Compute over 5 and 10 years. Then compare to the cost of capital:

INCREMENTAL ROCE VS WACC	INTERPRETATION
Well above	Reinvestment is creating value; retention justified; growth is genuinely valuable
Approximately equal	Growth is value-neutral; the company is running to stand still
Below	**Reinvestment is destroying value**; cash should be returned instead

This last case is common and rarely called out in research: a company growing revenue and profit in absolute terms while earning below its cost of capital on new investment is destroying value *through growth*. Growth is only good when it earns above the cost of capital.

Evaluating each allocation channel

1. Reinvestment in the core (capex)

- Split maintenance capex from growth capex where disclosable — maintenance is non-discretionary, growth is the decision being judged.
- Check **capex timing through the cycle**: capacity added at the cycle peak (when returns look best and costs are highest) is the classic value-destroying pattern in cyclicals and commodities. Counter-cyclical investment is the mark of a disciplined allocator.
- Compare **announced project returns to delivered returns**. Companies announce IRRs; few report whether they were achieved. Track the capacity that was commissioned and what happened to consolidated returns afterward.

2. Acquisitions

The highest-risk channel, and empirically the one where most value is destroyed. For each material deal, assemble:

QUESTION	WHERE TO FIND IT
Price paid, and multiple paid	Deal announcement
Synergies promised	Announcement and concall
Was the target's business actually growing?	Target's own filings pre-deal
Post-deal: did consolidated RoCE rise or fall?	Financials 2–3 years after
Has goodwill been impaired?	Balance sheet and notes

Serial acquirers who never impair goodwill despite underperforming acquisitions are signalling something. Impairment is an admission; its persistent absence alongside disappointing performance is a governance flag as much as an accounting one.

3. Debt repayment

Value-creating when leverage is genuinely excessive or refinancing risk is real. But note the mirror error: a company sitting on large net cash while insisting on further deleveraging is being conservative at shareholders' expense, since surplus cash earning treasury rates drags RoE.

4. Dividends

Assess **policy consistency** rather than yield level. Key questions: is there a stated payout policy? Is the payout ratio stable through the cycle, or cut opportunistically? Is the dividend covered by free cash flow, or funded by debt (a genuine warning sign)? A company borrowing to sustain a dividend is transferring risk, not returning value.

5. Buybacks

The most misunderstood channel. A buyback creates value **only if shares are repurchased below intrinsic value** — otherwise it is a transfer from continuing shareholders to exiting ones.

Assess: at what price and multiple did the company buy back, and what has happened since? Buying back at the peak of a re-rating is common and value-destructive. Also check whether the buyback is genuinely reducing share count or merely offsetting ESOP dilution — the latter is a compensation expense being routed through the buyback line, not a return of capital.

In India, mandatory **extinguishment** means the share-count reduction is permanent, which strengthens the case relative to treasury-share regimes.

The cash-deployment scorecard

A practical research output assembling a decade:

PERIOD METRIC	VALUE
Cumulative CFO (10 yr)	₹X
→ Growth capex	₹A (% of CFO)
→ Acquisitions	₹B
→ Debt repayment	₹C
→ Dividends	₹D
→ Buybacks	₹E
Incremental RoCE over period	Y%
Cost of capital	Z%
Value created / destroyed	(Y − Z) × capital deployed

This table, more than any narrative, tells a client whether management can be trusted with retained earnings — and it is directly usable in the management-quality assessment.

Signals of a disciplined allocator

- A **stated, consistent capital-allocation framework**, articulated and adhered to over years.
- Willingness to **return cash** when reinvestment opportunities are inadequate — the hardest discipline, because it implicitly admits limited growth options.
- **Counter-cyclical** investment and acquisition timing.
- Acquisitions in **adjacent** areas with articulated, measurable synergies, subsequently reported against.
- **Honest reporting** on underperforming investments, including impairment.

- Management compensation tied to **returns on capital**, not to revenue or absolute profit growth (which reward empire-building).

Signals of poor allocation

- **Diversification into unrelated businesses** — the classic conglomerate value destroyer.
- Growth capex continuing while incremental RoCE sits below the cost of capital.
- Serial acquisitions with rising goodwill and falling consolidated returns.
- Large cash balances held indefinitely alongside costly debt.
- Buybacks concentrated at valuation peaks.
- Dividends funded by borrowing.
- Compensation linked to size rather than returns.

Common mistakes

- Judging management on **reported RoCE** rather than incremental RoCE on capital they deployed.
- Treating **growth as automatically good** without checking it earns above the cost of capital.
- Assessing dividends by yield rather than by policy consistency and free-cash-flow coverage.
- Assuming a buyback is shareholder-friendly regardless of the price paid.
- Not tracking **acquisitions against what was promised** at announcement.
- Ignoring **capex timing** relative to the cycle.
- Treating capital allocation as a governance topic rather than as a direct driver of forecast returns and therefore of valuation.

Interview angle

"How do you judge whether management is good at capital allocation?" Lead with the quantitative test: incremental RoCE over 5–10 years versus the cost of capital — this tells you directly whether retained earnings created value. Then walk the five channels with what you'd check in each: capex timing through the cycle and delivered versus announced returns; acquisitions against promised synergies with goodwill impairment as the honesty test; debt repayment versus excess conservatism; dividend policy consistency and free-cash-flow coverage; and buyback prices versus intrinsic value. Close with the point that carries the most weight: growth only creates value when it earns above the cost of capital, and a company growing below it is destroying value *through* growth.

Alternative Data in Equity Research

The Problem / Why this matters

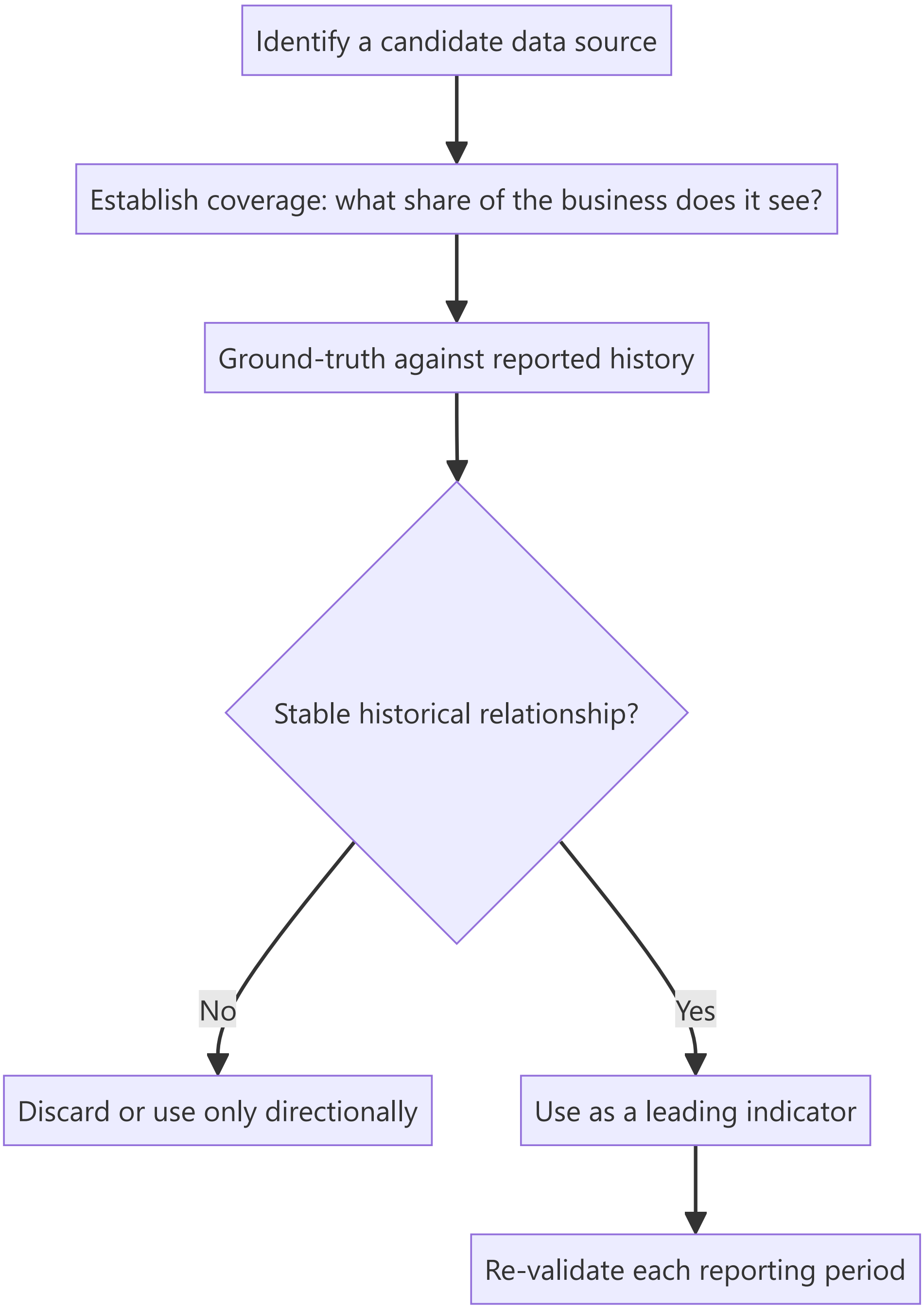
Company disclosure is quarterly, lagged, aggregated and identical for every analyst. Between reporting dates, the business generates continuous signals that are increasingly observable through non-traditional data sources — app downloads, satellite imagery, job postings, web traffic, payment data, shipping records. Analysts who can locate, validate and interpret these sources see inflections weeks or months before they appear in a filing. This is now a standard part of institutional research, and familiarity with both its power and its limits is expected at senior level.

Core Idea

Alternative data is any non-traditional dataset informative about a company's operations. Its value comes from **timeliness** (higher frequency than reporting) and **exclusivity** (fewer analysts using it), but it requires rigorous **ground-truthing** against reported figures before it can be trusted.

Why it works this way

A company reports quarterly what it did over three months. If footfall, hiring, downloads or shipping volumes are observable weekly, an analyst can construct a real-time proxy for the reported number. That proxy is only useful, however, if its historical relationship to the reported figure has been established — otherwise it is an uncalibrated signal that may be measuring something else entirely.


Identify a candidate data source
Establish coverage: what share of the business does it see?
Ground-truth against reported history
Stable historical relationship?
No
Yes
Discard or use only directionally
Use as a leading indicator
Re-validate each reporting period

Full technical content

The main categories

DATA TYPE	WHAT IT PROXIES	TYPICAL USE
App download / DAU-MAU rankings	User acquisition and engagement	Fintech, consumer internet, broking
Web traffic and search trends	Demand interest, brand momentum	E-commerce, consumer, travel
Satellite imagery	Parking-lot fill, construction progress, storage-tank levels, crop yields	Retail footfall, capex progress, commodities
Geolocation / footfall data	Store visits	Retail, QSR, malls, cinemas
Job postings and headcount	Expansion plans, functional priorities	Any — an early indicator of strategic direction
Shipping / customs / port data	Import-export volumes	Commodities, manufacturing, trade-exposed sectors
Payment / card-spend panels	Consumer spending by category and merchant	Consumer, retail
Pricing scrapes	Competitive pricing, discount intensity	E-commerce, airlines, hotels, financial products
App-store reviews / social sentiment	Product issues, satisfaction shifts	Consumer-facing businesses
Government/regulatory datasets	GST filings, e-way bills, electricity consumption, vehicle registrations	Sector-level demand

In the Indian context specifically, publicly available government data is unusually rich and under-used: **monthly vehicle registration data (VAHAN)**, **GST collections**, **e-way bill volumes**, **electricity generation**, and **monthly auto dispatch numbers** together give a genuinely high-frequency read on real economic activity, and are free.

Ground-truthing — the discipline that makes it research rather than noise

An alternative dataset is worthless until its relationship to the reported figure is established. The process:

1. **Assemble history** — the alternative series and the corresponding reported metric over as many periods as available.
2. **Measure the relationship** — correlation, and more importantly the *stability* of that correlation over time. A source that tracked well for eight quarters and then decoupled is telling you something changed.
3. **Establish coverage** — what share of the business does the data actually see? Card-spend panels capture card transactions, not cash; footfall data captures smartphone users with location enabled. Both have systematic, non-random coverage gaps.
4. **Check for structural breaks** — a change in the company's channel mix, or in the data provider's own panel composition, can break the relationship without either being obvious.
5. **Re-validate every period** — treat the relationship as an ongoing hypothesis, not a settled fact.

The systematic biases to correct for

- **Panel composition bias** — the panel is rarely representative. Smartphone-based footfall data skews urban and younger; card panels skew affluent.

- **Panel drift** — the provider's panel changes over time, creating apparent trends that are artefacts of composition change rather than of the underlying business.
- **Coverage mismatch** — the data may cover one channel or geography while reported revenue covers all.
- **Definitional mismatch** — app downloads are not customers; footfall is not sales; job postings are not hires.
- **Crowding** — once a dataset becomes widely used, its edge decays, and it stops being informative because it is already in the price.

From data to a research conclusion

The analytical chain must be explicit:

> *App downloads for the company's platform rose 34% QoQ (source: third-party ranking data, historical correlation to reported new-customer additions of 0.81 over 11 quarters). Applying that relationship implies new customer additions of ~2.4mn versus consensus ~1.9mn. Because ARPU on new customers is roughly 60% of the base, we model a 4% revenue upside to consensus for the quarter.*

That is usable research: source stated, historical relationship quantified, translation to a financial line explicit, and the conclusion sized. Compare with *"app downloads are strong, which is positive"* — which is an observation, not analysis.

Combining sources — the real edge

Single-source signals are fragile. The strongest alternative-data work **triangulates**: satellite-observed construction progress *plus* job postings for the new plant *plus* equipment-import customs records together give a far more confident read on a capacity expansion's timing than any one alone. Where multiple independent sources agree, confidence rises materially; where they conflict, that conflict is itself the finding worth investigating.

Alternative data also works best **paired with traditional primary research** — a channel check confirming what the data suggests, or vice versa.

Cost, compliance and practical constraints

- **Cost** — commercial alternative datasets are expensive, which is why they are more common at large institutions. Public datasets (government data, app rankings, job postings, web traffic estimates) are the accessible starting point and remain under-exploited.
- **Legality and terms of use** — web scraping must respect terms of service and applicable law; personal-data regulations constrain geolocation and transaction data.
- **Material non-public information** — alternative data is generally permissible under the mosaic theory because it aggregates non-material public observations. But data obtained from someone breaching a confidentiality duty is not permissible regardless of the format it arrives in. The source's *provenance* matters, not just its statistical form.
- **Vendor due diligence** — how was the data collected, with what consent, and is the panel disclosed? A vendor unwilling to explain collection methodology is a compliance and quality risk.

Common mistakes

- Using a dataset without **ground-truthing** it against reported history.
- Ignoring **coverage** — treating a partial-panel signal as representative of total business.

- Missing **panel drift**, and reading a data-composition artefact as a business trend.
- Confusing a proxy with the thing itself — downloads are not revenue.
- **Single-source dependence** without triangulation.
- Continuing to rely on a source after its historical relationship has broken.
- Assuming legality from availability — provenance and terms of use both matter.
- Presenting an observation without translating it into an estimate impact.

Interview angle

"How would you get an edge on a consumer internet company between quarters?" Name specific sources — app-download rankings, DAU/MAU estimates, web traffic, app-store review volume and sentiment, job postings signalling expansion, and payment-panel spend data. Then show the discipline that matters more than the list: establish the historical relationship between the proxy and the reported metric before trusting it, understand what share of the business the data actually sees, watch for panel drift, and triangulate across independent sources. Finish by translating it into a number — "this implies X% versus consensus" — because an observation that isn't converted into an estimate impact isn't yet research.

IPO Analysis — Valuing a Company With No Trading History

The Problem / Why this matters

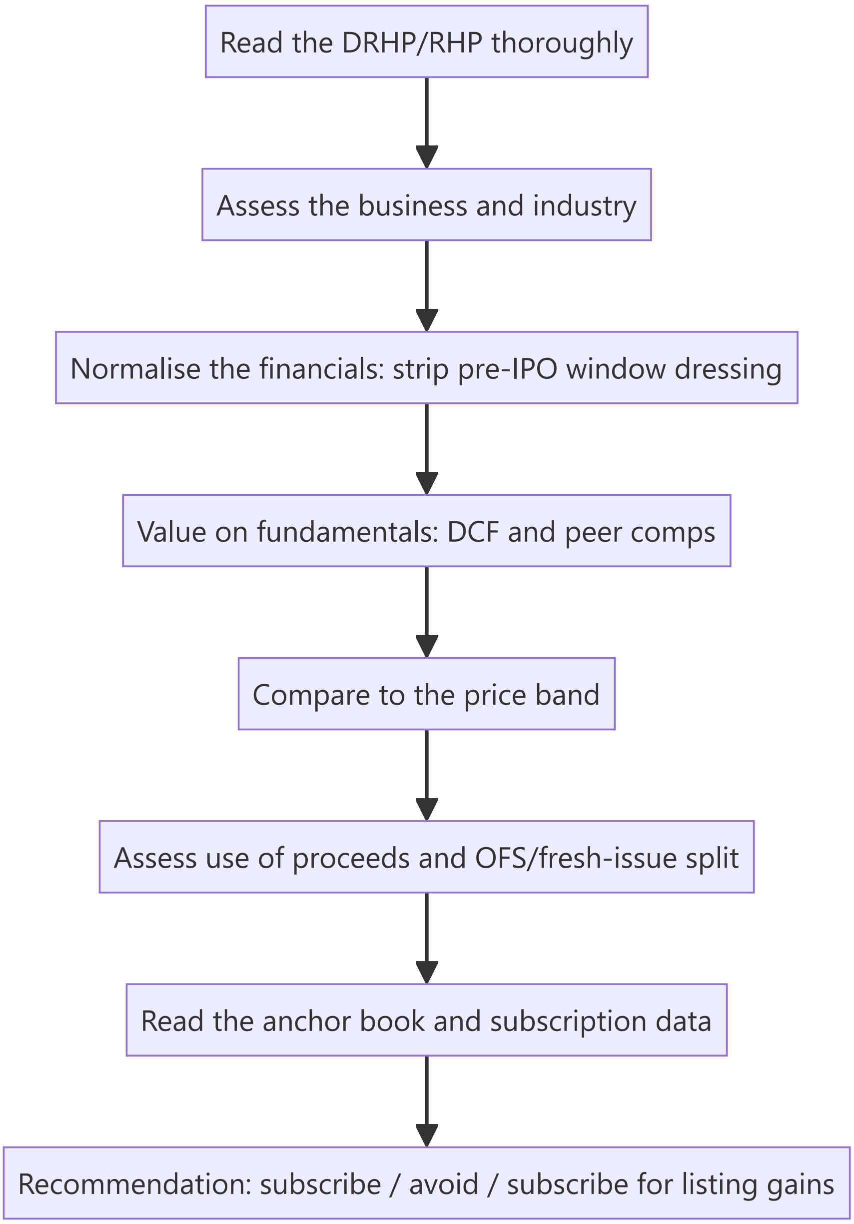
An IPO strips away the analyst's usual anchors. There is no historical share price, no consensus, no track record of management delivering against public guidance, and — critically — the seller controls the disclosure and chooses the timing. Companies list when *they* believe conditions are favourable, which introduces a systematic bias into the entire exercise. Assessing an IPO well requires a distinct discipline from covering a listed company.

Core Idea

IPO analysis is a **valuation problem with an adverse-selection overlay**. Value the business on fundamentals, then explicitly adjust for the structural fact that the issuer chose the timing, controls the narrative, and has priced the issue to sell.

Why it works this way

In secondary-market research, buyer and seller face the same public information. In an IPO, the seller has vastly superior information and chose both the moment and the framing. That asymmetry does not make every IPO a bad investment, but it means the default posture should be scepticism requiring evidence, rather than enthusiasm requiring disproof.



Full technical content

The prospectus is the primary source

The **DRHP** (Draft Red Herring Prospectus) and **RHP** are the most information-dense documents an analyst will encounter on a company, and most retail participants never read them. Priority sections:

SECTION	WHAT TO EXTRACT
Risk factors	Legally mandated candour — the company must disclose what could go wrong. Read every one; the material risks are here, not in the marketing
Objects of the issue	What the money is actually for (see below)
Financial statements + restated	3 years, restated to a consistent basis
Related-party transactions	Pre-IPO promoter dealings, often subsequently unwound
Litigation and contingent liabilities	Outstanding proceedings against the company and promoters
Management and promoter background	Prior ventures, any regulatory history
Capital structure history	Prices at which pre-IPO investors bought — the single most useful benchmark
Basis for issue price	The company's own justification, with its selected peer set

The pre-IPO placement price is the analytical goldmine. The capital-structure section discloses what earlier investors paid and when. If a private round three or six months before the IPO was priced at ₹180 and the IPO band is ₹420–460, the analyst must ask what changed to justify a 2.5× uplift in months. Sometimes there is a genuine answer; often the answer is simply that the market window opened.

Normalising pre-IPO financials

Companies frequently present unusually strong financials in the years immediately preceding a listing. Not necessarily improperly, but predictably. Check for:

- **Margin expansion in the pre-IPO year** that reverses historical patterns — often achieved through deferred spending on marketing, R&D or maintenance.
- **Working-capital tightening** immediately before listing, releasing cash flow that then reverses.
- **Related-party transactions restructured** just before the IPO — sales previously routed through promoter entities being brought in-house, flattering revenue and margin.
- **One-offs** included in the growth narrative.
- **Change in accounting policy** or a restatement improving the trend.
- **Promoter remuneration** reduced pre-IPO and restored afterward.

The corrective: build the forecast from **normalised** rather than peak-reported margins, and test whether the pre-IPO growth rate is achievable when the pre-listing incentives disappear.

Objects of the issue — where the money goes

Read the fresh-issue/OFS split (covered in the OFS chapter) and then the specific use of the fresh-issue proceeds:

USE OF PROCEEDS	SIGNAL
Capacity expansion / growth capex	Genuine growth funding — positive
Debt repayment	Balance-sheet repair; check whether the debt was productive
Working capital	Routine; assess whether it funds growth or plugs a gap
Acquisition (unspecified target)	Weak — the company is raising money without a defined plan
"General corporate purposes" in large proportion	Weak — regulation caps this, but a large allocation signals no specific plan
Repayment of promoter loans	Genuine concern — public money repaying insiders

The anchor book

Anchor investors are allotted the day before the issue opens, at the issue price, with a lock-in. Their identity is disclosed and is genuinely informative:

- **Marquee long-only domestic mutual funds and reputable global institutions** anchoring in size is a credibility signal, since they conducted their own diligence with full access.
- **An anchor book dominated by smaller, less-known entities**, or one that is thinly subscribed, is a negative signal.
- **Anchor concentration** matters: a book spread across many quality names is stronger than one dependent on two or three.
- Note the **lock-in expiry dates** (typically a portion at 30 days and the remainder at 90 days) as scheduled supply events post-listing.

Subscription data as it builds

During the bidding window, category-wise subscription is published daily:

- **QIB** subscription is the most informative — institutions with analytical resources voting with capital.
- **NII/HNI** subscription is often funded and leverage-driven, oriented to listing gains rather than fundamental conviction; very high NII subscription with weak QIB is a low-quality combination.
- **Retail** subscription reflects sentiment and marketing reach more than analysis.
- **The pattern matters more than the level**: strong QIB with weak retail suggests institutional conviction without retail hype; strong retail with weak QIB is the reverse and is generally the less encouraging configuration.

Grey market premium — use with care

The GMP is an unofficial, unregulated indication of expected listing gain. It is genuinely informative as a sentiment gauge but is thin, easily influenced, and has no regulatory standing. Treat it as one input on *listing-day* expectations and as essentially uninformative about **fundamental value** — the two questions are entirely separate, and conflating them is the most common retail error.

The two distinct recommendations

An IPO note should be explicit about which question it is answering:

1. **Subscribe for listing gains** — a short-horizon view driven by demand, GMP, subscription and market conditions. This is a trading call.
2. **Subscribe for the long term** — a fundamental view that the business at the issue price offers attractive returns over years.

These frequently diverge: an over-subscribed issue in a hot market may well deliver a listing pop while being expensive on fundamentals. Conflating them misleads the reader, and separating them explicitly is a mark of a well-constructed note.

Valuation without a trading history

- **Peer comparison** is the primary method, but scrutinise the company's own selected peer set in the "basis for issue price" section — issuers routinely select flattering comparables. Construct your own.
- **DCF** is usable but rests entirely on forecast assumptions with no track record of public guidance to calibrate against; widen the scenario range accordingly.
- Apply an **IPO discount** in principle: a listing company has no public track record, unproven governance under public scrutiny, and impending lock-in supply. Demanding a discount to listed peers is analytically defensible, and the absence of one is a reason for caution.

Common mistakes

- Not reading the **risk factors** and the **capital-structure history** in the prospectus.
- Extrapolating **pre-IPO peak margins** without normalising.
- Accepting the issuer's **selected peer set**.
- Treating **GMP** as a fundamental valuation signal.
- Conflating "**subscribe for listing gains**" with "good long-term investment."
- Ignoring **anchor lock-in expiry** as a scheduled post-listing supply event.
- Overlooking a large "**general corporate purposes**" allocation or promoter-loan repayment in the objects of the issue.
- Forgetting that the issuer chose the timing — the adverse-selection overlay applies to every IPO.

Interview angle

"How would you evaluate an IPO?" Structure: read the DRHP properly — risk factors, objects of the issue, related-party transactions, and especially the capital-structure history showing what pre-IPO investors paid and when; normalise the financials for pre-IPO window dressing; build your own peer set rather than accepting the issuer's; value on fundamentals and compare to the band; assess the fresh-issue/OFS split and where the money goes; read the anchor book quality and the QIB-versus-retail subscription pattern. Then close with the framing that shows judgement: separate the listing-gain call from the long-term investment call, and remember the issuer chose the timing — the information asymmetry runs one way.

The Buy-Side Analyst Role

The Problem / Why this matters

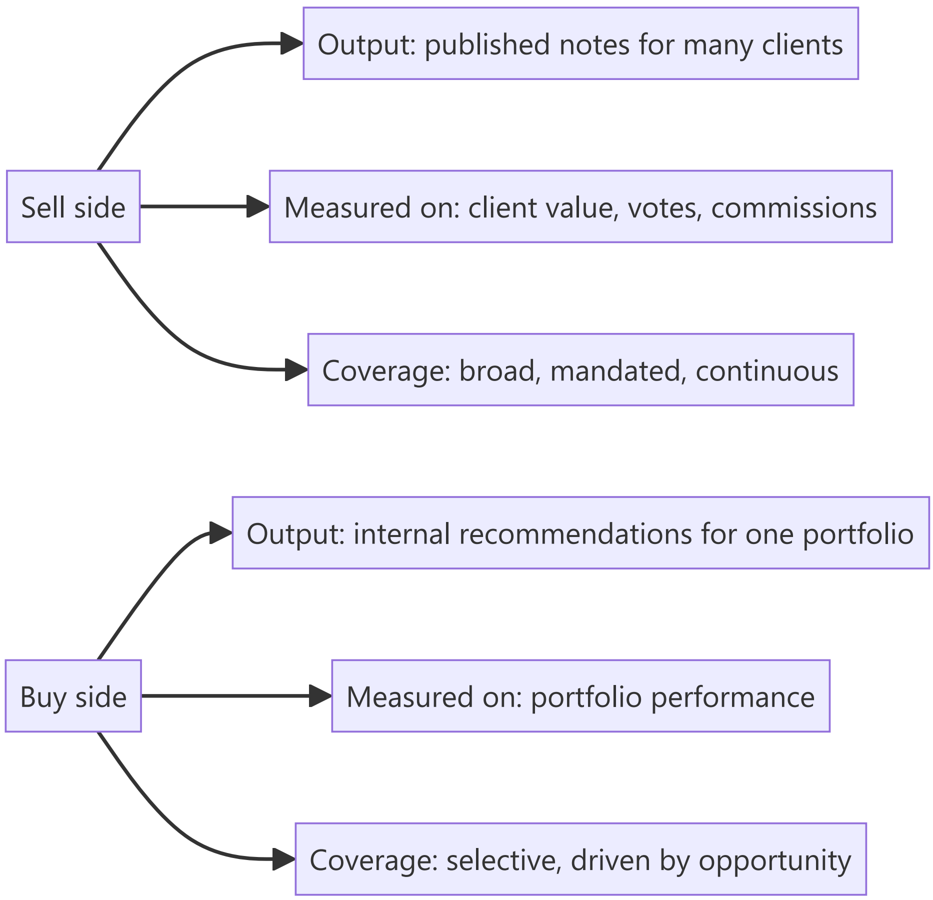
Sell-side and buy-side research look superficially similar — both analyse companies and form views — but the jobs differ in objective, output, incentives and success measurement. Candidates frequently prepare for one and interview for the other, and the mismatch shows immediately. Understanding the distinction matters both for choosing a career path and for answering "why buy side?" credibly.

Core Idea

The sell side produces **research as a product** distributed to many clients and monetised indirectly; the buy side produces **decisions on capital** for one portfolio, monetised by investment performance. Every difference in the role follows from that.

Why it works this way

A sell-side analyst is measured on whether clients value their work — commissions, broker votes, rankings. A buy-side analyst is measured on whether the portfolio makes money. Those two objectives overlap substantially but not entirely, and where they diverge, the behaviours diverge.



Full technical content

The core differences

DIMENSION	SELL SIDE	BUY SIDE
Employer	Broker, investment bank, independent research house	Mutual fund, hedge fund, PMS, insurance, pension, family office
Output	Published research notes, models, client calls	Internal recommendation memos, model, pitch to PM/IC
Audience	Many institutional clients	One PM or investment committee
Coverage	~10–25 stocks, mandated, must cover all continuously	Selective; can ignore uninteresting names entirely
Success measure	Client votes, commission share, rankings, accuracy	Contribution to portfolio return
Rating	Public Buy/Hold/Sell with target price	Position recommendation with size
Sizing	Not required	**Required** — how much of the portfolio
Time horizon	Often 12 months by convention	Whatever the fund's mandate is
Being wrong	Reputational cost	Direct financial cost
Access	Strong — sell side gets management access and hosts conferences	Varies; often relies on sell-side-arranged access

What changes in the analysis itself

1. Sizing replaces rating. A sell-side Buy says a stock will outperform. A buy-side recommendation must say *how much of the portfolio* — which forces explicit engagement with conviction level, downside, correlation to existing holdings, and liquidity. This is why buy-side analysts think in risk-reward and position size rather than target prices.

2. Selectivity replaces coverage obligation. A sell-side analyst must publish on every stock in their coverage regardless of whether anything interesting is happening. A buy-side analyst can say "nothing to do here" and move on — a genuine luxury that concentrates effort where returns are available.

3. Portfolio context matters. A buy-side analyst must consider what the fund already owns: adding a fifth private bank has different portfolio consequences from adding the first, even if the stock is equally attractive standalone. Correlation and factor exposure become part of the analysis.

4. Depth over breadth on fewer names. Because there is no publishing obligation, a buy-side analyst can spend six weeks on one idea. The work often goes deeper than sell-side research on the same name, particularly on primary research.

5. No requirement to be public or diplomatic. Buy-side views can be as negative as the evidence supports, with no corporate-access consequence and no management relationship to preserve. This is a real analytical freedom.

6. Different relationship with consensus. The sell side effectively *is* part of consensus. The buy side uses sell-side research as an input — often primarily to understand what the market believes, so as to identify where it may be wrong.

How the buy side actually uses sell-side research

Worth understanding because it explains what makes sell-side research valuable:

- **Models** — a well-built sell-side model saves the buy-side analyst days of work.
- **Consensus mapping** — understanding what the market expects, which is the baseline for finding differentiated views.
- **Access** — sell-side-arranged management meetings, conferences and expert calls.
- **Primary work** — channel checks and proprietary surveys the buy side cannot replicate at scale.
- **Sector education** — initiation notes as a fast route into an unfamiliar industry.

What the buy side generally does *not* use is the rating itself. This is why the most valued sell-side analysts are those whose work is deep, whose primary research is genuine, and who are available for detailed conversation — not those whose ratings have been most accurate.

Fund-type variation within the buy side

FUND TYPE	HORIZON	STYLE IMPLICATIONS
Long-only mutual fund	1–5 years	Benchmark-relative; constrained by mandate, sector limits, liquidity
Hedge fund (long/short)	Weeks to years	Can short; absolute-return focused; higher turnover; short ideas as valuable as longs
PMS / family office	Multi-year	Concentrated, less benchmark-constrained
Insurance / pension	Very long	Liability-driven, income-oriented, large size constrains small caps

The **long/short** distinction matters most analytically: a long-only analyst's downside work protects the portfolio, while a long/short analyst's downside work is itself a source of return — which means forensic accounting and moat-erosion analysis are directly monetisable rather than merely defensive.

The transition sell side → buy side

The common path, and what changes: - Shift from **communication** skills to **decision** skills. - Learning to size positions and think in portfolio rather than single-stock terms. - Adjusting to less external validation — no rankings, no client feedback, only performance over time. - Adjusting to a longer feedback loop: a call can look wrong for a year before being right, and the discipline is distinguishing "wrong" from "early."

Common mistakes

- Assuming buy-side work is simply sell-side work without the writing — the sizing and portfolio-context dimensions are genuinely different analytical problems.
- Interviewing for buy-side roles with a **target-price** frame rather than a risk-reward-and-sizing frame.
- Underestimating how much buy-side value comes from **saying no** — most ideas examined are rejected.
- Believing the buy side reads sell-side ratings; they mostly read the analysis and use the access.
- Ignoring **liquidity and correlation** in a buy-side pitch.

Interview angle

"Why do you want to be on the buy side rather than the sell side?" A credible answer engages with the actual differences: you want to be measured on decisions rather than on distribution; you want the selectivity to go deep on a few ideas rather than maintaining continuous coverage obligations; and you want to think in position sizing and portfolio context rather than in ratings. If asked to pitch, structure it the buy-side way: the idea, the differentiated insight, the risk-reward with a quantified bear case, the position size you would recommend and why, the liquidity constraint, and what would make you exit. Ending on the exit condition is what marks the answer as buy-side thinking rather than sell-side thinking with different vocabulary.

Shorting and Bear-Thesis Construction

The Problem / Why this matters

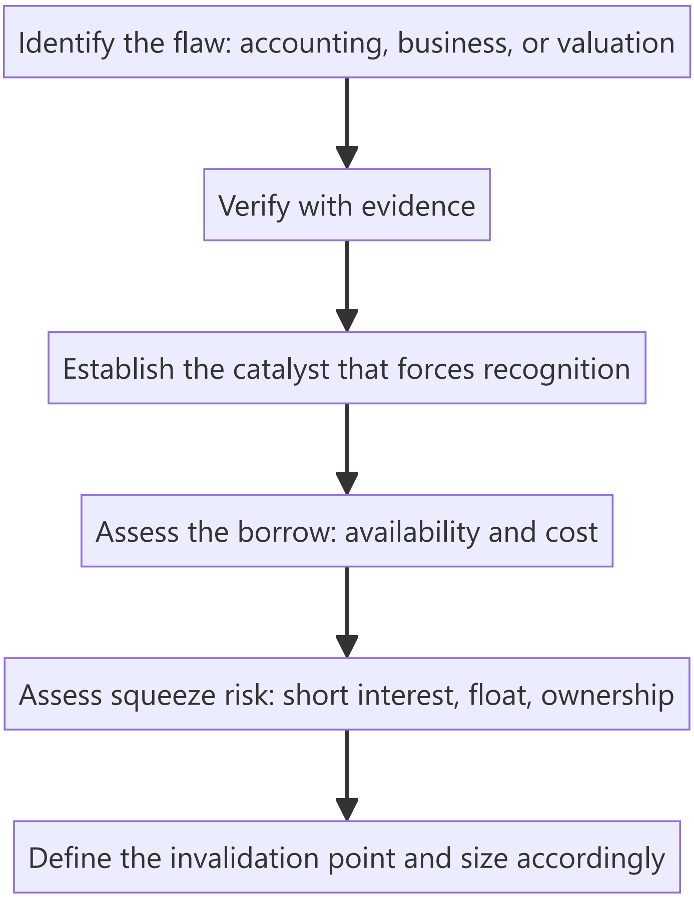
Almost all published equity research is bullish — typical sell-side rating distributions run heavily to Buy, with Sells often in low single-digit percentages. This creates a structural shortage of rigorous negative analysis, which is exactly why a well-constructed bear thesis is disproportionately valuable to the buy side. It is also genuinely harder: the asymmetry of short payoffs, the borrow constraint, and the career risk all work against it. Understanding how to build one is a differentiating skill even for an analyst who will never short.

Core Idea

A short thesis requires everything a long thesis requires **plus a catalyst and a timeline**, because the payoff structure is inverted — losses are theoretically unbounded, gains are capped at 100%, and time works against the position through borrow cost.

Why it works this way

A long position's worst case is losing the invested capital, and holding costs nothing. A short position's worst case is unbounded, and it costs borrow fees every day it is held. Being right eventually is sufficient for a long and insufficient for a short, which is why the analytical bar is higher.



Full technical content

The three families of short thesis

1. Accounting / fraud shorts — the reported numbers are not real. Built on the forensic-accounting toolkit: profit not converting to cash, receivables and inventory growth outrunning revenue, aggressive capitalisation, related-party flows, auditor resignations. These have the largest payoffs when correct and the highest evidentiary burden, and they typically require a catalyst — an auditor exit, a regulatory action, a short-seller report, a failed refinancing — to force recognition.

2. Business-deterioration shorts — the numbers are real but the business is structurally impaired. Technology substitution, moat erosion, a distribution channel being bypassed, a key patent expiring, permanent margin compression from a new low-cost entrant. These are the most analytically tractable and the most common productive short.

3. Valuation shorts — the business is fine but the price embeds impossible expectations. The most dangerous category, because an expensive stock can become far more expensive and there is no mechanism forcing correction. Valuation alone is rarely a sufficient short thesis; it needs a catalyst that changes the growth narrative.

The reverse-DCF — the analytical tool for valuation shorts

Instead of forecasting cash flows to derive a value, invert it: take the **current market price** and solve for the growth and margin the market must be assuming. Then assess whether those implied assumptions are achievable.

Example framing: "At ₹2,400, the market is implying 27% revenue CAGR for ten years with terminal EBIT margin of 24%. The company has never exceeded 17% margin, the addressable market implies a 55% share at that revenue level, and no company in this industry globally has sustained 27% for a decade."

That is a far stronger argument than "it trades at 68x, which is expensive," because it identifies the specific implied assumption that fails a plausibility test — and it is checkable by the reader.

The catalyst requirement

The single most common failure in short theses is being correct about the flaw and wrong about the timing. Acceptable catalysts:

CATALYST TYPE	EXAMPLES
Scheduled	Results, lock-in expiry, debt maturity, patent expiry, regulatory decision date
Financial	Refinancing requirement, covenant test, cash burn reaching a limit
Structural	Competitor capacity commissioning, technology launch, regulation taking effect
Disclosure	Auditor rotation, an accounting-standard change forcing disclosure, a mandated filing
Sentiment	Index exclusion, coverage initiation with a negative view, lock-in supply

A thesis with a **dated** catalyst is far more actionable than one relying on the market eventually noticing.

The mechanics that constrain the trade

Borrow availability and cost. A short requires borrowing the shares. In India this runs through the **Securities Lending and Borrowing (SLB)** mechanism, where availability is genuinely limited for many mid- and small-caps. Practical checks: is stock available to borrow, at what fee, and is the borrow **recallable**? A recall forces a buy-in at the worst possible moment.

Note the reflexive point: the stocks most attractive to short are often the ones hardest and most expensive to borrow, precisely because others have reached the same conclusion. High borrow cost is itself information — it tells you the trade is crowded.

Squeeze risk. Assess before entering: - **Short interest as % of free float** — high levels mean crowded positioning. - **Days-to-cover** (short interest ÷ average daily volume) — high days-to-cover means an exit stampede would be violent. - **Free float size** — a small float with concentrated promoter holding is squeeze-prone. - **Promoter/insider buying** — a promoter buying into a heavily shorted stock can force a squeeze.

Cost of carry — borrow fee plus any dividend the short must pay to the lender. A 12% annual borrow fee means the thesis must deliver more than 12% before it breaks even, and it means a thesis that takes three years to play out is uneconomic regardless of being correct.

In derivative markets, a short view can be expressed through single-stock futures or put options, which avoids the borrow problem but introduces expiry timing and, for options, time decay — trading one constraint for another.

Constructing the note

A rigorous bear thesis contains:

1. **The specific flaw**, stated plainly in the first line.
2. **Evidence** — this bar is higher than for a long thesis, because the claim is contrarian and the consequences of being wrong are asymmetric. Prefer primary and documentary evidence over inference.
3. **What the market believes and why it is wrong** — often via reverse-DCF for valuation shorts.
4. **The catalyst and its timing.**
5. **The bull case, presented fairly** — a bear thesis that does not engage seriously with the strongest counter-argument is not credible.
6. **Invalidation condition** — what specific evidence would make you cover. Stated in advance.
7. **Mechanics** — borrow availability and cost, squeeze risk, position size.

The asymmetry that governs sizing

Because losses are unbounded and gains capped, short positions are typically sized smaller than longs of equivalent conviction, with defined risk limits. Many practitioners set an explicit stop, not because the thesis has changed but because the position's risk profile requires it. The discipline that matters: **distinguish "the thesis is wrong" from "the position is too large"** — those require different responses, and confusing them is how short books blow up.

The professional and career dimension

Publishing a Sell has real consequences: loss of management access, hostility from the company, and occasionally legal pressure. This is precisely why so few exist, and why a well-evidenced Sell builds outsized credibility with institutional clients who are structurally starved of genuine negative research. Analysts who publish rigorous, fair Sells and are subsequently proven right build durable reputations from them.

Common mistakes

- Shorting on **valuation alone**, with no catalyst — expensive stocks routinely get more expensive.
- Correct thesis, **no timeline** — carry cost erodes the position while you wait.
- Not checking **borrow availability and cost** before committing to the idea.
- Ignoring **squeeze risk** in a crowded, high-days-to-cover name.

- Sizing a short like a long, despite the unbounded downside.
- Failing to state an **invalidation condition**, so the position is never re-examined.
- Not engaging with the strongest **bull argument**, making the thesis look like advocacy.
- Confusing "the market is irrational" with an investment thesis.

Interview angle

"Pitch me a short." Structure: state the specific flaw in one line — accounting, business deterioration, or expectations embedded in the price; give the evidence, ideally documentary or primary; if it is a valuation short, use a reverse-DCF to show precisely what the market must be assuming and why that is implausible; then the catalyst with timing, because a short without one is a carry-cost trap. Close on mechanics and risk: borrow availability and cost, short interest and days-to-cover for squeeze risk, the position size given unbounded downside, and the specific condition that would make you cover. Naming the invalidation condition unprompted is what marks a disciplined answer.

Behavioural Biases in the Analyst's Own Process

The Problem / Why this matters

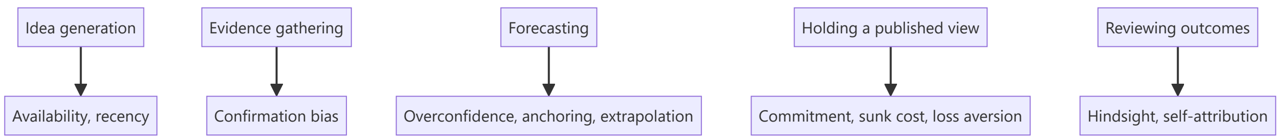
Behavioural finance is usually taught as an explanation for *market* mispricing — other people's irrationality creating opportunity. The more useful and less comfortable application is inward: the analyst is subject to the same biases, and an analyst who cannot identify their own is systematically less accurate. Interviewers ask about this because self-awareness about process failure is a genuine differentiator, and because a candidate who can describe a call they got wrong and *why* is demonstrating exactly the reflective capacity the job requires.

Core Idea

The analyst's process is vulnerable at specific, identifiable points — idea generation, evidence gathering, forecasting, and holding a published view. Each vulnerability has a known bias and a practical counter-discipline.

Why it works this way

Research is a sequence of judgements under uncertainty with delayed, noisy feedback — precisely the conditions under which cognitive biases operate most strongly and are hardest to detect. Markets punish these errors financially, but the feedback arrives slowly enough that the connection between the process failure and the outcome is easy to miss.



Full technical content

At idea generation

Availability and recency bias — ideas come disproportionately from what is salient: recent news, a sector that has just performed, a company that presented at a conference last week. The counter-discipline is a **systematic screen** run on defined criteria, so the idea funnel is not driven by attention.

Familiarity bias — over-covering what you already know and under-exploring unfamiliar sectors where mispricing may be greater precisely because fewer people have looked.

At evidence gathering — the dominant bias

Confirmation bias is the most consequential failure in research. Once a view forms — often within the first hours of looking at a company — subsequent evidence gets filtered: supporting data is accepted readily, contradicting data is scrutinised for flaws until a reason to discount it is found.

Practical counters: - **Write the bear case first**, before the bull case, when initiating on a company you like. - **Pre-commit to falsification conditions** — state in advance what evidence would change your mind, before you encounter it. This is the single most effective counter, because it removes the ability to redefine disconfirming evidence after the fact. - **Actively seek the strongest opposing view** — read the most credible bearish note on your bullish idea and engage with its best argument rather than its weakest. - **Red-team review** — have a colleague argue the opposite case internally.

Narrative bias — a compelling story about a company is more persuasive than it should be, and stories are easier to remember than base rates. The corrective is to check the story against the numbers: if the narrative says transformation, incremental RoCE should show it.

At forecasting

Overconfidence — forecast ranges are systematically too narrow. Analysts asked for 90% confidence intervals produce ranges containing the outcome far less than 90% of the time. The counter is to widen scenario ranges deliberately, and to build bear cases from coherent narratives rather than by mechanically shading the base case.

Anchoring — the first number encountered exerts disproportionate influence. Specific anchors in research: the current share price (which quietly shapes the target price), consensus (which makes a differentiated estimate feel risky), management guidance, and one's own previous forecast. The counter is to **build the forecast from drivers before looking at consensus or the current price**, then compare — rather than starting from consensus and adjusting.

Extrapolation — projecting recent trends indefinitely. This is the mechanism behind analysts being systematically late at cycle turning points in both directions. Counter with explicit **base rates**: how often do companies sustain 25%+ growth for five years? How frequently do margins at cyclical peaks persist? Historical frequency is a better starting point than the recent trend.

Conservatism / underreaction — revising estimates too slowly and too incrementally after genuinely new information, which is the documented mechanism behind post-earnings-announcement drift. When something materially changes, make the full revision at once rather than in cautious steps.

At holding a published view

Commitment and consistency bias — once a Buy is published with your name on it, admitting error is costly, so the incentive is to defend rather than to revise. This is the most professionally damaging bias because it is reinforced by external pressure.

Sunk-cost fallacy — six weeks of work on an initiation makes abandoning the thesis feel wasteful, even when the evidence has changed. The work is spent regardless; only the forward evidence matters.

Loss aversion — reluctance to downgrade a losing call and crystallise being wrong, versus willingness to take a small win by upgrading a stock that has performed.

Counters: **scheduled thesis reviews** independent of price moves, pre-committed invalidation conditions, and a culture where changing a view promptly is treated as competence rather than failure. An analyst who downgrades quickly on new evidence is more valuable than one whose ratings never change.

At reviewing outcomes

Hindsight bias — after the fact, the outcome feels as though it was predictable, which prevents genuine learning about what was actually knowable at the time.

Self-attribution bias — correct calls attributed to skill, incorrect ones to bad luck or unforeseeable events. This asymmetry prevents improvement entirely.

The single most effective counter across both: a **decision journal**. At the time of each recommendation, record the thesis, the key assumptions, the confidence level, the falsification conditions, and what you expected to happen. Reviewing this later — before looking at the outcome — gives an honest read on whether the reasoning was sound, separately from whether the outcome was good. Good process can produce bad outcomes and vice versa, and only a contemporaneous record lets you tell them apart.

Institutional biases worth naming

Beyond individual cognition, structural pressures push in predictable directions: - **Optimism bias in sell-side ratings**, from corporate access, banking relationships and management goodwill. - **Herding** — clustering near consensus because being wrong alone is more career-damaging than being wrong with everyone. - **Career risk asymmetry** — a missed opportunity costs less professionally than a visible loss, biasing toward caution on contrarian calls. - **Coverage pressure** — the obligation to publish continuously generates notes with no genuine view.

Naming these honestly is not cynicism; it is the precondition for resisting them.

Common mistakes

- Treating behavioural finance as something that only affects **other** market participants.
- Forming a view early and then gathering evidence — the sequence that guarantees confirmation bias.
- Building the model **after** looking at consensus and the current price, guaranteeing anchoring.
- Bear cases constructed by shading the base case rather than from a coherent alternative narrative.
- Defending a published view under evidentiary pressure because the rating carries your name.
- Reviewing calls by outcome rather than by process quality.
- No contemporaneous record, so every post-mortem is contaminated by hindsight.

Interview angle

"Tell me about a call you got wrong." The answer should demonstrate process reflection, not just narrate an outcome: state the thesis, what you assumed, what actually happened, and — critically — whether the **reasoning** was flawed or the reasoning was sound and the outcome adverse. Then name the specific bias if one was operating: did you anchor on consensus, extrapolate a trend past its base rate, or defend the view too long after contradicting evidence appeared? Finish with the process change you made. Candidates who can separate decision quality from outcome quality, and who describe a concrete counter-discipline like pre-committed falsification conditions or a decision journal, stand out sharply.

Dividend Policy and Shareholder Returns

The Problem / Why this matters

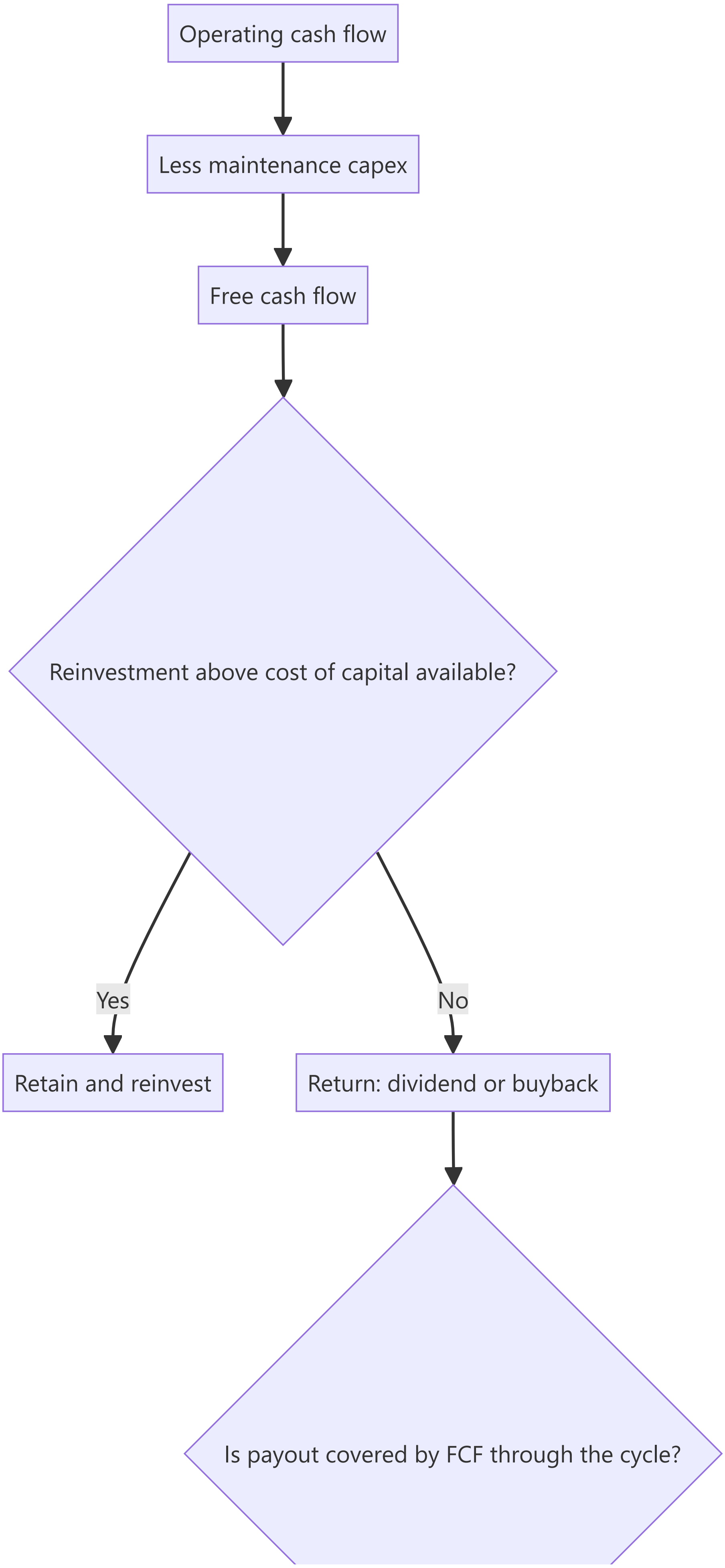
Dividend yield is one of the most widely quoted and least well-understood metrics in equity markets. A high yield can signal a mature, cash-generative, disciplined business — or a stock that has fallen sharply on a dividend the company can no longer afford. Distinguishing the two, and understanding how payout policy interacts with growth and valuation, is core to income-oriented equity research and to any long-horizon investment case.

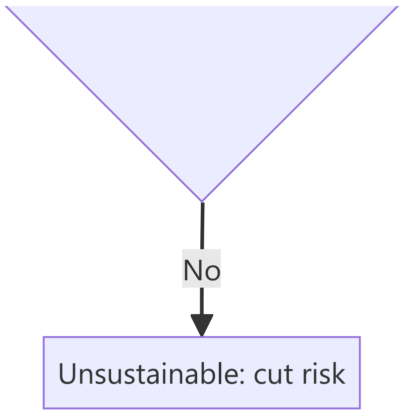
Core Idea

A company's payout policy is the visible output of its capital-allocation decision: what it cannot reinvest above the cost of capital, it should return. **Sustainability and coverage matter far more than the headline yield.**

Why it works this way

Dividends are paid from cash, not from reported profit. A payout policy is sustainable only if free cash flow supports it through the cycle, after funding the maintenance capex the business genuinely requires. A dividend funded by debt, by asset sales, or by underinvestment is a transfer of future value to present shareholders, not a return generated by the business.


Operating cash flow
Less maintenance capex
Free cash flow
Reinvestment above cost of capital available?
Yes
No
Retain and reinvest
Return: dividend or buyback
Is payout covered by FCF through the cycle?



Full technical content

The core metrics

METRIC	FORMULA	WHAT IT TELLS YOU
Dividend yield	DPS ÷ price	Current income return
Payout ratio	DPS ÷ EPS	Share of earnings distributed
FCF payout ratio	Total dividend ÷ FCF	**The sustainability test** — more reliable than the earnings-based ratio
Dividend cover	EPS ÷ DPS	Inverse of payout; below ~1.5 warrants scrutiny
Dividend CAGR	Growth in DPS over 5–10 yr	Policy consistency and management confidence
Total shareholder yield	(Dividends + net buybacks) ÷ market cap	The complete picture of cash returned

The FCF payout ratio is the metric that matters most and is the one most often skipped. A company paying out 60% of earnings sounds prudent; if it converts only half its earnings to free cash flow, that same dividend is consuming 120% of FCF and is being funded by the balance sheet.

Assessing sustainability — the checklist

1. **Is the dividend covered by free cash flow**, not just by earnings, averaged across a full cycle rather than in one favourable year?
2. **Is maintenance capex genuinely being funded?** A dividend sustained by deferring necessary asset replacement is borrowing from the future.
3. **What is the debt trajectory?** Rising debt alongside a maintained dividend is the clearest warning sign.
4. **How did the dividend behave in the last downturn?** Maintained, cut, or suspended? This is the single best evidence of both policy commitment and genuine capacity.
5. **Is there a stated policy?** A formal payout-ratio or minimum-DPS policy is more credible than ad hoc declarations.
6. **Cyclicality of earnings** — a cyclical business with a high payout ratio at the cycle peak has an inevitably unsustainable dividend.

The dividend-trap pattern

A high yield frequently reflects a falling price rather than a rising dividend — and the market is often correctly anticipating a cut. Diagnostic markers:

- Yield sharply above the company's own historical range and above sector peers.
- Payout ratio above ~80% of earnings, or above 100% of FCF.
- Deteriorating operating performance with the dividend held flat.
- Rising leverage.
- No stated policy, with the dividend maintained apparently to support the share price.

The analytical point: **a yield is only real if the dividend is paid.** Model the dividend forward from FCF rather than assuming the trailing DPS persists.

Dividends versus buybacks

Both return cash; they differ in mechanics and signalling:

	DIVIDENDS	BUYBACKS
Signal	Commitment — cutting is punished, so raising signals confidence	Flexible — can be paused without penalty
Value creation	Value-neutral in itself	Creates value **only if bought below intrinsic value**
Who benefits	All shareholders equally	Continuing shareholders, at the expense of sellers if underpriced
Effect on share count	None	Reduces it (permanently, in India, via mandatory extinguishment)
Best used when	Stable, predictable cash generation	Stock trades below intrinsic value; cash flow is lumpy

The buyback discipline test: examine the prices at which the company has historically repurchased. Buying back at valuation peaks — common, because that is when cash and confidence are highest — destroys value for continuing shareholders. Also check whether the buyback genuinely reduces share count or merely offsets ESOP issuance, in which case it is compensation expense routed through the buyback line.

The dividend-discount model

For stable, dividend-paying businesses, the DDM provides a valuation cross-check:

Value = $D_1 \div (Ke - g)$

Its practical limits: extremely sensitive to the spread between the cost of equity and growth (the same denominator problem as the perpetuity-growth terminal value), inapplicable to non-payers, and it values only distributed cash rather than the full cash-generating capacity. Most useful for utilities, mature financials and REITs; least useful for growth companies.

The **sustainable growth rate** relationship is the more broadly useful idea it embeds:

$g = RoE \times (1 - \text{payout ratio})$

This states formally that a company can only grow from retained earnings at the rate its returns and retention allow. A company forecasting 20% growth with a 15% RoE and a 50% payout is implicitly assuming external funding — which the model should then show explicitly. This is a fast, powerful consistency check on any forecast.

Policy interpretation as a signal

- **Initiating a dividend** typically signals a transition to maturity and confidence in sustainable cash generation.

- **Raising the payout ratio** can be positive (discipline, returning what cannot be reinvested well) or negative (an admission that growth opportunities have run out) — read it against the incremental-RoCE evidence.
- **Cutting** is severely punished, which is precisely why managements resist it and why a cut, when it comes, is credible information about genuine stress.
- **Special dividends** signal a one-off event rather than a change in ongoing capacity, and should not be capitalised into a forward yield.

The Indian context

Taxation of dividends in the hands of shareholders (rather than at the company level) changed the relative attractiveness of dividends versus buybacks for different investor classes, and payout behaviour shifted accordingly. Note also that **PSUs frequently carry elevated payout ratios** driven by government fiscal requirements rather than by an optimal capital-allocation judgement — a structural feature worth naming rather than reading as management discipline.

Common mistakes

- Screening on **dividend yield** without checking FCF coverage — the classic dividend trap.
- Using the **earnings** payout ratio rather than the FCF payout ratio for sustainability.
- Assuming the trailing dividend persists rather than modelling it forward from cash flow.
- Ignoring **maintenance capex** when assessing whether a payout is genuinely funded.
- Treating a buyback as automatically shareholder-friendly regardless of the price paid.
- Not checking whether buybacks merely offset **ESOP dilution**.
- Capitalising a **special dividend** into a forward yield.
- Applying the **DDM** to a company whose value comes from reinvestment rather than distribution.

Interview angle

"A stock yields 9% against a sector average of 2%. Interesting?" Treat it as a warning rather than an opportunity until proven otherwise: check whether the yield rose because the price fell; test the FCF payout ratio, not just the earnings payout; verify that maintenance capex is being funded and that debt is not rising to sustain the payment; check what happened to the dividend in the last downturn; and confirm whether a stated policy exists. If the dividend is genuinely covered by through-cycle free cash flow with stable leverage, it may be a genuine opportunity — otherwise the market is pricing an anticipated cut, and the yield is a number that will not be paid.

Thematic and Top-Down Research

The Problem / Why this matters

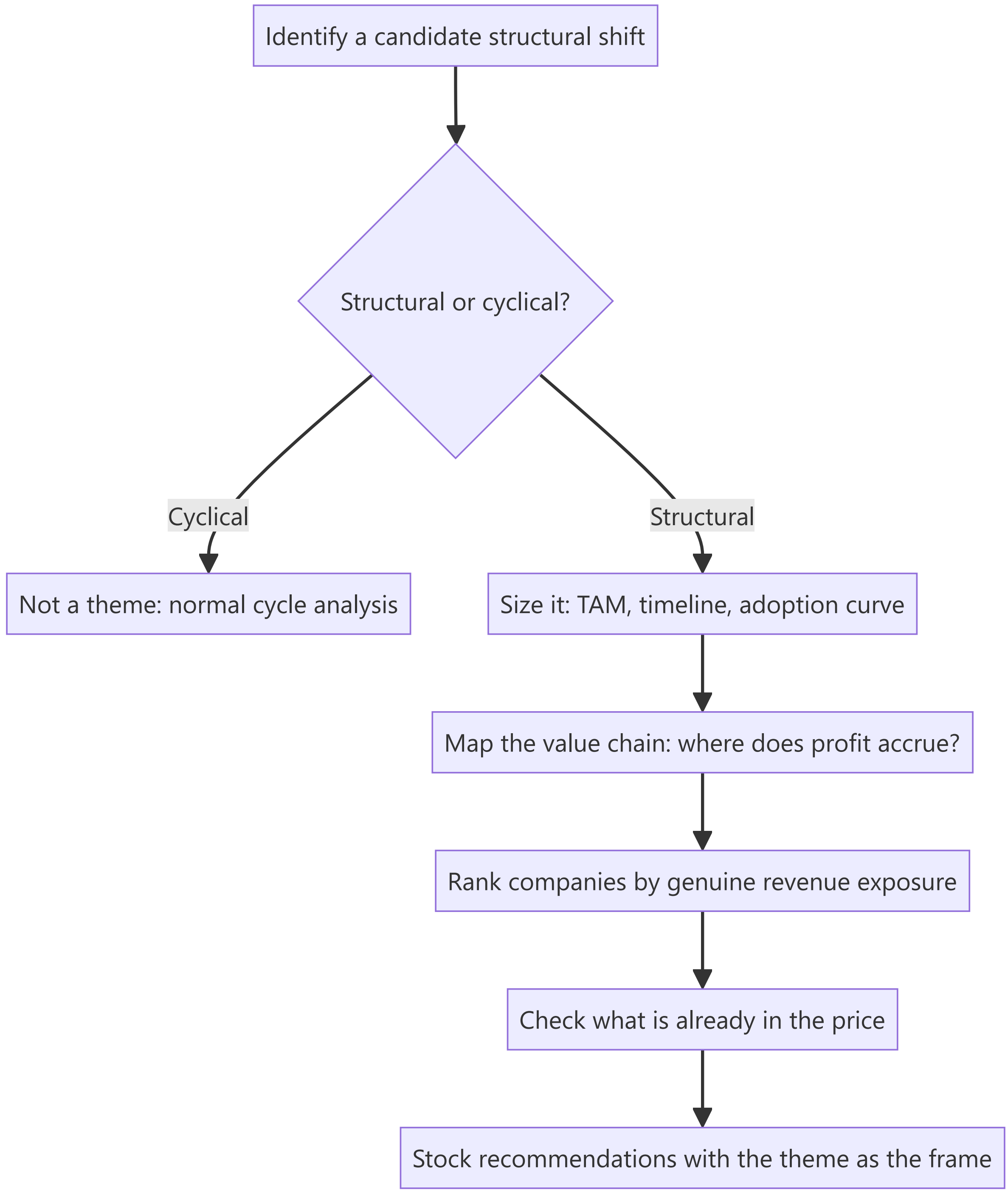
Single-stock research answers "is this company mispriced?" Thematic research answers a different and often more valuable question: "is this *shift* mispriced, and which companies are positioned for it?" Structural changes — energy transition, premiumisation, formalisation, import substitution, demographic shifts — play out over years and affect whole groups of companies simultaneously. A well-constructed thematic note can generate multiple actionable ideas from one body of work, which is why it is among the highest-leverage formats a research analyst produces.

Core Idea

Thematic research identifies a **structural change**, establishes that it is real and durable rather than cyclical, maps the **value chain** to find where the profit accrues, and then ranks companies by genuine exposure — distinguishing real beneficiaries from those merely associated with the narrative.

Why it works this way

Markets price cyclical changes reasonably efficiently because they are familiar and recur. Structural changes are harder: they unfold slowly, their magnitude is uncertain, and analysts anchored to historical patterns systematically underestimate them in the early stages and overestimate them once they become consensus. That asymmetry is the opportunity.



Full technical content

Step 1 — Distinguishing structural from cyclical

The most important and most frequently skipped test. A structural change alters the *level* or *nature* of demand permanently; a cyclical change alters its timing. Diagnostic questions:

- Is there an identifiable **driver that does not reverse**? Regulation, demographics, technology cost curves, income thresholds crossed.
- Has a similar transition **occurred in another market** already, giving an empirical template? (Cross-country analogues are the strongest available evidence — a category's penetration path in a comparable economy at a similar income level.)

- Does the change **survive a downturn**? Structural shifts continue through recessions; cyclical enthusiasm does not.
- Is the underlying **unit economics** changing, or only the volume?

Step 2 — Sizing and timing the theme

- **Total addressable market** and the realistic penetration path — build this bottom-up rather than citing an industry report's headline number.
- **The adoption curve.** Most structural shifts follow an S-curve: slow early adoption, rapid acceleration, then saturation. Identifying where on the curve a theme currently sits is the central analytical judgement, because returns are concentrated in the acceleration phase and the market usually prices the theme fully only after it has begun.
- **The timeline** — themes frequently take longer than expected to arrive and then move faster than expected once they do. Being early is functionally the same as being wrong for a position with a defined horizon.
- **The base rate check** — how quickly have comparable transitions actually occurred elsewhere? Analyst forecasts of adoption are systematically optimistic on timing.

Step 3 — Value-chain mapping — where the profit actually goes

This is where thematic research either creates or destroys value. A theme's existence does not tell you who profits from it. The disciplined approach maps the chain and asks, at each link, whether that participant has pricing power:

Example — an electrification theme: raw materials (lithium, copper) → cell manufacture → battery packs → vehicle assembly → charging infrastructure → power generation → grid equipment. Each link has different competitive intensity, different capital intensity, and different barriers. Historically in technology transitions, **assembly is often the most competitive and least profitable link**, while control of a scarce input or a proprietary component captures disproportionate profit.

The two questions at each link: 1. Is there a **structural barrier** at this link, or will competition compete returns away? (This is the moat framework applied to a value chain.) 2. Is the link's profit pool **growing faster than its capacity**? Capacity that arrives faster than demand destroys the economics regardless of theme strength.

The "picks and shovels" observation generalises this: in many transitions the enablers — the equipment, components or infrastructure everyone needs regardless of which end-player wins — capture more durable profit than the end-players competing with each other.

Step 4 — Ranking companies by genuine exposure

The most common failure in thematic investing is buying companies **associated with** a theme rather than **exposed to** it. Discipline:

TEST	QUESTION
Revenue exposure	What % of current revenue actually comes from the theme?
Forward exposure	What % of forecast incremental revenue?
Margin on theme revenue	Is the theme-related business as profitable as the base?
Competitive position within the theme	Is the company a leader or a marginal participant?
Investment required	What capex is needed, and does it earn above the cost of capital?

A company deriving 4% of revenue from a theme is not a theme play regardless of how often management mentions it on calls. Conversely, a company with 40% exposure whose management never uses the theme's vocabulary may be the better exposure precisely because it is less recognised.

Watch for narrative attachment — companies repositioning their communications to attach to a fashionable theme without changing the business. Compare the frequency of theme mentions on concalls against the actual segment revenue disclosure; a widening gap between the two is a real signal.

Step 5 — What is already priced in

A correct theme is not an investable idea if the market has already discounted it fully. Tests: - Have the identified beneficiaries **already re-rated** relative to their own history and to the market? - Use a **reverse-DCF** on the leading names: what growth does the current price imply, and is it plausible even if the theme plays out as expected? - Is the theme now **consensus**? Widely covered themes with universal analyst enthusiasm rarely offer attractive risk-reward at that stage.

The most valuable thematic work is typically done when a theme is **identifiable but not yet consensus** — which necessarily means it feels less certain at the time of writing.

Top-down versus bottom-up integration

Thematic research is top-down in origin but must terminate in bottom-up stock work. A theme note that identifies a genuine structural shift and then recommends companies without company-specific valuation, balance-sheet and governance analysis has done half the job. The failure mode is buying a poor company because it is in a good theme — a structurally attractive industry does not rescue a badly capitalised or badly governed participant.

Format

A thematic note typically runs 15–40 pages: the thesis and its evidence, sizing and timeline, value-chain map, the ranked beneficiary list with exposure quantified, what is priced in, risks to the theme itself, and specific stock recommendations with targets. Its distinguishing feature is that it should generate **several actionable ideas** from one research effort, which is what makes the format efficient.

Common mistakes

- Failing to test whether a shift is **structural or cyclical** — the foundational error.
- Sizing the theme from an **industry report headline** rather than a bottom-up build.
- Assuming theme growth translates to profit **without mapping the value chain**.
- Recommending companies **associated with** rather than **exposed to** the theme.
- Ignoring what is **already priced in** — being right about the theme and wrong about the entry.
- Underestimating the **timeline**, so a correct theme becomes a poor position over the relevant horizon.
- Abandoning company-level rigour because the theme is compelling.
- Ignoring that capacity often arrives faster than demand in a hot theme, destroying the economics.

Interview angle

"Pitch me a theme." Structure: name the structural shift and prove it is structural, not cyclical — the non-reversing driver, and ideally a cross-country analogue showing the same transition already completed elsewhere; size it bottom-up with an explicit view on where the adoption curve currently sits; map the value chain and identify **which link captures the profit and why** — usually where a barrier exists, often the enablers rather than the end-players; rank companies by actual revenue exposure rather than narrative association; and check what is already in the price using a reverse-DCF on the leaders. Naming a link that will *not* profit despite being in the theme is what demonstrates you have done the value-chain work rather than adopted a story.

Segment and Unit Economics Analysis

The Problem / Why this matters

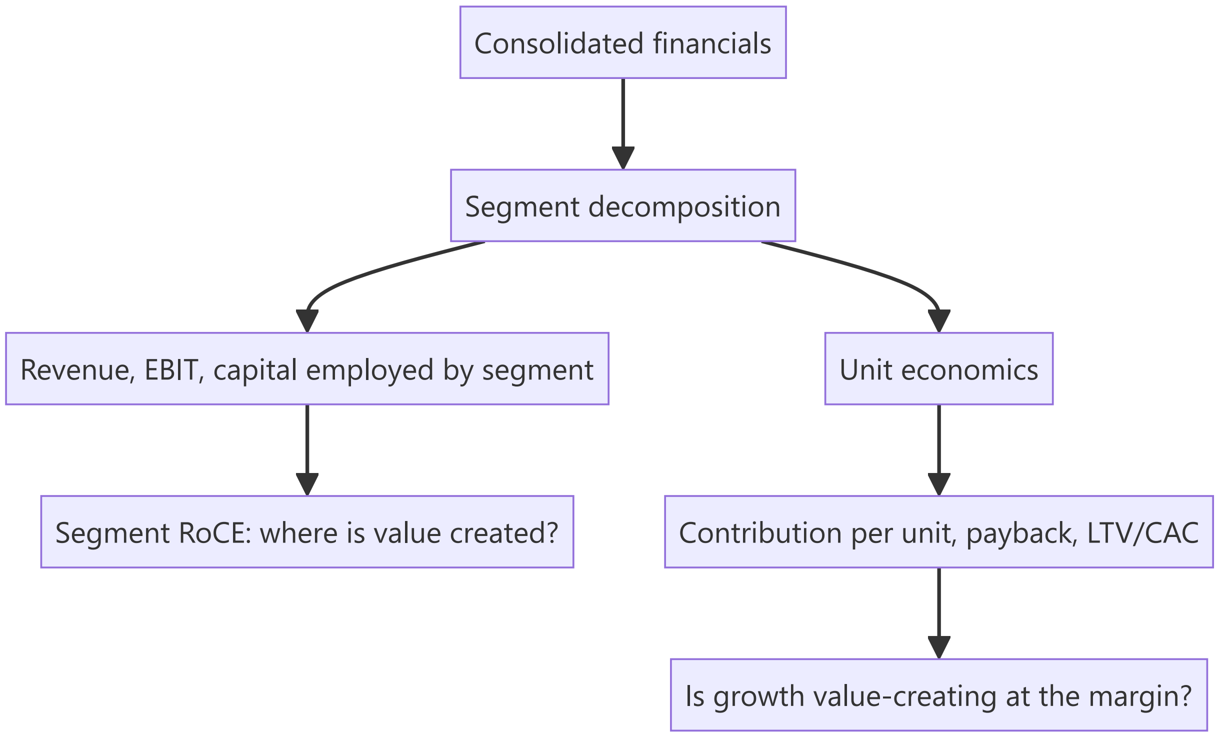
Consolidated financials average away the most important information in a multi-business company. A group reporting 12% EBIT margin might contain a 28%-margin core business subsidising a loss-making expansion — and the investment case depends entirely on which of those is growing. Similarly, a company's aggregate growth tells you nothing about whether it is acquiring customers profitably. Segment and unit-economics analysis is how an analyst gets underneath the consolidated numbers to the economics that actually drive value.

Core Idea

Decompose the business to the smallest level the disclosure supports — by segment, and where possible to a **repeatable unit** (a store, a customer, a plant, a subscriber) — because value is created or destroyed at the unit level, and consolidated numbers hide it.

Why it works this way

A company is a portfolio of activities with different economics. Aggregating them produces a number that describes none of them. Where a business is scaling, the consolidated picture is especially misleading: growth costs are expensed immediately while the returns accrue over years, so a business with excellent unit economics can report poor consolidated profitability precisely *because* it is growing well.



Full technical content

Segment analysis

What to extract. Indian accounting standards require segment disclosure of revenue, results and, usually, segment assets and liabilities. From that, build:

METRIC	DERIVATION	WHAT IT REVEALS
Segment revenue growth	YoY per segment	Where growth actually comes from
Segment EBIT margin	Segment result ÷ segment revenue	Which businesses are profitable
Segment RoCE	Segment EBIT ÷ segment capital employed	**Where value is created** — the key metric
Capital employed mix	Segment capital ÷ total	Where management is deploying money
Incremental capital by segment	Δ capital employed	Where management is *increasing* deployment

The critical cross-check: compare where capital is being deployed against where returns are highest. A company deploying most incremental capital into its lowest-RoCE segment is destroying value regardless of consolidated growth — and this is visible only through segment analysis. It is one of the most reliably valuable findings in fundamental research and one of the least frequently performed.

Practical complications: - **Unallocated corporate costs** sit outside segments; treat them explicitly (as in SOTP) rather than ignoring them. - **Inter-segment transfers** at internal prices can shift reported profit between segments; check the disclosure note. - **Segment definitions change**, breaking comparability. When a company re-segments, restate history where possible — and note that re-segmentation immediately before a weak period is worth a second look. - Some companies disclose **only revenue by segment**, not profit, which materially limits the analysis and is itself a disclosure-quality signal.

Unit economics

Where segment analysis decomposes by business line, unit economics decomposes to the repeatable transaction or asset. The relevant unit varies:

BUSINESS	UNIT	CORE METRICS
Retail / QSR	A store	Revenue per store, store-level EBITDA margin, capex per store, payback period, same-store sales growth
Subscription / fintech	A customer	CAC, ARPU, gross margin per user, churn, LTV, LTV/CAC, payback months
Manufacturing	A plant / tonne	Realisation per tonne, cost per tonne, capacity utilisation, EBITDA per tonne
Lending	A loan / borrower	Yield, cost of funds, credit cost, opex per account, RoA per product
Hotels	A room	ARR, occupancy, RevPAR, cost per available room
Airlines	A seat-kilometre	RASK, CASK, load factor, spread

The customer-economics chain

For subscription and consumer-internet businesses, the standard framework:

- **CAC (Customer Acquisition Cost)** = sales and marketing spend ÷ new customers acquired in the period.
- **Contribution margin per customer** = ARPU × gross margin %, less variable servicing cost.
- **Churn** — the annual or monthly rate at which customers leave; **customer lifetime ≈ 1 ÷ churn rate**.
- **LTV (Lifetime Value)** = contribution margin per period × lifetime, discounted.
- **LTV/CAC ratio** — a widely used benchmark, with roughly 3× often cited as a reasonable threshold, though the appropriate level varies by capital intensity and churn.
- **Payback period** = CAC ÷ contribution margin per month. Shorter payback means growth is self-funding sooner and the business needs less external capital.

Payback period is frequently more informative than LTV/CAC, because LTV depends on a churn assumption extrapolated over many years — the most uncertain input in the chain — whereas payback depends only on near-term, observable figures. A business claiming 5× LTV/CAC on a 40-month payback is making a much weaker claim than one showing 3× on an 11-month payback.

Cohort analysis — the honest version of unit economics

Aggregate metrics can improve simply because of mix shifts rather than genuine improvement. **Cohort analysis** — tracking each acquisition cohort separately over time — is the corrective, and it answers questions aggregates cannot:

- Is retention improving for **newer cohorts** versus older ones, or does the aggregate simply reflect a growing share of recent (not-yet-churned) customers?
- Does revenue per customer **expand** within a cohort over time (a strong signal) or decay?
- Is CAC rising as the company scales beyond its most accessible customer segment — the near-universal pattern, and one that aggregate CAC masks during rapid growth?

The last point matters enormously in practice: a fast-growing company's blended CAC is dominated by its cheapest early customers, and the *marginal* CAC of the newest cohort is what determines whether continued growth remains economic.

Same-store / like-for-like metrics

For any business adding capacity, separating growth from **new units** and growth from **existing units** is essential. Same-store sales growth (SSSG) isolates the underlying business's health from expansion. A company reporting 24% revenue growth of which 22 points come from new stores and 2 from SSSG is a very different proposition from the reverse — the first is buying growth with capital, the second is generating it.

The paired check: **new-store payback**. Expansion creates value only if new units earn above the cost of capital, so revenue growth from expansion must be assessed against capex per unit and time to maturity.

Connecting to valuation

Unit economics enter valuation directly: - A business with proven unit economics but consolidated losses due to growth investment can be valued on **mature-state unit economics** applied to a forecast unit count — the standard approach for scaling businesses, though it depends entirely on the maturity assumption being defensible from cohort evidence. - Segment RoCE justifies **different multiples for different segments**, which is the analytical basis for SOTP. - Deteriorating marginal unit economics is one of the earliest warnings that a growth story is ending, typically visible in cohort data well before it appears in consolidated results.

Common mistakes

- Analysing only **consolidated** figures for a genuinely multi-business company.
- Not computing **segment RoCE**, and therefore missing capital being deployed into the lowest-return business.
- Using **blended CAC** for a fast-growing company, masking rising marginal acquisition cost.
- Relying on **LTV/CAC** where LTV rests on an unvalidated long-horizon churn assumption; payback is more robust.
- Reading aggregate retention improvement that is actually a **cohort-mix artefact**.
- Ignoring **SSSG**, treating expansion-driven growth as organic strength.
- Missing **segment re-definition** that breaks historical comparability.
- Valuing a loss-making scaling business on mature unit economics without cohort evidence that maturity is achievable.

Interview angle

"A company is growing revenue 30% but losing money. How do you decide whether it's a good business?" Go to unit economics: is the *unit* profitable — contribution margin per customer or per store positive after variable costs? What is the CAC payback period, and is marginal CAC for the newest cohort rising or stable? Does cohort data show retention holding and revenue expanding within cohorts? Then separate growth investment (expensed now, returns later) from genuine operating losses, and check segment RoCE to see whether incremental capital is going into the highest-return part of the business. The conclusion follows: profitable units plus a reasonable payback plus stable marginal CAC means losses are growth investment; deteriorating marginal economics means they are not.

The Analyst Workflow and Coverage Management

The Problem / Why this matters

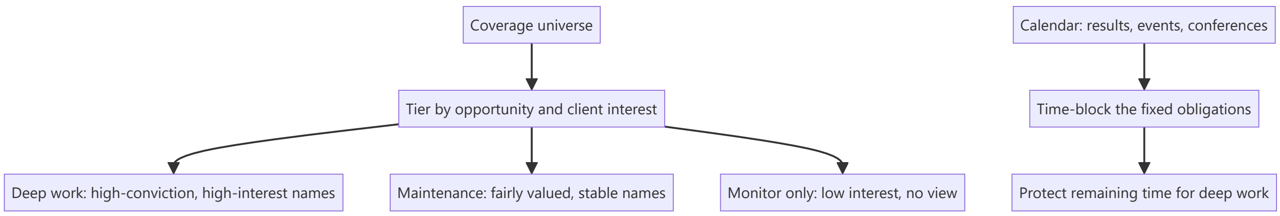
An equity research analyst typically covers 10–25 companies while fielding client calls, attending management meetings, updating models through earnings season, and producing thematic work — all with hard, externally-imposed deadlines. Analytical ability is necessary but not sufficient; analysts fail far more often from workflow collapse than from inability to value a company. How an analyst organises coverage, prioritises attention and maintains models is a genuine professional skill, and it is what makes the difference between covering 12 names well and covering 20 badly.

Core Idea

Coverage is a **portfolio of attention**. Not every stock deserves equal effort, and the analyst's core management decision is continuously reallocating time toward where the marginal research hour produces the most client value — which is rarely spread evenly.

Why it works this way

Research effort has sharply diminishing returns per name. The eleventh hour spent on a well-understood, fairly valued large-cap produces almost nothing; the first hour on an under-covered name where something has just changed can produce a genuine idea. Since total hours are fixed, allocation is the highest-leverage decision an analyst makes.



Full technical content

Tiering the coverage universe

Not a formal industry practice, but a discipline good analysts apply informally:

TIER	CHARACTERISTICS	EFFORT
Active ideas	High conviction, differentiated view, significant client interest	Deep, continuous — primary research, frequent updates
Core coverage	Large, widely held, must maintain a credible view	Full model maintenance, results updates, but limited primary work
Maintenance	Fairly valued, no differentiated view currently	Model updated at results; monitor for change
Watch	Not formally covered but relevant to the sector	Track key data; initiate if an inflection appears

The judgement is recognising when a name should **move tiers** — an inflection in a maintenance name deserves promotion to deep work, and an active idea whose thesis has played out should be demoted rather than defended.

The annual rhythm

Indian equity research runs on a predictable quarterly cycle, and the calendar largely determines workload:

PERIOD	DOMINANT ACTIVITY
Results season (roughly 4–6 weeks per quarter)	Previews, live result reactions, concalls, updates, revised estimates
Post-results	Management meetings, channel checks, deeper thematic work
Pre-results	Preview notes, consensus checks, positioning assessment
Conference season	Corporate access events, group meetings, client interaction
Annual report season	The deepest single source — full notes, related-party disclosures, accounting policy

The annual report deserves specific mention. It is the most information-dense document a covered company produces, and it arrives when analysts are least busy. Reading it properly — notes, related-party transactions, contingent liabilities, accounting policy changes, auditor's report, management discussion — is where most durable analytical edge on a covered name is actually built.

Model maintenance discipline

Models decay. Practical hygiene:

- **Update immediately after results**, while the detail is fresh. Deferred updates accumulate and become error-prone.
- **Keep one master file per company**, versioned by date, with a clear changelog of what was revised and why.
- **Maintain an assumptions log** — when you change a forecast driver, record the reason. Six months later, "why did I assume 14% margin?" is a question you must be able to answer.
- **Reconcile to reported figures** every quarter — actual versus your estimate, line by line. This is how forecasting accuracy improves; without it you never learn which lines you systematically get wrong.
- **Re-check the balance and sanity checks** after every update, since edits break links.

Information intake

The daily-to-weekly routine that keeps coverage current:

- **Exchange filings** for covered names — the primary source, and the only one that is both complete and timely.
- **Concall transcripts**, including those of competitors, customers and suppliers, which frequently contain more about your company than its own call does.
- **Sector and channel data** — monthly volumes, price data, government datasets.
- **Competitor research**, to know what consensus believes.
- **News and regulatory announcements.**

The discipline is having a **defined intake routine** rather than reacting to whatever surfaces, so that a material filing on a maintenance-tier name does not go unnoticed for a week.

Client interaction

For sell-side analysts, client-facing work is a substantial share of time and the primary channel through which research is valued: - **Inbound calls** — reactive, unpredictable, and the main mechanism by which clients form a view of an analyst's depth. - **Marketing** — roadshows and calls presenting recent work. - **Corporate access** — arranging and hosting management meetings, one of the most valued services. - **Bespoke work** — a specific model or analysis for a large client.

The practical tension: client work is externally scheduled and urgent, deep research is internally scheduled and important. Without deliberately **protecting blocks of time** for the latter, it is systematically crowded out — which is the most common way an analyst's work quality declines without any single visible failure.

The information-management problem

Over years of covering a company, an analyst accumulates concall notes, meeting notes, channel-check findings, and observations that will matter later but are impossible to recall on demand. A simple **per-company running file** — dated entries of what management said, what checks found, what changed — compounds enormously in value. Its highest use is checking what management said two years ago against what happened, which is the raw material for the guidance-accuracy assessment in management-quality work.

Judging your own accuracy

Few analysts systematically track their own record; those who do improve faster. Worth maintaining: - **Estimate accuracy** — your forecast versus actual, by line item, over time. Most analysts have systematic, correctable biases (typically over-forecasting revenue growth and under-forecasting cost inflation). - **Recommendation performance** — absolute and relative to benchmark and sector, over the stated horizon. - **Decision-quality review** — as covered in the behavioural material, assessing whether the reasoning was sound separately from whether the outcome was favourable.

Common mistakes

- Spreading effort **evenly** across coverage rather than allocating to where marginal research value is highest.
- Deferring model updates through results season until the detail is no longer fresh.
- No **assumptions log**, so past forecast decisions cannot be reconstructed or learned from.
- Letting client work crowd out **protected deep-research time**.
- Not reading the **annual report** properly because results-season material feels more urgent.
- No systematic **intake routine**, so filings on lower-tier names are missed.
- Never reconciling estimates to actuals, so systematic forecasting biases persist uncorrected.
- Defending tier assignment — keeping an idea "active" after its thesis has played out.

Interview angle

"How would you manage covering 15 companies?" Show that you understand it as an allocation problem rather than a scheduling one: tier the universe by where a differentiated view and client interest actually exist, giving deep primary work to a few names and disciplined maintenance to the rest; build the calendar around the fixed quarterly rhythm and protect blocks for deep work so client demands do not consume everything; maintain models immediately post-results with a logged assumptions trail; keep a running per-company file so historical management commentary is retrievable; and track your own estimate accuracy to find systematic biases. The point that lands well is recognising that the marginal research hour is worth far more on some names than others, and that reallocating it deliberately is the job.